

Stat 437 Project 1

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General rule and information

You must show your work in order to get points. Please prepare your report according to the rubrics on projects that are given in the syllabus. In particular, please note that you need to submit codes that would have been used for your data analysis. Your report can be in .doc, .docx, .html or .pdf format.

The project will assess your skills in K-means clustering, Hierarchical clustering, Nearest-neighbor classifier, and discriminant analysis for classification, for which visualization techniques you have learnt will be used to illustrate your findings.

Data set and its description

Please download the data set “TCGA-PANCAN-HiSeq-801x20531.tar.gz” from the website <https://archive.ics.uci.edu/ml/machine-learning-databases/00401/>. A brief description of the data set is given at <https://archive.ics.uci.edu/ml/datasets/gene+expression+cancer+RNA-Seq>.

You need to decompress the data file since it is a .tar.gz file. Once uncompressed, the data files are “labels.csv” that contains the cancer type for each sample, and “data.csv” that contains the “gene expression profile” (i.e., expression measurements of a set of genes) for each sample. Here each sample is for a subject and is stored in a row of “data.csv”. In fact, the data set contains the gene expression profiles for 801 subjects, each with a cancer type, where each gene expression profile contains the gene expressions for the same set of 20531 genes. The cancer types are: “BRCA”, “KIRC”, “COAD”, “LUAD” and “PRAD”. In both files “labels.csv” and “data.csv”, each row name records which sample a label or observation is for.

Task A. Clustering

For this task, you need to apply k-means and hierarchical clustering to cluster observations into their associated cancer types, and report your findings scientifically and professionally. Your laptop may not have sufficient computational power to implement k-means and hierarchical clustering on the whole data set, and genes whose expressions are zero for most of the subjects may not be so informative of a cancer type.

Please use `set.seed(123)` for random sampling via the command `sample`, random initialization of `kmeans`, implementing the gap statistic, and any other process where artificial randomization is needed.

(Task A1) Complete the following data processing steps:

- Filter out genes (from “data.csv”) whose expressions are zero for at least 300 subjects, and save the filtered data as R object “gexp2”.
- Use the command `sample` to randomly select 1000 genes and their expressions from “gexp2”, and save the resulting data as R object “gexp3”.
- Use the command `sample` to randomly select 30 samples and their labels from the file “labels.csv”, and save them as R object “labels1”. For these samples, select the corresponding samples from “gexp3” and save them as R object “gexpProj1”.
- Use the command `scale` to standard the gene expressions for each gene in “gexpProj1”, so that they have sample standard deviation 1. Save the standardized data as R object “stdgexpProj1”.

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
set.seed(123)  
gene_data = read.csv("data.csv", row.names = 1)  
gene_labels = as.data.frame(read.csv("labels.csv", row.names = 1))  
# filter genes who's expression are zero for at least 300  
# subjects (aka columns with more than 300 rows containing  
# 0)  
gexp2 = gene_data %>%  
  select_if(~sum(. == 0) <= 300)  
  
# sample 1000 genes and their expressions  
random_columns = sample(colnames(gexp2), 1000)  
gexp3 = gexp2 %>%  
  select(all_of(random_columns))  
  
# Sample 30 samples and their labels from the file  
# 'labels.csv'  
sample_indices = sample(1:dim(gexp3)[1], size = 30, replace = FALSE)  
gexpProj1 = gexp3[sample_indices, ]
```

```

gexpProj1 <- gexpProj1[order(rownames(gexpProj1)), ]
labels1 = gene_labels %>%
  filter(rownames(gene_labels) %in% rownames(gexpProj1))
print(gexpProj1)

```

```

##           gene_3429 gene_2105 gene_3873 gene_13883 gene_5487 gene_7795
## sample_119  9.810764 8.4846497 1.7559131 11.622038 11.49999 10.498021
## sample_275  9.617372 8.9457657 0.0000000 10.856208 11.82613  9.915779
## sample_29   11.985465 6.4745885 0.3952822 10.871174 11.96451 10.877706
## sample_295  10.439374 0.5506052 1.7327902 10.926244 11.85375 10.532999
## sample_298  9.394141 8.1027369 2.4089020 10.384050 11.29921 10.287077
## sample_404  8.597837 9.3833025 0.5285713 11.276502 11.11790 10.043602
## sample_464  9.718839 8.7709579 3.7480747 10.475571 12.10270 10.787984
## sample_468  9.538612 0.4978423 3.4449056 11.089053 11.56450  9.735292
## sample_531  9.505522 6.1750175 2.1979912 10.063152 11.64510 10.832400
## sample_57   9.028597 7.7503452 3.2629292  9.645181 11.31422 10.625070
## sample_578  6.921448 9.0573447 1.5483873 10.074784 10.89860 10.009605
## sample_579  9.174866 8.9842504 0.4596414 11.130751 10.95723 10.503488
## sample_603  9.372395 8.3302357 0.0000000 10.852022 11.95408 10.186931
## sample_627  8.647660 8.4910233 0.0000000 10.758223 10.78948 10.379617
## sample_638  6.437913 7.4915645 0.6050202 11.567756 11.31801 11.203201
## sample_673  8.575766 8.8804406 0.9656181 10.456529 11.21224 10.732625
## sample_68   9.244497 1.7029235 1.0893626 10.606655 11.56306 10.129502
## sample_69   9.427246 8.6558920 3.2614556  9.844903 11.37576 10.308032
## sample_691 10.566530 7.7071589 3.4360552 10.562061 11.21877 10.424460
## sample_701  7.485725 8.8517678 0.0000000 10.825412 11.63080 10.642178
## sample_735  9.178120 8.7790660 1.2250281  9.844729 11.77412  9.593989
## sample_74   9.683770 1.0453032 3.8643149 10.019910 11.85881 11.940604
## sample_740  9.357191 8.5906469 1.7714631 10.715224 11.17891 10.474913
## sample_75   8.135843 8.3569537 1.7006173 10.466648 11.29383 10.363850
## sample_762  7.176831 8.8952754 2.0293115 10.887929 11.49772 10.729340
## sample_768  8.141080 7.7316041 0.7163325 11.316706 11.31988 10.987676
## sample_793  8.100452 7.8228512 0.7893537 11.785211 11.64619 10.872113
## sample_800  7.652236 9.8712168 2.6869039 12.469151 11.17528 10.467493
## sample_82   9.314098 7.2409630 1.0259501 11.390760 11.17516 10.742764
## sample_95   8.274061 9.4710309 0.0000000 11.321426 11.38654  9.908480
##           gene_19324 gene_3190 gene_15034 gene_10795 gene_11905 gene_16177
## sample_119  10.906793 4.839829  8.209926  7.910037  0.3754009  5.9102098
## sample_275  10.699225 5.521453  8.309199 10.284304  2.1720714  5.6296364
## sample_29   9.438865 4.217843  6.199409  9.471895  3.7992428  0.9601410
## sample_295  11.244364 6.988048  8.861934  8.343950  0.0000000  6.8897047
## sample_298  12.846926 6.439450  7.875123  8.737298  0.0000000  2.3158850
## sample_404  10.600220 5.217882  8.392128  9.891958  4.6631340  7.5215455
## sample_464  11.159979 3.482229  7.390264  8.021829  6.8047377  6.7678004
## sample_468  11.125433 7.827190  8.622484  7.436687  0.4978423  8.4931670
## sample_531  10.824259 2.027048  7.410799  6.918315  6.1953456  3.5564417
## sample_57   10.544298 3.359071  7.696154  8.862213  6.3047885  6.7392811

```

## sample_578	12.981071	4.233090	8.474387	10.551247	0.0000000	0.9726927
## sample_579	10.807049	5.582083	7.799372	9.368589	2.0006851	6.9938328
## sample_603	11.220904	3.621454	8.712097	8.867520	0.0000000	2.0635720
## sample_627	11.148712	6.328190	8.950311	9.242445	0.8033924	6.0729397
## sample_638	10.185371	5.547354	9.913389	8.929605	0.0000000	5.2962261
## sample_673	12.061216	7.725046	8.379820	9.269316	0.5621814	2.8468343
## sample_68	11.093061	6.493418	8.152077	6.801236	0.4603756	8.4630339
## sample_69	11.094606	5.137868	7.999848	10.172753	0.0000000	1.7050490
## sample_691	12.844053	5.352409	8.857396	10.051671	1.6983518	3.5784316
## sample_701	11.470476	5.756724	8.523040	10.290595	0.0000000	5.0774225
## sample_735	12.520601	3.800185	7.922840	10.870303	1.2250281	3.9324581
## sample_74	10.451963	3.941266	7.238405	7.357833	3.7261773	1.9728764
## sample_740	10.987612	6.103882	7.494464	10.364901	0.0000000	2.4549918
## sample_75	12.415818	6.815755	7.634811	10.154767	2.3039561	0.6439485
## sample_762	12.007770	5.285850	8.367489	9.341510	0.0000000	7.6311916
## sample_768	10.826405	5.999554	9.150029	8.925201	0.0000000	3.7928241
## sample_793	12.182205	5.093353	8.512409	6.872447	0.0000000	6.9928364
## sample_800	10.781294	4.043371	8.478131	8.502677	7.1745363	6.0052068
## sample_82	11.345511	4.831006	8.755268	8.167649	1.3530409	4.8825990
## sample_95	10.439987	5.502439	8.094305	10.480608	6.3727467	7.6339978
##	gene_3329	gene_7114	gene_11268	gene_14877	gene_1825	gene_20199
## sample_119	6.122006	3.9321084	8.1356477	6.240150	7.560837	8.692717
## sample_275	5.448409	4.0175478	5.8057722	6.845089	8.788757	8.575460
## sample_29	2.738314	7.3261598	0.7051375	6.656861	10.294575	8.421846
## sample_295	6.355164	4.1854939	5.9185413	7.846217	7.992921	8.366471
## sample_298	5.665020	1.2180992	5.5675120	6.829609	8.458714	8.474939
## sample_404	5.427452	6.8803672	4.6118393	6.410592	8.489475	8.566861
## sample_464	4.377485	1.4307670	6.8478591	6.760287	9.193552	8.204331
## sample_468	6.167452	3.8677384	4.8590253	7.084787	8.223659	8.893936
## sample_531	5.693479	1.0180644	4.9416791	5.574365	8.605731	8.613307
## sample_57	5.543573	1.5774893	6.5071921	6.230478	8.597334	9.017591
## sample_578	4.923867	13.0935249	3.4732544	7.526296	8.394115	7.854931
## sample_579	3.931579	2.3582837	3.8588165	5.659097	7.854208	8.671555
## sample_603	3.814981	6.2989973	0.0000000	4.685660	6.386906	8.085717
## sample_627	5.078392	6.0966983	3.7726986	6.962271	7.054121	8.306399
## sample_638	5.207471	5.4926448	7.9092991	7.089583	8.082894	8.704875
## sample_673	6.501074	4.9986829	7.3779653	8.342190	8.865780	8.327288
## sample_68	6.444438	3.3251849	2.8857696	9.738106	8.414584	8.405014
## sample_69	6.274288	6.5703491	3.0414708	8.320647	8.201634	8.539768
## sample_691	6.652990	1.4243847	5.9778844	8.871298	8.293049	8.900399
## sample_701	5.209239	6.1432628	7.1612024	7.302858	8.462044	8.226749
## sample_735	7.860541	7.3940768	5.3304688	7.874317	8.851877	8.116063
## sample_74	6.746622	7.0192905	0.8463128	4.919116	8.755278	8.666469
## sample_740	5.244115	1.2505978	3.5040737	7.261409	7.211927	7.513649
## sample_75	6.826777	0.6439485	0.6439485	7.102532	8.965017	8.220756
## sample_762	6.660595	5.4643023	8.1885443	8.766761	8.383112	8.321897
## sample_768	6.434908	5.5035364	5.2830328	6.955720	8.271412	8.534171
## sample_793	4.860009	2.7711884	0.0000000	5.267128	8.254759	8.360847

## sample_800	7.104976	4.3727641	2.6869039	6.191211	8.494540	7.846299
## sample_82	5.298083	7.8161261	4.5410873	5.199810	8.320449	8.626895
## sample_95	6.667069	3.6485690	1.8174595	5.910423	8.113216	8.582300
##	gene_17079	gene_18234	gene_10955	gene_11668	gene_10673	gene_18337
## sample_119	0.9195308	7.971118	3.5111138	2.5247658	10.912365	4.995774
## sample_275	1.9716999	7.081340	5.5418679	4.1180856	10.816080	5.595166
## sample_29	1.1767057	7.534248	6.6747713	1.1767057	10.655280	6.722917
## sample_295	0.9482270	8.352180	2.4975868	2.9947788	9.294465	5.339946
## sample_298	4.7115500	8.811577	4.6172456	0.9962441	9.795509	4.495279
## sample_404	0.0000000	7.383091	5.9453151	4.4164344	11.304026	4.760551
## sample_464	3.8646319	7.845132	5.9421907	2.9604190	10.552525	5.004097
## sample_468	0.4978423	8.059550	0.8671848	3.9850176	9.997927	5.068404
## sample_531	3.4248952	8.318945	6.7221244	3.5564417	10.883010	4.735046
## sample_57	3.5340367	7.586232	7.0868581	3.2629292	10.203470	4.847115
## sample_578	2.1271710	8.171407	4.4067883	0.0000000	10.215727	6.827921
## sample_579	1.0879380	6.578321	2.5553254	2.3582837	10.539207	4.989557
## sample_603	2.1806571	6.678255	4.3277808	7.5648278	11.163165	6.198459
## sample_627	1.5174773	8.074832	3.0141767	3.4177850	10.306381	4.945154
## sample_638	2.2949883	8.154550	5.0687869	3.8625605	10.349337	5.276969
## sample_673	1.7580473	8.829976	4.1043450	2.1161318	10.677024	4.815524
## sample_68	1.7029235	8.148258	4.2795530	3.9691689	9.844582	5.074968
## sample_69	2.7817793	7.968943	3.9510749	1.8923910	9.882010	5.799647
## sample_691	3.6446869	8.926432	5.2282838	2.3013828	10.095265	5.420169
## sample_701	1.9141052	7.307064	1.9141052	4.8880613	10.739620	5.227121
## sample_735	2.8564479	7.286946	4.4269692	4.7945666	10.333703	6.036785
## sample_74	2.3937731	7.842080	6.1213677	0.8463128	10.041194	6.454539
## sample_740	1.9103105	7.218626	3.5892122	2.6259484	10.533826	6.534995
## sample_75	3.7110103	7.750513	3.5119613	1.0875986	10.327676	5.788939
## sample_762	2.8407855	7.921413	4.8449129	2.6174866	10.669629	6.049563
## sample_768	2.1815479	8.305528	4.5349891	5.0368418	10.475977	5.352794
## sample_793	1.4961044	9.910490	2.9853547	4.5141223	9.938067	4.823367
## sample_800	2.9633076	7.160890	3.7266460	3.5702808	10.838660	4.324623
## sample_82	0.6023619	7.768925	4.2582023	5.5895568	11.173971	5.805437
## sample_95	0.7834986	6.958181	3.5183967	3.1593708	10.869301	5.079779
##	gene_3450	gene_9209	gene_4610	gene_16022	gene_9764	gene_19849
## sample_119	10.301748	0.0000000	8.629167	4.406537	4.465948	0.0000000
## sample_275	10.349093	0.6640277	10.475288	4.165848	6.507887	1.7388975
## sample_29	9.642590	0.3952822	8.732011	2.868529	4.921712	6.5541590
## sample_295	4.942195	0.0000000	10.071382	5.827021	4.985168	0.9482270
## sample_298	12.502305	6.3962571	9.345605	6.121836	3.765588	1.2180992
## sample_404	10.151372	1.8698319	9.642467	4.446025	5.803932	1.6836515
## sample_464	10.238954	1.4307670	8.992080	4.023513	2.134780	0.0000000
## sample_468	3.783907	0.0000000	9.735989	5.584253	5.282099	0.4978423
## sample_531	10.052745	2.3507801	9.014612	6.068851	1.018064	0.0000000
## sample_57	11.211341	1.5774893	9.147793	6.230478	2.653496	1.2159892
## sample_578	13.459278	2.7601784	9.580594	4.647407	6.363315	0.0000000
## sample_579	10.542848	1.5241640	10.569096	3.023167	5.821065	1.3224474
## sample_603	8.100652	1.4682184	7.929294	5.004295	5.605945	0.4365883

## sample_627	8.954650	2.7199938	9.025469	2.946057	4.498857	2.6360097
## sample_638	9.432684	4.6114734	9.450521	4.078149	4.999986	1.2032012
## sample_673	9.333587	3.5800128	9.180886	2.116132	5.531849	6.8258874
## sample_68	4.511247	0.0000000	9.190958	4.940129	4.101432	0.8089204
## sample_69	12.279497	2.8736360	10.029701	5.333814	5.501560	0.0000000
## sample_691	10.970746	5.3324466	11.541615	5.971445	5.938813	0.6427478
## sample_701	5.817012	3.9788363	8.954647	4.721411	5.154215	0.0000000
## sample_735	10.951802	0.9197596	9.625278	6.254630	6.167822	0.5319678
## sample_74	9.776976	0.0000000	8.443254	5.063796	4.544757	0.8463128
## sample_740	8.946778	0.0000000	9.114135	5.106482	6.070660	0.4274989
## sample_75	12.556453	5.9220622	9.704237	5.284466	5.176391	0.3576083
## sample_762	10.161724	6.2768279	9.689039	4.818897	5.577266	0.0000000
## sample_768	8.848795	0.7163325	10.301005	5.329913	5.386445	1.5504082
## sample_793	7.037503	1.9683857	7.921496	4.089439	5.871306	3.5311693
## sample_800	9.909458	2.5259185	11.229870	5.942791	6.164110	0.0000000
## sample_82	6.230699	1.3530409	10.124238	3.294077	4.048323	0.0000000
## sample_95	10.139142	1.4871261	9.606015	1.817460	5.893018	0.0000000
##	gene_12993	gene_7166	gene_19936	gene_9921	gene_16292	gene_65
## sample_119	2.281936	1.1300742	6.234467	0.3754009	8.802413	12.011398
## sample_275	4.423303	2.1720714	5.577619	1.1169635	9.084245	11.318481
## sample_29	5.009025	2.8048796	6.509941	4.8913949	8.005114	10.111892
## sample_295	1.921589	1.5155106	8.227986	0.5506052	8.719690	9.433796
## sample_298	4.265857	0.4131608	6.418799	0.4131608	8.933018	8.749574
## sample_404	3.694535	2.0346735	6.497940	0.5285713	9.120891	11.070497
## sample_464	4.337704	3.3283626	5.548085	3.6860635	9.331591	9.312479
## sample_468	3.550150	1.4050666	8.763219	0.0000000	8.871265	10.194892
## sample_531	5.880842	1.8331330	6.035415	2.0270479	9.184384	9.070703
## sample_57	4.847115	0.7324863	6.558718	1.8662750	8.993454	10.336373
## sample_578	3.342469	2.6538855	7.552423	2.4144332	9.036344	9.631374
## sample_579	2.555325	3.2102642	6.338514	1.0879380	9.552850	10.829532
## sample_603	1.468218	2.3896494	6.037037	0.7712095	9.275491	10.758598
## sample_627	1.517477	3.0141767	7.148070	1.0824982	9.634746	10.247014
## sample_638	4.548628	3.0060287	7.149046	0.8330925	8.969136	10.063530
## sample_673	6.349155	3.1082903	6.071031	2.1161318	9.012750	11.763578
## sample_68	2.462393	1.5259185	7.556261	0.0000000	8.670270	9.945938
## sample_69	3.566803	0.9291850	6.463547	4.1478684	9.008468	9.683747
## sample_691	3.192068	1.4243847	6.267901	0.3569326	9.271876	9.487462
## sample_701	4.413093	3.9359783	6.606541	3.2182232	8.806389	9.653451
## sample_735	6.760979	4.9483205	7.863307	0.9197596	8.519904	10.440786
## sample_74	4.370227	1.3761237	7.963879	4.9811722	10.594521	10.238620
## sample_740	3.366126	1.0247452	4.728976	1.2505978	9.045178	8.959825
## sample_75	4.702392	1.5700257	6.406428	2.1294479	9.133448	10.717659
## sample_762	5.395546	3.6797165	7.004412	3.6206920	9.211292	10.789615
## sample_768	2.541614	2.2803328	6.346477	3.3678323	9.435731	9.378345
## sample_793	1.496104	2.0967679	5.372339	1.4961044	9.843447	9.133954
## sample_800	5.042456	2.1373396	6.931021	0.7483754	8.784308	10.980861
## sample_82	1.353041	4.0483234	6.095114	0.0000000	10.194190	10.361494
## sample_95	3.424600	2.2036710	6.504660	1.2884176	9.001144	10.933764

##	gene_17376	gene_8565	gene_8444	gene_19134	gene_13198	gene_7790
## sample_119	9.244005	9.643515	10.128729	10.415002	12.27188	2.3675109
## sample_275	12.690308	11.666473	9.277336	10.307098	11.31881	7.2007809
## sample_29	5.369480	9.959483	7.663217	10.573486	12.42717	2.5178804
## sample_295	8.533201	10.043068	11.408993	10.209161	11.64669	7.1619884
## sample_298	11.007167	8.198671	9.583043	9.719808	11.94300	5.8505569
## sample_404	13.808512	11.249487	10.054455	10.107400	11.21069	5.6343784
## sample_464	8.763471	9.387455	7.643726	9.662652	11.67161	3.9724721
## sample_468	9.208239	10.768242	12.472668	9.519784	11.32447	3.1423154
## sample_531	9.381322	8.225675	8.316630	10.627005	11.44190	4.9890209
## sample_57	9.396027	8.275719	9.211959	9.863736	11.15679	3.8308333
## sample_578	7.298292	8.930758	7.320178	9.998097	11.46397	1.2890672
## sample_579	13.738451	11.400981	10.420602	10.118526	11.31647	5.9492548
## sample_603	12.014753	8.262128	12.821467	9.693234	11.97491	0.4365883
## sample_627	11.496469	11.110327	12.186374	9.307497	12.29068	1.0824982
## sample_638	9.323125	9.936202	11.790560	9.762752	11.58327	0.8330925
## sample_673	8.927790	9.314280	9.741924	9.969430	11.60269	2.2665767
## sample_68	8.475275	9.687040	11.208716	9.794770	11.18678	1.0893626
## sample_69	7.966338	8.054843	8.623183	10.578363	10.84837	1.4896971
## sample_691	8.756149	8.077644	8.692978	9.585051	11.20747	4.6191547
## sample_701	10.910883	11.399796	11.495135	9.928795	10.87411	3.4182979
## sample_735	6.975527	9.713734	3.704739	10.113898	10.53528	0.5319678
## sample_74	7.385629	9.841129	10.125362	10.499387	12.18118	1.3761237
## sample_740	6.829951	10.871582	10.292966	10.770664	11.27723	1.2505978
## sample_75	8.592797	7.533314	9.774034	10.329875	11.39589	1.4264258
## sample_762	7.238347	9.095497	8.211226	10.519695	10.86148	2.2002846
## sample_768	9.808443	11.427784	11.650150	10.077123	11.90093	0.9741622
## sample_793	6.087603	7.594549	10.530201	10.211389	11.86540	0.7893537
## sample_800	11.230687	11.780327	10.103825	9.477455	11.38903	6.6767469
## sample_82	8.698819	12.417117	9.247389	9.218817	11.50294	0.6023619
## sample_95	14.428812	11.918282	9.595291	10.114367	11.23987	6.7564095
##	gene_13697	gene_10059	gene_11718	gene_11987	gene_14650	gene_7070
## sample_119	1.3137683	0.9195308	9.759349	11.88470	7.558566	11.65445
## sample_275	0.6640277	4.6750820	10.008737	11.47597	10.467565	11.87727
## sample_29	1.5315686	6.0901950	9.786998	11.22426	6.705591	10.90974
## sample_295	0.5506052	1.7327902	10.304488	11.61882	6.411386	12.62094
## sample_298	2.6566108	6.4682640	9.372558	11.66561	8.741774	11.72235
## sample_404	0.0000000	4.0814419	9.661511	12.16565	8.937756	12.25416
## sample_464	2.6058736	2.3094673	10.431957	11.06000	8.196012	11.38767
## sample_468	3.4984807	2.5716525	9.044509	12.62791	5.632986	13.24872
## sample_531	1.8331330	3.9404957	10.194548	11.76473	7.924563	11.42586
## sample_57	4.8135709	1.8662750	10.014355	11.94258	7.168732	12.11899
## sample_578	1.9588056	2.1271710	9.644444	12.09583	11.809780	12.35314
## sample_579	1.3224474	2.8834253	10.084051	12.43122	8.697868	11.87064
## sample_603	0.4365883	1.6415923	9.513100	12.16829	6.120004	11.13691
## sample_627	0.4569113	4.5456410	8.928415	12.77265	7.941247	11.64803
## sample_638	1.7417905	1.3578335	9.677216	12.63436	7.590430	11.62540
## sample_673	0.0000000	1.5389851	10.463851	11.80489	8.617144	11.73784

## sample_68	3.5268576	2.6468999	9.679784	11.77138	6.077700	12.63003
## sample_69	0.5381408	0.5381408	9.534315	12.71798	9.685332	12.69436
## sample_691	1.8180325	1.8180325	9.653522	12.54977	7.929395	12.38172
## sample_701	0.5474498	1.7256539	10.441844	11.98778	4.616758	11.37017
## sample_735	0.5319678	3.8456306	10.237867	10.40849	9.215455	12.05937
## sample_74	1.2202061	1.5168724	10.346879	12.29199	9.028671	11.71837
## sample_740	0.0000000	1.7714631	10.033877	11.48937	7.572336	11.44669
## sample_75	1.9308887	0.3576083	9.474247	12.23325	8.663562	11.77315
## sample_762	1.0196308	4.2848398	10.308612	11.59351	7.514446	11.48987
## sample_768	0.9741622	4.5148418	9.612032	12.16245	8.315756	11.92892
## sample_793	0.0000000	2.0967679	10.176460	12.30367	7.207102	10.95517
## sample_800	0.7483754	2.1373396	10.180208	12.51980	10.245505	12.04715
## sample_82	0.0000000	0.6023619	10.199918	12.15525	6.210650	11.85217
## sample_95	0.0000000	1.4871261	10.180978	12.36234	6.295361	12.23137
##	gene_861	gene_7561	gene_8239	gene_11266	gene_4594	gene_17408
## sample_119	11.313694	3.9037421	6.782199	1.4766927	0.3754009	4.0139981
## sample_275	12.332658	2.5047729	6.606365	1.4612142	1.9716999	5.5776192
## sample_29	5.986445	0.0000000	7.003647	2.2575840	4.4669990	7.9380976
## sample_295	9.457830	0.0000000	5.306455	0.0000000	0.0000000	2.6116681
## sample_298	7.685968	1.5799525	4.915526	1.2180992	1.5799525	3.8003408
## sample_404	10.631941	2.5526968	5.848928	0.0000000	2.6575714	3.6945348
## sample_464	7.181391	0.0000000	6.185708	0.0000000	7.2310963	5.0540497
## sample_468	4.752781	1.7959331	4.838392	0.0000000	2.2353288	1.1610492
## sample_531	5.679117	0.5970305	5.527036	1.3436352	7.2931313	5.0121217
## sample_57	6.280442	1.8662750	6.453756	1.5774893	8.3122068	4.7792282
## sample_578	13.547160	1.2890672	6.560088	1.5483873	1.2890672	1.5483873
## sample_579	10.768466	4.1301319	6.145547	0.8077671	1.7010611	6.6768457
## sample_603	6.783810	0.7712095	5.997688	0.7712095	1.9361762	3.2449784
## sample_627	11.256126	1.0824982	6.143069	1.3163196	6.8172779	4.8364695
## sample_638	11.975228	2.5081751	5.509246	3.4112364	2.8056020	3.0060287
## sample_673	14.198537	2.1161318	6.589386	2.4028040	1.9481523	6.0295412
## sample_68	3.270021	0.8089204	5.527095	0.8089204	4.0033864	0.4603756
## sample_69	13.900206	0.5381408	6.095078	1.8923910	0.5381408	8.5432716
## sample_691	11.051569	2.7853204	6.422233	1.8180325	0.6427478	2.5288963
## sample_701	9.021505	4.7464310	5.719304	2.8071694	1.7256539	4.4742738
## sample_735	14.550326	2.8564479	6.421516	4.2353594	0.9197596	5.7080851
## sample_74	11.079858	1.7628165	7.696154	0.3402774	0.0000000	2.4650080
## sample_740	10.652935	2.5430009	6.914863	2.8494789	0.0000000	3.5892122
## sample_75	8.807278	4.3903857	6.602922	0.6439485	1.0875986	2.1294479
## sample_762	10.671904	6.3783087	5.497491	2.2002846	2.0293115	5.8195941
## sample_768	10.044585	2.0754984	5.667642	1.8366917	1.9610306	5.8635792
## sample_793	11.027934	0.7893537	6.040489	2.6084071	0.0000000	4.8233672
## sample_800	11.380223	4.9816655	5.718992	1.6039289	1.8951475	7.0168417
## sample_82	12.934028	4.8044976	6.289222	3.5063100	0.0000000	3.5063100
## sample_95	11.175412	3.6894669	6.799087	1.2884176	2.5967205	0.4443490
##	gene_10921	gene_3712	gene_6422	gene_11322	gene_17351	gene_11791
## sample_119	1.3137683	10.732150	7.521710	5.385207	9.448676	8.739608
## sample_275	4.0686880	10.756314	9.557992	4.636619	10.765634	6.460802

## sample_29	0.0000000	8.627009	8.768962	7.510479	9.429664	7.758683
## sample_295	7.5316622	10.922086	8.642936	5.137508	8.740219	5.754356
## sample_298	5.2780405	11.243971	9.672521	5.619525	8.854582	6.014931
## sample_404	2.9330438	10.809760	8.192751	4.344729	10.529421	8.050714
## sample_464	3.4073392	10.954822	7.990104	4.689892	11.526005	4.784959
## sample_468	9.3406691	10.202405	8.907618	4.720678	8.486553	6.882998
## sample_531	2.9381164	10.371864	6.942561	4.296780	10.987236	7.362181
## sample_57	1.2159892	10.388103	8.711006	5.226809	9.955019	7.456412
## sample_578	0.0000000	10.202834	8.410532	4.255433	10.519440	4.619436
## sample_579	1.5241640	10.613237	9.383255	4.505326	10.755680	6.981031
## sample_603	0.0000000	10.840093	5.819592	5.715652	9.629659	7.160366
## sample_627	1.5174773	11.395197	7.451162	5.058204	7.724562	7.574722
## sample_638	0.3339961	10.710789	7.860001	7.508650	10.018951	7.349011
## sample_673	0.0000000	11.552664	7.002466	7.823921	10.213615	3.793355
## sample_68	8.2762314	10.671090	8.610471	5.361912	9.562744	5.925392
## sample_69	0.0000000	9.988912	7.786015	5.380279	8.609892	5.487091
## sample_691	0.3569326	11.112316	5.836240	5.478548	9.549837	5.622752
## sample_701	0.0000000	10.695689	7.443358	6.225994	9.961895	6.075774
## sample_735	0.5319678	10.115811	7.633177	5.241112	10.906943	3.492533
## sample_74	0.0000000	10.681247	7.512930	7.507858	8.047473	7.240944
## sample_740	0.0000000	10.994092	7.306107	2.649133	6.055779	6.405997
## sample_75	1.8203219	11.390336	6.190048	4.332543	9.433158	5.908614
## sample_762	0.0000000	10.935585	8.030303	5.452335	8.811985	4.591883
## sample_768	0.0000000	10.929170	8.970287	8.242726	8.953545	6.729662
## sample_793	0.4480552	11.866630	5.562215	5.082230	9.534011	5.816426
## sample_800	2.3446581	11.233386	8.117503	4.657954	10.595994	7.968817
## sample_82	0.0000000	10.746304	3.438985	4.587539	10.271825	6.326958
## sample_95	2.5967205	10.676248	8.175734	4.179240	10.670081	7.221374
##	gene_6876	gene_10892	gene_14825	gene_9022	gene_13549	gene_18631
## sample_119	13.839903	3.654195	8.283320	0.0000000	6.971739	9.118497
## sample_275	8.276943	4.165848	9.025594	0.0000000	8.544180	6.632097
## sample_29	2.517880	4.315385	8.807020	0.3952822	7.519455	10.771688
## sample_295	6.636980	3.709644	9.106563	4.2920087	11.195458	9.356124
## sample_298	5.567512	3.108792	9.641712	1.2180992	5.808498	6.829609
## sample_404	7.685569	3.789927	8.879311	0.0000000	8.487699	6.532830
## sample_464	5.238172	4.337704	9.858980	1.0912598	7.002151	6.196869
## sample_468	5.370973	3.498481	7.960396	2.6683461	11.129154	9.084482
## sample_531	5.204908	3.840785	9.364590	1.8331330	7.780638	6.995688
## sample_57	5.118347	3.534037	8.833213	2.7977186	8.489948	8.676437
## sample_578	10.351050	3.198636	9.284741	0.0000000	7.913697	10.506129
## sample_579	5.886584	3.742297	8.048705	1.8586176	8.802203	8.077045
## sample_603	14.146624	3.536140	9.298342	2.2889196	5.126077	8.859858
## sample_627	13.236596	2.350045	7.610921	1.0824982	10.375984	9.510547
## sample_638	14.193964	2.958472	7.476876	4.5486277	9.154654	9.153995
## sample_673	13.432816	3.580013	8.448335	4.5771750	8.721793	9.653361
## sample_68	6.354507	3.270021	10.136748	2.5575816	10.647800	7.949774
## sample_69	8.790127	2.781779	10.476341	1.7050490	8.728421	8.332838
## sample_691	5.964978	2.725349	9.419313	2.1269398	6.230643	7.894084

## sample_701	14.554937	3.537768	9.195104	0.9432837	4.561442	11.186418
## sample_735	12.840377	3.434215	11.206843	0.9197596	7.074291	9.663060
## sample_74	5.538498	5.097809	7.474615	0.6153221	8.555609	6.717745
## sample_740	13.699562	3.317058	8.287278	1.2505978	5.106482	8.540752
## sample_75	6.190048	2.901070	9.339797	1.5700257	6.660224	7.533314
## sample_762	13.333547	3.620692	8.538674	1.6110307	9.367548	8.374205
## sample_768	13.597226	3.688762	7.646199	0.4021764	8.936606	10.780474
## sample_793	14.070440	3.283922	7.775683	5.1086624	9.440321	9.716187
## sample_800	6.470994	3.298482	8.930187	2.1373396	7.941147	8.422040
## sample_82	14.043890	4.177878	7.341185	0.0000000	9.099879	9.547272
## sample_95	6.780271	2.973134	7.153562	1.4871261	9.004187	6.625517
##	gene_1553	gene_12210	gene_783	gene_14387	gene_18202	gene_18790
## sample_119	10.311226	8.938836	6.789964	10.503647	8.451965	2.914163
## sample_275	9.448812	10.449592	6.866933	10.792026	8.301204	3.852239
## sample_29	10.284766	9.220743	2.987957	10.335971	8.034892	3.894042
## sample_295	8.619263	9.490821	8.170967	9.601979	8.689152	5.087310
## sample_298	8.737298	8.954394	8.148975	9.178986	8.399026	5.175876
## sample_404	9.701927	10.741147	6.744094	10.958190	8.524569	5.672425
## sample_464	8.555943	9.542041	7.555440	9.309911	7.783542	1.936025
## sample_468	8.790326	8.620975	8.967033	10.955025	8.152320	3.271052
## sample_531	9.398957	8.993001	9.000242	10.443213	6.868551	1.609046
## sample_57	9.046756	8.602253	9.013903	10.168246	8.031357	3.449205
## sample_578	9.819845	11.710936	5.155778	9.087012	5.956905	6.468977
## sample_579	9.563620	10.966779	5.210330	10.423358	8.139812	3.476550
## sample_603	9.215873	10.153755	5.051176	9.394645	7.847552	6.558975
## sample_627	9.797616	9.518632	5.140664	8.998399	8.673231	4.945154
## sample_638	9.675839	8.472821	5.806474	8.996618	9.331316	5.643114
## sample_673	8.373618	9.441332	5.826485	8.661956	7.902297	3.580013
## sample_68	8.596525	9.389199	8.864304	10.169148	8.452233	3.270021
## sample_69	9.219016	9.308136	7.803634	9.513049	7.208732	6.056399
## sample_691	8.933903	9.267283	7.485596	10.459391	7.600627	4.568865
## sample_701	9.372308	8.856083	5.410480	9.589213	8.042070	6.724555
## sample_735	7.863307	9.218699	9.235871	10.248485	6.939297	6.737065
## sample_74	10.176323	8.321401	7.880043	10.658113	7.853740	5.888614
## sample_740	8.500109	9.976263	6.711522	9.759209	7.749239	5.106482
## sample_75	9.079709	9.078210	7.123056	9.011364	7.402705	4.182605
## sample_762	9.450386	9.584810	7.104735	9.205028	7.714699	5.014993
## sample_768	9.730329	9.730874	6.380277	10.429647	8.661044	5.781638
## sample_793	9.468035	9.265259	5.409897	8.670334	8.724746	7.273124
## sample_800	8.853799	10.035376	8.378490	10.576210	7.749340	5.385241
## sample_82	9.874002	9.300517	6.687607	10.084862	8.355109	4.932416
## sample_95	9.233253	10.447094	5.444833	10.969494	8.427694	2.680279
##	gene_5760	gene_19297	gene_18276	gene_19353	gene_8439	gene_5440
## sample_119	8.721441	9.149026	5.074227	12.39249	9.393009	7.857396
## sample_275	9.116505	8.759875	5.879238	11.96867	9.835918	8.818946
## sample_29	6.679213	11.304289	8.380149	12.05217	8.268097	7.007184
## sample_295	7.966361	8.978170	8.116443	12.99689	10.644839	9.796225
## sample_298	7.531647	8.552162	8.993400	12.10358	11.219060	8.400444

## sample_404	9.459630	8.750359	5.659853	11.86572	10.224701	8.318055
## sample_464	7.667991	8.398094	4.693364	13.53824	10.592681	7.385397
## sample_468	8.025726	8.478244	8.109909	12.32905	10.875926	9.324997
## sample_531	7.715276	8.889203	5.790201	12.59785	10.593410	6.535425
## sample_57	8.126627	8.999059	5.062329	12.69258	10.812915	8.009302
## sample_578	9.323183	8.854815	5.437907	11.80105	10.021008	8.620898
## sample_579	9.701341	8.672885	5.957989	12.36719	10.196270	8.622952
## sample_603	8.384637	8.603574	4.449436	12.01340	9.405210	8.453147
## sample_627	7.292211	8.127675	6.180605	12.86720	8.717320	7.853328
## sample_638	9.698632	8.514525	5.227672	12.30898	10.184727	8.856083
## sample_673	8.414482	8.020841	6.763982	12.58903	9.949434	7.769983
## sample_68	8.111470	8.709666	7.234808	12.99993	11.391625	10.668282
## sample_69	8.856354	8.356905	6.395450	11.56162	10.658863	8.857763
## sample_691	9.019096	8.533844	6.164442	11.35697	11.358673	8.202481
## sample_701	8.951961	8.052139	4.771019	11.91620	9.945785	8.118318
## sample_735	9.226236	8.159458	5.670690	11.17826	11.188218	7.992689
## sample_74	8.081867	9.238228	4.772403	11.77179	8.438838	7.917020
## sample_740	9.292922	8.155638	4.854664	12.86674	9.523348	7.267882
## sample_75	8.863251	7.950777	6.768409	11.65278	10.363543	7.589172
## sample_762	8.010321	9.220018	5.710605	12.67090	10.937514	8.396369
## sample_768	7.399333	9.759517	5.764675	11.75236	9.668979	8.271412
## sample_793	7.881952	8.977031	5.253418	13.12479	8.923937	6.465973
## sample_800	9.256343	8.045072	4.981665	12.31579	10.272711	8.464309
## sample_82	8.600173	9.251073	4.572847	12.25314	8.084718	6.896538
## sample_95	9.569501	8.661532	5.592322	11.99000	9.479800	8.082836
##	gene_17120	gene_17221	gene_183	gene_11486	gene_2552	gene_18882
## sample_119	1.989793	8.655985	12.01692	10.410960	7.061139	9.040854
## sample_275	3.526094	9.342428	11.87014	9.318857	7.725366	5.989514
## sample_29	1.365021	8.510543	10.76300	10.212265	9.458152	10.567823
## sample_295	2.815903	9.278359	12.62651	9.108992	7.791886	5.729300
## sample_298	5.022426	9.366772	10.86510	9.578044	8.201923	8.561055
## sample_404	2.316812	9.736013	11.42800	9.960835	7.906067	6.083515
## sample_464	1.091260	9.179556	10.91919	8.719652	9.206463	9.100050
## sample_468	3.985018	8.321680	12.79419	9.887405	7.929028	5.032528
## sample_531	0.000000	9.283841	11.11148	8.677935	8.384590	9.148347
## sample_57	0.000000	8.333075	11.47816	9.347046	8.524048	8.814256
## sample_578	3.198636	8.915022	11.15358	9.699756	5.136171	9.413507
## sample_579	3.322736	9.792675	11.96811	10.196737	7.922739	6.733910
## sample_603	6.746313	9.141096	11.43664	10.419486	7.910097	7.393502
## sample_627	6.217190	9.545494	10.88387	10.371123	6.087399	7.520131
## sample_638	3.006029	9.043667	10.75237	11.502931	5.745646	9.437571
## sample_673	4.998683	9.804759	10.75498	12.634167	7.895757	7.402543
## sample_68	4.975039	9.299947	12.63006	9.500459	7.503436	5.550430
## sample_69	2.206831	9.363644	11.22442	8.234487	7.768171	8.576110
## sample_691	4.651741	9.625603	11.17507	8.137821	8.452727	5.372095
## sample_701	7.578373	9.577062	11.21088	10.381240	7.633053	6.951285
## sample_735	7.943892	10.312111	11.19429	8.627917	7.020813	5.720339
## sample_74	1.516872	8.094553	10.38283	9.875848	8.404022	11.903524

## sample_740	4.921860	9.049878	12.13546	12.187510	5.207288	8.667296
## sample_75	3.913100	9.459913	11.47841	9.514199	8.379023	7.983216
## sample_762	8.849418	10.250784	10.45138	10.399203	6.826485	8.893717
## sample_768	5.145661	8.953545	10.38757	12.206590	7.781969	7.077478
## sample_793	5.788182	9.355444	11.46989	10.047860	7.088587	7.103938
## sample_800	5.809221	9.267529	11.23635	8.836738	7.615696	7.140175
## sample_82	3.632233	10.387953	10.88306	10.607478	7.530984	6.858391
## sample_95	1.817460	9.779439	11.92687	10.191244	7.710510	5.079779
##	gene_17446	gene_10002	gene_14386	gene_19363	gene_13215	gene_8939
## sample_119	8.636726	7.831282	5.644843	4.647965	7.642492	9.420680
## sample_275	9.769846	8.616858	2.774945	3.597484	9.027209	9.916320
## sample_29	7.354223	8.596451	7.414102	6.277682	5.574211	9.654260
## sample_295	9.903725	7.911722	1.259544	8.126075	11.696229	8.842787
## sample_298	9.517325	6.320467	7.952479	5.756351	9.557240	8.309945
## sample_404	9.509530	8.833870	1.869832	3.538476	9.133451	10.372005
## sample_464	10.255064	6.930406	8.907993	2.851639	10.505613	9.207139
## sample_468	9.449258	7.157377	1.405067	7.820128	12.299634	9.673677
## sample_531	9.381322	7.225555	9.136568	3.201947	9.020319	8.979308
## sample_57	9.814775	7.168732	8.762458	3.449205	9.378930	9.759366
## sample_578	9.838923	7.476487	6.834029	6.916799	13.055049	9.113047
## sample_579	9.396971	8.642048	3.023167	2.883425	9.359552	9.939063
## sample_603	9.748803	7.737200	3.190172	5.745512	9.641835	9.706615
## sample_627	9.524481	8.263311	2.122076	4.996570	8.626395	9.468335
## sample_638	9.199452	7.699316	6.339385	3.181977	8.662839	9.446728
## sample_673	9.709449	8.537983	5.199107	1.758047	6.225998	9.677697
## sample_68	10.354547	7.345662	1.324119	7.202036	13.304095	9.572708
## sample_69	9.854094	8.093560	2.578432	4.790392	6.943230	9.560896
## sample_691	10.551035	7.261155	9.122965	6.130167	11.974522	8.886931
## sample_701	9.945110	7.574881	4.444004	3.648086	10.146492	8.523311
## sample_735	10.781360	8.650503	3.704739	5.670690	10.272606	9.942402
## sample_74	8.953268	7.899454	1.045303	6.340797	4.083843	8.924427
## sample_740	10.262506	7.661044	3.213596	8.600935	1.617816	8.523942
## sample_75	9.503056	6.569130	8.257025	7.859373	10.331141	9.056261
## sample_762	10.372060	8.347152	6.333064	3.494864	8.965149	9.405197
## sample_768	9.229446	8.042529	3.275052	7.091150	4.631710	9.561446
## sample_793	9.795184	8.604372	2.608407	5.562215	10.010654	9.575655
## sample_800	9.723569	8.555774	5.100788	4.464479	9.175530	9.895757
## sample_82	8.821210	7.673281	0.000000	5.442599	10.190998	10.209734
## sample_95	8.974498	8.492694	0.444349	4.454038	6.789703	10.209587
##	gene_2234	gene_6149	gene_16153	gene_7131	gene_4909	gene_9071
## sample_119	12.81696	7.120881	10.158673	9.210384	9.122564	10.169887
## sample_275	12.37629	7.763199	9.844199	9.791891	9.334823	5.448409
## sample_29	12.68144	8.388314	9.892177	7.211197	8.740830	11.196049
## sample_295	12.74801	8.254570	9.346886	8.729199	7.503381	6.728342
## sample_298	12.41609	7.584940	9.980668	9.699814	7.340571	9.846283
## sample_404	12.03697	7.885074	9.338647	10.264677	10.333077	5.803932
## sample_464	12.13757	7.284385	9.162580	9.696476	7.504477	9.326519
## sample_468	11.95016	7.833225	8.929729	8.656768	7.984401	6.216243

## sample_531	12.17727	7.149452	9.362345	10.202099	8.131502	10.271953
## sample_57	11.95224	7.220843	9.632012	9.314440	7.469235	10.727121
## sample_578	12.52613	7.507343	8.583380	10.538228	7.974638	9.450694
## sample_579	12.26893	7.865857	8.991712	10.676856	10.468492	7.683016
## sample_603	13.44285	7.950060	8.920031	9.648201	8.405516	8.579787
## sample_627	13.10326	8.313173	12.519292	9.812636	7.475320	9.186350
## sample_638	12.93352	7.966009	9.943473	8.477046	8.829596	10.307246
## sample_673	12.16156	8.026125	9.652508	9.943861	8.817505	7.105888
## sample_68	12.32811	7.407667	9.110823	9.188171	8.143603	6.796364
## sample_69	13.08070	7.992185	11.222668	9.769940	8.068246	9.200288
## sample_691	12.77086	8.392794	9.244100	10.342630	7.937709	6.440997
## sample_701	12.42868	7.466129	10.192737	9.568319	8.577923	7.506962
## sample_735	12.04148	7.577013	9.823589	10.495725	8.545389	7.166736
## sample_74	13.33195	8.304511	9.749155	8.754390	8.074752	9.915721
## sample_740	12.39369	7.121927	10.343230	9.960376	10.622848	9.672245
## sample_75	12.93976	7.580718	8.912084	10.349303	8.267349	8.444593
## sample_762	12.19974	7.574003	9.222503	10.539246	7.286983	8.915369
## sample_768	12.02152	8.338861	9.781819	8.750988	7.526961	7.067144
## sample_793	12.74829	8.570224	11.317610	10.130326	9.195385	7.402969
## sample_800	12.39405	7.850556	8.922121	10.018451	9.210703	9.062661
## sample_82	12.96165	8.243607	11.198034	10.709222	6.112721	6.658297
## sample_95	12.19599	7.618195	9.382594	10.409709	9.885274	3.424600
##	gene_12169	gene_11657	gene_5503	gene_18762	gene_7677	gene_1118
## sample_119	9.253167	9.249652	12.40345	11.611933	9.423830	0.6730136
## sample_275	7.368341	5.646569	12.31406	11.986859	9.181465	1.7388975
## sample_29	12.595819	6.374020	12.17337	11.169643	9.296877	0.9601410
## sample_295	9.771123	8.511318	12.67423	12.621042	8.262696	0.5506052
## sample_298	3.537420	8.970696	12.53467	12.354208	8.797068	0.4131608
## sample_404	7.784432	4.937627	12.22432	12.894490	9.037341	0.0000000
## sample_464	5.632704	7.002151	12.28865	10.378436	9.647491	0.6464392
## sample_468	6.403455	8.090414	12.69776	13.523011	8.207644	0.8671848
## sample_531	9.347663	7.598023	12.20103	10.183139	8.897029	0.5970305
## sample_57	4.076867	7.135176	12.21927	10.666180	8.501264	1.8662750
## sample_578	4.877386	10.197512	12.00095	11.548682	8.467129	0.0000000
## sample_579	8.212788	4.883513	12.38905	12.402913	8.615766	0.0000000
## sample_603	6.688180	8.221713	12.15672	10.638056	8.583899	1.0427144
## sample_627	9.615688	7.033753	12.30733	11.607090	9.159859	0.0000000
## sample_638	9.153334	6.932309	12.18210	11.987403	9.541697	1.2048293
## sample_673	9.402245	8.523111	12.06331	12.526438	8.537983	2.8468343
## sample_68	8.723849	9.522015	12.74660	10.196676	8.278064	0.4603756
## sample_69	7.788966	8.426801	12.37259	10.885734	8.858239	0.0000000
## sample_691	5.104286	10.043260	12.56114	13.084632	8.855529	0.3569326
## sample_701	7.327939	8.183893	11.78350	11.888903	8.360250	0.0000000
## sample_735	10.477343	9.391469	12.08388	12.049239	9.194747	0.9197596
## sample_74	7.116510	5.970442	12.77685	9.563865	10.614756	0.6153221
## sample_740	5.914457	7.107562	12.49490	12.645138	9.613970	1.0247452
## sample_75	4.605696	8.274164	12.30911	9.771714	8.715660	0.8827604
## sample_762	8.365242	8.953239	11.80068	11.538766	9.233500	1.3453964

## sample_768	8.948858	7.413002	11.51421	11.719457	9.476303	0.7163325
## sample_793	8.878891	6.649864	11.86287	11.674002	8.277269	0.0000000
## sample_800	8.478131	6.714603	11.44990	12.337750	8.791521	0.7483754
## sample_82	7.594250	8.533143	11.94458	11.471690	8.978879	0.0000000
## sample_95	6.385250	3.424600	12.33843	12.663651	8.951844	0.4443490
##	gene_13385	gene_13489	gene_19827	gene_17059	gene_13645	gene_3590
## sample_119	9.823541	8.666576	7.407659	8.047647	9.933863	8.336016
## sample_275	8.949182	9.384667	7.363232	6.881310	9.762098	9.357930
## sample_29	9.661454	8.835166	7.403396	8.132094	9.687395	8.263668
## sample_295	9.300856	10.887480	7.771470	7.468714	8.828403	8.180242
## sample_298	10.023255	8.953428	8.354412	7.709015	10.510121	7.895357
## sample_404	9.329489	9.709572	7.263551	7.472073	10.528991	9.502506
## sample_464	9.694243	8.969481	7.479788	9.289205	9.187971	9.230679
## sample_468	9.578510	11.333250	7.136407	7.226499	8.827441	7.790896
## sample_531	9.670738	8.357790	7.283663	9.628382	10.000282	9.229437
## sample_57	9.679628	9.017591	7.107761	9.010206	11.083925	9.026770
## sample_578	9.927854	8.667743	7.392214	8.436811	9.382637	9.372199
## sample_579	8.834332	9.258877	7.100242	10.049726	10.740894	7.958002
## sample_603	9.357847	7.840897	7.580892	9.400692	9.760605	8.176617
## sample_627	9.378078	9.836349	7.739727	9.368148	10.154312	7.916984
## sample_638	10.060574	9.262001	7.024685	8.579825	9.585295	8.608029
## sample_673	9.323967	9.092984	7.634717	8.354844	10.867147	9.098011
## sample_68	9.772618	10.774276	7.176014	6.597899	9.174157	8.174776
## sample_69	10.151118	9.516616	6.169454	7.466456	10.014648	9.742793
## sample_691	10.063071	9.617894	7.852642	8.243141	9.454945	8.027314
## sample_701	10.169224	7.953527	6.692734	8.824125	9.370297	7.794695
## sample_735	10.278833	7.992689	6.643648	6.755034	10.020994	9.406713
## sample_74	9.483602	9.493217	8.029961	8.266388	9.757603	7.329482
## sample_740	9.594577	8.325004	6.682756	7.999154	9.758060	8.598369
## sample_75	10.399502	8.179357	7.565803	9.273145	10.301222	7.742956
## sample_762	9.942759	8.944241	7.248127	10.012597	10.326609	8.621916
## sample_768	10.023796	10.193414	8.079005	8.454435	9.259270	7.992412
## sample_793	9.958804	7.319907	7.653139	9.801084	9.594384	7.283283
## sample_800	9.528407	9.411316	7.559607	8.274625	9.821133	8.782084
## sample_82	9.272229	6.574500	7.748213	7.255718	11.360946	7.621077
## sample_95	9.090708	8.887083	7.309294	6.871967	10.467443	8.907816
##	gene_15621	gene_18008	gene_14560	gene_3977	gene_4553	gene_15416
## sample_119	9.328455	10.818622	11.023637	6.738700	8.881811	10.273970
## sample_275	9.146571	10.633867	10.494116	7.727839	11.414278	9.836841
## sample_29	8.686385	10.013713	10.277101	6.953172	8.041812	10.189083
## sample_295	8.396630	9.931454	10.622052	7.886440	9.796979	10.569628
## sample_298	8.518641	10.388695	9.141359	7.438193	9.998083	8.936929
## sample_404	9.378516	10.610462	9.761242	8.462584	12.217237	9.684966
## sample_464	8.931568	10.398509	11.051576	7.691855	7.887074	10.061817
## sample_468	8.797023	10.084250	10.419865	8.811651	8.052845	9.652766
## sample_531	8.848069	10.727691	9.767248	7.859181	4.867615	9.402240
## sample_57	8.582473	10.303518	9.896146	7.649386	7.014891	9.407315
## sample_578	8.117425	10.213979	9.704758	7.456864	9.770619	9.884441

## sample_579	8.593824	9.720718	9.150649	7.703938	11.530514	9.696129
## sample_603	9.601061	10.450417	10.180357	6.966107	9.593163	10.152412
## sample_627	9.057096	9.971515	9.969372	6.456325	12.391268	9.245106
## sample_638	8.235335	10.662027	9.905182	8.143960	10.731752	9.776542
## sample_673	8.323001	9.937567	10.019980	6.873567	11.416576	10.229732
## sample_68	8.558191	10.229564	10.621878	8.476799	8.962156	9.891018
## sample_69	9.400783	10.517856	9.453828	8.199417	9.803900	9.620524
## sample_691	8.764553	9.950955	9.109102	7.045148	10.840566	9.872352
## sample_701	9.300220	10.179586	8.838741	6.507576	6.432483	10.265392
## sample_735	8.304867	10.160174	9.760465	7.943892	10.122013	9.992868
## sample_74	8.445454	11.831343	10.061088	7.641474	10.205207	10.777025
## sample_740	8.220461	10.139539	9.783123	7.355554	9.355021	9.772922
## sample_75	8.253039	10.382203	9.198440	8.122243	9.670272	9.483348
## sample_762	8.245814	10.604961	9.278468	7.855747	10.220052	10.834542
## sample_768	8.468905	10.916029	9.928008	7.763969	10.654627	9.574809
## sample_793	8.495040	10.670577	10.462185	7.096283	8.898840	10.149417
## sample_800	9.701353	10.118020	9.560367	7.046818	10.604840	8.920097
## sample_82	8.395628	10.984048	10.344340	6.460282	9.652264	10.275449
## sample_95	9.296678	10.224243	9.696585	7.991743	10.190356	9.813953
##	gene_8961	gene_3191	gene_4418	gene_20359	gene_15378	gene_11703
## sample_119	9.415159	6.612936	10.53458	7.572147	0.6730136	8.493003
## sample_275	10.430170	8.665428	10.80653	7.889467	0.0000000	8.921627
## sample_29	10.509132	6.915999	9.35888	7.042108	5.2310675	8.263668
## sample_295	8.568952	9.587237	11.63731	7.990286	0.0000000	8.584752
## sample_298	9.105332	8.354412	12.25544	4.080351	0.9962441	8.946658
## sample_404	10.357618	7.749259	11.15845	9.156005	0.0000000	9.482120
## sample_464	9.897112	7.730769	12.19671	7.456765	0.6464392	8.055944
## sample_468	8.068434	7.710868	11.96748	8.043864	0.0000000	8.734195
## sample_531	10.202736	8.656099	12.24733	7.635704	2.0270479	8.657933
## sample_57	10.214768	8.209561	12.35248	5.935344	0.0000000	9.071800
## sample_578	9.759449	8.721366	12.12361	7.110102	1.9588056	8.089662
## sample_579	8.898369	7.533252	11.16345	8.583982	0.0000000	9.272947
## sample_603	10.628108	7.620293	10.95177	7.236713	0.0000000	8.963569
## sample_627	8.987849	7.140462	11.96838	7.033753	0.4569113	8.642626
## sample_638	10.320530	7.125021	11.13521	5.370782	7.7681711	9.084391
## sample_673	9.384870	8.460091	12.50685	5.561265	1.5389851	9.370404
## sample_68	8.337872	8.067666	12.05743	6.695326	0.0000000	9.018670
## sample_69	10.568982	8.977752	11.92729	4.072346	1.7050490	9.011001
## sample_691	8.357429	9.623038	13.48392	7.813582	2.5973881	8.816670
## sample_701	8.816763	8.074538	11.71796	5.296523	0.0000000	8.870380
## sample_735	8.942265	10.855102	11.94308	5.196548	1.8778620	9.182620
## sample_74	10.535771	5.052281	10.57422	6.764381	1.0453032	8.564759
## sample_740	8.890231	8.227115	11.90401	7.488933	1.4458856	10.431978
## sample_75	9.034766	8.348259	12.25077	7.114294	8.1236972	8.902788
## sample_762	10.216940	8.805373	12.30716	6.546305	1.3453964	9.048686
## sample_768	10.210732	5.918291	11.12127	8.466285	3.7242901	9.326551
## sample_793	8.942210	6.854395	11.15061	5.253418	1.8274939	9.100846
## sample_800	9.526186	9.031133	11.89456	8.277790	0.7483754	8.229583

## sample_82	10.862033	6.716319	11.06049	8.737866	0.6023619	9.541265
## sample_95	9.362183	8.078994	10.96533	8.951559	0.4443490	9.287548
##	gene_3519	gene_12769	gene_13454	gene_10247	gene_17717	gene_17882
## sample_119	8.767423	9.444990	11.510220	4.6307607	8.945034	10.078858
## sample_275	9.216273	9.302924	9.570245	1.9716999	8.961293	9.931407
## sample_29	9.236658	6.938415	10.434545	5.6025578	9.355316	9.394966
## sample_295	10.500503	10.432270	9.382481	0.9482270	8.836874	9.080791
## sample_298	10.816880	11.511431	10.498590	0.0000000	8.775001	8.801964
## sample_404	9.191375	9.669371	10.357618	2.0346735	9.159324	9.921222
## sample_464	8.873632	8.198794	9.696476	2.7339833	8.846393	9.661657
## sample_468	10.378035	12.380126	9.929469	0.0000000	8.970854	8.687376
## sample_531	8.860851	8.200756	10.529304	5.3749421	8.390289	9.301528
## sample_57	9.274935	8.460452	10.051698	1.8662750	8.145387	9.606531
## sample_578	9.397127	10.538228	9.934254	2.2779550	9.110173	10.173665
## sample_579	9.529598	9.068335	10.635936	3.7012718	9.298250	10.343530
## sample_603	9.123837	10.086428	10.654591	4.1949144	7.049097	9.393886
## sample_627	10.126692	9.948309	11.265328	3.7328227	8.691276	9.044935
## sample_638	9.484495	11.872529	10.915566	5.3339854	8.353601	8.783738
## sample_673	10.268706	8.723422	10.832882	3.5800128	8.474813	10.395577
## sample_68	9.997250	11.922101	9.636148	0.0000000	9.475786	9.564894
## sample_69	9.227743	10.577117	9.875081	0.0000000	9.105222	10.187921
## sample_691	9.452058	12.252592	10.574064	0.3569326	9.389528	9.607031
## sample_701	11.116175	9.268186	10.414791	2.8992723	8.480653	10.448054
## sample_735	9.323820	7.079058	9.647906	4.8401913	8.443159	10.472376
## sample_74	9.599913	9.057762	10.756990	0.0000000	9.165977	9.116315
## sample_740	9.497425	9.065750	10.590419	6.5668334	8.371981	10.594931
## sample_75	8.971500	11.010444	10.979625	0.0000000	9.228287	9.914798
## sample_762	10.138899	8.286733	9.999620	5.8584311	9.073218	10.448292
## sample_768	10.424597	9.849744	10.771745	3.2750518	8.361501	9.558909
## sample_793	11.189621	8.515282	11.273720	3.3368832	8.477074	9.682057
## sample_800	9.800381	10.057640	10.778504	3.8677186	9.332003	9.916580
## sample_82	9.306433	8.678892	10.779801	3.5706451	9.420200	10.369990
## sample_95	9.250443	8.824192	10.478790	1.4871261	8.625822	10.331913
##	gene_20116	gene_17497	gene_7026	gene_2336	gene_13199	gene_13631
## sample_119	5.022399	10.477718	4.595730	4.5042327	10.372343	7.663793
## sample_275	6.079542	9.983136	5.820769	1.7388975	10.029053	7.245923
## sample_29	5.701987	7.617644	6.665848	7.0660569	11.927563	8.601143
## sample_295	4.897908	9.070851	9.539384	0.0000000	10.249919	7.386949
## sample_298	5.007631	8.633914	2.932023	1.4103411	8.948595	11.289938
## sample_404	5.608443	9.969775	5.485597	1.2187192	9.902669	7.114898
## sample_464	2.605874	9.593619	9.457570	5.2596190	9.776764	12.026817
## sample_468	3.985018	9.456892	10.102750	0.0000000	9.527063	9.891793
## sample_531	2.027048	9.309708	8.738058	4.7895415	9.472632	11.476311
## sample_57	3.159920	9.437908	9.463817	5.2495667	9.904136	11.338397
## sample_578	5.175121	8.486051	2.277955	0.5667665	10.111644	7.027364
## sample_579	4.883513	10.551170	4.555454	0.0000000	10.464168	6.901760
## sample_603	5.697201	9.511005	1.042714	0.4365883	10.105529	9.544159
## sample_627	7.385172	9.721983	6.406504	1.0824982	10.521306	9.670145

## sample_638	6.223459	9.096114	4.396338	4.2058304	9.956770	9.198811
## sample_673	5.964981	9.765808	7.164565	0.5621814	9.158206	7.658726
## sample_68	3.703776	9.393238	9.826776	1.3241193	9.437713	10.236780
## sample_69	5.879368	8.670373	4.881714	0.9291850	9.002123	7.815268
## sample_691	4.328047	9.817762	5.006904	3.3592260	8.620759	10.658649
## sample_701	7.564371	10.422086	3.144846	6.0953823	9.910248	7.376820
## sample_735	5.732489	9.232666	5.826170	0.9197596	9.060231	6.176742
## sample_74	3.208627	7.982206	3.941266	0.0000000	10.795066	7.305433
## sample_740	5.838238	10.111031	3.366126	0.4274989	9.552794	8.250175
## sample_75	4.310827	9.423736	2.219339	1.5700257	8.917130	10.451664
## sample_762	6.257586	10.021174	5.285850	1.6110307	10.268916	7.452332
## sample_768	7.021458	10.285102	4.066356	0.0000000	10.621457	8.560207
## sample_793	5.077880	9.813983	4.536929	0.0000000	10.087993	10.198531
## sample_800	6.343925	9.878531	3.798901	1.6039289	9.685954	8.063525
## sample_82	5.737005	9.867614	4.956689	0.0000000	9.812148	8.248530
## sample_95	4.355813	10.404726	3.729248	0.7834986	10.093682	7.783175
##	gene_1221	gene_5748	gene_1472	gene_221	gene_483	gene_5448
## sample_119	11.98985	7.387319	4.5042327	6.587305	8.174746	4.630761
## sample_275	11.75061	8.025123	2.5047729	10.265498	8.722435	6.028788
## sample_29	11.62261	6.643279	2.8685287	3.185296	8.874450	5.505808
## sample_295	11.97066	8.513155	2.9081020	8.580888	9.293396	6.279505
## sample_298	11.35932	8.949564	3.8998427	8.177828	8.787961	4.850689
## sample_404	11.05738	13.934253	1.6836515	9.187560	7.821589	5.568646
## sample_464	12.33564	7.470626	4.2967951	7.771384	10.544897	4.784959
## sample_468	11.40801	7.419421	4.0220748	8.889166	7.178057	6.301622
## sample_531	12.13663	7.397700	5.8166842	6.757623	10.274343	4.121339
## sample_57	11.62763	6.977520	4.0768669	7.378260	8.974607	4.744048
## sample_578	12.42510	10.858098	5.5897845	6.407086	9.272446	5.646327
## sample_579	11.50653	7.977463	2.5553254	8.992638	8.199957	4.622919
## sample_603	11.45282	6.246250	4.0174142	5.597335	9.510307	2.879784
## sample_627	11.47189	8.231476	3.0792245	5.061431	8.570937	4.718926
## sample_638	11.24344	7.605946	2.1330703	5.269875	9.306602	5.694819
## sample_673	12.49439	7.135525	4.5771750	3.941642	9.893437	5.751908
## sample_68	11.87075	6.563906	3.6615330	8.861006	7.784955	5.404406
## sample_69	11.40665	7.783057	4.1842009	5.587353	9.795851	6.315816
## sample_691	11.49972	10.485044	5.3915168	7.455829	10.220342	4.818861
## sample_701	11.62222	7.151869	2.4892089	4.826274	9.853807	5.313366
## sample_735	11.59238	8.039895	4.7232964	5.746216	10.520903	6.718855
## sample_74	12.33183	8.253871	1.5168724	5.439676	8.982386	3.573411
## sample_740	12.29299	7.417979	3.8180938	7.368646	8.869106	5.077303
## sample_75	11.77384	8.377844	5.3456235	7.775972	9.855143	5.083698
## sample_762	12.01584	7.868489	5.3955461	6.480273	10.014746	5.060117
## sample_768	11.64943	9.051201	3.0130692	6.977497	8.388146	5.823191
## sample_793	11.96614	6.016343	1.6712934	6.718170	9.025220	4.947741
## sample_800	11.39451	7.253791	2.9633076	12.630747	8.384374	5.773804
## sample_82	11.70790	5.650357	1.6195069	5.839257	8.961276	3.506310
## sample_95	11.32122	6.593548	0.7834986	10.195101	8.694560	5.468528
##	gene_12922	gene_18495	gene_5603	gene_11390	gene_20208	gene_10556

## sample_119	1.3137683	12.53083	3.720738	0.0000000	10.471543	1.9897930
## sample_275	1.1169635	10.85205	6.015814	1.1169635	9.969502	0.0000000
## sample_29	0.7051375	11.66762	4.746479	0.7051375	9.426040	0.7051375
## sample_295	2.4975868	11.02603	3.227325	0.0000000	8.387548	4.2218771
## sample_298	1.7316612	11.14027	1.218099	0.4131608	10.879507	5.0942023
## sample_404	0.5285713	10.65416	5.456819	0.9145645	9.857808	0.5285713
## sample_464	0.6464392	11.28682	2.851639	0.6464392	6.912171	0.6464392
## sample_468	6.5374887	10.68362	3.826437	0.4978423	8.945757	2.3563410
## sample_531	2.1979912	10.73849	1.833133	1.8331330	6.823546	1.0180644
## sample_57	1.2159892	10.95619	2.797719	1.2159892	8.079602	0.0000000
## sample_578	0.0000000	10.98132	1.768163	0.5667665	9.087089	1.2890672
## sample_579	1.3224474	11.04926	4.883513	0.0000000	9.349785	0.0000000
## sample_603	0.0000000	11.40228	1.271067	0.0000000	10.177955	1.0427144
## sample_627	2.7993567	11.88845	3.141449	0.4569113	9.585124	2.4517782
## sample_638	1.4974846	11.64535	3.577864	0.8330925	9.228379	2.8056020
## sample_673	0.0000000	10.58959	2.846834	1.2806000	9.885480	4.3931347
## sample_68	5.6935399	11.24667	4.904359	0.0000000	8.991063	3.2126783
## sample_69	0.9291850	10.81137	4.415732	0.9291850	10.901184	3.9510749
## sample_691	0.6427478	10.85804	2.381256	0.0000000	10.084237	6.5616408
## sample_701	1.9141052	11.20148	2.230111	0.0000000	9.225205	0.0000000
## sample_735	0.0000000	10.30297	4.365888	2.0430645	9.454303	3.9324581
## sample_74	0.6153221	12.11936	2.239826	0.0000000	9.828651	1.0453032
## sample_740	1.2505978	11.48045	1.910311	1.6178157	10.552467	2.0369253
## sample_75	0.0000000	10.60069	2.303956	0.3576083	10.548639	4.1600812
## sample_762	1.6110307	10.81159	3.427593	0.0000000	8.712149	2.4913657
## sample_768	0.7163325	11.30171	4.431322	0.9741622	9.039306	2.2803328
## sample_793	0.4480552	11.45363	1.827494	0.0000000	8.726981	1.8274939
## sample_800	0.7483754	11.43928	5.718992	0.0000000	9.985529	1.8951475
## sample_82	0.0000000	11.73699	2.744118	0.0000000	9.613749	0.0000000
## sample_95	0.4443490	10.84738	3.216455	0.0000000	9.571760	0.4443490
##	gene_17750	gene_4381	gene_4934	gene_17821	gene_4099	gene_13359
## sample_119	11.393154	2.7956631	14.807908	5.180900	10.663914	11.346331
## sample_275	10.615712	1.7388975	10.494706	3.700961	9.897330	10.859838
## sample_29	8.858864	1.3650206	11.606077	5.730012	11.492474	9.719029
## sample_295	12.096330	2.3736760	11.199740	2.943396	9.933512	10.998485
## sample_298	10.079031	1.9943263	11.165504	1.994326	9.974027	10.674272
## sample_404	10.307121	3.0142839	10.278624	4.173823	9.628825	11.367497
## sample_464	7.964214	2.3094673	9.952486	3.972472	10.230837	10.400324
## sample_468	12.007595	3.6482468	10.814534	3.358860	9.769752	10.864496
## sample_531	9.954798	3.1192566	10.467443	1.018064	9.975518	10.407172
## sample_57	9.311440	2.7977186	10.320789	1.866275	9.543987	10.489597
## sample_578	8.533665	2.5391092	9.608432	3.703854	10.014983	10.468400
## sample_579	10.789110	3.0231666	10.081057	3.534697	10.016140	11.534536
## sample_603	11.866452	3.2977637	10.634212	4.720300	10.181239	10.680298
## sample_627	9.352611	2.4517782	11.194160	2.254140	10.030777	10.729068
## sample_638	9.559628	2.8583789	11.396546	3.290822	9.653612	10.347898
## sample_673	9.304529	2.1161318	10.884705	3.460218	10.158749	10.349956
## sample_68	11.001844	3.0255427	10.598629	3.429536	9.634101	10.566806

## sample_69	9.608878	1.7050490	10.150534	5.484325	10.694332	10.620018
## sample_691	8.822940	1.6983518	11.292253	3.941247	10.023546	10.990189
## sample_701	9.783029	1.2536263	10.611440	3.425755	10.173265	10.252902
## sample_735	9.349684	0.9197596	9.092406	4.815048	9.801413	9.898290
## sample_74	8.527809	2.1562024	11.770259	4.433741	11.912410	10.250642
## sample_740	10.066264	2.5430009	10.827359	3.937203	9.611849	10.535839
## sample_75	10.625362	1.4264258	10.363850	2.531569	9.816509	10.346181
## sample_762	7.984219	5.1463416	10.347810	2.077516	9.605887	9.626660
## sample_768	9.679763	2.6191547	11.733282	5.134085	9.487046	10.814094
## sample_793	9.616639	2.7711884	10.975927	4.743833	9.455368	12.277563
## sample_800	10.194511	2.6869039	10.188638	3.083894	9.232083	10.995343
## sample_82	10.080031	4.2582023	9.817957	5.478254	9.651335	11.033031
## sample_95	10.334340	2.6802793	10.045255	4.629718	9.805837	11.651150
##	gene_7618	gene_5438	gene_19914	gene_7850	gene_7489	gene_709
## sample_119	7.808514	8.792777	10.661003	1.8775872	2.6665748	8.032988
## sample_275	7.979082	9.888462	11.068180	2.1720714	7.8127934	6.874133
## sample_29	3.386094	9.200793	1.939753	2.5951456	0.0000000	7.908128
## sample_295	7.333916	10.915872	2.088549	2.8159033	12.2578239	6.900963
## sample_298	6.597385	11.083885	4.992682	1.5799525	4.0517897	7.825372
## sample_404	5.837810	9.696552	10.536315	2.9330438	7.4863845	6.490861
## sample_464	6.334604	9.603088	9.089609	1.4307670	7.9345224	6.861943
## sample_468	2.356341	11.127291	1.405067	2.3563410	13.1498852	7.738970
## sample_531	5.100999	10.207746	8.670692	2.6149925	2.7309442	7.185470
## sample_57	3.896388	10.310306	7.921496	1.5774893	0.0000000	7.605961
## sample_578	7.732961	11.445667	10.495585	1.5483873	4.8535716	6.515252
## sample_579	5.512635	10.235416	11.428151	1.0879380	2.3582837	6.764102
## sample_603	8.598321	9.790270	11.264243	0.4365883	0.0000000	7.160366
## sample_627	6.831510	10.501797	11.303684	1.3163196	8.1674834	8.524980
## sample_638	6.534443	10.129090	10.879912	0.6050202	0.3339961	8.606102
## sample_673	6.567810	10.561240	10.442767	0.5621814	2.5272455	7.297228
## sample_68	1.324119	11.439083	3.703776	0.8089204	11.9503335	7.284357
## sample_69	6.750754	10.746892	5.996368	2.2068307	10.3187913	6.927245
## sample_691	5.520611	11.628842	4.201720	0.8812731	7.3382461	6.388799
## sample_701	6.564956	10.634657	10.962838	0.5474498	0.0000000	7.506962
## sample_735	8.238261	11.987530	10.873390	1.8778620	3.1009776	7.000856
## sample_74	2.393773	9.750938	9.084864	0.6153221	4.0378654	8.080450
## sample_740	7.778990	10.043342	9.508397	4.2944222	1.2505978	7.801469
## sample_75	7.616314	11.190356	3.401084	1.2669364	2.2193389	6.691967
## sample_762	8.918434	10.594931	11.301010	2.8407855	6.0383330	8.118277
## sample_768	6.685281	9.661552	10.796989	1.1928254	1.5504082	8.451788
## sample_793	7.436712	10.310749	11.572255	1.4961044	0.4480552	7.283283
## sample_800	7.468388	10.076829	10.286130	2.8317151	6.8483100	7.083447
## sample_82	5.831928	10.495425	10.759247	0.0000000	1.8443053	6.933702
## sample_95	5.537368	9.730667	11.513417	1.9579889	4.1366836	6.636018
##	gene_18075	gene_10016	gene_11979	gene_627	gene_16516	gene_4144
## sample_119	4.698380	5.161505	9.250355	7.590056	10.90254	5.035532
## sample_275	3.371740	7.295126	9.298915	7.812793	10.40302	8.163891
## sample_29	3.428933	3.666972	7.238558	8.048694	11.14977	3.386094

## sample_295	4.221877	6.875841	10.344740	7.274523	10.30184	8.536814
## sample_298	5.765174	4.595641	9.001822	9.507499	10.01575	6.500324
## sample_404	3.694535	7.158892	9.735092	7.542243	10.48290	6.677043
## sample_464	3.864632	8.005136	8.058024	6.140181	10.80084	6.659011
## sample_468	4.879368	7.098474	11.122750	7.614055	10.14851	7.344589
## sample_531	4.391788	4.461928	7.966563	6.995688	10.01107	6.375382
## sample_57	4.426667	6.639429	7.941048	5.787688	10.11158	6.243132
## sample_578	6.371723	6.155310	8.823972	6.680043	10.70473	6.950865
## sample_579	3.322736	7.948128	8.510179	7.239303	10.31492	7.488740
## sample_603	4.561619	5.145649	8.757400	7.493071	10.65617	4.922607
## sample_627	5.404501	7.974088	8.862963	8.319916	10.48801	5.839438
## sample_638	4.143402	5.275029	9.024309	7.251255	11.03214	6.827019
## sample_673	3.690629	6.298442	10.041673	8.808671	10.01402	8.849480
## sample_68	4.438073	6.985739	10.915177	7.322135	10.55315	8.055489
## sample_69	5.028817	8.093682	8.873143	6.036666	10.11921	6.911080
## sample_691	4.804446	6.705149	10.869193	7.427975	10.52845	7.430327
## sample_701	4.503921	5.191859	7.918183	8.553483	10.39070	9.311789
## sample_735	4.514009	5.723143	9.048130	7.282820	10.23734	8.929128
## sample_74	5.745764	6.373420	6.030374	8.560700	11.13516	4.593252
## sample_740	4.439051	5.592526	9.717269	7.961722	10.83555	6.524225
## sample_75	4.428578	5.548618	9.627302	6.594547	10.43505	6.019117
## sample_762	5.592705	6.140611	9.233624	7.608587	11.36899	7.198533
## sample_768	5.611830	5.893615	8.242193	7.892500	11.04411	6.402378
## sample_793	3.484989	5.027225	8.515282	8.330155	10.94668	5.816426
## sample_800	5.042456	7.448025	9.308325	6.618045	10.52654	6.686304
## sample_82	5.558170	4.723062	8.042442	6.883938	11.29494	4.693320
## sample_95	3.159371	6.599999	9.633742	7.344935	10.34317	6.915724
##	gene_18687	gene_11263	gene_15850	gene_7872	gene_2555	gene_2645
## sample_119	9.832648	9.525632	11.45873	0.0000000	10.047615	5.972987
## sample_275	9.174566	9.005456	11.76858	0.6640277	8.034804	4.998918
## sample_29	7.563387	8.515432	12.43192	1.1767057	8.353985	4.245199
## sample_295	8.903132	8.230198	11.30549	1.5155106	7.718341	2.717408
## sample_298	9.336013	9.578401	11.60021	1.8689633	8.446426	7.713579
## sample_404	8.936455	9.564344	11.91544	1.2187192	11.099585	4.355369
## sample_464	9.621841	8.742623	11.19275	2.1347795	9.113365	5.171847
## sample_468	8.449619	8.214504	11.67668	1.9577289	7.273357	5.187384
## sample_531	9.939063	9.039363	11.21647	1.3436352	9.034491	5.410155
## sample_57	9.648753	9.031299	11.56065	1.8662750	8.751427	6.588772
## sample_578	8.696481	9.725932	11.01456	2.7601784	7.878725	3.342469
## sample_579	8.971365	9.665386	11.67093	0.0000000	9.272947	3.895661
## sample_603	9.134763	9.823924	11.89127	1.4682184	9.814251	5.890055
## sample_627	8.616843	9.840312	11.28159	0.4569113	10.591616	4.450512
## sample_638	9.298867	9.551091	12.43311	1.7417905	9.963561	6.684482
## sample_673	8.972265	8.754821	10.61086	0.5621814	9.943861	6.763982
## sample_68	8.093708	8.555210	11.56900	0.0000000	8.099653	6.581002
## sample_69	10.382721	11.013965	11.40458	2.9599927	7.569620	3.391452
## sample_691	8.941335	9.369150	11.04178	2.0311481	6.504811	7.418536
## sample_701	8.598834	9.443513	11.29756	0.0000000	10.955149	5.313366

## sample_735	9.376062	9.152102	11.04637	0.5319678	9.042042	3.023983
## sample_74	9.193708	10.381197	11.44306	0.3402774	9.440074	3.208627
## sample_740	9.667358	9.639420	11.82583	3.7823771	9.450554	2.704363
## sample_75	9.862137	10.117656	11.57108	1.0875986	7.919275	6.903352
## sample_762	8.713032	9.379870	10.69083	2.9406930	9.651751	4.429422
## sample_768	8.871748	9.272648	11.23034	1.8366917	10.022451	3.858250
## sample_793	8.977031	9.208605	11.74441	1.4961044	9.632612	6.501199
## sample_800	9.432763	9.925292	11.17045	1.8951475	8.779854	5.100788
## sample_82	7.698691	9.883063	10.88750	1.0259501	9.307613	1.353041
## sample_95	8.725618	9.601682	11.85926	0.7834986	8.556276	4.165968
##	gene_8883	gene_18690	gene_15353	gene_10860	gene_11037	gene_9032
## sample_119	11.216328	7.483888	7.761870	10.68073	7.048236	7.184161
## sample_275	10.243162	9.496287	9.382142	11.48364	7.229165	5.195808
## sample_29	11.223688	8.238295	8.315965	11.15934	10.543960	8.351200
## sample_295	9.663626	8.008625	7.567454	11.41121	8.607279	6.270847
## sample_298	9.587395	7.662547	9.500250	10.92737	6.188947	5.484467
## sample_404	10.886817	10.185210	8.888771	11.19879	7.482816	6.664511
## sample_464	9.868608	8.719652	9.906491	10.53779	6.804738	6.128572
## sample_468	9.989182	8.037091	6.815985	11.10614	7.450345	7.710868
## sample_531	9.917047	8.592382	9.893914	10.89121	6.679466	5.942271
## sample_57	10.213955	8.487289	9.986411	10.99367	7.897682	6.899732
## sample_578	10.882941	7.732961	8.884497	10.45462	8.446798	4.305095
## sample_579	10.421781	11.093576	9.004417	11.71429	6.595217	6.655529
## sample_603	10.320507	7.501535	8.578456	11.40135	7.094753	5.864033
## sample_627	10.286719	7.493535	8.311485	11.32185	8.388891	6.245804
## sample_638	10.203018	8.750965	7.882472	11.12610	6.913727	7.261089
## sample_673	9.868651	7.744565	8.760351	11.08469	6.672609	4.063814
## sample_68	9.752638	7.664483	7.765144	11.37334	7.263335	7.079058
## sample_69	10.623333	6.815883	9.023585	11.86364	7.531350	4.766627
## sample_691	10.515237	8.857396	8.703703	11.31817	6.953382	4.602588
## sample_701	9.853807	7.332072	8.574590	10.47122	9.085403	5.378827
## sample_735	10.013546	7.409298	9.210033	11.15363	7.772084	3.243623
## sample_74	10.411966	5.554914	7.240944	11.87684	5.808725	7.580635
## sample_740	9.593927	5.032405	9.560061	10.92808	8.146889	6.834319
## sample_75	9.996120	8.117866	8.947041	11.45881	6.530141	6.467319
## sample_762	9.967414	7.638646	8.548602	10.17461	8.243374	4.792397
## sample_768	10.680017	7.565247	7.763969	11.41680	9.108621	5.847559
## sample_793	10.153577	8.167915	8.381474	11.65686	6.726286	5.889147
## sample_800	10.364091	10.227616	9.544692	11.41225	7.803033	6.890252
## sample_82	10.561718	7.996728	7.877812	11.67123	5.353987	6.613194
## sample_95	10.540525	10.596712	9.060818	11.25953	6.962364	5.592322
##	gene_1938	gene_17759	gene_18065	gene_9411	gene_10519	gene_1121
## sample_119	6.090896	10.806227	5.462399	12.61054	9.769529	9.042480
## sample_275	7.106034	10.580899	5.728367	12.28834	8.514722	9.772740
## sample_29	7.014244	8.914573	3.044167	12.07110	7.728322	11.185352
## sample_295	6.387559	9.747137	8.661269	12.04442	8.689442	8.967494
## sample_298	7.572427	10.345239	4.711550	12.23939	9.598652	10.448467
## sample_404	6.518976	9.238636	6.566896	12.25990	8.410723	9.035882

## sample_464	7.203691	11.280278	4.254692	12.34977	8.163871	9.196337
## sample_468	6.642380	9.407848	8.435954	12.51543	8.043503	9.085576
## sample_531	7.927606	11.252098	2.027048	12.21587	9.272800	10.617146
## sample_57	7.337542	11.276654	3.614144	12.54373	9.317361	11.031550
## sample_578	7.197836	10.388308	6.230927	11.20869	8.212686	9.712226
## sample_579	6.384596	9.903088	5.648342	12.01025	7.913135	8.185153
## sample_603	6.731998	10.133027	4.988321	12.46307	8.038145	8.570433
## sample_627	5.659722	9.492005	7.189162	12.54101	8.424464	8.379235
## sample_638	7.263542	9.619618	9.917670	12.83141	8.355695	7.989860
## sample_673	6.357563	9.671177	7.634717	11.89103	8.776663	8.845004
## sample_68	6.815755	9.881275	7.518307	12.50031	8.956370	8.182265
## sample_69	6.838851	10.541677	6.798401	11.79022	8.962665	10.577117
## sample_691	6.548723	9.970465	5.342462	10.85134	8.269716	9.098061
## sample_701	5.874881	9.726782	7.450989	11.91241	8.374222	8.319279
## sample_735	5.848671	10.471015	6.246180	11.07738	8.001431	9.203502
## sample_74	7.923357	9.183186	7.424124	12.47282	9.463612	10.407172
## sample_740	7.832358	11.046660	4.591284	12.29981	9.621387	9.173070
## sample_75	6.869045	10.256138	3.194954	12.33345	7.756903	9.554155
## sample_762	5.779680	10.256857	8.188544	11.68683	8.579595	7.115460
## sample_768	6.563068	9.419566	7.323172	11.89717	9.534451	8.580944
## sample_793	6.148119	9.447864	4.981391	11.96141	7.893732	8.073552
## sample_800	6.831181	10.533174	6.230933	12.13852	8.011294	9.691897
## sample_82	3.748032	9.707849	5.239925	12.05885	8.762292	9.556236
## sample_95	5.978019	9.951680	5.267083	12.35600	8.083697	9.396558
##	gene_12834	gene_20503	gene_13249	gene_11030	gene_15035	gene_15074
## sample_119	10.434378	7.175285	4.257599	6.5364819	10.856348	5.308630
## sample_275	4.212071	6.978116	6.852411	1.9716999	10.183499	8.088078
## sample_29	3.894042	7.262443	3.470550	1.9397531	11.267875	10.380212
## sample_295	4.185494	6.411386	6.065934	0.5506052	9.879729	7.075158
## sample_298	6.846343	6.277946	6.065807	6.2467862	10.151067	7.680753
## sample_404	4.663134	6.923078	7.186758	0.9145645	10.333580	6.497940
## sample_464	5.102331	6.833624	5.280752	9.1250957	10.409953	10.543805
## sample_468	4.405421	6.556563	6.683444	0.0000000	9.771793	8.323203
## sample_531	6.693835	7.278914	5.803503	12.6758711	10.319029	9.493291
## sample_57	6.507192	6.528023	5.623176	9.0841600	10.745582	10.603747
## sample_578	7.851930	6.437283	6.739727	11.2497475	10.690513	4.439158
## sample_579	3.375860	6.808243	7.461348	0.0000000	10.140766	7.488137
## sample_603	6.177481	7.022090	6.469764	6.1633256	10.247322	6.858901
## sample_627	7.352653	7.101860	6.112321	7.5831580	10.518014	6.640002
## sample_638	7.314008	6.983700	6.727675	7.4010758	10.676565	8.523542
## sample_673	7.760507	6.446948	7.070529	8.7761139	9.696251	8.526789
## sample_68	5.842753	6.796364	6.053405	0.4603756	9.489184	8.624031
## sample_69	3.818677	7.065131	4.692896	10.7780525	10.856838	4.415732
## sample_691	6.565920	6.672963	7.307902	4.9159463	10.002787	6.313975
## sample_701	7.599117	7.206985	7.646408	5.0774225	10.346890	5.378827
## sample_735	7.069520	6.611116	8.197207	8.7314310	9.575560	7.386414
## sample_74	3.166507	7.004434	3.289333	4.6246975	11.035033	5.617916
## sample_740	7.174536	7.143200	7.548614	8.5178623	10.533154	4.046020

## sample_75	7.520108	6.723026	5.413848	1.0875986	10.370295	5.406499
## sample_762	7.099337	6.986991	6.060709	9.3506210	9.864644	6.295819
## sample_768	6.572846	7.936709	5.790045	8.4464796	10.652908	6.948227
## sample_793	7.326474	7.316616	7.266318	6.3612244	10.219834	4.514122
## sample_800	2.686904	7.104976	6.597933	1.2386646	11.013965	6.750754
## sample_82	7.792719	7.056714	5.870772	5.5260845	10.569618	4.370883
## sample_95	3.037927	6.577293	7.612883	0.4443490	10.244114	10.346325
##	gene_1315	gene_2557	gene_16199	gene_6374	gene_2868	gene_14488
## sample_119	6.689705	10.406545	10.352143	9.191442	6.2570653	9.272014
## sample_275	9.649023	9.951132	9.395975	7.633322	5.3064773	9.183293
## sample_29	7.259480	9.610377	9.341301	10.016349	4.4874092	8.566781
## sample_295	5.815160	9.592457	7.649881	8.735505	8.0622890	10.836689
## sample_298	7.162422	8.341199	8.981145	9.058549	7.3412208	9.943252
## sample_404	9.257930	10.217206	9.889942	8.444373	6.8307624	9.879813
## sample_464	11.719316	9.458729	9.197725	8.715777	4.0728090	9.053228
## sample_468	4.859025	9.924584	7.115127	10.157903	6.2940675	11.973113
## sample_531	12.418754	8.789106	9.698595	9.135255	4.4937882	9.658893
## sample_57	13.204132	9.459757	9.271848	9.048560	5.1722755	9.696691
## sample_578	6.337791	9.914248	9.117329	7.530063	4.2330904	9.852505
## sample_579	8.572780	10.287458	10.097940	8.147470	6.4479742	9.638885
## sample_603	8.325476	9.143802	9.786812	7.138825	3.3486998	10.573278
## sample_627	7.853328	10.315263	9.430237	8.361369	7.2714817	10.267220
## sample_638	8.774329	9.276618	9.606662	8.536958	6.2234592	9.880675
## sample_673	7.798051	9.168993	9.604750	8.173877	4.3261778	10.135106
## sample_68	5.311859	9.498961	8.172897	8.920231	7.8731918	11.229870
## sample_69	6.905652	9.220110	9.292621	7.586690	3.3279173	10.315218
## sample_691	6.599712	9.353370	9.308735	8.133502	3.0029905	10.281814
## sample_701	7.773555	9.725209	9.708598	7.791690	3.7002399	10.373735
## sample_735	6.749078	8.600006	10.119693	7.397897	0.9197596	10.130069
## sample_74	6.615806	9.203505	7.074431	10.047737	7.0998212	8.662686
## sample_740	8.335824	10.060210	9.484287	7.409221	3.9844891	9.826509
## sample_75	7.657097	7.878468	9.934422	8.698170	2.8456908	10.288301
## sample_762	6.115179	8.335725	9.155208	8.149930	3.4275928	9.051484
## sample_768	8.154195	9.993165	8.761714	9.005004	6.9178613	9.932287
## sample_793	7.262903	10.046865	9.465064	8.087145	2.6920924	9.901730
## sample_800	8.162698	11.079345	9.938737	8.236105	4.7091031	9.477483
## sample_82	7.322622	10.401285	9.321700	8.619310	2.6280036	10.502146
## sample_95	9.360601	9.715276	10.243686	9.111550	8.0439350	9.548115
##	gene_7176	gene_16314	gene_17829	gene_7988	gene_18832	gene_7643
## sample_119	10.481920	7.256039	4.577888	13.79774	3.987212	9.279279
## sample_275	11.974970	6.978116	6.517123	13.28245	5.327630	8.691890
## sample_29	11.047574	5.593171	6.035791	13.97570	4.566298	9.245555
## sample_295	9.404416	8.390646	5.126225	12.79140	2.717408	7.979700
## sample_298	10.705546	8.460071	4.314755	13.19207	3.108792	8.529981
## sample_404	11.500354	6.707911	5.913895	13.32640	4.611839	8.517425
## sample_464	11.636951	7.996507	5.728106	13.54324	4.561424	8.378624
## sample_468	8.590456	8.059550	4.838392	12.85904	3.331361	7.600240
## sample_531	10.938705	7.397700	5.558760	13.12384	5.918012	7.960639

## sample_57	11.364332	7.832187	5.543573	13.36582	4.879897	8.196219
## sample_578	10.964074	7.768423	5.864597	13.81078	4.305095	8.326474
## sample_579	11.072870	8.187015	5.267667	13.16899	5.439673	8.032112
## sample_603	11.237252	7.760893	5.846421	13.87498	4.743897	8.690567
## sample_627	11.005203	6.914062	3.606122	13.66782	3.515031	8.846230
## sample_638	10.531762	6.913727	5.379836	14.00793	4.890213	8.741787
## sample_673	10.601613	7.556537	5.862375	13.32198	4.143769	8.733144
## sample_68	9.480859	8.490438	4.603377	13.43665	3.270021	7.674870
## sample_69	10.788784	8.177056	6.046567	13.38818	4.504608	7.812363
## sample_691	10.378663	8.236440	3.826915	13.11283	3.436055	8.328635
## sample_701	10.628783	10.643838	4.020449	13.79458	4.282343	7.380626
## sample_735	10.503726	8.020208	4.541893	13.06777	4.747446	8.391703
## sample_74	10.194437	7.934381	5.334915	14.22321	4.477457	9.369157
## sample_740	11.745695	8.045083	5.971072	13.63391	7.118349	7.472220
## sample_75	10.706289	7.584955	4.428578	13.97260	5.550005	8.092250
## sample_762	10.702909	8.411680	5.752446	13.44964	2.617487	8.441637
## sample_768	11.023990	9.025621	5.685778	13.49756	4.198541	9.396187
## sample_793	11.168089	7.927985	5.211495	13.79916	4.895744	8.699330
## sample_800	11.975637	11.196823	4.782220	13.17138	5.100788	8.480879
## sample_82	11.850753	10.564197	5.239925	13.58846	4.664130	9.031268
## sample_95	11.483927	7.360856	5.559602	13.01670	5.225965	8.496678
##	gene_8931	gene_10553	gene_5733	gene_16020	gene_6205	gene_14146
## sample_119	8.942327	6.581945	6.9160227	1.4766927	4.426614	9.717185
## sample_275	9.428155	9.227568	7.4669615	3.6655178	5.409262	10.150496
## sample_29	9.844508	6.427824	6.4383478	2.1592417	4.039384	11.592303
## sample_295	10.298647	8.657954	7.6532177	3.5470796	4.942195	9.861825
## sample_298	9.644704	6.087075	3.8674320	1.8689633	5.665020	10.756214
## sample_404	10.076495	8.993929	8.0840324	1.8698319	5.038190	10.022285
## sample_464	9.386238	9.019054	3.9195594	2.7339833	4.898682	11.974884
## sample_468	10.636643	7.182156	8.4665252	2.2353288	5.068404	9.954849
## sample_531	9.674362	9.573142	3.4248952	4.5262441	3.940496	11.357601
## sample_57	10.025389	8.906635	5.2242603	2.6534957	5.716876	11.594381
## sample_578	9.010998	8.058749	7.5035237	2.8591745	8.074286	10.233140
## sample_579	9.879718	9.140132	8.3732219	2.0006851	4.687369	10.716331
## sample_603	8.525486	8.209712	7.4616022	2.4837771	8.757563	10.945978
## sample_627	10.057762	7.404333	6.3149502	1.3163196	6.585075	8.542386
## sample_638	9.255251	7.998703	5.6355745	1.3578335	6.315990	11.044462
## sample_673	9.352637	11.570738	4.4571280	3.3295542	7.541856	9.602098
## sample_68	10.290595	8.014127	7.4108583	2.8857696	6.499427	9.627260
## sample_69	9.671801	8.826498	5.3175501	2.3416158	5.397091	9.910752
## sample_691	8.577954	7.809974	7.1936042	1.8180325	6.491372	10.185656
## sample_701	9.338772	8.553483	7.9318780	2.2301108	8.113518	9.942406
## sample_735	9.181513	9.918371	4.9273371	5.0682108	7.879792	9.065042
## sample_74	9.616628	8.529887	6.6390903	2.8312692	5.579195	10.080351
## sample_740	8.958825	6.474738	5.9946702	1.7714631	6.017891	10.614102
## sample_75	9.301720	7.274895	0.8827604	0.6439485	3.771695	9.347294
## sample_762	9.051484	7.912213	5.3778352	1.6110307	9.516638	10.477181
## sample_768	10.206428	7.582293	7.9592485	2.8295046	8.012306	10.197757

## sample_793	9.297345	7.306700	5.4704002	2.3236007	7.442769	10.337789
## sample_800	9.646112	9.659516	8.3121433	1.8951475	5.910541	10.692214
## sample_82	9.028666	6.159278	9.1295432	2.5016939	7.159649	10.412845
## sample_95	9.846532	9.145023	8.9282519	1.2884176	4.828307	10.115421
##	gene_16975	gene_14973	gene_2097	gene_3577	gene_11058	gene_7277
## sample_119	10.655960	10.130648	2.914163	8.886349	2.0938644	6.217283
## sample_275	10.541842	9.152510	4.165848	8.759875	4.2120712	5.327630
## sample_29	9.797765	9.789054	4.221019	8.630121	0.0000000	4.659200
## sample_295	11.049672	9.485220	5.554868	8.054745	0.0000000	2.815903
## sample_298	10.241959	9.627871	5.693030	8.109684	0.0000000	5.577565
## sample_404	10.326137	9.327761	3.879392	8.478778	3.5923853	7.044110
## sample_464	10.576758	9.327788	3.482229	7.070486	5.9288654	3.328363
## sample_468	11.057322	9.987420	5.770026	7.962786	0.0000000	4.161485
## sample_531	9.770639	8.886081	5.143467	7.491091	5.1642392	6.311638
## sample_57	10.312554	9.743087	4.779228	7.802575	5.3465705	6.011285
## sample_578	8.968667	10.245196	4.439158	7.548722	3.2723517	5.372378
## sample_579	10.282683	9.256225	3.966624	7.988162	4.2769805	5.135301
## sample_603	10.192170	9.806301	4.138143	8.498957	3.4915068	3.953042
## sample_627	10.694131	10.197069	3.732823	8.565270	1.0824982	5.990798
## sample_638	10.575766	10.287228	3.698563	8.289189	1.6248028	4.516148
## sample_673	9.967399	9.570804	5.561265	7.950334	0.5621814	6.672609
## sample_68	11.200445	9.309383	5.378564	8.099653	0.4603756	4.132692
## sample_69	10.319808	8.956902	3.391452	8.424905	0.5381408	4.321012
## sample_691	10.203960	9.558453	6.486864	7.318082	0.0000000	7.340740
## sample_701	10.667777	8.576337	4.616758	7.636407	5.5160877	3.648086
## sample_735	10.083985	8.780530	6.465855	7.245153	3.6545387	5.935318
## sample_74	10.776811	10.307451	4.192762	8.801589	0.0000000	3.754888
## sample_740	10.618853	8.837915	5.518268	7.623691	2.0369253	5.612801
## sample_75	10.004445	8.545791	6.025361	7.524518	0.0000000	6.783221
## sample_762	10.510072	9.272487	4.818897	7.969513	3.8435417	6.351332
## sample_768	10.878580	10.507547	3.496680	9.307647	5.6118304	4.120692
## sample_793	10.612030	9.488658	4.823367	8.769980	1.4961044	4.624733
## sample_800	10.262365	9.739720	5.385241	8.131559	4.7091031	5.877553
## sample_82	9.682600	9.102600	3.045146	8.596316	5.2980827	5.573950
## sample_95	10.335547	9.653949	4.523549	8.429206	3.0999179	4.430098
##	gene_9657	gene_12258	gene_4846	gene_18471	gene_8553	gene_8199
## sample_119	7.015638	5.136491	5.574610	10.96092	6.182287	4.426614
## sample_275	7.000744	7.433202	7.624803	11.60690	5.218630	6.028788
## sample_29	5.166069	10.839259	9.822582	11.21070	5.897715	2.738314
## sample_295	5.297786	9.592457	8.978170	10.57843	3.901311	8.729199
## sample_298	5.978596	3.215648	6.854632	11.16383	7.072878	8.472248
## sample_404	6.513745	7.704056	6.701785	11.76954	4.386225	5.237476
## sample_464	5.636152	3.621267	7.542382	11.69320	4.871085	5.004097
## sample_468	5.704728	6.877953	7.288654	10.97288	5.297298	10.007196
## sample_531	5.355207	4.678404	7.190536	11.57875	4.391788	4.707000
## sample_57	5.573341	5.298882	7.121534	12.11190	3.830833	4.633053
## sample_578	6.153556	8.164163	6.574730	12.28828	6.183955	3.992605
## sample_579	6.683500	10.031219	7.851868	12.22200	3.476550	5.701324

## sample_603	5.770848	3.885662	5.846421	11.80167	5.819592	3.397885
## sample_627	6.898099	6.522136	5.013310	11.64232	4.739195	5.801391
## sample_638	6.876689	6.752294	5.627995	11.48836	6.727675	4.596017
## sample_673	5.883372	6.140885	6.545907	11.85956	6.819809	7.369676
## sample_68	6.254265	9.823786	7.688341	10.81398	5.661752	9.965600
## sample_69	7.208820	6.169454	6.315816	11.13837	4.588331	4.288012
## sample_691	7.419269	5.127637	8.045820	11.55809	6.304221	6.486864
## sample_701	6.201687	6.303562	4.865399	11.60073	6.207735	5.530564
## sample_735	6.714699	5.791749	7.860541	11.35492	5.997048	6.807664
## sample_74	6.053376	5.521892	6.030374	10.28364	6.989412	3.438346
## sample_740	6.653747	4.506272	5.270133	12.05020	5.872612	4.439051
## sample_75	6.720689	4.290233	5.541317	11.85791	7.653067	5.460677
## sample_762	6.214723	8.631290	5.920924	11.57054	7.796981	5.752446
## sample_768	6.206715	3.724290	5.294896	11.24895	5.171631	4.612877
## sample_793	6.343024	4.343465	2.917432	10.76660	5.966802	4.290756
## sample_800	6.561218	9.499332	6.848310	11.43333	5.129093	5.910541
## sample_82	5.644690	9.694377	4.804498	11.79736	3.802628	5.372151
## sample_95	6.608289	8.660246	7.023632	11.79016	3.842396	4.917093
##	gene_7522	gene_12313	gene_6299	gene_18241	gene_13023	gene_16180
## sample_119	8.334684	7.593458	2.7325731	10.721415	3.9168588	9.596192
## sample_275	7.438077	4.542425	1.1169635	8.599571	5.2026920	8.116521
## sample_29	4.746479	7.409662	0.7051375	11.724880	2.8553918	6.830699
## sample_295	6.162697	3.840836	0.9482270	8.258957	7.8820438	7.979700
## sample_298	2.932023	6.640123	0.7339617	9.683035	2.5786731	8.167564
## sample_404	5.803932	4.156024	1.8698319	8.838060	4.2586395	8.370631
## sample_464	8.170766	7.810205	1.7054029	9.222510	0.0000000	7.947795
## sample_468	5.854649	3.705867	0.0000000	8.864901	3.6083362	8.714744
## sample_531	6.470189	7.781714	0.0000000	9.864068	0.5970305	7.726286
## sample_57	4.554006	7.142291	0.0000000	9.514278	2.4932374	8.552289
## sample_578	3.703854	7.633642	0.5667665	8.606720	4.2422900	7.733226
## sample_579	4.532242	5.264289	0.0000000	8.874345	3.9137320	8.238075
## sample_603	10.484622	5.479567	1.4682184	8.811888	4.4808209	9.111081
## sample_627	8.833810	6.296057	1.3163196	10.100005	7.3717679	8.882674
## sample_638	10.395823	7.089477	0.3339961	9.932746	6.1140421	9.111326
## sample_673	2.266577	4.982902	0.0000000	9.406303	3.4788295	9.127463
## sample_68	4.689970	4.503381	0.4603756	9.260698	4.7700732	8.234037
## sample_69	7.417701	6.423988	0.5381408	8.806243	2.2717543	7.367240
## sample_691	7.827908	4.478758	0.0000000	8.092556	4.2261540	7.827908
## sample_701	11.074175	6.146368	1.2536263	8.636124	2.6238201	8.477070
## sample_735	4.596088	6.451594	1.4769001	7.866067	5.4081174	7.860541
## sample_74	8.432187	8.069810	1.3761237	11.628641	0.8463128	8.222433
## sample_740	6.758237	6.753444	1.7714631	9.434824	2.1159654	8.604839
## sample_75	7.407489	5.610730	0.0000000	9.276422	2.9161900	7.974466
## sample_762	6.104448	7.337122	2.3531539	9.094144	4.5103417	7.627038
## sample_768	7.844875	4.987694	0.7163325	9.593541	5.0917337	8.880630
## sample_793	5.280708	5.949647	0.7893537	9.538257	5.0333349	9.051201
## sample_800	7.322811	6.176369	1.8951475	9.548627	7.9441791	7.730986
## sample_82	7.313255	4.873193	1.0259501	8.918658	5.5023656	8.881383

## sample_95	7.404503	2.401794	1.8174595	8.418603	4.1992012	8.712706
##	gene_20470	gene_13572	gene_1760	gene_18896	gene_4396	gene_18597
## sample_119	8.824332	8.641318	9.302209	8.898976	5.392675	9.020358
## sample_275	7.598946	8.061436	9.565801	8.688093	7.541267	9.431821
## sample_29	9.409079	8.955446	10.617293	10.855305	3.662308	7.214261
## sample_295	7.718341	6.753417	10.210854	9.871163	5.962211	9.435733
## sample_298	9.300796	6.567434	9.333790	9.189676	6.816945	9.756773
## sample_404	7.866049	7.895836	8.914134	9.065971	6.689467	9.346514
## sample_464	8.431548	7.284385	8.871893	9.491953	4.814243	8.733131
## sample_468	7.497165	7.012960	9.913530	9.158130	7.062726	9.733897
## sample_531	8.240424	4.121339	9.507596	10.032404	4.243502	8.828682
## sample_57	8.260242	6.558718	9.242214	9.349973	3.449205	9.646373
## sample_578	8.114924	6.249297	9.353217	9.398156	7.488137	8.480233
## sample_579	8.007224	8.923114	9.530332	9.172390	7.356249	8.052785
## sample_603	8.417806	7.523868	9.759429	9.482784	5.981643	9.038192
## sample_627	9.772638	7.928814	9.189113	8.826723	7.148070	9.571159
## sample_638	8.972133	6.983700	9.902829	9.630566	5.267244	11.064258
## sample_673	9.838186	7.556537	9.682933	9.456087	5.486573	9.023358
## sample_68	6.858454	5.730988	10.019896	9.058887	7.004996	8.920231
## sample_69	7.707062	7.069992	8.827933	10.000085	6.205252	8.705166
## sample_691	8.948729	6.587132	9.972477	8.804044	8.463324	9.295264
## sample_701	8.978599	7.099432	9.831329	8.701553	8.854647	10.136479
## sample_735	7.702761	7.246722	9.087683	9.081760	7.826942	9.844729
## sample_74	8.780802	7.658669	10.905801	10.413099	6.445763	9.292469
## sample_740	8.714252	6.730382	9.097885	9.580351	8.833561	9.046120
## sample_75	8.088348	8.050583	9.615453	9.874726	7.515676	9.328480
## sample_762	8.899937	5.957160	9.908486	9.232392	8.418194	9.193683
## sample_768	10.455646	5.925940	9.987705	9.542210	8.414212	10.085287
## sample_793	10.122336	6.925276	9.976636	8.872179	8.699694	8.773584
## sample_800	7.912931	8.007448	9.261148	9.349507	7.188045	9.603877
## sample_82	9.455490	7.494144	10.040838	7.779168	5.896097	8.736111
## sample_95	7.628744	9.093568	9.433145	8.796374	6.544224	9.135802
##	gene_19431	gene_11847	gene_9539	gene_13007	gene_6376	gene_18128
## sample_119	9.620825	7.324514	4.066124	7.488668	10.370720	11.064851
## sample_275	9.575779	9.158425	7.289696	6.535418	8.285074	10.039495
## sample_29	9.371818	9.305820	4.694747	7.348657	9.978539	9.430256
## sample_295	9.502885	10.014829	9.340696	11.965139	8.835760	11.422002
## sample_298	9.131157	4.674562	5.265658	8.727172	11.268378	11.401130
## sample_404	9.525210	8.975023	7.142270	6.325593	7.863307	11.016042
## sample_464	9.550768	6.163124	5.712637	7.711440	9.389889	10.690679
## sample_468	9.862716	7.453740	6.794728	10.593046	9.538300	11.554934
## sample_531	9.451660	4.867615	6.101530	7.384482	9.340836	10.792489
## sample_57	9.174464	7.007498	7.029574	8.626603	9.127460	10.721979
## sample_578	8.974199	5.485172	5.155778	8.926483	9.306834	9.512835
## sample_579	9.502987	9.208525	8.668899	6.769071	8.007224	10.993045
## sample_603	9.449629	3.397885	3.814981	8.242064	9.119263	10.198691
## sample_627	9.826318	8.135345	5.876507	8.222505	8.750148	11.580301
## sample_638	9.285680	6.932309	5.333985	8.139966	12.881129	10.591195

## sample_673	9.086678	8.850968	6.914026	8.513739	11.643879	10.099203
## sample_68	9.626574	8.822985	7.752628	11.228061	9.139955	11.690823
## sample_69	9.866723	8.342915	6.617609	9.163230	8.095946	9.851929
## sample_691	9.325872	6.450287	5.391517	9.464145	10.919199	11.166828
## sample_701	9.688567	4.695949	4.020449	7.977497	10.506427	9.215404
## sample_735	10.264783	10.533106	7.912195	9.608894	8.037448	9.369270
## sample_74	9.304048	9.469058	2.776336	8.370918	9.650919	11.184119
## sample_740	8.959825	5.938991	4.819857	8.693971	8.429712	9.445567
## sample_75	8.571624	3.858190	5.047342	8.604316	10.650073	10.680544
## sample_762	9.697989	7.836423	7.083032	9.045885	9.782880	9.950285
## sample_768	9.684281	7.042721	4.888149	8.383985	10.213080	10.607247
## sample_793	9.372983	6.328912	4.089439	7.355633	8.883349	11.520972
## sample_800	9.823299	7.875829	7.247358	5.974337	9.311417	10.795707
## sample_82	9.830328	5.137463	4.777499	7.874631	6.883938	10.230729
## sample_95	9.750132	8.172127	5.893018	5.016888	7.322361	10.126627
##	gene_14913	gene_7097	gene_20028	gene_12774	gene_9732	gene_55
## sample_119	1.8775872	5.412944	7.258849	0.3754009	1.7559131	7.447505
## sample_275	6.6405718	7.112148	7.165982	2.6461857	0.6640277	9.258184
## sample_29	4.8759913	8.207864	9.360103	1.3650206	1.1767057	5.666617
## sample_295	1.7327902	4.985168	6.956010	4.3258397	2.0885487	8.187159
## sample_298	7.1987004	6.612131	8.074623	2.4089020	0.9962441	7.643539
## sample_404	6.5259185	7.019713	7.379266	1.2187192	0.0000000	9.815383
## sample_464	2.4652693	6.460330	7.954382	5.5127958	1.0912598	6.833624
## sample_468	0.0000000	4.957980	7.359143	4.1278809	0.4978423	7.747293
## sample_531	3.2019475	6.302295	7.555410	5.3204630	1.0180644	7.739706
## sample_57	2.6534957	6.791892	6.791892	6.6939051	0.0000000	5.838604
## sample_578	1.9588056	2.760178	6.174372	0.5667665	1.5483873	7.499694
## sample_579	5.4145077	5.349609	6.205461	1.0879380	0.0000000	8.631166
## sample_603	1.7963901	6.924753	7.717957	0.4365883	0.0000000	5.932408
## sample_627	2.7199938	4.400483	6.996355	0.0000000	0.4569113	6.224397
## sample_638	3.0060287	5.665501	7.710125	0.0000000	1.9506550	7.383177
## sample_673	2.4028040	7.193032	7.434687	0.0000000	0.5621814	7.519558
## sample_68	0.8089204	4.848973	7.000766	4.4128357	0.0000000	7.907480
## sample_69	3.7716955	2.683696	8.035085	0.9291850	0.0000000	7.182027
## sample_691	6.3545721	7.011026	7.160023	1.8180325	0.8812731	7.380167
## sample_701	2.3654686	6.679804	7.481113	2.7087611	1.2536263	8.728485
## sample_735	2.1912468	6.140728	7.424359	2.0430645	1.4769001	7.228088
## sample_74	2.7192053	3.208627	9.125277	0.0000000	0.6153221	8.708349
## sample_740	2.0369253	2.916916	7.480596	0.4274989	0.7567671	6.906602
## sample_75	5.1074368	6.920198	6.458116	1.2669364	0.3576083	7.876744
## sample_762	3.1218609	4.870473	8.123599	5.6963834	1.0196308	7.262677
## sample_768	3.4549918	6.270407	6.818633	0.0000000	0.7163325	9.523425
## sample_793	2.0967679	5.871306	6.386564	0.0000000	0.4480552	6.386564
## sample_800	1.6039289	4.324623	7.698302	0.0000000	0.7483754	7.660017
## sample_82	1.0259501	4.804498	6.694838	0.0000000	0.0000000	6.673019
## sample_95	6.3790109	6.911404	7.338522	1.4871261	0.0000000	9.282544
##	gene_17155	gene_15912	gene_20310	gene_14653	gene_8777	gene_9779
## sample_119	13.09857	9.528636	11.64658	9.685774	9.617554	8.679026

## sample_275	13.24099	9.164314	11.05001	9.891128	7.037086	8.887955
## sample_29	12.76697	7.844266	11.85344	10.469276	6.265913	9.016229
## sample_295	12.03869	6.355164	10.89701	9.755694	10.190343	7.023577
## sample_298	13.62824	8.593615	10.66193	8.967831	7.094035	7.607271
## sample_404	13.16619	10.750807	11.38216	9.345534	6.016068	7.947047
## sample_464	13.33131	6.413262	10.89078	9.418774	7.723067	9.361396
## sample_468	12.03179	6.396415	10.85361	9.672570	9.309151	8.482171
## sample_531	13.45816	4.917408	10.21895	9.470549	7.274140	7.925353
## sample_57	12.91708	5.480847	10.95717	9.075343	7.428988	8.830826
## sample_578	15.16377	7.777939	11.45460	8.839739	7.986678	7.668495
## sample_579	13.90418	11.203024	11.16227	9.726480	7.626607	8.452871
## sample_603	13.13474	9.240522	11.72423	9.182044	12.196906	7.989786
## sample_627	13.73330	8.589194	11.66551	9.120274	10.299998	7.621195
## sample_638	12.56684	10.128420	11.73475	9.285076	10.009857	8.278338
## sample_673	14.52896	8.400312	11.10512	9.459020	8.113107	8.613499
## sample_68	12.43941	4.830032	10.77922	9.483892	8.701871	8.164605
## sample_69	14.15211	8.601518	10.94978	9.367598	8.763358	8.794435
## sample_691	13.83453	7.142934	10.32696	9.308735	7.453608	9.049568
## sample_701	13.75255	7.788686	10.90811	9.152455	12.180810	8.384063
## sample_735	13.80736	8.017727	10.93317	8.957874	9.247557	10.239754
## sample_74	12.63349	10.527741	12.20900	10.492744	6.662006	6.230762
## sample_740	13.48743	10.033877	10.70543	9.223876	8.024214	8.603519
## sample_75	13.71788	7.876744	10.80667	8.699146	5.666444	8.511618
## sample_762	13.73225	7.183318	11.35492	9.366425	7.930554	8.790990
## sample_768	13.20083	9.322935	11.64538	9.176724	7.923399	7.940924
## sample_793	14.01909	8.892221	11.46650	9.418991	11.592135	6.906482
## sample_800	13.56879	11.181705	10.60106	8.768651	6.050310	9.519707
## sample_82	13.90274	10.617036	11.53176	10.446453	5.920986	8.012703
## sample_95	13.48392	10.856449	11.09468	9.465889	5.169256	9.351120
##	gene_13928	gene_5342	gene_15672	gene_6543	gene_19836	gene_3729
## sample_119	2.524766	9.521653	8.781130	8.893578	7.897507	4.198345
## sample_275	5.559853	10.509815	8.633940	9.352783	7.011171	4.089642
## sample_29	2.595146	9.869998	7.274243	9.737790	9.942950	5.982784
## sample_295	5.815160	8.795585	7.889273	13.627830	7.992921	5.067454
## sample_298	6.014931	8.946658	8.768346	10.209197	8.838996	4.290513
## sample_404	5.095325	8.985136	8.923366	8.939059	8.312039	5.485597
## sample_464	7.996507	10.129644	8.507466	9.516659	7.033610	3.972472
## sample_468	7.483945	8.247329	7.921692	14.424888	6.505130	2.571653
## sample_531	4.391788	9.614298	8.297953	8.812322	7.288405	4.319553
## sample_57	5.698616	9.565557	8.963179	9.519489	5.459317	3.959093
## sample_578	4.674845	10.039591	7.824628	10.532151	5.012573	3.806664
## sample_579	5.056159	9.197502	8.134041	9.288612	7.326205	4.578293
## sample_603	2.389649	9.952896	8.537882	10.327272	7.847552	2.063572
## sample_627	3.417785	8.639934	7.662241	10.500683	5.839438	3.258413
## sample_638	2.805602	8.023472	7.464162	9.763629	6.353244	3.535381
## sample_673	3.793355	9.053426	8.926376	10.312066	6.953382	2.527246
## sample_68	5.977969	8.959982	8.332838	13.922798	8.153983	3.969169
## sample_69	4.353281	8.755836	8.502490	9.276936	6.556558	5.008698

## sample_691	5.663265	9.739401	9.188277	10.798893	5.139175	3.700562
## sample_701	4.842375	9.441925	8.804902	9.613213	5.600838	4.138552
## sample_735	5.248235	9.510020	9.509138	9.027323	5.791749	4.747446
## sample_74	2.465008	8.123733	5.256784	10.093404	9.777413	2.984571
## sample_740	4.549404	8.949800	9.564668	10.068120	4.710096	3.317058
## sample_75	3.991735	9.467958	9.274457	10.750070	7.899012	2.728116
## sample_762	3.559149	9.313831	7.522534	10.121041	6.115179	3.121861
## sample_768	3.688762	8.987750	7.492422	9.708875	8.314298	5.307017
## sample_793	2.214653	9.541784	8.797396	8.858662	5.658971	2.608407
## sample_800	4.885384	8.725274	9.367439	8.743697	6.839771	4.220609
## sample_82	1.619507	9.796262	7.100957	9.737927	4.296751	2.628004
## sample_95	5.140042	9.677015	8.827632	8.788157	8.274816	4.632449
##	gene_14556	gene_6002	gene_13912	gene_2741	gene_311	gene_6207
## sample_119	8.587868	0.6730136	11.82899	9.629511	2.524766	12.834084
## sample_275	7.068832	3.0024145	10.38232	8.119102	7.723852	9.912283
## sample_29	8.865702	2.0536674	10.41603	9.536995	4.171415	11.087615
## sample_295	8.071988	2.3736760	10.60666	7.958292	2.611668	9.899519
## sample_298	8.466688	4.4739103	11.34155	9.290238	3.899843	10.344496
## sample_404	7.279945	3.5384761	11.10278	8.286507	6.364851	10.320665
## sample_464	7.967474	0.0000000	11.12498	9.281559	6.976467	11.001936
## sample_468	8.003984	0.0000000	10.82079	7.620110	2.571653	9.646103
## sample_531	7.466219	0.0000000	11.00282	9.297233	6.357455	10.739418
## sample_57	8.199923	0.0000000	10.91332	9.002784	5.322923	10.259049
## sample_578	7.261709	3.7038538	10.79424	9.538268	1.958806	7.812325
## sample_579	7.027419	1.5241640	10.92451	9.610772	7.685646	10.085539
## sample_603	8.139137	1.0427144	11.09208	8.717375	8.847737	9.815383
## sample_627	8.463741	1.9929859	11.28571	9.589456	4.798341	10.639910
## sample_638	9.361808	1.6248028	11.37859	8.665620	2.958472	12.584998
## sample_673	7.136355	5.0410590	10.65630	8.771404	5.777199	10.923446
## sample_68	7.885629	0.8089204	11.29089	7.887184	4.922355	9.510905
## sample_69	7.264302	2.6836965	10.08720	9.823468	5.638786	10.524072
## sample_691	8.185460	4.9686207	11.47490	9.227382	6.214375	9.424628
## sample_701	7.595683	5.7689835	10.33199	8.376690	2.807169	10.039851
## sample_735	8.203769	4.8401913	10.06249	8.397435	6.056248	9.263347
## sample_74	9.553501	0.6153221	11.01811	8.804167	1.871686	10.799160
## sample_740	7.723108	2.9813230	10.34515	8.504223	6.767390	9.788196
## sample_75	8.429875	3.2386035	11.29291	9.479437	2.303956	10.014676
## sample_762	8.345809	8.8574306	10.82926	9.183524	3.559149	10.770226
## sample_768	8.184439	2.2803328	11.02577	8.462347	3.858250	10.913144
## sample_793	7.176881	5.0142794	10.66638	9.694196	3.960382	11.443488
## sample_800	8.381638	4.0564014	10.87932	7.640975	7.001611	10.179362
## sample_82	7.130653	1.6195069	11.63173	8.611685	3.438985	9.979211
## sample_95	7.048487	2.7592411	10.81113	7.872472	5.830387	10.226858
##	gene_18545	gene_13457	gene_4739	gene_2845	gene_1386	gene_14005
## sample_119	11.01599	9.940860	9.489806	11.274075	9.815145	2.1909625
## sample_275	13.77177	10.365808	10.122026	11.445449	10.931099	3.6655178
## sample_29	10.79898	9.067313	12.486372	9.236658	9.669082	1.1767057
## sample_295	13.01520	10.515237	10.081936	9.070851	10.340095	3.5470796

## sample_298	12.40508	9.456662	9.722075	7.384810	10.507160	2.4962834
## sample_404	12.50594	10.614130	9.846919	11.472990	10.929673	3.0142839
## sample_464	14.07454	10.251719	9.185170	9.100050	10.583149	2.4711614
## sample_468	12.76780	10.583393	9.910445	7.300014	9.933730	1.6137202
## sample_531	12.92513	10.210866	9.062427	7.820447	10.389094	3.0315187
## sample_57	12.90255	9.824279	9.826383	9.831629	11.017804	2.7977186
## sample_578	13.00957	11.019973	8.501459	9.249815	10.722500	1.9588056
## sample_579	14.09091	11.214689	9.161087	12.270795	10.795829	1.8646121
## sample_603	12.80261	10.551026	8.750479	8.635972	10.692057	1.0427144
## sample_627	11.57219	10.030267	11.790646	10.529333	10.411543	2.3542547
## sample_638	12.09473	10.795520	10.276055	9.855337	10.035871	2.5081751
## sample_673	12.25634	10.851234	10.170726	9.178513	10.566682	3.6906294
## sample_68	13.97008	10.469560	9.792325	7.997242	10.118811	2.2507797
## sample_69	13.66389	10.576692	9.081316	7.355342	10.654421	2.4648773
## sample_691	10.74654	10.164995	8.340037	8.544729	10.626485	3.4729816
## sample_701	13.13840	10.770441	9.930857	9.022787	10.824306	1.2536263
## sample_735	12.55082	10.860303	8.740482	9.055402	11.058506	4.4615417
## sample_74	11.30475	9.873395	10.731268	6.930867	9.320345	2.5976980
## sample_740	13.02996	10.586540	9.764367	9.106032	10.818015	2.2610493
## sample_75	10.56636	10.232049	8.544706	6.068329	10.623049	2.2193389
## sample_762	12.02670	10.324878	8.942734	9.616312	10.506506	2.6174866
## sample_768	13.08323	10.026012	10.524023	9.220920	10.425279	2.4596677
## sample_793	12.82967	10.201217	11.260931	11.319458	10.471462	0.7893537
## sample_800	12.35646	10.215290	9.424223	10.543728	11.096017	2.9633076
## sample_82	10.42044	10.688985	9.492284	10.426485	10.667546	1.3530409
## sample_95	12.95294	10.669762	9.996177	11.977673	10.759539	2.6802793
##	gene_1897	gene_14645	gene_3003	gene_18785	gene_2696	gene_6920
## sample_119	9.417667	10.689317	9.914002	7.517032	7.175285	8.497756
## sample_275	10.072776	8.872099	8.648721	8.190111	9.228975	7.611932
## sample_29	9.387518	10.692302	6.960489	9.682977	3.924309	7.448348
## sample_295	9.186986	10.028376	9.226696	7.333916	10.039879	8.816555
## sample_298	9.928110	11.451196	10.369303	7.237258	7.018567	8.069289
## sample_404	9.809704	9.347493	8.565582	8.597244	8.709770	8.231500
## sample_464	10.236935	9.482864	9.179556	8.187649	7.845132	8.182042
## sample_468	9.248869	9.893044	9.279803	6.917826	8.209673	9.145588
## sample_531	9.410958	9.846503	9.535767	8.136504	7.571055	8.144368
## sample_57	8.868350	9.703575	9.440657	9.050360	7.349295	8.231956
## sample_578	10.814430	11.012939	9.353217	8.737721	5.194213	7.105070
## sample_579	10.758498	9.828243	8.392678	8.650154	9.352944	8.147470
## sample_603	9.801739	10.318916	9.093365	8.068730	4.939321	8.426726
## sample_627	9.243333	11.253262	9.314286	8.600318	7.752247	9.324381
## sample_638	9.589204	10.765178	8.897652	7.314008	7.483195	8.502140
## sample_673	9.578018	9.901366	8.556360	7.617313	7.698982	8.969525
## sample_68	9.318755	10.064635	9.844582	7.227587	8.607693	8.382438
## sample_69	9.973726	10.064837	8.447501	8.897966	7.045519	7.367240
## sample_691	10.836358	11.860237	10.292506	8.174726	7.615151	8.521777
## sample_701	10.684863	10.734947	9.372308	7.070003	6.902037	7.431840
## sample_735	10.054048	8.770254	9.060231	8.346961	6.501791	6.571084

## sample_74	9.595455	11.153970	9.411623	6.316931	6.927719	9.043290
## sample_740	9.722579	10.910613	9.000223	8.451195	6.121567	7.241821
## sample_75	10.389793	11.678138	9.860390	7.578599	6.396827	8.621935
## sample_762	11.078177	10.099598	9.954772	7.693375	7.387423	8.576046
## sample_768	10.352992	10.978739	8.667903	6.424146	8.615648	8.554048
## sample_793	10.343064	10.639956	9.334439	8.171562	6.000016	9.593025
## sample_800	10.102934	10.161485	8.932197	8.601949	8.903807	8.455956
## sample_82	9.814642	10.343197	9.170338	7.056714	5.805437	7.789345
## sample_95	10.444477	9.965276	8.685744	8.725618	7.881328	7.249000
##	gene_20264	gene_6118	gene_15916	gene_5263	gene_2226	gene_6544
## sample_119	3.903742	4.898508	8.775271	8.888159	1.755913	5.636248
## sample_275	4.889386	4.972307	8.967826	9.170180	4.342484	7.093740
## sample_29	6.679213	6.810584	7.331822	8.888908	2.257584	3.200708
## sample_295	4.828809	8.374579	8.297824	9.437669	3.547080	6.784151
## sample_298	2.578673	5.065921	9.368945	9.699239	2.867956	6.510854
## sample_404	4.558659	4.003980	7.195495	8.721916	4.783912	6.790954
## sample_464	3.061569	5.280752	8.640118	9.059355	2.605874	5.301580
## sample_468	4.058213	9.470364	8.940660	8.961082	4.405421	5.823495
## sample_531	3.940496	7.362181	9.027422	9.366834	2.027048	6.302295
## sample_57	2.493237	6.899732	10.097716	8.920490	1.577489	5.919667
## sample_578	5.136171	7.392214	8.875657	9.004254	5.075695	5.911488
## sample_579	6.655529	3.210264	7.253564	8.395898	5.023242	6.091002
## sample_603	3.621454	6.298997	7.962376	8.708446	4.352942	4.853756
## sample_627	4.057303	6.748958	8.343279	8.819604	1.082498	5.903699
## sample_638	4.548628	5.276969	8.789697	9.117246	3.052068	6.872632
## sample_673	5.604291	8.579904	9.267086	9.677906	3.026287	6.516174
## sample_68	5.442320	8.524710	8.989091	8.895387	3.270021	6.132842
## sample_69	3.723133	6.274288	8.906145	9.195847	2.873636	6.738592
## sample_691	3.798102	4.683601	8.047353	9.430507	3.708022	7.253214
## sample_701	4.138552	7.151869	10.784095	8.674108	3.891808	6.890860
## sample_735	5.859791	6.296151	10.762233	8.943573	3.889697	9.957069
## sample_74	4.840947	3.208627	6.788881	9.152500	1.645102	4.106298
## sample_740	3.413594	5.405410	8.822612	8.148644	4.747618	6.540351
## sample_75	5.788939	4.351155	9.392073	9.007428	3.939791	6.668218
## sample_762	5.207791	6.703890	8.335725	8.857431	3.034128	7.213590
## sample_768	5.408454	5.963587	10.133977	9.119501	3.013069	6.742707
## sample_793	6.262029	8.249588	9.121746	9.364240	4.207893	6.495387
## sample_800	6.459894	4.114184	8.280947	8.330540	5.237067	6.881958
## sample_82	3.294077	1.025950	9.741423	9.736174	2.744118	6.116820
## sample_95	5.755315	4.136684	8.358045	8.715753	4.304066	6.185270
##	gene_13037	gene_2239	gene_739	gene_1198	gene_3593	gene_15637
## sample_119	3.023646	4.926972	10.210610	4.8985083	9.000406	7.420230
## sample_275	5.646569	6.002725	10.102225	1.7388975	8.712420	6.632097
## sample_29	4.171415	4.196434	10.834542	1.8161081	10.133913	6.597060
## sample_295	6.683388	4.292009	9.575861	4.5136613	6.753417	6.235691
## sample_298	6.462851	5.867044	9.851471	6.5522218	8.293205	3.266247
## sample_404	5.826604	5.270925	10.057450	1.4698339	8.585661	5.220454
## sample_464	4.925767	5.665211	10.330469	1.4307670	8.885751	5.548085

## sample_468	5.915009	4.730564	10.240326	2.3563410	7.561525	4.433207
## sample_531	3.940496	5.649955	10.137913	2.1979912	8.366777	4.557986
## sample_57	5.118347	5.090612	10.331197	0.7324863	9.146110	4.286999
## sample_578	4.674845	6.905255	9.498508	8.1568820	8.076864	6.230927
## sample_579	5.322808	4.666200	10.096952	1.7010611	7.379188	7.158246
## sample_603	3.778293	5.391953	10.106916	3.7406578	8.558314	7.501535
## sample_627	5.312814	5.583305	10.135799	4.4004833	8.108331	5.594471
## sample_638	4.626766	5.467384	10.301599	3.3387952	9.660131	3.338795
## sample_673	5.575751	5.501824	9.388977	3.1858824	8.232191	4.765858
## sample_68	9.081087	3.784713	9.837486	2.4623927	7.960697	4.992184
## sample_69	4.415732	4.836778	9.953478	5.5300707	8.236650	6.265838
## sample_691	4.833138	6.879645	9.737031	9.4863323	7.328199	6.495865
## sample_701	3.593975	5.706611	9.357160	6.6862764	8.553483	6.956661
## sample_735	4.840191	5.756487	9.191452	4.8844221	8.319035	3.309991
## sample_74	3.810638	3.166507	11.194480	1.8716860	10.613992	9.440626
## sample_740	4.871760	2.981323	10.124910	7.9577289	9.168752	4.344260
## sample_75	3.939791	6.768409	10.591569	9.8936216	8.571624	7.605931
## sample_762	5.060117	6.177940	9.279659	5.8711487	9.517648	7.924468
## sample_768	5.887282	4.009894	9.817724	3.7589640	9.684281	6.164469
## sample_793	3.484989	4.624733	9.514030	1.2967217	8.292110	3.050206
## sample_800	5.773804	5.582363	9.884769	4.1697406	8.566164	5.773804
## sample_82	3.802628	5.526084	10.248663	6.7304769	8.576870	10.442695
## sample_95	5.253505	6.111937	9.759767	1.2884176	7.967001	4.951131
##	gene_11506	gene_11762	gene_3541	gene_4009	gene_6581	gene_16211
## sample_119	7.870285	3.654195	8.672892	1.3137683	10.751075	7.417726
## sample_275	8.590849	13.156241	6.221898	0.6640277	10.692110	9.467991
## sample_29	9.712541	9.289929	2.159242	0.0000000	11.798593	8.840136
## sample_295	8.016418	3.153886	0.000000	0.5506052	10.506972	6.778064
## sample_298	8.608403	3.992922	13.899621	0.7339617	11.229462	6.664938
## sample_404	8.742929	14.866328	6.218225	0.5285713	10.478426	8.285689
## sample_464	7.533602	4.755069	12.962699	0.0000000	11.344412	7.488901
## sample_468	8.361992	4.022075	0.000000	6.7622292	9.739472	5.934583
## sample_531	8.690518	2.488926	14.809160	0.0000000	10.347123	8.271869
## sample_57	8.455023	3.896388	13.761707	0.0000000	10.584634	7.635058
## sample_578	9.838165	3.855442	14.117667	0.5667665	9.817539	8.360711
## sample_579	8.551166	12.741604	6.583975	0.0000000	9.727757	8.958364
## sample_603	9.537314	4.449436	12.140890	0.4365883	9.641835	7.765588
## sample_627	9.026501	8.546705	10.761011	1.3163196	12.445949	7.463394
## sample_638	9.240415	8.994410	12.236382	0.8330925	10.299506	6.706240
## sample_673	8.874594	5.236052	7.216329	0.9656181	10.057883	8.004878
## sample_68	8.180397	5.058784	0.000000	0.0000000	10.037025	6.061549
## sample_69	8.491657	2.058178	7.681871	3.1185923	10.548225	9.414229
## sample_691	8.327490	4.110898	15.774321	6.2198171	10.459964	8.732544
## sample_701	8.857521	4.444004	8.832912	4.9539822	10.218103	7.574881
## sample_735	8.360729	7.818416	8.005265	1.2250281	10.244780	7.658868
## sample_74	9.967226	3.783038	3.208627	1.5168724	11.284755	9.325139
## sample_740	8.949800	2.704363	3.589212	0.4274989	10.322018	6.457857
## sample_75	8.484795	8.762193	16.203608	2.2193389	9.550370	7.459145

## sample_762	8.660887	6.125831	14.172873	0.0000000	10.347810	6.852373
## sample_768	9.063390	6.871080	8.392292	2.8295046	10.463228	7.985108
## sample_793	9.007975	5.689833	6.170173	1.2967217	11.480361	8.321973
## sample_800	8.117503	4.056401	10.202026	0.0000000	9.751147	9.354009
## sample_82	9.158612	3.570645	8.473405	0.0000000	11.196123	8.213702
## sample_95	8.118842	13.571575	3.805654	0.7834986	10.323820	8.702076
##	gene_1355	gene_5068	gene_11578	gene_13104	gene_6221	gene_1982
## sample_119	7.616600	1.2100771	10.980697	5.3086303	7.929513	9.129257
## sample_275	8.324851	1.5702686	8.906557	5.4092619	8.137186	6.937415
## sample_29	6.920627	1.7661711	9.454918	8.5401552	5.935351	9.035607
## sample_295	7.735522	2.0469450	8.463398	6.4502429	8.433944	5.289410
## sample_298	7.542390	1.3135361	7.620454	7.3293748	10.321128	9.374720
## sample_404	7.286142	2.5807003	9.902669	7.3869144	7.996756	4.416434
## sample_464	7.220688	1.5187869	7.061657	3.5534219	8.002883	9.317600
## sample_468	8.502852	1.4836739	7.459259	1.7959331	9.626163	7.097864
## sample_531	6.535425	1.6960829	7.740118	1.8331330	8.286151	8.175395
## sample_57	7.036866	1.9600297	8.764500	3.0489859	7.937156	7.840539
## sample_578	6.686795	0.6129658	8.710317	0.5667665	8.402347	9.071610
## sample_579	7.866463	1.5939752	8.247329	7.1002418	6.168307	6.708256
## sample_603	7.433719	1.1568496	10.200727	0.4365883	6.802206	7.079762
## sample_627	7.207463	1.3961047	9.718082	2.3500446	8.559584	8.419042
## sample_638	6.465606	1.7618810	10.865045	1.3578335	8.008955	9.333647
## sample_673	9.061166	1.0072673	8.209546	8.2882386	7.458333	7.864452
## sample_68	8.899913	1.5993654	8.689512	5.6617524	10.191121	7.527156
## sample_69	6.451226	3.0540844	9.084202	9.9554509	7.994744	9.364634
## sample_691	7.093180	1.9178518	7.705881	9.3334237	8.953638	8.170516
## sample_701	8.062138	1.5907218	8.255005	6.4324834	7.202467	8.458263
## sample_735	8.647394	0.5408208	6.203262	5.6452088	8.683735	8.414503
## sample_74	6.600070	2.1843438	11.022007	5.4447068	7.323460	8.856426
## sample_740	5.289089	3.1727915	8.166434	3.1021374	6.851662	9.212180
## sample_75	6.788777	2.1813888	8.909641	0.6439485	9.639057	7.385819
## sample_762	8.155137	1.4193231	8.075233	5.4303355	8.024624	8.648012
## sample_768	6.183739	1.6169222	9.675788	2.7627528	8.157448	7.273078
## sample_793	8.813473	1.7561693	8.330661	2.9853547	6.779864	7.910619
## sample_800	7.979116	0.8168045	9.265601	4.8518090	7.234386	8.366650
## sample_82	5.945456	1.9467683	9.612784	0.6023619	6.425917	6.877585
## sample_95	7.655731	2.2849286	9.340124	7.8227811	7.773693	4.864478
##	gene_4599	gene_9088	gene_6650	gene_638	gene_2405	gene_18643
## sample_119	8.4260215	0.0000000	7.445048	1.9897930	7.707525	9.997744
## sample_275	2.1720714	0.6640277	7.393639	1.9716999	7.646011	8.984567
## sample_29	1.1767057	0.0000000	5.942763	1.1767057	7.125878	10.987484
## sample_295	0.5506052	4.4228049	6.575037	0.9482270	7.849130	9.129489
## sample_298	0.0000000	0.4131608	5.825468	6.6218790	7.569916	8.870001
## sample_404	1.8698319	0.5285713	8.134883	1.2187192	7.641495	10.017365
## sample_464	0.0000000	0.6464392	7.135730	0.6464392	7.046011	9.852820
## sample_468	0.0000000	2.2353288	5.780880	1.4050666	7.592651	9.141384
## sample_531	0.0000000	0.0000000	6.320918	0.5970305	7.732764	9.790818
## sample_57	0.0000000	0.7324863	6.684664	1.8662750	7.654135	10.665585

## sample_578	7.2894472	2.0471195	7.064150	8.4645948	7.507343	10.310079
## sample_579	6.2629856	0.8077671	7.485717	0.8077671	7.601741	10.277020
## sample_603	8.5677470	0.0000000	6.637880	1.4682184	7.763239	8.893477
## sample_627	7.4169748	1.6939887	5.659722	1.5174773	7.484557	10.087211
## sample_638	9.7653574	0.3339961	7.868069	1.8499993	8.050659	10.514901
## sample_673	4.7144159	0.5621814	7.270763	1.2806000	7.060264	9.576218
## sample_68	0.4603756	3.8613401	5.298141	0.8089204	7.245572	8.707073
## sample_69	0.0000000	1.8923910	6.364122	2.7817793	7.492045	9.705002
## sample_691	0.3569326	2.0311481	7.239875	16.5684015	7.689614	9.972075
## sample_701	10.5241695	0.0000000	7.514264	0.5474498	7.867773	10.287932
## sample_735	4.6987352	3.4925326	8.650503	3.9739877	8.172892	8.781992
## sample_74	0.8463128	2.0674666	5.595158	4.0143196	7.734493	10.357442
## sample_740	7.9355423	0.2589410	7.653648	1.6178157	6.894102	10.586868
## sample_75	0.0000000	1.8203219	6.749668	10.0399609	7.385819	10.917618
## sample_762	6.4221758	4.4294218	7.981293	5.1038753	7.893635	10.380980
## sample_768	7.3883015	1.1928254	6.992610	1.3826673	7.985108	10.119875
## sample_793	6.6602955	0.5230601	7.701896	0.7893537	8.187867	10.259378
## sample_800	4.5919726	0.7483754	7.451673	2.8317151	7.240887	9.466431
## sample_82	9.3170191	1.6195069	6.680338	1.0259501	7.709449	9.495393
## sample_95	1.4871261	0.0000000	8.244254	1.6617040	7.315847	9.788293
##	gene_14567	gene_17816	gene_13617	gene_11205	gene_9983	gene_7314
## sample_119	8.235756	2.3675109	7.224851	10.922064	8.907038	6.245811
## sample_275	8.817076	1.1169635	9.393470	10.785681	9.914905	5.679846
## sample_29	6.205584	8.2016976	8.501789	10.195298	8.218587	4.527389
## sample_295	9.169627	3.3599287	9.009731	9.921755	10.081936	3.227325
## sample_298	11.653781	2.1096614	7.280975	9.795837	9.356591	5.817007
## sample_404	8.324041	2.9330438	8.422212	9.752322	9.185906	5.203229
## sample_464	10.278891	1.9274414	7.273880	10.644496	8.707998	4.023513
## sample_468	9.130818	5.4546925	8.501395	9.099711	9.290286	3.907900
## sample_531	9.797482	0.5970305	7.978339	10.522228	9.774866	5.942271
## sample_57	9.898148	0.0000000	7.909641	10.039920	8.790746	6.939144
## sample_578	8.724650	2.4144332	7.169183	9.768169	9.404316	3.855442
## sample_579	6.687397	0.0000000	8.463402	9.723159	9.353771	4.827418
## sample_603	9.603686	1.0427144	8.517162	10.446174	9.246408	6.486917
## sample_627	7.825162	0.4569113	8.510313	8.631810	8.982549	6.280791
## sample_638	8.721889	0.3339961	7.802245	9.527283	8.616674	5.207471
## sample_673	9.700389	0.5621814	7.327499	9.597492	9.631861	4.393135
## sample_68	9.131276	5.3490609	8.182265	10.109426	9.270391	4.334347
## sample_69	9.999436	0.5381408	7.915915	9.953446	8.950326	4.903670
## sample_691	11.167180	0.3569326	6.992066	9.636184	8.960969	5.639096
## sample_701	9.812203	0.0000000	8.215591	10.172740	9.858118	3.418298
## sample_735	12.177117	4.0143553	7.974902	10.543882	9.822878	2.666893
## sample_74	8.665524	0.8463128	9.015384	9.260244	9.009442	5.488097
## sample_740	8.561971	2.7787553	7.334130	10.090562	9.295298	5.417100
## sample_75	9.841495	0.3576083	7.066649	10.116825	8.790880	6.050071
## sample_762	10.378750	7.9275113	7.518496	9.587588	9.784561	2.840785
## sample_768	8.658751	0.4021764	8.263889	9.153121	8.945098	3.920827
## sample_793	9.616639	0.0000000	8.081334	9.858721	9.745249	5.333778

## sample_800	9.995569	3.3948845	7.538655	10.577457	8.918073	5.453985
## sample_82	8.109287	8.2307746	8.784383	9.689724	9.655976	7.084521
## sample_95	7.135176	0.0000000	8.335672	10.282185	9.467361	5.307057
##	gene_1617	gene_6305	gene_19961	gene_3335	gene_3193	gene_18096
## sample_119	11.698848	9.655792	7.272770	1.7559131	8.741551	3.0236457
## sample_275	9.802316	8.137186	5.577619	0.0000000	9.616082	2.3480057
## sample_29	10.898344	8.342821	7.387190	0.0000000	9.216906	2.3496767
## sample_295	9.811975	8.474091	5.420429	1.9215887	8.671124	3.5470796
## sample_298	9.543844	8.230875	5.969846	1.4103411	9.494274	2.9320234
## sample_404	10.702978	8.134883	6.841935	2.0346735	9.381370	0.5285713
## sample_464	10.149163	8.402923	5.632704	2.7339833	9.256202	6.4227998
## sample_468	10.580287	8.330939	5.120601	3.0733575	8.944223	2.5716525
## sample_531	9.028840	8.369017	7.631977	3.9878756	10.371896	2.1979912
## sample_57	9.945372	8.882573	6.637513	3.2629292	9.393150	3.3590713
## sample_578	10.280794	8.511005	6.468977	1.2890672	9.445704	4.0773028
## sample_579	10.261672	8.062942	5.648342	0.0000000	10.468756	2.0006851
## sample_603	10.768275	8.113596	6.962028	1.0427144	9.081180	0.7712095
## sample_627	11.289581	8.976157	5.429659	1.6939887	9.182625	2.2405586
## sample_638	11.474953	8.812447	6.755769	0.8330925	8.867273	3.1819771
## sample_673	11.091006	8.604835	4.886960	0.0000000	9.029790	5.6999597
## sample_68	10.086760	8.723849	5.775035	4.6255160	9.549024	3.9691689
## sample_69	9.777390	9.263203	5.515886	3.2614556	9.509474	3.4523055
## sample_691	10.067609	8.564115	5.984293	2.5288963	9.263915	4.2662693
## sample_701	10.227580	7.466129	4.910373	0.0000000	9.179056	1.9141052
## sample_735	9.663853	8.539054	4.165389	0.5319678	9.146709	3.3099910
## sample_74	11.264466	9.163247	7.572602	2.3188663	9.366403	4.3515506
## sample_740	10.555145	7.929436	6.877278	3.4595497	10.011381	4.5280024
## sample_75	9.624735	8.227539	6.411204	0.0000000	9.617515	3.5119613
## sample_762	10.881657	8.689890	5.060117	0.0000000	9.158933	4.4966157
## sample_768	11.791228	8.876689	6.466720	0.4021764	8.762761	3.8898726
## sample_793	10.953316	8.042010	5.788182	1.0652276	8.916333	2.3236007
## sample_800	10.611753	8.088963	6.525250	1.6039289	10.002238	4.4644788
## sample_82	11.015080	8.248530	6.417196	0.0000000	7.751618	1.6195069
## sample_95	10.509102	7.700544	6.560854	0.0000000	10.472214	0.4443490
##	gene_7892	gene_17366	gene_19763	gene_7092	gene_6824	gene_9620
## sample_119	9.382127	8.407200	3.932108	5.253501	8.313282	12.74433
## sample_275	5.505208	9.507914	3.287886	8.746192	8.356271	12.16505
## sample_29	9.697287	9.282646	3.550334	6.230019	5.475474	10.80206
## sample_295	5.766725	9.304041	4.571204	6.778064	10.060750	11.71988
## sample_298	9.517325	9.511436	3.729977	7.711302	9.978768	12.28218
## sample_404	5.541490	8.988912	4.224897	5.792460	8.451685	12.21818
## sample_464	9.139628	9.782332	1.705403	5.078392	10.039522	11.12059
## sample_468	4.685066	9.456892	6.842124	8.099111	10.376245	11.72140
## sample_531	9.604831	9.564376	2.488926	4.282025	9.539748	11.45933
## sample_57	8.941992	9.901144	3.614144	3.896388	9.827435	11.27048
## sample_578	10.114783	9.366952	2.760178	6.652702	8.495703	12.63928
## sample_579	7.267686	8.670228	4.865058	6.122422	8.714716	11.88545
## sample_603	7.378260	10.733363	2.879784	7.845333	9.663997	12.52769

## sample_627	7.688537	9.217381	3.811492	7.109664	9.520096	12.46927
## sample_638	8.987022	9.781674	4.988194	8.278338	8.917996	11.77069
## sample_673	7.075629	9.907821	2.846834	7.331768	9.152176	11.18228
## sample_68	6.185956	9.187241	4.975039	3.898189	10.179474	12.39116
## sample_69	8.478919	10.424607	4.110605	7.559285	7.849599	12.97636
## sample_691	5.971445	9.353989	2.951569	6.827133	9.836484	12.83505
## sample_701	6.055898	10.090801	3.144846	8.142082	9.600405	12.28400
## sample_735	6.807664	11.259225	3.309991	5.124618	9.696782	12.44751
## sample_74	9.712582	10.836989	2.067467	7.891395	8.492025	11.51485
## sample_740	8.548749	8.491841	2.543001	4.802136	8.527332	11.79903
## sample_75	8.779841	8.732924	2.531569	4.226624	9.693983	11.20212
## sample_762	7.678987	9.285610	2.029312	4.496616	10.125698	11.91006
## sample_768	6.164469	9.514574	4.473690	8.750988	7.816350	11.61346
## sample_793	6.741089	9.335252	3.283922	8.526726	7.017376	12.66943
## sample_800	7.811792	9.596292	3.933393	6.525250	10.657408	12.05938
## sample_82	6.354627	8.111991	3.506310	2.501694	6.939803	12.15967
## sample_95	4.381000	8.571400	4.589871	7.615541	9.104006	12.16449
##	gene_19291	gene_6415	gene_8690	gene_19825	gene_5096	gene_20287
## sample_119	9.000406	9.742431	7.567629	10.178939	4.595730	8.724715
## sample_275	9.113463	8.704357	7.398641	8.870587	6.028788	8.918177
## sample_29	9.261317	9.575048	5.091349	8.220282	8.757450	9.144533
## sample_295	8.641261	5.436002	7.771470	8.484670	6.450243	8.301825
## sample_298	9.333050	8.366060	8.532917	7.505558	8.008059	6.008628
## sample_404	8.605465	9.389027	7.549076	9.327802	5.996206	8.675692
## sample_464	9.964780	9.079093	7.897374	7.715331	6.840765	6.976467
## sample_468	8.868721	5.668485	8.852096	8.809996	8.822198	7.912841
## sample_531	9.639552	10.577042	9.042929	7.833472	7.756903	8.257440
## sample_57	10.225159	10.135530	8.587444	8.547296	7.488652	7.728016
## sample_578	10.222143	8.533665	8.000299	8.814086	5.888234	7.857925
## sample_579	9.069342	7.971003	5.931624	8.440321	5.626594	8.539907
## sample_603	8.967650	9.418679	7.359758	9.393970	4.138143	8.955449
## sample_627	9.364076	8.272033	6.738930	9.122264	5.722149	8.938145
## sample_638	8.123889	10.377579	6.830319	9.472187	4.245998	8.415045
## sample_673	8.263847	8.678829	8.348529	8.342190	7.369676	8.560734
## sample_68	8.761827	5.467054	9.096033	9.248490	7.442380	7.411401
## sample_69	9.109927	8.509664	8.156629	8.584681	6.604263	7.731020
## sample_691	9.628675	9.015970	9.516500	7.754801	7.292487	7.629255
## sample_701	8.394407	7.404597	6.095382	9.140707	4.100254	8.405227
## sample_735	7.504477	7.116625	11.163474	7.809845	4.396755	7.907570
## sample_74	9.988912	9.490019	7.912255	10.059547	2.659719	8.991499
## sample_740	8.863777	9.157471	6.829951	8.765167	2.261049	8.466648
## sample_75	10.132525	9.576120	9.464515	7.649924	7.684973	7.732900
## sample_762	8.895275	8.532683	8.971069	8.344873	3.282810	9.577070
## sample_768	9.101884	8.094357	6.625468	9.077908	6.357832	9.657463
## sample_793	8.935787	8.483344	5.853245	8.810504	5.225607	9.280571
## sample_800	8.876934	7.247358	9.477483	9.145828	5.184090	8.562613
## sample_82	9.391259	7.373100	6.408421	7.692378	4.857035	9.512209
## sample_95	9.231524	7.941828	5.919047	8.770274	6.469863	8.568180

##	gene_4227	gene_228	gene_8890	gene_16475	gene_18500	gene_5644
## sample_119	11.13211	11.933503	11.713859	11.589712	11.804147	5.924397
## sample_275	10.77577	15.130639	11.294850	10.905831	11.466163	4.641696
## sample_29	10.96605	10.830729	12.099147	10.882055	11.784226	6.683640
## sample_295	10.97436	14.255324	9.839312	10.070134	10.819357	8.806108
## sample_298	10.59732	13.247899	11.213031	10.167381	12.864095	3.617545
## sample_404	11.27057	13.370292	11.479043	10.972233	11.588602	3.835348
## sample_464	10.47786	13.428870	11.383402	10.841855	12.221931	3.864632
## sample_468	11.61170	14.150112	10.863536	10.600721	11.309789	4.685066
## sample_531	10.76089	11.456067	11.709713	10.323077	12.469158	4.707000
## sample_57	11.12039	12.577273	12.205793	10.997878	11.708558	7.337542
## sample_578	10.86705	12.798364	11.184125	10.592747	11.977037	4.532111
## sample_579	11.31201	15.145805	11.104192	10.804930	11.264830	5.512635
## sample_603	11.78197	11.592018	11.592919	11.194172	11.686833	3.702004
## sample_627	11.90635	12.369352	11.016662	11.127246	11.603469	5.094215
## sample_638	11.81047	12.200145	11.344596	11.977559	11.715657	5.500968
## sample_673	10.30230	13.116303	11.935585	10.686229	11.163819	5.726169
## sample_68	11.11107	13.497577	11.027014	10.733126	11.522572	5.618247
## sample_69	10.53202	14.161628	11.936001	10.328462	10.607950	8.509664
## sample_691	10.99396	11.472980	10.404886	9.428745	11.812065	5.127637
## sample_701	11.00556	11.692724	10.958161	10.601195	11.477030	3.288048
## sample_735	10.67875	13.306418	11.069020	10.143779	9.834902	6.391182
## sample_74	11.76122	10.713069	11.898106	11.792278	11.645744	9.400879
## sample_740	10.95690	13.323885	11.276630	10.842885	11.664341	4.506272
## sample_75	11.00729	12.014962	10.824688	10.363226	13.033540	3.913100
## sample_762	11.05119	13.944712	11.504740	10.834542	11.705144	4.792397
## sample_768	11.41934	14.033182	11.168215	11.086110	11.765746	4.514842
## sample_793	11.66222	11.467621	10.799978	11.489713	11.369565	3.702381
## sample_800	11.13754	13.756369	11.157858	10.624238	11.442519	4.372764
## sample_82	11.24253	9.535052	11.520604	11.227014	11.275164	2.501694
## sample_95	11.20767	13.910052	11.163442	10.833254	11.240726	4.277486
##	gene_12016	gene_11264	gene_13584	gene_1475	gene_1564	gene_6480
## sample_119	2.7325731	7.589322	5.767967	2.6282603	11.455548	3.683854
## sample_275	2.6461857	7.211694	5.467593	5.2601461	10.917215	4.017548
## sample_29	2.4362152	6.426344	6.919769	3.2498094	10.490610	6.718608
## sample_295	0.9482270	5.684894	5.106897	4.7874063	12.478059	6.553784
## sample_298	5.5974626	6.207126	5.484467	4.3411888	12.129760	3.693465
## sample_404	3.7972932	7.205998	4.224897	5.0915856	10.519450	2.846935
## sample_464	0.0000000	6.289130	5.649049	3.5492502	10.260390	3.807530
## sample_468	0.8671848	5.176742	5.441072	6.1931922	12.860160	4.433207
## sample_531	0.0000000	6.669764	5.977907	5.2081153	10.329415	4.078208
## sample_57	0.7324863	6.395776	6.962295	3.6915342	9.969372	3.896388
## sample_578	4.9465624	7.004681	6.193472	5.1637402	11.171540	6.074891
## sample_579	4.6460186	5.971769	4.749062	4.4620130	10.306825	2.358284
## sample_603	2.7341568	7.988764	6.170420	2.1442096	10.759189	2.655443
## sample_627	2.9460571	6.638546	6.165709	8.6958721	10.829667	4.979632
## sample_638	6.6330432	7.010612	5.458864	3.0732204	11.086030	5.977264
## sample_673	0.0000000	5.856692	6.306368	3.6906294	11.997585	5.041059

## sample_68	1.3241193	5.102221	5.755088	7.5735942	12.813755	4.069487
## sample_69	2.8745216	7.169264	6.315816	1.9726927	10.695907	4.881714
## sample_691	2.3812557	6.361826	6.100969	2.8613996	11.946479	2.898247
## sample_701	0.5474498	6.280452	6.004973	1.5661819	12.163065	3.935978
## sample_735	3.4365883	5.611004	5.848671	4.9405943	11.078278	3.973988
## sample_74	0.0000000	8.180819	7.082947	1.6957266	10.705987	7.552300
## sample_740	0.4385058	5.653398	6.417684	0.4311415	10.489176	3.782377
## sample_75	0.6439485	6.629479	5.682337	1.7936462	10.598657	2.728116
## sample_762	0.5980793	7.339547	6.228230	4.2568144	11.213530	4.560825
## sample_768	3.8617569	6.234453	6.814525	3.1982898	12.114409	4.840378
## sample_793	4.3703875	5.793308	6.024435	3.3132894	11.164888	2.692092
## sample_800	2.8317151	7.517984	5.237067	4.0780462	10.482657	3.298482
## sample_82	4.0573290	7.522958	6.687607	2.2262002	11.766574	2.744118
## sample_95	2.8341246	6.677536	5.225965	3.6389890	10.684380	1.957989
##	gene_6568	gene_15700	gene_18398	gene_16947	gene_3810	gene_1643
## sample_119	6.426141	10.317627	8.134123	10.398273	9.932110	9.045730
## sample_275	9.410918	10.776729	10.165259	12.375525	7.620528	9.289514
## sample_29	7.645161	10.136735	8.989837	6.027032	10.587759	8.993411
## sample_295	9.879016	10.809302	9.606287	11.072488	7.676345	8.929498
## sample_298	9.421516	9.955832	10.280168	11.017261	9.983036	9.218871
## sample_404	9.428471	10.695681	10.136145	12.448790	7.827222	9.916607
## sample_464	8.984048	11.084244	9.577699	8.496083	10.325395	8.854379
## sample_468	8.901413	9.773832	9.671842	10.611873	6.600109	9.291235
## sample_531	8.523711	11.323094	9.738967	8.135899	10.828335	8.529727
## sample_57	8.751427	10.025389	9.677008	10.106877	9.436530	9.023106
## sample_578	7.545011	11.280777	9.386792	10.584540	10.176672	8.430791
## sample_579	9.222186	11.043417	9.876842	12.297959	8.220069	9.716862
## sample_603	9.303998	10.174639	9.608918	9.625948	8.808482	9.340629
## sample_627	8.625033	9.911946	9.326059	10.089689	8.139162	9.894545
## sample_638	9.741848	9.945379	10.446794	10.757715	9.313706	9.338300
## sample_673	9.374550	11.384795	9.923478	10.522003	9.152176	9.539740
## sample_68	9.063964	10.390954	9.842220	11.181631	7.978013	8.975149
## sample_69	8.165741	10.650774	9.827063	10.274763	9.247194	8.340905
## sample_691	8.608362	10.656648	10.264466	9.750260	8.496215	10.332618
## sample_701	9.625066	11.005239	9.352077	9.945292	9.494558	8.354280
## sample_735	8.422027	11.411273	9.833493	11.849687	8.935713	8.888987
## sample_74	8.469467	10.082828	9.906955	9.800169	11.002485	9.001977
## sample_740	10.208442	10.401413	9.078690	11.687126	10.212217	8.448348
## sample_75	9.649705	10.498420	9.594084	8.609290	9.133448	11.168484
## sample_762	8.058376	11.111044	9.059839	10.328821	9.994537	8.870155
## sample_768	8.484529	10.078004	10.465526	10.181711	9.030951	9.022057
## sample_793	9.655593	10.998188	10.555557	10.722586	9.548115	8.854130
## sample_800	9.305227	11.004923	9.495259	11.947549	9.236966	9.128984
## sample_82	7.594250	10.976643	8.892154	10.354062	7.826046	8.950737
## sample_95	9.854423	11.070691	9.335855	12.586793	6.498919	9.787703
##	gene_13388	gene_19884	gene_3025	gene_10371	gene_3008	gene_10767
## sample_119	6.778301	11.240094	10.077670	9.433875	4.066124	7.737964
## sample_275	6.244318	12.671817	9.467991	9.301590	3.909476	7.671046

## sample_29	9.137616	9.885463	10.714391	10.136735	3.044167	8.743727
## sample_295	6.473064	11.454140	10.120989	9.460687	5.526056	8.106746
## sample_298	4.765312	12.285344	10.278240	9.880182	5.655561	8.273241
## sample_404	7.246836	13.162640	9.665446	8.598894	4.829525	7.691757
## sample_464	6.760287	12.846811	10.747287	10.219931	3.919559	6.956916
## sample_468	7.144832	11.754817	10.040317	8.464844	6.985079	7.315023
## sample_531	7.215640	12.747096	10.395149	10.217097	5.224824	8.262264
## sample_57	5.996414	13.193875	10.217170	9.151154	5.298882	7.271127
## sample_578	5.485172	11.428638	10.568260	10.033643	4.195694	8.185782
## sample_579	7.886593	11.767431	8.475620	8.968125	3.782199	8.320006
## sample_603	6.503869	12.532349	9.464860	9.350033	4.048559	7.710676
## sample_627	5.956586	11.372604	9.735918	9.280110	2.636010	8.110274
## sample_638	5.343273	11.082741	10.753685	9.458021	4.078149	7.924848
## sample_673	5.407801	11.725111	10.670709	9.090464	3.889250	8.202163
## sample_68	5.784904	11.967907	10.114419	8.716130	5.734861	7.541778
## sample_69	7.264940	12.783187	10.907965	10.114510	5.048655	8.121932
## sample_691	5.139175	11.506278	10.607404	9.974486	6.844850	8.744867
## sample_701	6.234507	11.157562	9.591805	9.393251	4.247715	9.148939
## sample_735	6.263031	12.667744	9.274814	10.397568	4.235359	8.886270
## sample_74	6.535358	11.706591	10.206184	11.098355	4.956670	8.053263
## sample_740	6.758237	11.081743	9.449130	9.908517	5.451617	7.894315
## sample_75	5.846496	10.345250	9.893195	9.459337	5.460677	8.344522
## sample_762	7.099337	11.481603	10.298647	9.818608	4.429422	8.426835
## sample_768	6.276404	12.781241	10.034881	9.191079	4.964408	7.957386
## sample_793	5.307487	11.407772	9.029249	9.334439	4.026357	8.777183
## sample_800	7.543921	12.247435	9.429923	9.331349	5.755764	7.717073
## sample_82	4.002414	10.908273	9.531196	8.906247	4.218603	8.469190
## sample_95	6.692441	10.615078	9.498626	8.515625	3.216455	8.299799
##	gene_6241	gene_12987	gene_7282	gene_1705	gene_10150	gene_3491
## sample_119	9.150537	7.769745	2.5269952	6.550273	3.707679	7.445048
## sample_275	9.916651	6.837741	2.3496484	7.782474	6.141768	7.972382
## sample_29	10.549438	6.484778	3.1009104	8.251781	5.740936	12.981676
## sample_295	7.513435	10.069503	2.8159033	6.244560	4.653427	8.360364
## sample_298	8.847330	6.791254	4.1420785	7.465517	6.677945	8.137129
## sample_404	10.124173	8.139418	2.0378039	7.824411	6.790616	7.685569
## sample_464	9.854582	9.076073	3.4073392	7.252647	5.238172	9.022193
## sample_468	7.774701	11.967482	3.1496984	6.831725	5.029254	8.854666
## sample_531	10.066601	7.559339	2.0342862	7.846387	5.164239	8.557475
## sample_57	9.279548	8.487289	3.8983051	7.343426	7.042447	8.063821
## sample_578	9.616415	7.863901	0.9726927	8.282602	7.818921	8.289272
## sample_579	9.419280	7.955824	2.0006851	8.444878	5.039792	7.497748
## sample_603	8.701112	7.882533	1.9388127	7.356654	2.389649	8.076308
## sample_627	8.245714	7.801269	2.3510912	8.401660	5.575999	8.178655
## sample_638	8.468583	8.871412	2.1390765	7.976495	6.868995	8.632857
## sample_673	9.950824	6.617661	3.0271364	7.434687	6.838435	7.905170
## sample_68	8.129010	10.846744	3.3273425	7.098980	4.987576	8.452233
## sample_69	10.059966	7.413882	2.3416158	9.465868	6.497076	8.885608
## sample_691	9.085446	6.578685	3.2357117	8.471330	6.338499	8.372169

## sample_701	9.436170	7.525153	0.9467309	7.302858	4.631605	7.567880
## sample_735	9.829964	6.220531	2.6668929	8.672740	6.963000	8.341025
## sample_74	8.969433	4.577266	3.1275342	7.876781	5.488181	10.784226
## sample_740	9.532783	6.855965	1.4531751	6.790055	6.607759	8.532714
## sample_75	10.022507	6.775828	2.7305530	8.193289	3.103481	9.276422
## sample_762	9.765105	7.839657	2.0293115	7.783588	8.474201	8.454563
## sample_768	8.632108	8.092662	1.7006173	7.121440	5.939076	8.476738
## sample_793	8.437103	7.997473	1.0677075	8.745705	5.756949	7.936585
## sample_800	9.714256	7.024397	2.9720860	7.512029	7.033632	8.030143
## sample_82	10.080031	7.574775	2.5016939	6.399595	5.081378	8.292110
## sample_95	10.124768	7.950270	3.6073185	7.628744	4.673805	7.700544
##	gene_14526	gene_10064	gene_1531	gene_12195	gene_18073	gene_524
## sample_119	10.986084	3.0830431	9.059455	10.037039	8.135648	12.35058
## sample_275	10.386736	4.4998331	8.351120	9.953686	7.658583	12.27224
## sample_29	10.681265	0.7051375	7.494464	8.080956	5.942763	11.61509
## sample_295	10.767605	5.0061813	7.382935	8.876274	9.792450	11.49838
## sample_298	10.766413	2.5786731	6.058648	9.466839	9.406790	11.40431
## sample_404	10.541890	5.1502875	8.031329	10.025347	7.486385	12.24578
## sample_464	10.398509	3.3283626	8.831288	9.349204	8.507466	11.65249
## sample_468	10.994021	3.3313613	6.600109	9.628185	10.392264	11.31912
## sample_531	10.167155	5.2444647	6.620563	8.722384	9.864862	11.53556
## sample_57	10.390696	0.8269247	7.493951	9.287204	8.841558	11.30710
## sample_578	10.966851	3.0388751	8.223437	9.097187	7.771602	12.78785
## sample_579	10.762556	1.9228448	6.793688	9.887740	7.158246	12.82357
## sample_603	10.704941	5.8014251	9.283693	9.191666	6.683233	12.10870
## sample_627	10.799217	4.3220363	7.903581	9.408255	8.217095	11.89129
## sample_638	10.383780	5.1237159	7.695695	9.534509	8.206365	11.64102
## sample_673	9.990558	5.6183532	7.018456	9.520214	9.041733	11.81828
## sample_68	10.582453	2.3604486	7.231202	9.267753	10.485155	11.71538
## sample_69	9.953478	1.3093508	6.535621	9.696380	8.614890	11.65336
## sample_691	10.042494	4.9556336	6.676930	10.030543	9.004192	12.38876
## sample_701	10.589576	1.2536263	7.094563	9.350037	7.499623	11.94705
## sample_735	9.751537	1.5966013	9.377998	9.889499	8.627917	12.55863
## sample_74	10.828152	3.2818614	6.076589	7.896236	9.206756	10.88890
## sample_740	9.881343	0.4531224	7.919191	10.066264	8.320340	12.31932
## sample_75	10.317537	1.8203219	7.154727	9.289452	8.173747	11.77512
## sample_762	10.322570	3.2045790	9.237313	9.258036	9.023275	13.44294
## sample_768	10.737653	5.0922751	8.970287	8.825468	9.435731	11.21521
## sample_793	10.511249	2.7711884	8.646933	9.017131	7.899659	11.48770
## sample_800	10.350652	6.1364908	7.495823	10.328709	8.658872	12.22015
## sample_82	10.419960	2.8515991	8.357389	9.614616	7.744726	12.92011
## sample_95	10.453415	1.2141248	8.271454	10.363850	8.540988	12.38158
##	gene_293	gene_8171	gene_17213	gene_19058	gene_8166	gene_20049
## sample_119	6.290314	0.6730136	11.538456	10.783784	11.18720	9.561916
## sample_275	5.577619	0.6640277	9.825733	10.363248	10.85666	9.363062
## sample_29	6.329484	1.6808194	10.608292	9.462903	12.83799	9.576239
## sample_295	4.875239	3.8557306	10.743951	11.292741	10.59500	9.601117
## sample_298	5.985031	1.2180992	10.485306	12.112361	10.07726	8.450531

## sample_404	4.081442	1.2187192	10.078858	10.474395	10.55511	9.552669
## sample_464	9.509963	1.9360254	9.979397	9.236098	10.58899	8.478033
## sample_468	2.758943	2.2353288	11.369428	11.698865	10.64111	9.573837
## sample_531	9.236808	2.7309442	9.894690	11.364354	10.66029	9.097950
## sample_57	9.202271	2.3129265	10.201046	10.506744	11.55179	8.751427
## sample_578	6.230927	3.1209657	10.135914	9.659976	10.79424	9.063848
## sample_579	4.220020	1.5241640	10.346857	10.379335	10.47980	9.681700
## sample_603	4.495363	1.7963901	11.614213	10.636779	10.73636	9.331954
## sample_627	4.425714	1.3163196	11.280962	12.089056	11.07264	10.093127
## sample_638	8.093940	1.0299829	11.420187	11.677962	11.53609	9.524847
## sample_673	7.450494	1.5389851	9.816241	10.593625	10.63095	9.464868
## sample_68	3.378304	2.8857696	11.098762	11.361642	10.50694	9.568471
## sample_69	6.838851	1.2364620	9.435108	10.170488	11.13867	8.626487
## sample_691	6.608039	0.8812731	10.376842	11.909169	10.00593	9.212370
## sample_701	4.842375	0.0000000	11.633577	11.854977	10.77729	9.701417
## sample_735	8.005265	0.0000000	9.523193	12.053936	10.68501	9.104143
## sample_74	8.974782	2.1562024	10.358036	8.887544	11.11056	9.473381
## sample_740	7.466603	1.0247452	10.919117	11.170977	10.78935	8.801582
## sample_75	8.130957	0.8827604	10.147981	11.637956	11.30306	9.174761
## sample_762	8.186000	2.3531539	10.776581	11.613039	10.85470	9.897649
## sample_768	6.105066	2.8295046	11.716567	10.890355	11.08312	10.796210
## sample_793	4.290756	0.7893537	10.802339	11.513629	10.96467	10.009129
## sample_800	8.128056	0.7483754	10.888667	11.172384	10.70460	9.635051
## sample_82	2.744118	1.0259501	11.306791	11.486619	10.52058	10.747119
## sample_95	2.680279	0.0000000	10.512661	10.613412	10.50767	9.824895
##	gene_8566	gene_1681	gene_11286	gene_13492	gene_106	gene_11358
## sample_119	10.529108	3.823036	8.062154	5.4028449	2.0938644	8.2200252
## sample_275	5.050715	3.453281	5.696203	2.8931680	2.5047729	2.3480057
## sample_29	10.413025	3.284292	9.222266	3.2007084	2.8685287	7.1707463
## sample_295	6.676846	0.972325	4.942195	3.8557306	3.4275391	0.0000000
## sample_298	10.140510	2.469756	7.729444	2.3158850	2.1096614	1.2180992
## sample_404	5.903270	4.433080	5.076533	4.1549558	1.2187192	0.9145645
## sample_464	10.169837	2.690439	7.589719	1.7054029	1.7054029	0.0000000
## sample_468	6.815985	2.014748	4.796224	1.7959331	2.3563410	0.0000000
## sample_531	10.126717	2.860824	7.371141	2.9381164	2.4889263	0.5970305
## sample_57	10.890371	3.648040	8.035002	3.6141444	3.1599198	0.0000000
## sample_578	10.200470	4.077303	7.975688	0.0000000	2.5391092	16.3717001
## sample_579	7.162029	2.958472	5.830608	3.3758596	2.1299753	5.6375088
## sample_603	9.779870	3.769349	6.642989	0.4365883	1.4682184	1.9361762
## sample_627	8.634532	2.565110	7.914762	2.7199938	1.6939887	4.2103265
## sample_638	10.195667	3.517112	7.980962	2.6345933	1.6248028	3.4802134
## sample_673	9.285929	2.942026	6.160239	0.5621814	2.2665767	13.3352787
## sample_68	7.426709	1.714575	5.009123	1.3241193	1.5259185	0.0000000
## sample_69	11.125588	5.869585	8.126606	2.0581779	1.4896971	7.0003832
## sample_691	6.837792	2.467984	4.955634	2.8428984	1.4243847	0.3569326
## sample_701	7.348507	1.088142	5.731884	0.0000000	2.2301108	9.4787818
## sample_735	7.511547	1.893518	4.927337	3.0239827	2.4485838	13.0742736
## sample_74	10.974156	4.674681	8.733809	4.5772657	0.0000000	2.3188663

## sample_740	9.864364	3.508023	3.952706	0.4274989	1.7714631	6.1144024
## sample_75	9.742640	3.108675	7.293122	2.6655292	1.2669364	0.0000000
## sample_762	9.048686	4.050014	7.642362	2.3531539	2.6174866	6.9455024
## sample_768	7.487261	3.708408	7.017889	2.4596677	1.1928254	4.4944092
## sample_793	7.339503	1.082703	6.899103	1.2967217	0.7893537	4.6458746
## sample_800	8.433431	3.988812	4.851809	5.6620146	1.6039289	10.5077946
## sample_82	7.039774	3.583796	6.451768	1.0259501	1.6195069	15.0924805
## sample_95	3.375318	3.480213	5.371318	2.4135399	2.0860704	1.6617040
##	gene_3718	gene_7369	gene_2091	gene_5230	gene_15192	gene_19567
## sample_119	3.5844214	4.140934	11.11868	6.1464231	4.406537	10.149264
## sample_275	8.6381752	3.665518	10.70125	2.8931680	6.792816	10.994049
## sample_29	2.8048796	6.042706	10.90120	5.0913486	4.951406	11.593629
## sample_295	2.0885487	7.718341	10.58237	0.0000000	4.257373	10.630112
## sample_298	0.9962441	5.384944	10.95245	1.5799525	4.729694	10.043205
## sample_404	6.0740708	4.355369	10.95337	5.4422104	5.555132	11.140600
## sample_464	2.1347795	3.407339	10.64811	3.4822287	6.007030	9.808033
## sample_468	2.1031955	5.854649	10.25937	0.4978423	4.460467	10.550468
## sample_531	0.5970305	3.940496	10.86607	5.5898024	6.012687	10.096636
## sample_57	2.9288251	2.797719	10.48561	2.3129265	4.943302	10.515246
## sample_578	8.1422143	5.231647	10.45412	2.4144332	5.012573	10.460046
## sample_579	5.1657833	3.150543	10.89297	0.4596414	5.056159	11.398776
## sample_603	1.2710670	4.138143	10.76476	1.9361762	5.706992	11.146168
## sample_627	6.6560816	4.891832	11.29921	9.0929738	4.996570	11.008338
## sample_638	1.3578335	4.785085	10.18473	7.0504255	3.938060	10.854759
## sample_673	1.9481523	4.886960	10.12775	4.7144159	5.358389	11.018958
## sample_68	2.3604486	5.916441	10.17853	0.0000000	4.487268	10.491202
## sample_69	6.9002027	4.692896	11.00128	9.7435537	5.397091	11.258784
## sample_691	4.4254862	5.864312	10.07627	0.8812731	5.679159	11.650163
## sample_701	3.4792694	4.444004	10.44089	2.3654686	4.670036	11.009444
## sample_735	4.3020997	5.106059	10.30042	7.3073921	7.405507	11.146550
## sample_74	6.7177450	5.418031	11.39871	1.7628165	5.205717	10.388405
## sample_740	0.0000000	3.547277	10.43378	3.7081872	4.938178	9.779731
## sample_75	2.1294479	4.932434	10.83783	0.3576083	6.043933	10.769499
## sample_762	2.2002846	6.247867	10.03960	8.1680804	4.463407	10.388738
## sample_768	2.9544383	5.493276	11.04390	7.8827042	5.271071	10.847958
## sample_793	4.4434409	4.236041	10.97123	6.2132967	4.860009	10.827708
## sample_800	4.9816655	3.394885	10.73197	4.3727641	5.210821	10.809993
## sample_82	1.0259501	4.441397	11.47380	1.0259501	5.316960	11.545075
## sample_95	4.9511308	3.648569	10.65111	2.4135399	6.178103	11.188626
##	gene_12183	gene_5420	gene_4323	gene_2788	gene_15834	gene_1022
## sample_119	10.183586	9.327121	12.91114	8.245676	11.638938	7.612139
## sample_275	8.330312	9.509072	12.35617	8.460485	10.965243	8.448463
## sample_29	11.360359	10.346835	11.48234	9.628298	10.997307	7.633758
## sample_295	8.667878	9.250007	12.45149	9.656167	9.919042	8.297824
## sample_298	9.847842	9.305341	12.40722	8.636320	10.391909	7.243583
## sample_404	8.283644	9.618305	12.82194	9.090769	11.190893	8.931234
## sample_464	9.816198	9.856343	12.30422	8.220828	10.708825	8.473454
## sample_468	8.415205	9.166432	11.90235	9.065745	10.336797	7.736185

## sample_531	9.840062	9.021743	11.68948	8.386802	9.995216	8.773980
## sample_57	9.673505	9.287204	11.74988	9.002784	10.873682	8.092070
## sample_578	9.621711	8.925056	11.84285	8.734466	10.773766	8.507196
## sample_579	8.938642	9.401780	12.66574	8.568382	11.134606	8.731370
## sample_603	9.599091	9.984945	12.68932	8.565056	11.178640	8.245429
## sample_627	10.808538	9.230863	13.27066	8.735089	11.302833	7.994908
## sample_638	9.011258	9.600866	13.03722	8.608029	10.736825	8.250223
## sample_673	8.420516	9.311038	12.96295	10.102330	10.529030	8.430511
## sample_68	8.727689	9.227606	12.39499	9.805113	10.086760	8.115387
## sample_69	9.593765	8.870359	11.69448	9.622182	10.184565	8.460521
## sample_691	8.969070	8.426135	12.24001	8.950366	9.505956	8.964214
## sample_701	8.615902	9.554220	12.96663	8.662651	10.762797	8.031930
## sample_735	7.633177	9.098285	12.42686	9.011208	10.342152	9.831375
## sample_74	11.177519	10.524875	12.18942	9.637493	10.484199	6.952706
## sample_740	9.611849	9.602920	12.78423	8.487687	10.843152	8.673401
## sample_75	9.315163	8.510618	11.92583	8.779841	10.391297	7.895612
## sample_762	9.102257	9.000304	13.16450	8.704173	10.937514	8.471362
## sample_768	9.144304	9.770180	13.20334	8.111230	11.388184	7.354690
## sample_793	10.069530	9.756833	12.75742	8.211105	10.719389	8.523872
## sample_800	9.377796	8.808575	12.54053	8.985418	11.029536	8.612010
## sample_82	8.741362	9.877184	13.29003	8.289724	11.011828	8.359666
## sample_95	8.152148	9.723299	12.74169	8.495583	10.966376	9.133024
##	gene_18578	gene_5676	gene_5261	gene_11052	gene_11168	gene_14329
## sample_119	9.244005	0.0000000	7.097379	1.9897930	9.248945	9.289249
## sample_275	10.188564	0.0000000	7.344473	0.0000000	9.814541	8.661269
## sample_29	8.592924	2.7383139	7.457323	1.1767057	8.626673	7.398017
## sample_295	8.350130	0.9482270	7.090483	0.0000000	9.982723	6.134344
## sample_298	10.006480	0.9962441	7.831643	1.4103411	8.933018	6.149056
## sample_404	9.917266	0.5285713	7.003883	0.0000000	9.377555	8.172982
## sample_464	9.098564	0.0000000	7.443797	1.7054029	9.162580	8.646238
## sample_468	8.340144	0.0000000	7.019891	0.4978423	9.676207	4.257855
## sample_531	9.714527	0.0000000	7.041659	0.0000000	8.772291	8.462134
## sample_57	10.050801	1.2159892	7.616549	1.2159892	8.936158	7.909641
## sample_578	10.186238	0.5667665	7.160860	1.2890672	8.981085	8.968667
## sample_579	10.080058	0.8077671	7.484388	0.4596414	9.327892	7.593354
## sample_603	9.682108	0.0000000	7.511990	0.0000000	9.265587	10.957051
## sample_627	9.186350	0.0000000	7.782297	1.3163196	9.026501	8.647994
## sample_638	8.761395	0.6050202	7.813749	0.3339961	8.630016	9.794429
## sample_673	8.467879	0.0000000	7.524863	0.9656181	9.004743	10.221696
## sample_68	8.861974	0.0000000	7.059312	0.4603756	9.620385	5.755088
## sample_69	10.490921	1.7050490	7.409399	2.3416158	9.012264	7.665904
## sample_691	9.407438	0.6427478	7.272434	0.6427478	9.263994	10.551035
## sample_701	9.338772	1.2536263	7.657233	0.5474498	9.712570	11.783121
## sample_735	10.918953	1.4769001	6.749427	0.0000000	9.836311	10.034551
## sample_74	8.164424	0.8463128	7.358089	2.9845711	8.061900	5.195281
## sample_740	10.846666	0.0000000	7.565498	4.1336622	10.465444	9.653834
## sample_75	9.141374	0.0000000	7.523131	0.8827604	9.652734	10.371526
## sample_762	9.194949	1.0196308	7.623699	3.6797165	9.830865	7.485797

## sample_768	7.748951	0.4021764	7.900287	1.3826673	9.870702	12.843841
## sample_793	9.376149	0.4480552	8.147993	0.4480552	10.351536	12.137779
## sample_800	9.921646	0.0000000	6.660837	0.7483754	9.151582	9.107806
## sample_82	8.953757	0.0000000	7.202858	1.3530409	8.576870	5.094346
## sample_95	10.194806	0.7834986	7.330532	0.0000000	9.673834	8.317847
##	gene_11740	gene_13753	gene_9811	gene_19677	gene_10410	gene_1151
## sample_119	10.613725	10.630222	5.061444	9.091293	5.3301030	9.485626
## sample_275	10.643306	11.470328	6.562435	7.438418	1.7388975	9.084245
## sample_29	10.765535	10.465097	6.417223	6.865461	1.9397531	9.349688
## sample_295	10.667830	11.170952	5.754356	7.478745	0.5506052	8.603834
## sample_298	10.491041	10.894067	6.574830	7.508001	0.4131608	8.675460
## sample_404	9.877888	11.118948	7.401033	7.049511	1.4698339	8.747391
## sample_464	10.297833	11.360940	5.728106	6.729118	1.9360254	9.449427
## sample_468	10.129012	11.244887	5.723979	8.019007	0.0000000	8.802361
## sample_531	10.745985	11.934752	6.143981	7.703869	1.3436352	9.370188
## sample_57	10.272292	11.088755	5.172275	7.590894	2.9288251	8.799340
## sample_578	10.634303	11.508255	4.804725	7.485934	0.0000000	8.074286
## sample_579	10.277311	11.426972	7.052101	7.858621	0.0000000	8.723709
## sample_603	10.046592	10.890667	6.090383	8.158766	0.7712095	8.815287
## sample_627	9.914895	10.601687	6.032454	7.688250	3.0792245	8.971883
## sample_638	10.135838	10.916148	5.565880	7.605020	5.1662013	8.903159
## sample_673	10.889512	11.398867	5.604291	6.093452	7.3862074	8.618892
## sample_68	10.318147	11.344851	5.025866	7.722644	0.4603756	8.800049
## sample_69	10.724957	11.341786	5.811309	7.688558	1.7050490	7.992185
## sample_691	10.580551	11.434946	5.166266	6.966488	0.8812731	8.975521
## sample_701	10.871397	11.339160	6.692734	7.953847	7.6760064	8.473328
## sample_735	11.111612	12.319836	5.213985	6.877229	3.4342145	7.529063
## sample_74	10.146887	9.758490	5.444707	8.494184	0.8463128	8.385910
## sample_740	10.233811	11.192823	5.618077	9.103060	3.6299744	7.894315
## sample_75	10.370775	11.768619	4.310827	7.191365	2.6655292	9.046325
## sample_762	10.986368	10.728473	3.736410	8.069176	8.9562251	8.394167
## sample_768	9.965168	9.833568	7.094542	7.423822	0.7163325	9.041552
## sample_793	10.682327	11.252559	6.706296	8.411469	4.6872455	9.138623
## sample_800	10.282498	11.378750	6.175283	8.149564	0.7483754	8.881102
## sample_82	10.199452	10.725204	5.509769	7.558604	2.2101395	8.976195
## sample_95	10.301245	11.339410	6.799087	7.632952	0.0000000	8.598499
##	gene_11812	gene_15811	gene_1382	gene_3225	gene_17328	gene_7860
## sample_119	8.866277	7.598931	11.81846	3.752898	5.519595	2.3675109
## sample_275	7.924599	7.229165	11.19120	4.068688	6.392187	0.0000000
## sample_29	10.323257	4.921712	11.58862	4.039384	9.168760	0.3952822
## sample_295	8.552973	9.258790	10.78101	3.945505	6.296663	1.2595437
## sample_298	7.709015	10.344496	11.25337	6.738471	4.577725	3.8342764
## sample_404	8.161878	7.991749	10.73853	7.673111	6.518976	0.0000000
## sample_464	7.305159	8.658401	11.45196	8.368790	5.102331	1.0912598
## sample_468	8.571828	9.691388	10.04371	4.774666	5.520925	0.8671848
## sample_531	6.855840	9.469505	11.60653	10.337444	5.374942	2.0270479
## sample_57	6.791892	8.803618	10.80227	8.526640	6.165488	0.7324863
## sample_578	6.809479	7.328845	12.13557	7.230568	8.723009	6.5600882

## sample_579	8.345197	6.384596	11.64836	1.087938	6.906277	0.0000000
## sample_603	8.817550	7.124287	10.90430	5.716719	3.297764	2.2889196
## sample_627	8.306399	7.759708	10.92784	2.122076	7.711798	0.4569113
## sample_638	8.527825	8.683096	10.82834	5.458864	4.323125	2.0447438
## sample_673	9.330388	8.713631	12.01878	1.948152	5.061785	2.4028040
## sample_68	9.616937	10.854674	10.59548	9.510163	4.957692	0.4603756
## sample_69	10.318791	8.100704	11.43825	5.028817	8.403877	4.3848442
## sample_691	8.946269	9.439282	11.46791	9.104696	5.850344	0.3569326
## sample_701	9.114521	6.767959	11.82134	2.899272	5.362740	1.2536263
## sample_735	10.053437	8.931766	12.30097	2.764792	8.346961	0.5319678
## sample_74	7.833700	8.022606	11.23691	2.659719	5.716873	0.0000000
## sample_740	7.847559	7.553921	11.37470	2.261049	4.854664	5.6429817
## sample_75	7.675075	7.960587	11.28235	11.992776	5.584244	0.0000000
## sample_762	8.704173	8.668194	12.93725	3.121861	6.554213	5.5138034
## sample_768	8.253285	8.183209	10.90712	2.954438	4.964408	1.1928254
## sample_793	8.117726	7.053415	11.75619	2.323601	5.573293	1.8274939
## sample_800	7.816152	8.142015	11.50629	2.525919	7.104976	0.0000000
## sample_82	7.562960	6.574500	11.00129	0.000000	4.541087	0.6023619
## sample_95	7.861329	6.409936	11.04260	2.596721	6.625517	0.0000000
##	gene_9160	gene_3539	gene_19524	gene_13169	gene_3721	gene_15198
## sample_119	5.6101538	1.4766927	10.643414	7.267217	10.282254	13.28257
## sample_275	0.6640277	2.1720714	10.004642	7.062543	10.644442	14.52189
## sample_29	3.4289329	0.0000000	9.955826	4.622749	9.582183	12.45222
## sample_295	2.6116681	0.9482270	10.025804	4.626533	10.078845	13.28862
## sample_298	1.7316612	4.9623130	10.088722	8.240400	10.528054	12.86928
## sample_404	0.5285713	1.2187192	10.192872	6.092897	10.918833	12.96899
## sample_464	4.1204681	2.6058736	10.176909	4.166611	9.873833	14.41520
## sample_468	2.1031955	3.0733575	9.540696	6.524633	10.318757	12.21150
## sample_531	1.0180644	3.7882263	9.756173	5.184717	10.741147	12.74806
## sample_57	2.1068173	2.6534957	9.966511	6.083426	10.303518	13.28207
## sample_578	6.2029270	2.7601784	9.616415	8.726293	10.701809	13.19155
## sample_579	2.2486259	1.5241640	9.789007	7.135319	11.098276	12.61978
## sample_603	7.7514637	0.0000000	10.239754	7.444145	10.441284	14.11434
## sample_627	6.1806054	0.4569113	10.100005	9.027530	10.480134	12.26558
## sample_638	2.4405536	1.9506550	9.823754	9.170393	10.204608	11.92431
## sample_673	2.2665767	3.0262866	9.667800	9.056012	11.163819	13.24831
## sample_68	1.7029235	1.8606053	9.331439	4.810839	10.279738	12.51224
## sample_69	4.5046077	2.8736360	10.613375	10.143064	11.080218	13.22071
## sample_691	0.8812731	4.9425239	8.681189	7.921067	11.724198	13.19769
## sample_701	4.6959493	0.0000000	9.920590	10.206892	10.799120	13.68076
## sample_735	3.4342145	4.5692420	9.753775	9.221933	11.195095	14.21039
## sample_74	2.1562024	0.0000000	11.972653	6.968114	8.667413	13.51244
## sample_740	5.7392622	2.2610493	10.237174	8.663622	10.084250	13.51353
## sample_75	2.5315686	2.8456908	9.418407	8.085366	10.634330	13.14947
## sample_762	3.9906463	1.8352773	9.476629	9.537729	10.356320	13.71063
## sample_768	3.3221878	0.4021764	10.354062	10.832273	9.928008	12.73892
## sample_793	3.7809190	1.4961044	10.091845	7.792367	10.181438	13.26514
## sample_800	3.6505808	3.0838941	9.920634	7.272947	10.300982	13.04837

## sample_82	5.5097687	1.6195069	10.373942	7.955156	11.765717	13.21733
## sample_95	0.4443490	0.7834986	9.844270	5.875397	11.106059	13.20220
##	gene_10706	gene_9643	gene_6768	gene_4667	gene_14720	gene_7999
## sample_119	7.574397	6.8981600	10.372343	10.524463	10.612537	0.3754009
## sample_275	8.050790	7.1182247	10.759322	11.053179	10.094526	0.6640277
## sample_29	6.661365	14.4033117	10.338781	9.487181	10.180494	0.7051375
## sample_295	7.325575	4.4228049	11.128755	11.038556	10.924864	0.9482270
## sample_298	6.495030	12.3749797	10.805696	11.651254	10.149798	0.0000000
## sample_404	7.813101	6.3015163	10.787029	10.746355	9.773693	0.0000000
## sample_464	6.833624	7.1757641	10.532843	10.667803	10.377211	1.0912598
## sample_468	6.767695	0.8671848	10.560342	10.786874	10.011814	0.0000000
## sample_531	6.339301	3.0315187	9.915516	11.142190	9.862481	1.0180644
## sample_57	6.977520	5.8877860	9.751969	11.085685	9.733029	1.2159892
## sample_578	6.962052	7.0589550	10.628272	11.718576	9.087012	0.5667665
## sample_579	7.825856	7.2641611	10.547916	11.107871	9.766188	0.0000000
## sample_603	7.306007	6.6832333	9.602374	10.418369	9.883890	0.7712095
## sample_627	7.388379	4.6348836	10.801628	11.180326	10.235476	4.3220363
## sample_638	6.796909	3.6394753	10.875511	10.520295	10.275449	0.6050202
## sample_673	7.230126	5.6866770	11.094150	11.288047	9.856839	0.0000000
## sample_68	6.700537	3.1529459	10.622564	11.131040	10.367327	0.0000000
## sample_69	6.624235	5.8682447	10.965856	11.135478	9.482376	6.8558402
## sample_691	7.665883	12.4244839	11.552578	12.238500	9.901855	5.0685332
## sample_701	7.206985	5.4260467	10.329404	11.147618	10.274565	0.9432837
## sample_735	6.934045	5.8374639	10.804922	12.400815	10.540467	0.5319678
## sample_74	6.703101	6.6197123	9.643261	9.955294	10.012499	9.5694322
## sample_740	7.221964	6.2707812	9.872880	11.452205	10.205500	0.0000000
## sample_75	7.081702	12.0158924	9.885922	11.335301	9.490137	0.3576083
## sample_762	7.413738	3.4948636	10.484954	11.094196	10.061952	0.0000000
## sample_768	7.793538	8.7380112	10.748746	10.331555	10.111097	1.1928254
## sample_793	9.889595	4.6247326	10.784578	10.643243	10.612703	0.0000000
## sample_800	7.495823	6.5877715	11.145353	10.913293	9.482975	0.0000000
## sample_82	8.638193	1.3530409	10.272432	11.076949	10.237557	0.0000000
## sample_95	8.237373	6.5386390	11.088484	11.213602	9.684622	0.4443490
##	gene_19103	gene_5704	gene_4026	gene_18412	gene_322	gene_17428
## sample_119	11.137638	10.460476	9.611538	8.120321	13.82902	10.003167
## sample_275	9.786172	11.058330	9.395975	6.571328	12.36722	9.366834
## sample_29	11.625348	11.944503	10.049331	5.675542	13.07320	9.632886
## sample_295	10.601483	10.979347	9.689028	5.962211	12.32222	9.125894
## sample_298	10.039577	10.712123	9.079037	9.667236	12.99204	10.265439
## sample_404	9.821040	11.388367	9.181515	6.317613	12.89172	9.669952
## sample_464	9.927579	10.031756	9.494218	6.163124	12.39847	9.289205
## sample_468	10.954720	11.563997	9.093217	7.457126	12.66252	9.499401
## sample_531	9.691457	10.030998	9.389063	7.972463	12.54351	8.843246
## sample_57	9.627193	11.064453	9.426858	8.549742	12.72916	8.582473
## sample_578	9.815230	9.210511	8.940681	6.421169	12.54159	9.315578
## sample_579	9.924415	10.961356	9.611464	4.331777	12.48491	9.723282
## sample_603	10.177282	10.973367	9.174122	6.127315	13.89212	9.943608
## sample_627	10.208466	13.108560	9.367334	6.905122	14.13874	9.473221

## sample_638	11.283036	11.149449	8.992936	7.299767	14.29032	10.073967
## sample_673	9.391027	10.595845	9.331454	6.659068	11.78983	9.511763
## sample_68	10.533894	11.388798	8.682201	7.659125	12.59907	9.596063
## sample_69	9.636182	9.536952	7.888658	6.178488	13.56154	8.665560
## sample_691	10.016920	9.917432	9.468154	7.196371	12.90106	9.513734
## sample_701	10.913301	10.949097	8.866107	5.886182	13.66516	9.288555
## sample_735	9.990969	9.693679	7.806975	5.453010	12.76992	8.059317
## sample_74	10.654072	11.055730	9.408803	6.222461	13.85877	10.065672
## sample_740	11.405285	11.233476	8.530024	5.642982	13.15215	10.549515
## sample_75	9.406496	9.002688	9.344174	6.703696	13.97715	10.217461
## sample_762	10.921171	9.783722	9.791269	4.920274	12.70048	9.572209
## sample_768	11.144849	11.240928	9.845212	6.834977	13.42013	10.143842
## sample_793	10.753869	10.703722	9.840586	5.595193	13.09220	9.885713
## sample_800	10.373017	10.788523	8.619512	6.993935	13.23919	9.232083
## sample_82	11.187494	11.094493	10.633595	7.255718	13.25384	9.851521
## sample_95	9.966837	11.644501	9.565386	5.716258	12.87178	9.344613
##	gene_13090	gene_5449	gene_15839	gene_12019	gene_7359	gene_20128
## sample_119	11.042870	7.933030	11.658037	2.1909625	5.952362	6.380556
## sample_275	10.868884	8.195874	11.349834	2.1720714	4.675082	5.369023
## sample_29	11.736389	8.606983	11.169449	2.8685287	3.341929	4.812688
## sample_295	10.444352	8.111605	11.776396	5.2894103	7.055131	4.805024
## sample_298	11.179181	7.915312	11.190078	3.8998427	6.795572	5.135617
## sample_404	10.964269	8.440703	12.058834	3.0142839	5.672425	4.663134
## sample_464	11.106236	7.852205	10.495475	2.6058736	5.259619	5.665211
## sample_468	10.962187	7.995767	11.883590	4.1614846	7.284847	6.537489
## sample_531	11.575360	7.998726	10.841698	1.0180644	6.664966	5.057247
## sample_57	10.892884	7.246209	10.512642	2.1068173	6.217712	5.522966
## sample_578	9.627868	8.373326	10.726193	1.2890672	6.230927	4.035571
## sample_579	10.282254	8.369951	12.133441	2.3582837	3.476550	3.659045
## sample_603	10.655862	7.786518	11.833448	3.2977637	5.265129	5.096578
## sample_627	10.408776	8.226099	11.986820	3.8861014	5.830019	3.957042
## sample_638	10.802056	7.671937	11.943614	4.1219769	5.826480	5.112896
## sample_673	11.324422	8.590643	10.952523	4.8397329	6.262272	3.793355
## sample_68	10.274483	7.794760	10.729935	3.1529459	7.284357	4.580887
## sample_69	9.991139	8.958213	10.393423	2.8736360	6.528573	4.353281
## sample_691	10.157915	8.318638	11.430552	4.0632527	7.549669	5.679159
## sample_701	10.262142	8.517599	10.310124	6.1896052	5.456691	4.138552
## sample_735	9.896941	8.519904	9.960947	5.0489075	7.951704	2.666893
## sample_74	9.690927	8.328581	11.443757	0.6153221	7.197168	7.034007
## sample_740	10.089729	8.222128	10.527770	6.3338173	6.156860	5.001676
## sample_75	10.272641	8.372952	10.786825	5.5237815	6.401636	6.115916
## sample_762	10.036778	8.570209	10.083971	4.5290775	7.627446	3.843542
## sample_768	10.777305	8.790283	11.788082	4.7223458	7.458546	5.985714
## sample_793	10.402106	7.991295	11.403901	6.6127683	6.649864	4.666705
## sample_800	10.794611	8.315227	12.215824	1.6039289	6.136491	5.184090
## sample_82	10.265158	7.669615	12.349397	1.6195069	6.210650	6.240618
## sample_95	10.359903	8.137442	12.242260	2.2036710	3.375318	5.253505
##	gene_8561	gene_9067	gene_3786	gene_691	gene_18569	gene_10007

## sample_119	6.380092	10.186535	9.040040	12.323818	8.222896	2.856129
## sample_275	3.964519	5.646569	8.809578	11.010493	9.573568	3.198856
## sample_29	6.277682	10.813725	8.153010	11.722462	8.707207	0.960141
## sample_295	4.599133	6.181295	8.860491	11.013155	8.718098	4.985168
## sample_298	5.149162	9.391201	8.783627	10.852498	8.221287	5.516173
## sample_404	5.012314	5.934917	8.697520	11.372805	8.558826	3.789927
## sample_464	3.864632	9.589391	9.218408	10.310704	8.583842	2.309467
## sample_468	5.170979	5.620956	8.491516	11.392167	9.088854	5.468183
## sample_531	4.362245	9.595300	8.592382	11.198347	8.833551	4.078208
## sample_57	5.298882	9.694389	7.953882	11.484768	9.237480	3.534037
## sample_578	4.619436	10.526978	9.164067	10.897573	8.821442	2.127171
## sample_579	5.840091	7.139173	9.013873	11.345682	8.353495	2.644618
## sample_603	5.979543	8.206262	8.585109	11.778640	9.723098	1.936176
## sample_627	6.754099	9.174304	8.713483	11.556769	9.570452	4.267259
## sample_638	7.128912	9.618663	8.275915	11.159506	7.915539	3.577864
## sample_673	6.160239	7.955882	8.159528	10.660985	8.923547	3.521352
## sample_68	4.487268	6.546604	8.628398	11.292621	8.210004	4.511247
## sample_69	4.059104	9.903960	8.850715	11.270184	9.880627	2.058178
## sample_691	4.367818	7.258510	9.330283	10.568735	9.525867	6.794676
## sample_701	5.952187	6.599693	9.331557	10.150598	9.686145	1.253626
## sample_735	4.569242	8.263598	9.324822	9.702969	9.674913	5.966514
## sample_74	5.069406	9.944431	8.010752	11.946157	7.556383	3.402518
## sample_740	5.217618	8.515144	7.985478	10.975060	9.020483	3.102137
## sample_75	4.932434	8.749990	8.877125	10.872659	9.835733	4.654854
## sample_762	6.823940	8.307934	9.849461	10.500892	9.188611	5.341740
## sample_768	6.902435	6.890763	8.771301	11.721518	8.658751	3.614969
## sample_793	6.800887	7.724432	9.304840	10.894643	9.325458	3.437294
## sample_800	4.981665	8.172932	9.241835	11.265375	8.145484	4.169741
## sample_82	6.464606	6.230699	9.047154	11.968285	7.336560	5.094346
## sample_95	6.073822	3.424600	9.059842	11.597256	8.340496	3.037927
##	gene_17858	gene_16972	gene_345	gene_9084	gene_5656	gene_3210
## sample_119	9.178921	9.123886	3.932108	7.897507	10.724966	8.255529
## sample_275	9.506756	9.351491	4.383455	7.616240	10.154211	9.793792
## sample_29	8.482848	5.434004	10.804349	8.281313	9.689598	10.058316
## sample_295	9.252209	9.031674	6.262138	7.457315	9.901623	8.922605
## sample_298	8.554708	8.684791	3.163209	7.866945	10.151904	8.731682
## sample_404	9.556065	9.395676	5.095325	7.618936	10.397921	9.162704
## sample_464	9.097073	7.927837	2.309467	8.153684	9.461046	9.964780
## sample_468	9.584716	8.764586	4.161485	6.985079	9.743638	8.667044
## sample_531	8.898532	9.688771	2.350780	7.865535	8.930714	8.615196
## sample_57	9.169502	9.419910	3.690048	7.881585	9.182697	9.034070
## sample_578	9.255505	8.652296	2.127171	8.716422	9.543859	9.693902
## sample_579	9.929974	9.577149	4.384561	7.610073	10.378923	8.610541
## sample_603	9.374818	9.315245	3.012050	7.559431	10.251092	8.437057
## sample_627	9.551952	8.927310	5.214906	7.654257	10.135325	8.328302
## sample_638	9.157289	10.435170	4.564598	7.815652	9.809165	10.023283
## sample_673	8.711990	8.583493	5.082221	8.813265	10.273714	8.266079
## sample_68	9.242009	8.723849	2.002703	7.476284	10.241268	8.815780

## sample_69	8.621528	8.262367	7.623530	7.707062	9.674199	10.480911
## sample_691	8.746753	10.121313	3.398145	7.323154	10.096373	7.399479
## sample_701	9.798063	9.162945	4.020449	8.170090	10.450438	7.985404
## sample_735	9.361468	7.439257	5.780090	7.603715	9.583967	9.762689
## sample_74	8.715670	6.572115	3.754888	9.146421	9.525333	8.886733
## sample_740	9.794932	8.495979	2.543001	8.536741	8.843340	9.260030
## sample_75	8.566279	8.847477	3.711010	8.472281	10.318803	9.172653
## sample_762	9.596344	8.196145	4.591883	8.270024	9.571237	9.293899
## sample_768	9.328718	10.082721	4.009894	8.408763	9.684281	7.170566
## sample_793	9.061071	8.654776	2.692092	8.136520	10.964406	7.510606
## sample_800	9.429923	9.309872	8.145484	7.194757	10.034441	8.634397
## sample_82	9.844221	7.829336	4.441397	8.165048	9.613662	7.506526
## sample_95	9.733115	9.477615	5.079779	7.385957	10.043287	8.587721
##	gene_11292	gene_8932	gene_12026	gene_8668	gene_582	gene_3615
## sample_119	9.231221	12.22052	0.0000000	6.0331454	3.1254197	11.109001
## sample_275	8.181426	12.03587	2.1720714	3.7926472	2.8042188	9.653218
## sample_29	10.085353	11.32216	3.2007084	5.2669327	5.3430631	9.019054
## sample_295	8.003411	11.35447	0.9482270	5.0061813	3.6376682	9.385491
## sample_298	7.994353	10.34117	1.8689633	11.2574762	2.1074535	10.382624
## sample_404	7.874231	12.04877	0.5285713	3.3634515	3.2425415	9.111689
## sample_464	9.266873	11.25280	1.4307670	7.3205479	3.3283626	9.928416
## sample_468	7.429808	10.97111	0.8671848	2.1031955	5.0684043	9.613082
## sample_531	8.600028	11.05798	1.6090457	8.6376501	0.0000000	10.277334
## sample_57	8.738079	11.42771	1.2159892	7.2272212	2.3129265	10.532746
## sample_578	9.266819	10.93742	1.9588056	2.4144332	0.0000000	11.802217
## sample_579	8.686087	11.97747	1.8586176	2.3582837	2.2486259	9.814482
## sample_603	9.259776	12.32861	0.0000000	2.9474417	0.0000000	9.696924
## sample_627	10.144849	11.33190	0.8033924	3.6496040	1.0824982	10.378002
## sample_638	8.588636	11.86660	2.5727684	6.5545735	0.0000000	10.753685
## sample_673	8.310063	11.58334	3.7429009	6.3063680	0.0000000	11.696189
## sample_68	7.927665	11.30489	0.0000000	3.0255427	4.2652343	8.319315
## sample_69	9.156387	11.15392	3.7716955	3.1185923	2.5903864	11.661239
## sample_691	7.623391	10.40638	2.1269398	8.0686224	1.2650167	11.018374
## sample_701	8.454472	11.56228	2.0807600	5.0774225	1.9141052	11.677530
## sample_735	7.687236	11.12367	2.6668929	6.3601957	2.3256455	11.517123
## sample_74	10.242114	11.89589	0.3402774	1.5168724	0.0000000	8.276325
## sample_740	8.972759	12.29735	4.6519127	1.7714631	0.4274989	11.685642
## sample_75	8.266281	10.95755	1.2669364	8.1751503	0.3576083	10.391297
## sample_762	9.061225	11.30629	2.2002846	5.6820198	3.7364100	11.520717
## sample_768	8.887504	11.96997	0.4021764	0.7163325	0.0000000	10.962556
## sample_793	9.697326	11.78134	1.0652276	3.9937468	0.0000000	10.201658
## sample_800	9.139459	12.15539	2.1373396	0.7483754	0.0000000	9.222261
## sample_82	8.090213	12.38127	1.0259501	4.6039230	0.0000000	10.324664
## sample_95	7.773693	12.10502	1.9579889	2.5080483	2.6802793	9.050061
##	gene_10804	gene_13324	gene_2049	gene_13587	gene_971	gene_15061
## sample_119	9.447449	1.6230240	3.906034	5.528912	5.7036102	11.314702
## sample_275	4.801412	3.2878860	3.527558	7.831421	6.9646759	9.529753
## sample_29	10.272327	7.0421080	2.058039	6.915999	3.9539450	9.509185

## sample_295	6.172025	5.9835614	4.391197	7.089996	0.9482270	10.065766
## sample_298	9.073721	9.5824186	6.165544	7.933036	4.5777249	10.813428
## sample_404	5.185795	2.6575714	3.918339	7.685569	7.5388175	9.645658
## sample_464	9.117776	3.8646319	3.972472	6.745170	4.8710847	9.517772
## sample_468	5.235520	3.6000318	6.046746	6.531076	1.7959331	9.771113
## sample_531	9.213330	5.7357280	4.924974	7.729288	5.0348584	9.171617
## sample_57	9.169502	5.3698364	4.943302	6.883657	5.2242603	9.246935
## sample_578	9.588731	3.5344476	2.442811	7.100021	5.6183326	10.763021
## sample_579	6.472567	0.8077671	2.963326	7.944876	6.8604290	10.413469
## sample_603	7.458710	1.4682184	3.012050	5.846421	6.7745367	10.307690
## sample_627	8.428431	1.0824982	2.242816	8.442404	4.7189260	11.590802
## sample_638	8.949506	1.3578335	3.846714	7.167468	6.1083342	10.468451
## sample_673	7.905170	4.7658583	3.112266	7.090843	0.9656181	10.061749
## sample_68	6.463001	3.4790236	5.216618	6.237183	1.5259185	9.919300
## sample_69	9.456617	2.3416158	4.645730	8.722254	2.3416158	10.034070
## sample_691	5.898653	6.6204369	5.636738	8.703703	5.9121660	11.023054
## sample_701	7.484831	3.6480857	2.411236	6.962005	4.8880613	10.085300
## sample_735	6.687998	1.2250281	5.510585	9.621198	3.7047392	9.514424
## sample_74	10.630786	2.0674666	5.428226	4.577266	2.0674666	9.696948
## sample_740	9.840685	0.4274989	4.811553	6.744376	4.9543173	9.709601
## sample_75	8.461958	7.6928386	4.885013	7.991216	6.1676409	10.573041
## sample_762	9.870989	2.2002846	4.765402	8.841363	3.4275928	10.810813
## sample_768	6.917861	1.7006173	4.397611	6.707635	5.9857145	10.309431
## sample_793	7.149056	0.0000000	3.341545	6.655087	4.2078929	11.713807
## sample_800	8.078113	1.8951475	5.831097	8.545308	7.8291455	10.408096
## sample_82	5.818743	4.5086188	3.703964	4.572847	6.1060090	10.378436
## sample_95	3.037927	1.8174595	1.824809	8.141131	7.1277468	9.541846
##	gene_10155	gene_8127	gene_13718	gene_11375	gene_19270	gene_17576
## sample_119	0.000000	15.50792	10.33499	8.873607	8.802413	6.052653
## sample_275	1.969528	14.21463	10.77433	8.224344	6.964676	6.845089
## sample_29	0.000000	15.09795	10.58302	8.680859	7.138845	5.423450
## sample_295	3.101734	13.95109	10.71752	8.362404	6.696369	5.451406
## sample_298	1.077790	14.93563	10.54448	10.223278	7.000000	5.278041
## sample_404	1.685626	14.27345	10.71632	8.919418	7.133204	6.670798
## sample_464	2.978122	15.54741	11.80021	10.087529	6.262076	3.407339
## sample_468	0.000000	13.70019	11.23801	8.260962	5.885145	6.556563
## sample_531	1.503451	14.57274	12.28389	10.547203	7.445156	3.424895
## sample_57	0.000000	14.54901	12.78038	10.286049	6.842665	4.813571
## sample_578	0.000000	14.54901	10.47330	9.806727	6.945245	4.077303
## sample_579	3.008433	14.53805	10.91444	7.362838	6.946333	5.721985
## sample_603	1.518888	14.97501	10.91542	9.312842	8.402824	3.919750
## sample_627	1.863106	15.35161	11.61961	8.502927	7.678072	5.939172
## sample_638	1.420940	14.48550	11.13046	10.622281	8.665620	5.458864
## sample_673	0.000000	13.85343	10.64990	10.498001	9.152176	7.002466
## sample_68	2.426506	13.96624	10.73025	9.294777	6.045216	5.042421
## sample_69	2.593641	14.71273	10.95308	9.011001	7.780087	5.180395
## sample_691	1.421371	14.15421	11.00976	9.083953	7.460521	5.606223
## sample_701	4.420880	15.08767	11.38294	9.093959	7.907131	3.700240

## sample_735	3.263861	13.60763	10.08636	8.095044	7.658868	7.162250
## sample_74	0.000000	15.52508	11.03082	9.655683	6.237795	5.618776
## sample_740	2.747967	15.44686	11.25664	7.814788	7.318624	4.392434
## sample_75	1.660062	13.73762	10.07800	9.117461	7.253772	4.017022
## sample_762	0.000000	15.85387	10.71971	7.746111	7.218578	4.765402
## sample_768	1.089769	14.89154	11.03795	9.158983	7.415725	4.934384
## sample_793	2.553090	15.48718	11.26734	9.422824	8.298800	4.369117
## sample_800	2.587581	14.95833	11.32470	8.661297	6.742357	6.881958
## sample_82	1.786973	14.18201	11.11755	9.354047	6.493849	5.589557
## sample_95	2.267746	14.09859	10.64105	7.883535	6.746756	6.042590
##	gene_4989	gene_13313	gene_3498	gene_20179	gene_3830	gene_3230
## sample_119	7.574397	1.1300742	11.009773	9.611538	7.123956	0.0000000
## sample_275	7.075105	5.2411003	11.153945	6.722698	8.783028	1.1169635
## sample_29	7.522181	2.9879575	12.780860	7.003647	10.081577	0.9601410
## sample_295	6.884036	2.9947788	10.617789	7.239455	5.951418	0.5506052
## sample_298	6.273993	1.9943263	10.167381	6.977394	5.738541	2.3158850
## sample_404	5.270925	4.0039800	10.761884	6.619781	7.631868	2.0346735
## sample_464	7.551101	3.0615688	11.858035	8.065104	4.628902	6.6509254
## sample_468	7.186243	2.2353288	10.924761	6.338822	6.028647	0.0000000
## sample_531	6.271035	2.4889263	11.970390	8.242864	5.244465	0.0000000
## sample_57	6.698788	1.8662750	11.247904	9.227962	3.762157	6.1787226
## sample_578	7.484267	0.0000000	9.825978	9.219856	4.532111	1.7681631
## sample_579	5.849509	5.6483418	10.247049	7.570751	7.913787	2.0006851
## sample_603	5.667418	0.4365883	11.898987	8.338135	4.194914	0.0000000
## sample_627	6.146867	1.3163196	11.376049	9.414571	7.599876	0.0000000
## sample_638	6.140654	1.7417905	11.634412	9.817937	7.843507	0.0000000
## sample_673	5.561265	0.9656181	10.761817	7.828613	7.394394	0.9656181
## sample_68	7.439257	2.1320832	10.192971	6.620123	3.703776	0.4603756
## sample_69	6.669792	0.0000000	9.822748	9.990487	6.240183	0.9291850
## sample_691	6.979900	1.4243847	10.321274	7.060404	6.364434	2.1269398
## sample_701	6.095382	0.0000000	10.555030	9.225448	3.537768	1.2536263
## sample_735	6.254630	1.6912661	11.134580	8.290558	8.940957	2.7647920
## sample_74	5.587200	0.8463128	10.626914	7.068725	3.289333	0.6153221
## sample_740	5.947079	0.0000000	9.989054	8.968791	6.283611	0.7567671
## sample_75	6.498444	0.6439485	10.936174	7.935943	3.829931	2.6655292
## sample_762	6.737890	2.4913657	10.925755	9.248321	6.413509	2.2002846
## sample_768	6.492706	2.5416140	11.730012	9.105256	8.392292	0.0000000
## sample_793	7.599972	1.0652276	11.341091	7.699371	6.546867	0.4480552
## sample_800	5.662015	2.1373396	10.003237	6.331798	7.892440	0.7483754
## sample_82	6.285776	0.6023619	10.995075	7.138579	3.691288	0.6023619
## sample_95	5.345957	4.9342098	10.439239	6.609621	7.090059	0.4443490
##	gene_6277	gene_13218	gene_844	gene_5406	gene_308	gene_3930
## sample_119	8.904358	8.023133	6.251448	5.061444	8.032988	8.884537
## sample_275	10.248710	10.445367	7.620528	5.962725	6.792816	9.101232
## sample_29	8.721280	9.302945	5.009025	3.588637	7.577738	8.301629
## sample_295	7.358634	7.378919	7.453493	7.378919	9.403425	8.914292
## sample_298	10.710695	7.323406	7.619413	5.452048	7.614533	8.605946
## sample_404	10.097044	10.406885	7.398308	4.291611	6.252120	8.849452

## sample_464	9.546412	9.909038	9.771174	2.605874	7.027375	8.383519
## sample_468	7.165681	7.384274	7.911853	7.827997	9.364314	9.085576
## sample_531	7.578810	8.386802	8.369017	4.589050	6.419251	7.542103
## sample_57	7.615725	8.704160	9.776114	1.577489	7.937156	8.338981
## sample_578	11.087257	8.084553	8.394115	3.902624	9.618182	8.652296
## sample_579	10.676530	10.377709	7.123728	5.983881	6.832206	9.026385
## sample_603	11.853699	10.104599	7.328379	4.401992	6.760500	9.384002
## sample_627	10.345738	10.292299	8.024115	5.478703	7.980082	9.751835
## sample_638	9.733925	8.795627	7.640259	2.369606	7.966009	9.021419
## sample_673	8.964029	8.294818	7.705584	2.641823	8.288239	8.708694
## sample_68	6.834888	8.152077	8.816984	6.546604	12.021983	8.904469
## sample_69	11.300753	8.579523	7.992185	2.206831	7.963728	8.287634
## sample_691	9.145306	7.105207	9.197273	6.267901	7.617218	8.852002
## sample_701	9.832790	10.537218	9.555105	7.070003	7.626330	9.809238
## sample_735	10.551228	8.982246	9.844729	3.492533	7.674701	9.064659
## sample_74	8.291097	9.213879	6.853347	4.655472	8.902857	8.353429
## sample_740	9.985629	9.073159	8.871234	1.617816	7.508191	9.114567
## sample_75	8.207088	7.004793	8.236985	3.103481	9.137778	8.473422
## sample_762	9.193683	8.741012	10.899221	7.732235	7.902954	10.088059
## sample_768	9.695512	9.754694	6.955720	5.271071	9.787078	9.514460
## sample_793	9.531882	6.644623	7.938727	3.619307	5.669333	9.974429
## sample_800	9.596292	9.843726	8.547932	6.695799	6.035430	8.975710
## sample_82	8.481791	7.111803	6.210650	4.721925	4.932416	7.822743
## sample_95	9.350276	11.124121	6.999436	4.568099	5.645264	8.820466
##	gene_5729	gene_6887	gene_2364	gene_3179	gene_15392	gene_14429
## sample_119	12.145477	7.877941	4.982273	6.421146	5.242218	7.201742
## sample_275	10.695167	8.935778	6.266393	4.499833	5.864840	7.062543
## sample_29	12.549623	4.927849	6.563803	7.288857	8.856905	8.481577
## sample_295	3.076525	7.115938	4.626533	6.715633	5.951418	6.085813
## sample_298	11.793754	7.022834	4.782796	6.562381	7.470919	7.145627
## sample_404	10.892209	8.358387	6.235273	5.287365	5.709495	6.518976
## sample_464	12.048926	5.831973	4.561424	5.171847	5.817585	7.687908
## sample_468	0.000000	8.379127	5.632986	6.491980	4.318707	6.989763
## sample_531	11.066621	4.875195	4.391788	6.185218	5.966126	6.375382
## sample_57	11.638449	5.536392	5.224260	6.721304	6.962295	6.730314
## sample_578	12.219631	8.876987	5.250010	5.231647	6.095344	6.899454
## sample_579	12.613138	8.808231	6.066978	4.555454	5.488721	5.349609
## sample_603	11.903520	8.089207	5.485952	6.052481	5.605945	7.753825
## sample_627	13.326415	6.790537	4.425714	5.956586	6.064932	6.896114
## sample_638	11.314957	7.768184	4.164513	7.773330	5.989148	7.657118
## sample_673	10.832131	8.884708	4.219517	4.886960	6.357563	7.065400
## sample_68	3.325185	6.385327	5.584737	5.814120	4.307210	6.575326
## sample_69	11.718875	7.616571	6.178488	5.162295	9.017307	6.528573
## sample_691	10.816064	7.056995	5.139175	5.420169	7.343239	6.034574
## sample_701	11.346082	8.287135	4.561442	4.413093	7.175085	6.592813
## sample_735	10.084583	8.684093	4.723296	3.492533	8.248872	7.949103
## sample_74	13.219346	4.758293	6.326525	9.052712	6.836656	5.174183
## sample_740	11.674726	10.085831	4.690959	5.711046	7.648696	8.160865

## sample_75	10.990643	8.192288	5.274016	6.453493	8.397230	7.120134
## sample_762	11.532273	7.587740	4.709853	5.323348	6.351332	6.963428
## sample_768	10.994509	7.613126	4.668652	6.492706	5.036842	6.802103
## sample_793	12.765897	8.929572	4.913287	6.064238	5.239581	7.684124
## sample_800	12.210866	7.040575	5.877553	5.974337	5.042456	7.559607
## sample_82	12.158856	7.733605	3.294077	5.969514	6.785721	6.597841
## sample_95	11.286442	8.436478	5.706329	4.791205	5.793339	6.469863
##	gene_4102	gene_10565	gene_15857	gene_7287	gene_12132	gene_1935
## sample_119	10.82194	6.400994	11.089656	2.3675109	10.274308	6.967088
## sample_275	11.23803	7.731808	10.345206	0.0000000	10.177694	8.644426
## sample_29	10.70842	10.790039	11.257512	1.8161081	10.249718	6.820958
## sample_295	11.32184	9.786389	9.932826	2.7174077	9.690653	8.532804
## sample_298	11.33304	8.690588	11.146632	4.2407723	10.696063	8.216940
## sample_404	10.63395	8.469805	10.331107	1.4698339	10.903249	8.618727
## sample_464	11.13626	4.755069	11.146728	0.6464392	10.183933	7.700072
## sample_468	10.80027	9.529477	9.766343	2.9247080	10.361867	8.763969
## sample_531	11.09056	5.244465	10.908625	5.1223870	10.719380	8.194934
## sample_57	11.05730	4.911946	9.864760	4.0768669	10.542384	8.769428
## sample_578	10.87113	5.888234	11.570074	2.5391092	9.622592	8.376464
## sample_579	10.99411	8.371585	10.681151	0.0000000	11.076882	9.009815
## sample_603	10.60309	5.404045	11.315546	1.2710670	10.948046	7.671251
## sample_627	10.89802	7.416975	11.196909	2.2405586	11.797305	8.528931
## sample_638	11.08551	6.124585	11.459421	3.3387952	12.103914	8.070083
## sample_673	11.08437	6.712472	10.860877	2.4028040	10.122828	7.906398
## sample_68	10.91406	9.174157	10.390545	3.4790236	9.863336	8.320638
## sample_69	11.38876	10.134388	12.193553	2.3416158	9.967226	7.922489
## sample_691	11.20986	8.322437	10.933344	3.5441391	11.160829	9.208466
## sample_701	11.34787	4.349195	11.940460	4.5329652	10.754345	7.931033
## sample_735	11.60627	6.706586	11.304944	2.8564479	9.193649	8.441492
## sample_74	10.64260	12.105185	11.677839	2.8312692	11.035761	7.225718
## sample_740	10.33479	5.986848	10.716708	2.7787553	10.005119	7.578826
## sample_75	11.02045	8.584365	11.695372	2.3838840	10.430348	8.737626
## sample_762	11.46885	6.530358	11.579288	3.2045790	11.010458	8.628260
## sample_768	10.57630	7.374387	10.650890	2.9544383	11.499701	8.920788
## sample_793	11.03846	6.262029	11.254710	1.9683857	11.423158	8.564824
## sample_800	10.23984	10.275089	10.645821	2.6869039	10.839196	8.527981
## sample_82	10.41010	6.345464	11.052521	1.8443053	10.555279	8.034766
## sample_95	10.98112	8.542386	10.500145	0.0000000	11.194412	9.248570
##	gene_1751	gene_7018	gene_12736	gene_4555	gene_11921	gene_3323
## sample_119	0.9195308	7.061139	9.083221	4.811101	8.055727	6.217283
## sample_275	0.6640277	7.703772	9.745131	6.729226	8.667505	6.244318
## sample_29	1.5315686	5.036991	10.611043	2.171399	9.352474	4.011460
## sample_295	0.0000000	9.889658	10.327654	4.199068	9.522162	7.656546
## sample_298	0.0000000	7.631526	9.651672	2.474073	9.347807	7.179023
## sample_404	1.2187192	6.767721	9.352454	6.419271	8.846406	6.325593
## sample_464	1.4307670	6.698816	9.073820	4.557968	8.782117	4.296795
## sample_468	0.0000000	10.117812	9.842732	6.018320	9.441583	7.355518
## sample_531	2.1979912	7.297833	9.195301	2.504646	8.948788	6.843004

## sample_57	1.8662750	7.395371	10.576730	4.027986	8.685737	6.204833
## sample_578	0.0000000	7.178794	10.421613	7.280993	8.640169	5.922978
## sample_579	0.0000000	6.168307	9.854672	5.494979	9.366130	3.782199
## sample_603	1.0427144	5.677415	10.544568	8.907176	8.869699	2.655443
## sample_627	0.4569113	7.781858	9.988969	6.905471	9.245106	5.429659
## sample_638	1.0299829	7.004333	9.063400	5.513693	9.875482	5.934889
## sample_673	0.0000000	9.304529	10.593158	6.121270	9.105518	6.493466
## sample_68	0.8089204	9.369654	10.190047	5.804931	9.414057	7.179760
## sample_69	0.0000000	7.756156	10.684582	3.718690	8.212667	6.178488
## sample_691	0.0000000	7.845615	10.828724	5.972155	8.209336	7.363048
## sample_701	0.0000000	5.828774	9.706423	5.519432	9.633473	5.077422
## sample_735	0.5319678	9.894240	10.733066	11.207570	7.877057	10.102829
## sample_74	1.3761237	7.848754	8.791094	2.812272	9.971055	7.057244
## sample_740	0.0000000	6.405997	8.995911	5.912971	11.469153	5.581972
## sample_75	0.3576083	7.945853	10.061668	5.319542	8.850999	6.959863
## sample_762	0.5980793	9.010399	10.019452	7.289834	8.400760	7.600968
## sample_768	1.9610306	7.317367	8.995250	5.968337	9.937027	4.993819
## sample_793	0.4480552	7.290010	8.976301	5.327253	9.958804	3.531169
## sample_800	0.0000000	8.128056	9.076425	4.733273	9.185671	7.625658
## sample_82	1.3530409	7.695089	10.001929	7.053904	8.138763	5.072299
## sample_95	0.0000000	5.821213	9.301791	3.027101	8.945239	4.864478
##	gene_11149	gene_4633	gene_623	gene_9409	gene_10759	gene_5267
## sample_119	8.531748	12.81909	10.654896	9.141453	10.279146	4.630761
## sample_275	8.938909	11.60392	8.249497	8.811458	10.229179	4.972307
## sample_29	8.892740	11.06317	8.015694	10.340184	11.124742	5.290362
## sample_295	9.185834	11.21984	8.741753	8.840155	10.050665	5.929583
## sample_298	8.748465	11.59453	11.525942	7.607197	9.953421	1.579952
## sample_404	8.050714	11.88273	8.410892	8.784056	10.626302	3.300065
## sample_464	8.789508	11.09659	8.648271	7.631446	10.226761	6.056883
## sample_468	8.120632	11.31959	8.383497	8.356558	10.353709	5.759089
## sample_531	8.701220	11.43310	8.561395	8.120647	10.204608	6.012687
## sample_57	7.921496	11.99566	9.026770	9.034070	11.414838	6.528023
## sample_578	8.874179	11.52626	8.912144	8.480233	10.017658	2.277955
## sample_579	7.888883	12.11684	7.799372	9.143965	10.430045	5.626594
## sample_603	8.540624	12.29522	12.294813	9.432026	10.407151	2.063572
## sample_627	8.233260	12.46490	9.969905	9.783656	10.101477	4.120468
## sample_638	7.275715	11.38310	10.043014	8.574904	9.899290	2.750821
## sample_673	7.936373	11.86012	8.973620	9.039122	9.591446	4.863542
## sample_68	8.703173	10.85204	7.990858	7.541778	10.095700	4.251348
## sample_69	9.053353	12.31539	8.251710	8.042524	9.905322	6.169454
## sample_691	9.216460	11.22059	12.904817	6.780008	9.089167	3.544139
## sample_701	7.676006	12.66568	8.946573	10.435128	10.239193	8.074538
## sample_735	9.042042	11.28935	8.737474	9.126180	9.184833	6.564302
## sample_74	8.883489	11.75923	7.639551	5.444707	10.554004	0.000000
## sample_740	8.183288	12.57128	8.326555	10.868390	9.846642	2.036925
## sample_75	8.408024	11.91871	10.502623	6.734506	10.183313	0.000000
## sample_762	8.528673	11.56346	9.469328	9.040265	10.283864	3.679717
## sample_768	8.476738	12.15576	10.059872	9.702209	10.014397	4.739810

## sample_793	8.605720	12.54757	9.976636	9.096054	9.309815	7.126674
## sample_800	8.393154	12.11815	7.744773	8.396070	10.082269	2.831715
## sample_82	8.865155	11.97220	6.016463	7.522880	10.045787	5.831928
## sample_95	8.319473	12.20561	6.002572	8.499898	10.374083	4.545999
##	gene_3639	gene_8855	gene_3465	gene_10984	gene_535	gene_16399
## sample_119	7.353641	8.687526	8.613502	10.240338	8.134123	3.654195
## sample_275	9.210591	6.957915	10.303461	10.358761	9.883114	3.526094
## sample_29	5.545292	8.300188	9.779790	10.590709	8.388314	3.511000
## sample_295	10.119188	7.321387	9.056090	9.017513	12.689124	8.386647
## sample_298	9.728856	8.438172	9.063716	10.536131	9.435399	4.338596
## sample_404	8.267161	6.969657	9.714587	10.688355	9.149275	4.475014
## sample_464	8.040536	9.598888	10.187649	10.118682	10.624549	4.377485
## sample_468	11.888302	9.738776	8.813807	9.608521	13.575232	6.046746
## sample_531	8.165128	9.709237	9.127605	10.979932	11.313246	3.424895
## sample_57	8.183536	8.955510	9.789925	10.801733	9.471903	4.594071
## sample_578	9.728684	7.271584	9.701509	9.512835	7.476487	4.701776
## sample_579	7.983894	6.130173	9.710939	10.085539	9.435008	4.410253
## sample_603	6.820358	5.278327	9.639995	10.441647	4.705331	4.624972
## sample_627	8.796822	6.238703	8.953166	9.991196	9.913618	3.811492
## sample_638	7.221345	6.647474	9.033698	9.890606	8.071763	5.432999
## sample_673	9.135163	5.659733	9.766345	9.992584	9.493759	5.964981
## sample_68	11.881710	8.759326	9.635412	9.849294	13.414963	5.842753
## sample_69	8.356905	8.232320	10.745658	11.013958	7.576469	5.087539
## sample_691	11.352181	9.883967	9.446856	8.891902	9.536267	5.056419
## sample_701	6.550824	5.313366	9.208332	9.420710	7.585323	3.288048
## sample_735	9.846827	8.397435	10.150268	10.105162	9.051772	3.434215
## sample_74	7.334246	7.991278	9.912789	10.550939	8.601771	4.254375
## sample_740	7.218626	3.102137	10.127530	10.315387	7.121927	4.986060
## sample_75	9.216290	5.442127	10.259767	10.278206	8.194673	4.892002
## sample_762	8.608687	4.765402	9.732295	9.307998	9.628533	5.207791
## sample_768	8.340291	6.258342	9.571277	9.980825	9.804818	5.158705
## sample_793	7.374753	4.841802	9.644404	9.211329	7.658354	2.985355
## sample_800	8.711299	7.316671	9.865266	10.170876	9.919622	3.298482
## sample_82	7.404324	5.933273	9.659595	10.092414	7.812806	3.632233
## sample_95	6.885233	6.697468	9.528263	10.385409	8.159448	3.159371
##	gene_9730	gene_10605	gene_11728	gene_14921	gene_6565	gene_17646
## sample_119	2.5974120	7.684700	10.123746	10.331667	10.206270	5.687063
## sample_275	5.1958082	8.353905	10.889907	9.702626	9.637424	5.195808
## sample_29	3.6259367	8.923749	9.737256	10.087025	11.218545	3.831543
## sample_295	2.2381448	7.989150	10.277183	11.673623	10.076976	7.708760
## sample_298	1.9943263	8.731489	10.355120	10.544800	9.659566	4.765312
## sample_404	3.2337500	8.226219	10.358574	9.867566	10.042398	5.366864
## sample_464	5.6652108	8.623486	10.966498	9.930925	10.074730	4.814243
## sample_468	1.6137202	8.602910	10.135363	9.973296	10.487679	7.685106
## sample_531	3.6179684	8.330478	10.829548	9.632116	10.495885	4.649230
## sample_57	4.2869994	8.244597	10.776918	10.249671	10.880785	5.543573
## sample_578	4.8535716	8.865272	10.156121	10.881840	9.263436	4.532111
## sample_579	3.5243897	8.568009	10.466851	8.905158	9.435789	5.072337

## sample_603	3.5794336	7.785479	9.886014	9.219297	9.777548	4.401992
## sample_627	0.0000000	7.786734	10.290146	9.731499	10.709299	6.165709
## sample_638	2.6345933	7.867390	9.889414	10.036586	10.970163	5.605014
## sample_673	2.5272455	9.098011	11.212138	10.132654	10.037094	7.193032
## sample_68	2.2507797	8.939159	10.252866	10.259967	10.411405	7.423553
## sample_69	1.2364620	8.653723	10.793555	10.523180	10.320834	5.317550
## sample_691	2.8428984	8.305989	11.209173	11.135139	9.482376	5.836240
## sample_701	1.7256539	8.657715	10.728669	9.826938	9.706208	4.975305
## sample_735	2.0430645	8.327050	11.574139	11.379514	9.141070	6.944519
## sample_74	2.1562024	8.189612	9.832314	8.844790	11.121443	4.729656
## sample_740	3.7457644	7.329554	11.003680	10.389223	10.064864	5.405410
## sample_75	4.0903284	8.938371	10.273947	10.564226	10.674113	4.409608
## sample_762	0.5980793	7.855747	10.328910	10.102291	10.302753	6.585416
## sample_768	2.2803328	7.915736	10.087821	9.409370	10.348739	5.145661
## sample_793	2.3236007	8.660385	10.717505	9.037270	10.253800	5.333778
## sample_800	2.5259185	8.207722	10.811607	9.340455	9.809156	5.262843
## sample_82	2.2101395	7.613465	10.582330	10.211949	9.518002	5.298083
## sample_95	4.0132300	7.982800	10.206734	8.518464	9.421053	4.405761
##	gene_5455	gene_4975	gene_10909	gene_16248	gene_8244	gene_7094
## sample_119	12.65828	0.0000000	6.458725	6.0908958	11.454937	10.115655
## sample_275	13.16792	2.1720714	8.277185	0.0000000	10.818886	9.151027
## sample_29	13.82817	0.3952822	8.452748	7.2019671	10.455420	10.075693
## sample_295	13.70130	1.7327902	6.615600	0.5506052	11.149855	9.709194
## sample_298	14.49221	0.9962441	9.154851	2.9933845	12.491710	10.243936
## sample_404	13.18605	1.2187192	8.351664	0.0000000	13.693912	9.444135
## sample_464	13.14814	1.0912598	7.158741	5.0783921	10.999147	10.321579
## sample_468	13.67469	0.4978423	7.318751	0.0000000	12.405003	9.137168
## sample_531	13.57761	0.0000000	8.240424	5.2638419	12.815788	11.087164
## sample_57	13.07505	0.0000000	7.576265	3.2629292	12.191176	10.310306
## sample_578	13.30997	1.2890672	6.821799	7.3963393	13.187760	9.958181
## sample_579	13.05077	1.3224474	7.576212	0.0000000	10.715361	8.679501
## sample_603	13.31327	0.0000000	6.746313	10.5284248	10.744699	9.546206
## sample_627	13.31417	0.0000000	6.571065	6.3283913	9.633393	9.150857
## sample_638	13.67237	0.0000000	8.374331	4.4829193	11.784287	9.773968
## sample_673	13.50299	2.7479674	7.538166	1.9481523	12.066674	9.415391
## sample_68	14.04168	0.4603756	7.270370	0.4603756	12.668947	9.517581
## sample_69	13.11573	0.9291850	7.848241	5.1982702	12.083350	10.484561
## sample_691	13.77786	2.0311481	7.623947	3.8828876	11.753580	9.666038
## sample_701	13.28926	1.9141052	7.342697	3.2182232	11.099887	9.538191
## sample_735	13.78247	1.6912661	7.008619	2.3256455	13.266832	9.478806
## sample_74	13.09198	0.3402774	8.877845	2.1562024	10.766057	10.320722
## sample_740	14.67683	0.7567671	6.218289	1.0247452	11.341719	9.679543
## sample_75	13.22104	1.7006173	7.134652	3.1499422	11.514522	10.854798
## sample_762	13.26887	2.2002846	5.592705	5.3047191	11.509829	9.922299
## sample_768	13.70389	1.1928254	8.501268	2.9544383	10.718233	9.723762
## sample_793	14.00845	1.4961044	7.170666	2.3236007	10.740422	9.782115
## sample_800	12.89443	0.7483754	7.333558	1.6039289	10.919385	9.506093
## sample_82	12.90965	0.0000000	6.902797	3.5063100	9.901620	9.822093

## sample_95	12.50720	3.0379269	7.508175	0.0000000	10.407034	9.292536
##	gene_12232	gene_4988	gene_14477	gene_20015	gene_1414	gene_4500
## sample_119	5.736148	2.367511	9.136891	5.760080	9.375689	1.1300742
## sample_275	4.118086	4.641696	10.036297	5.263222	9.651122	1.7388975
## sample_29	7.337452	1.680819	8.417861	5.819362	8.458497	0.3952822
## sample_295	3.547080	2.497587	10.055093	5.741882	9.925261	0.0000000
## sample_298	6.988741	2.408902	9.783617	6.756276	9.614507	0.7339617
## sample_404	3.482461	5.185795	9.284398	5.095325	9.670938	1.8698319
## sample_464	5.458188	2.733983	8.711891	6.937085	8.887473	0.0000000
## sample_468	5.770026	3.331361	9.481218	6.159157	10.015262	1.4050666
## sample_531	6.024096	2.027048	9.533774	6.511305	8.854476	0.5970305
## sample_57	7.461831	3.534037	9.730784	6.363919	9.585569	0.0000000
## sample_578	2.951812	2.653885	8.400290	3.593151	9.035019	1.9588056
## sample_579	3.858817	3.742297	9.143008	3.210264	8.831019	0.4596414
## sample_603	5.238370	5.890055	8.063023	5.792257	9.290222	1.2710670
## sample_627	5.670316	1.693989	8.579391	2.946057	8.880177	0.0000000
## sample_638	5.424274	4.265669	8.772611	3.888168	9.951465	1.2032012
## sample_673	4.998683	4.022119	9.343144	1.538985	9.596781	4.6879957
## sample_68	6.148218	2.730988	9.762038	5.745011	9.470193	0.4603756
## sample_69	3.620821	2.959993	9.653293	3.510683	8.951643	0.0000000
## sample_691	4.480931	3.578432	9.280365	3.359226	9.683870	0.0000000
## sample_701	7.323784	3.593975	9.464717	3.067553	9.304438	3.6480857
## sample_735	4.091844	2.448584	10.795910	3.434215	8.953987	0.0000000
## sample_74	5.563054	3.783038	8.311771	4.744048	9.199594	0.0000000
## sample_740	1.910311	2.916916	8.280896	3.158935	9.396952	0.7567671
## sample_75	3.858190	3.438998	8.441093	4.447301	8.270230	0.3576083
## sample_762	5.623094	4.737892	8.876477	4.496616	9.126325	7.7835876
## sample_768	5.592735	3.175892	8.697249	5.022643	9.873178	2.1815479
## sample_793	4.026357	2.424922	8.055895	2.608407	8.946476	0.7893537
## sample_800	4.632082	3.485246	8.952138	3.726646	9.388080	2.8317151
## sample_82	2.501694	3.045146	9.042910	1.619507	9.197330	0.6023619
## sample_95	4.165968	5.821213	9.367707	4.165968	9.611346	0.4443490
##	gene_20345	gene_6427	gene_6117	gene_4135	gene_18206	gene_3273
## sample_119	2.914163	6.000034	6.685548	2.448293	11.13402	10.949287
## sample_275	5.306477	5.893493	6.984783	5.195808	10.65546	10.799500
## sample_29	1.365021	5.920415	5.745011	3.249339	10.42531	9.183772
## sample_295	6.004600	5.754356	7.990286	8.116443	10.99717	11.612136
## sample_298	5.665020	6.000059	7.259282	6.135510	10.98868	10.513027
## sample_404	4.416434	7.948472	10.047833	4.081442	10.97033	11.536592
## sample_464	2.134780	4.166611	6.567398	4.337704	10.46926	10.145385
## sample_468	1.795933	6.734588	9.584716	6.636416	11.18920	11.492234
## sample_531	4.282025	6.292894	6.384264	3.617968	10.04716	10.136389
## sample_57	5.004043	6.721304	7.014891	4.132338	10.75445	10.507269
## sample_578	5.231647	5.286042	5.530938	4.853572	10.64961	9.973611
## sample_579	3.427083	6.841671	6.963786	3.858817	11.31095	11.053043
## sample_603	3.348700	5.957234	4.425909	3.579434	10.75506	10.949396
## sample_627	2.350045	6.845603	6.032454	5.078392	11.97472	11.247803
## sample_638	3.608998	6.429348	7.216300	4.851684	11.00606	11.057484

## sample_673	3.329554	5.885813	6.953382	6.617661	10.94139	10.185619
## sample_68	4.132692	5.693540	8.632505	7.381889	10.45185	11.098716
## sample_69	3.771695	5.690219	6.006547	5.250602	10.83379	10.156134
## sample_691	6.197922	6.130167	8.167699	8.307178	11.55231	11.194154
## sample_701	3.354636	5.863492	4.643643	3.700240	11.02582	10.953134
## sample_735	8.934399	4.129085	8.516397	7.674701	10.27417	9.982252
## sample_74	5.829116	6.423585	7.253573	3.402518	11.70659	11.026198
## sample_740	2.981323	5.357672	6.967503	4.319560	11.15797	9.987918
## sample_75	6.444202	7.060577	7.087686	5.423336	10.94175	10.529294
## sample_762	8.319577	5.561659	7.631192	5.806412	11.17815	10.670807
## sample_768	4.739810	6.288322	6.959457	4.431322	11.65058	11.508330
## sample_793	4.624733	6.716319	4.394315	3.484989	11.29521	11.145276
## sample_800	6.035430	6.970658	8.103304	3.867719	10.78463	10.409582
## sample_82	5.278955	6.381775	5.239925	2.851599	12.71415	11.224876
## sample_95	4.934210	7.594160	7.178914	3.375318	11.07215	11.840086
##	gene_10029	gene_9196	gene_3612	gene_11434	gene_4490	gene_17339
## sample_119	7.662056	5.740806	8.329366	10.952508	8.923907	10.530153
## sample_275	6.041644	6.169093	6.002725	14.846058	6.815460	9.296329
## sample_29	7.362181	1.680819	8.701757	9.618499	10.104481	10.779826
## sample_295	6.361539	6.709249	6.575037	12.854777	12.486848	9.728298
## sample_298	6.535868	6.842174	6.786923	12.507408	8.123475	10.118552
## sample_404	6.349590	5.913895	6.701785	13.130114	7.713215	9.326420
## sample_464	5.881371	4.169789	6.093171	13.489296	8.590205	9.254507
## sample_468	6.445738	6.502641	6.847295	12.165268	9.909827	10.004319
## sample_531	6.648182	5.721783	6.410582	11.410997	9.313200	9.176343
## sample_57	7.621165	5.480847	6.907696	12.439356	8.731360	9.944743
## sample_578	5.975892	5.096135	5.956905	12.509797	8.657465	8.493527
## sample_579	6.142869	3.931579	6.578321	14.853378	7.556445	9.359866
## sample_603	7.567492	5.733680	7.617695	10.605702	8.390724	9.854940
## sample_627	7.813384	4.425714	7.973686	11.474725	9.931893	9.956831
## sample_638	7.562097	5.134455	9.178849	11.571715	9.007613	10.439052
## sample_673	6.243431	6.271200	5.618353	13.270049	9.570804	8.231389
## sample_68	6.661051	6.145539	6.481330	12.229005	10.597931	9.910018
## sample_69	6.625358	4.836778	6.178488	12.561169	10.099137	9.582460
## sample_691	6.579871	6.906903	6.135937	11.902100	9.404454	8.158943
## sample_701	5.587056	3.846253	7.113919	12.238987	9.378323	9.206360
## sample_735	6.671308	9.632590	5.124618	12.956204	8.518153	7.404333
## sample_74	7.758916	6.690152	10.055689	9.984376	8.064770	11.320423
## sample_740	6.860330	3.547277	7.472220	12.447680	9.808866	9.234257
## sample_75	6.513003	6.476463	6.953323	11.894935	8.129510	9.178548
## sample_762	7.062553	8.420360	7.188411	13.004982	9.808726	9.038025
## sample_768	7.977635	5.771825	7.822513	12.416449	9.389453	10.227604
## sample_793	7.110415	4.467384	6.890276	11.272659	8.339917	9.685998
## sample_800	6.370368	6.489890	6.914852	13.091710	7.698302	8.822921
## sample_82	7.257718	5.239925	6.658297	9.351759	6.909029	10.151461
## sample_95	5.893804	5.267217	6.422122	13.920520	5.211997	9.101831
##	gene_1561	gene_19052	gene_20097	gene_976	gene_16198	gene_2997
## sample_119	7.614327	10.856807	7.114711	10.387145	3.7330506	9.403176

## sample_275	7.624803	10.143804	7.735773	10.178416	6.2828382	8.718437
## sample_29	6.254046	10.857756	8.367812	11.921012	5.0164874	9.580402
## sample_295	9.115051	10.263621	7.669778	11.098802	2.7295745	9.569698
## sample_298	7.159084	10.464127	6.915342	9.687684	3.2161599	10.005540
## sample_404	6.847483	10.273481	7.804557	10.849257	4.0719857	9.094276
## sample_464	7.294805	9.993858	6.683023	10.095015	1.5073126	9.416388
## sample_468	8.770043	10.570548	7.254018	10.675692	5.2060258	8.853381
## sample_531	6.650318	10.035225	7.339529	10.052745	0.5970305	10.525795
## sample_57	7.714445	10.544935	7.378260	10.646073	1.2763180	9.604081
## sample_578	8.248763	10.388825	6.911044	10.063152	5.2947347	10.074141
## sample_579	8.266515	11.053641	7.455196	9.980154	4.2939149	9.168632
## sample_603	7.033808	10.892808	7.387423	10.239754	5.1272412	9.326405
## sample_627	8.682419	11.210355	7.025527	9.886617	2.0731004	9.812039
## sample_638	8.863365	11.151600	8.121192	11.232391	3.2021356	10.359387
## sample_673	7.450494	10.141813	7.044722	10.150306	1.3113869	9.924893
## sample_68	9.419613	10.346658	7.216688	10.702528	4.3886993	8.764324
## sample_69	8.002393	11.067515	6.590794	10.213420	1.0506062	9.970480
## sample_691	6.819975	10.470567	6.644921	9.509296	3.2574633	9.766153
## sample_701	7.940038	11.438932	7.376820	9.813683	2.8298090	9.258434
## sample_735	7.493807	9.408608	7.240906	9.907696	3.3190398	9.174851
## sample_74	7.001442	10.574281	8.520513	11.529977	2.8018622	10.030226
## sample_740	7.716497	10.151841	7.749239	9.258406	3.8788724	8.923399
## sample_75	6.322714	10.667644	6.797882	9.361551	2.7225972	9.189432
## sample_762	8.860621	10.183561	7.188411	10.439249	3.0057952	9.842214
## sample_768	9.975318	11.251997	7.856930	10.475004	2.8757408	10.082295
## sample_793	6.691129	11.319458	7.544771	10.350320	2.1403206	9.666625
## sample_800	7.341078	10.271918	7.446058	9.592485	4.7461246	9.543378
## sample_82	5.764768	11.178491	7.598112	10.770416	4.4601921	10.601715
## sample_95	7.003490	10.446339	7.367170	10.934030	4.6861867	8.417080
##	gene_4145	gene_12167	gene_19119	gene_17017	gene_17947	gene_13552
## sample_119	10.53287	2.597412	8.173262	7.9469940	8.493003	8.960515
## sample_275	11.42474	6.451196	8.539655	5.1726155	14.004501	9.482207
## sample_29	11.12657	3.924309	9.529017	11.7820317	8.473925	10.160766
## sample_295	11.90836	7.362715	8.754260	0.0000000	11.713301	8.644613
## sample_298	11.59731	4.557551	8.576174	0.0000000	14.501253	6.866982
## sample_404	11.05828	7.096357	8.526300	4.1549558	13.464329	9.213051
## sample_464	11.13880	4.925767	9.754276	3.6212665	11.970012	11.483917
## sample_468	10.85554	4.226447	7.919239	0.0000000	13.180562	9.522219
## sample_531	11.39872	5.079288	9.705698	2.1979912	12.996330	10.850531
## sample_57	10.93639	5.392733	9.563036	4.3816018	13.022455	12.504034
## sample_578	10.72415	4.619436	9.924285	0.0000000	12.439169	8.890362
## sample_579	11.45982	6.703059	8.660894	7.7347570	13.766767	9.168632
## sample_603	10.65364	4.166804	8.608806	0.7712095	12.452923	8.521330
## sample_627	11.24288	4.239077	7.698920	6.0966983	12.454595	7.789168
## sample_638	10.73925	3.480213	7.584572	6.8070200	10.154704	8.362961
## sample_673	10.99625	3.460218	9.199712	6.6590683	10.149087	7.234702
## sample_68	10.84793	3.661533	8.374287	1.3241193	13.550194	7.122528
## sample_69	11.16418	5.598973	8.245272	11.8004955	10.837005	8.835090

## sample_691	11.38737	4.307749	8.248473	1.2650167	13.031460	5.092470
## sample_701	10.67757	4.695949	8.801922	2.9858647	8.597114	4.532965
## sample_735	11.10100	6.515919	8.491589	9.9009830	11.998104	7.442960
## sample_74	11.81759	3.166507	9.898136	12.2394702	10.943702	10.280562
## sample_740	10.79859	4.105033	7.423780	3.2135961	10.582010	8.174706
## sample_75	10.84270	4.137200	8.755635	1.5700257	12.035840	3.005813
## sample_762	10.40850	4.765402	8.595336	4.9918891	12.562366	8.538674
## sample_768	10.46945	5.119215	8.450461	6.2883216	9.937027	8.732571
## sample_793	11.53945	3.387955	8.960254	6.0082023	9.801084	3.855402
## sample_800	10.80012	4.372764	8.232848	3.8677186	13.682600	8.070840
## sample_82	11.24345	1.619507	9.329855	1.3530409	9.742296	10.506774
## sample_95	10.87264	6.199500	8.707066	2.7592411	11.714915	10.021924
##	gene_5838	gene_16507	gene_3833	gene_10453	gene_4692	gene_10770
## sample_119	11.545790	11.329107	6.354595	7.653182	7.464595	9.353577
## sample_275	8.669576	9.191970	7.142301	8.859448	7.536915	8.663352
## sample_29	8.520481	12.458127	6.306691	9.032139	7.109507	11.637386
## sample_295	9.361231	9.781825	6.488081	8.449367	8.087351	9.116261
## sample_298	9.855608	11.247868	6.058648	7.804125	10.164630	10.728856
## sample_404	8.299936	8.842240	7.128644	8.273362	8.024431	8.789837
## sample_464	8.664440	11.233626	9.431821	7.559775	8.868406	9.462202
## sample_468	9.790710	9.384022	6.524633	8.351107	7.771985	8.880122
## sample_531	9.395667	10.784144	8.445259	8.240424	9.425012	9.676172
## sample_57	8.928347	11.264677	9.516886	8.333075	9.154507	8.949733
## sample_578	10.365327	11.157631	6.468977	7.659318	8.960329	8.737721
## sample_579	8.487736	9.295521	6.666231	8.692643	7.683016	8.728818
## sample_603	9.043033	10.369499	5.291401	8.913703	7.770267	9.091496
## sample_627	10.505871	10.784668	5.442078	8.905056	7.616409	9.878074
## sample_638	9.011988	11.713979	5.916340	8.201266	7.997326	8.787997
## sample_673	9.853127	9.904237	7.344518	7.785649	8.369466	8.036635
## sample_68	9.374566	9.419613	6.406581	8.270973	7.539190	8.699267
## sample_69	9.140296	10.586483	7.899623	8.861974	9.430388	8.925053
## sample_691	10.161812	10.390169	6.708242	8.727778	9.119330	8.588512
## sample_701	9.378323	9.835708	6.143263	8.106291	8.493787	9.391269
## sample_735	9.749298	9.766386	9.138789	7.974902	7.450337	8.199398
## sample_74	9.119769	12.855680	5.914289	10.160981	9.808982	9.855283
## sample_740	9.018565	10.623909	6.078044	8.063800	8.030066	8.136320
## sample_75	9.301720	11.902590	6.794234	8.621935	9.342299	9.435504
## sample_762	9.881241	11.149398	6.998624	8.175789	8.473450	9.020424
## sample_768	8.790283	10.183871	6.014699	9.065123	8.771301	9.632630
## sample_793	9.270372	10.330469	5.915499	9.273772	6.706296	8.781963
## sample_800	8.950156	9.799280	6.865226	8.524150	7.423385	7.554397
## sample_82	8.849036	11.273055	5.957534	8.929430	6.723422	8.904686
## sample_95	8.507069	8.152148	6.609621	8.274816	7.703038	8.594466
##	gene_9886	gene_2242	gene_13472	gene_8763	gene_2198	gene_5925
## sample_119	3.8452893	6.773443	7.092895	2.3675109	7.783430	7.590056
## sample_275	0.6640277	8.372547	8.479505	3.7304769	9.398477	9.252668
## sample_29	0.9601410	6.031488	8.389670	3.7662135	9.762080	8.853958
## sample_295	0.0000000	7.045006	4.920222	2.0885487	8.496542	8.535011

## sample_298	1.4103411	7.079783	7.645925	2.8679558	8.711282	9.488977
## sample_404	0.5285713	8.098011	8.111991	4.6377146	8.384507	10.527692
## sample_464	1.7054029	7.431096	8.523068	3.4073392	9.632156	9.139628
## sample_468	0.0000000	6.629408	5.791650	0.4978423	7.886952	9.014935
## sample_531	1.3436352	6.253694	5.392657	2.1979912	9.270730	10.115421
## sample_57	1.8662750	6.826027	8.001875	3.7621574	8.706448	9.279548
## sample_578	1.9588056	8.227703	7.358765	3.4093366	8.753895	6.164728
## sample_579	0.4596414	8.660641	9.065306	2.2486259	8.839317	9.076380
## sample_603	2.0635720	8.639428	4.166804	1.9361762	8.843306	4.624972
## sample_627	3.0792245	7.304940	6.909593	3.0792245	8.291037	7.097937
## sample_638	3.4112364	7.725796	6.509910	3.0520680	8.007493	6.562548
## sample_673	3.3295542	7.566655	6.091336	2.2665767	8.646602	7.606775
## sample_68	0.0000000	6.880992	5.042421	2.0027025	8.332838	8.582447
## sample_69	4.5046077	8.226518	7.545389	3.2614556	9.101640	7.510044
## sample_691	2.7853204	8.555498	7.558283	1.8180325	9.459553	8.222954
## sample_701	4.5893680	9.324678	4.175868	2.8992723	8.242221	6.447817
## sample_735	4.4565696	8.303036	5.393042	2.5618638	8.722323	6.304314
## sample_74	0.6153221	6.138843	6.750191	1.3761237	9.691391	10.351591
## sample_740	2.1533514	7.428904	4.484215	2.8494789	9.168752	5.592322
## sample_75	2.7281164	8.210501	7.632763	2.5315686	9.207393	8.389993
## sample_762	6.0936362	8.223751	6.927351	3.9906463	8.411680	6.675167
## sample_768	5.1058080	7.188767	4.807628	3.2750518	9.022948	7.894448
## sample_793	3.9262367	8.153223	3.818646	0.7893537	8.769980	5.627435
## sample_800	0.7483754	7.231778	9.421364	2.9633076	8.755095	7.972727
## sample_82	2.3632552	8.277487	7.819438	2.3632552	9.253526	5.694331
## sample_95	0.0000000	8.557781	8.036025	3.8782153	8.294837	10.642674
##	gene_8741	gene_9948	gene_18763	gene_9522	gene_15537	gene_17964
## sample_119	6.820639	9.768717	9.504046	1.3137683	11.036146	9.194373
## sample_275	8.750114	10.404269	10.772875	3.1039843	10.550862	8.992868
## sample_29	8.072535	8.907017	9.339899	0.3952822	9.890263	9.813515
## sample_295	11.989086	8.343950	9.982723	6.3220273	9.795472	9.807492
## sample_298	7.574942	8.198671	10.224472	1.7316612	10.490971	8.648293
## sample_404	8.508880	10.638762	10.592793	1.6836515	10.575747	9.352386
## sample_464	8.752170	9.545320	9.702361	0.6464392	10.202736	8.571032
## sample_468	12.293371	7.620110	9.603943	7.9793680	9.842784	9.543884
## sample_531	8.760401	10.554598	10.189491	2.4889263	10.517236	9.580888
## sample_57	10.279158	9.882056	10.251234	2.1068173	10.132140	8.492598
## sample_578	4.501783	8.836707	10.035624	7.1787943	10.738844	8.626180
## sample_579	9.682359	9.660451	11.096794	0.4596414	10.909668	9.250902
## sample_603	4.762795	9.218442	9.350813	1.4682184	11.470532	8.972730
## sample_627	6.412827	9.592203	10.035913	2.9460571	11.146110	9.957529
## sample_638	7.489479	9.546621	8.850396	4.4315962	11.246070	9.768378
## sample_673	8.365303	10.982730	10.219398	7.6242983	10.146059	9.265971
## sample_68	11.293914	9.042524	9.670497	5.0909677	10.112948	8.965415
## sample_69	7.600195	10.014020	11.518476	2.6836965	12.118094	8.671973
## sample_691	7.523499	7.503555	10.797078	4.8044460	11.110444	9.352752
## sample_701	6.724555	8.477070	10.180742	1.7256539	11.453245	8.883126
## sample_735	5.986943	9.892210	11.629398	5.6322973	11.089239	8.229713

## sample_74	6.115847	9.136909	9.639418	0.0000000	11.392156	9.459280
## sample_740	6.457857	8.854133	9.272152	3.5472771	10.561412	8.284088
## sample_75	8.052112	8.116406	10.433877	0.8827604	11.573500	8.667559
## sample_762	4.818897	10.826405	9.847854	3.9906463	10.316191	8.620033
## sample_768	6.890763	9.918455	8.791329	1.3826673	10.323978	9.422948
## sample_793	5.062243	9.765914	9.709299	5.5842831	11.111038	9.941811
## sample_800	6.492941	8.639326	9.628870	1.6039289	11.165567	8.444737
## sample_82	10.088298	9.066805	10.803841	4.4753962	11.030888	9.489166
## sample_95	10.027920	9.875070	11.254426	0.7834986	10.773576	9.174137
##	gene_17132	gene_19189	gene_2205	gene_7459	gene_12263	gene_5464
## sample_119	7.262085	9.046540	3.8452893	0.9195308	8.502490	9.608246
## sample_275	6.383628	9.643762	3.8522386	2.6461857	8.982902	10.386736
## sample_29	7.450972	10.242233	0.0000000	2.5178804	7.309067	9.099169
## sample_295	7.206340	12.259193	5.3232980	2.3736760	10.601910	9.675154
## sample_298	8.372408	9.447098	3.3624138	0.9962441	8.465497	9.757878
## sample_404	6.099522	9.399462	3.5923853	3.6443639	8.750359	10.185210
## sample_464	8.042612	9.502112	1.0912598	1.4307670	9.708222	9.773040
## sample_468	4.536469	11.329325	7.7583232	1.4050666	10.237377	8.863626
## sample_531	7.838132	9.007445	1.0180644	1.3436352	10.303313	9.909377
## sample_57	7.706379	10.270681	3.0489859	0.7324863	10.083705	9.615074
## sample_578	9.265885	10.339304	0.5667665	0.9726927	8.971434	9.627868
## sample_579	5.859520	8.489242	1.8586176	2.2486259	9.234820	9.991890
## sample_603	7.855672	11.336992	0.4365883	1.0427144	8.317508	10.087370
## sample_627	8.252126	9.890587	4.2103265	0.4569113	9.056088	9.235330
## sample_638	8.454649	9.851274	5.1556966	1.4974846	8.630962	9.486069
## sample_673	8.459936	9.374550	4.0638135	3.5213523	9.720083	9.720083
## sample_68	7.074206	11.077837	6.3004807	2.2507797	10.263504	9.037371
## sample_69	10.673654	11.054692	6.1512155	2.4648773	9.958731	9.364634
## sample_691	8.142561	10.579099	6.7651899	2.7853204	8.956086	9.240088
## sample_701	8.401473	10.206318	4.7710193	1.2536263	8.773297	9.494558
## sample_735	10.449272	10.765560	4.6224443	2.3256455	10.904597	10.270003
## sample_74	7.951658	9.415562	1.5168724	3.1665066	7.223075	10.122957
## sample_740	9.404680	10.477708	3.8869695	1.6178157	8.476556	10.050869
## sample_75	8.892039	9.640583	4.0418738	1.4264258	8.270230	9.754904
## sample_762	8.890097	10.464648	5.3955461	2.4913657	10.031839	10.083971
## sample_768	8.063708	10.516754	4.4526613	1.1928254	9.054694	9.348062
## sample_793	6.666146	8.888902	5.7976589	6.7312376	8.905429	9.781519
## sample_800	7.327463	8.847427	3.7266460	1.2386646	8.909938	9.441255
## sample_82	5.663100	8.397846	3.5706451	1.0259501	9.998337	10.761834
## sample_95	5.808364	8.065459	2.9053124	2.5967205	8.139290	10.595705
##	gene_19193	gene_5200	gene_15714	gene_20521	gene_17261	gene_10641
## sample_119	0.9195308	7.1959086	7.684784	6.647561	10.710712	9.538142
## sample_275	7.2950709	1.4612142	7.136314	5.505208	11.040831	9.632248
## sample_29	8.3595300	4.7121941	3.732399	6.129693	11.887441	9.127386
## sample_295	2.0885487	0.0000000	7.777597	6.814256	9.962025	9.733403
## sample_298	0.4131608	2.4962834	8.517335	5.931168	10.505196	10.591017
## sample_404	8.1905655	1.6836515	6.546553	6.252120	11.403997	9.634748
## sample_464	0.6464392	7.7460440	7.158741	5.582532	11.490169	9.575199

## sample_468	3.0009555	0.0000000	8.090414	7.769263	9.754002	8.791199
## sample_531	1.0180644	6.8494364	7.683254	5.868237	11.621626	9.321906
## sample_57	1.5774893	3.6900483	7.986906	6.783260	11.270482	9.802644
## sample_578	0.0000000	4.1956941	7.005916	5.033921	11.049427	9.703225
## sample_579	5.0397920	0.4596414	5.224878	6.009316	11.185935	9.650248
## sample_603	0.0000000	3.2977637	5.096578	5.687341	11.104409	9.313148
## sample_627	1.3163196	2.9460571	9.625930	5.722149	10.709618	9.672867
## sample_638	1.3578335	7.5568969	8.683096	6.012622	10.613670	10.674016
## sample_673	0.5621814	7.1058876	7.426718	4.932694	10.535528	9.439729
## sample_68	2.3604486	0.4603756	8.244392	6.393737	10.442923	8.928891
## sample_69	6.4031774	0.9291850	7.616902	5.333814	11.446690	11.826604
## sample_691	0.8812731	1.0858325	8.816670	5.080551	9.779575	10.124522
## sample_701	1.2536263	1.9141052	6.369461	5.244788	10.632568	10.367131
## sample_735	0.9197596	5.9561891	10.873736	4.129085	11.609774	10.025679
## sample_74	9.6283107	1.6451021	8.791231	6.093543	11.035218	10.069369
## sample_740	0.0000000	5.5715464	7.975624	5.485327	10.696759	10.000324
## sample_75	1.0875986	4.7178957	7.765508	4.997396	10.932857	10.390878
## sample_762	1.3453964	5.3955461	8.846202	5.146342	10.992747	8.866729
## sample_768	0.9741622	2.1815479	7.434612	5.993015	10.281014	10.608920
## sample_793	0.0000000	2.6920924	7.436712	5.941379	10.502941	9.349069
## sample_800	1.6039289	2.6869039	9.535470	6.403075	10.661653	9.507912
## sample_82	0.6023619	6.5901020	6.381775	5.933273	10.998005	7.349321
## sample_95	8.1411313	1.2884176	5.154725	6.042590	11.040379	9.135478
##	gene_4386	gene_5183	gene_8922	gene_4924	gene_3057	gene_15771
## sample_119	10.80740	3.5111138	11.646716	10.376212	3.511114	12.93274
## sample_275	12.02039	1.4612142	9.911407	9.784260	5.100721	12.78725
## sample_29	11.54783	0.3952822	11.320028	9.305102	3.296384	13.14689
## sample_295	11.18081	1.5155106	10.317164	9.756470	7.828625	12.69277
## sample_298	11.68978	1.5799525	10.539992	9.023128	6.683682	13.54319
## sample_404	11.84238	2.0346735	10.105175	10.055052	3.014284	12.48236
## sample_464	11.66594	1.4307670	10.485839	9.479441	4.120468	12.73701
## sample_468	10.54948	1.1610492	10.220536	9.439016	1.405067	12.95050
## sample_531	11.16792	2.1979912	10.031701	9.156144	3.031519	13.22572
## sample_57	10.94835	0.7324863	10.606831	9.486610	4.335076	12.66463
## sample_578	11.99185	0.5667665	10.182655	8.677952	4.233090	12.43152
## sample_579	11.56146	1.3224474	11.149455	10.206392	5.604510	12.54493
## sample_603	11.23460	1.9361762	11.246937	10.353224	6.717539	12.66299
## sample_627	10.81242	1.6939887	12.486229	9.982708	3.366602	12.65782
## sample_638	10.13369	1.8499993	11.151105	10.494206	2.216300	13.06142
## sample_673	11.55001	1.5389851	9.908535	9.765808	2.527246	12.49951
## sample_68	11.33672	1.8606053	10.077243	9.312797	2.557582	12.32335
## sample_69	11.19274	1.2364620	10.073338	10.143064	4.254231	14.06856
## sample_691	11.72209	1.8180325	9.941926	8.813974	3.826915	12.88932
## sample_701	10.79632	0.5474498	10.571317	10.666135	1.253626	12.85217
## sample_735	12.15931	2.7647920	9.158059	9.215455	2.043064	13.36930
## sample_74	10.88094	2.0674666	10.450046	11.225738	2.659719	11.65672
## sample_740	10.67582	2.5430009	11.852748	10.136004	3.158935	13.19060
## sample_75	11.64315	1.4264258	10.707264	9.691042	4.748408	13.08960

## sample_762	11.44913	2.3531539	10.483403	9.441888	2.491366	11.53802
## sample_768	10.87756	3.0694100	11.268636	10.583572	2.075498	12.19011
## sample_793	10.77279	1.4961044	12.549850	10.449087	1.496104	12.52822
## sample_800	11.51928	1.2386646	11.363565	9.966455	1.238665	11.88984
## sample_82	11.48114	0.0000000	11.339466	10.437315	1.353041	11.85963
## sample_95	12.10162	1.6617040	10.156563	10.280968	4.405761	12.72311
##	gene_5139	gene_17459	gene_19790	gene_2384	gene_20525	gene_17145
## sample_119	10.323989	8.989482	10.720398	10.800948	10.265287	2.3675109
## sample_275	10.173315	9.892902	11.062282	10.453806	9.761127	3.9645188
## sample_29	10.989054	8.620909	9.921542	10.238847	10.432239	1.3650206
## sample_295	11.567609	8.672818	9.705183	10.435181	10.291712	5.5970335
## sample_298	10.353665	10.214040	9.963997	10.642304	8.895339	7.6859682
## sample_404	10.290906	9.351407	11.308316	10.736664	9.767115	0.0000000
## sample_464	9.778623	9.598888	10.089040	10.677297	9.646472	10.3132685
## sample_468	12.252529	8.698864	9.681286	10.344596	10.540128	0.0000000
## sample_531	9.560465	8.898532	10.448519	10.446401	8.857667	10.4239668
## sample_57	10.323764	9.161190	10.737222	11.147987	8.922457	11.6616133
## sample_578	10.373202	8.869733	11.085021	11.091396	9.453666	2.1271710
## sample_579	10.065982	9.805436	11.106157	10.625353	10.005119	2.8081379
## sample_603	9.712807	9.335904	11.859337	10.592233	9.857756	0.0000000
## sample_627	10.602031	9.690454	11.123882	10.342008	10.258472	0.8033924
## sample_638	11.739376	8.526804	10.648510	10.270295	10.048705	0.0000000
## sample_673	10.437814	9.577117	10.121585	11.116617	9.616181	0.0000000
## sample_68	10.793652	8.419341	10.050393	10.594782	10.035486	2.1320832
## sample_69	9.783324	8.020085	11.070248	11.263299	9.024838	0.9291850
## sample_691	9.465293	10.419739	11.150357	10.519381	9.775423	4.1341470
## sample_701	10.085300	10.270249	10.993858	11.354745	8.998260	0.0000000
## sample_735	9.153549	10.387522	11.148820	11.782597	8.770254	0.9197596
## sample_74	11.027332	9.554201	9.733917	9.399744	9.822604	0.8463128
## sample_740	9.325081	10.473168	11.530650	11.063031	8.661165	0.0000000
## sample_75	11.666326	9.767064	11.275001	10.083626	9.330372	5.8394255
## sample_762	10.140216	9.932951	10.198138	11.118403	9.001752	3.5591491
## sample_768	10.581455	9.045945	9.901096	10.408648	10.156905	0.4021764
## sample_793	9.737884	10.046374	10.903995	10.643243	9.926775	0.4480552
## sample_800	9.565553	8.110421	11.646293	10.337823	10.102934	4.2748207
## sample_82	10.768705	10.025140	12.259711	9.514939	9.509822	0.0000000
## sample_95	10.731658	9.728215	10.912328	10.613743	9.705961	1.4871261
##	gene_9609	gene_9845	gene_10681	gene_7496	gene_19589	gene_2152
## sample_119	7.368969	4.301741	5.061444	1.4766927	10.549852	11.545862
## sample_275	6.815460	5.172616	4.068688	1.1169635	9.934857	12.752968
## sample_29	7.898632	3.831543	6.514922	3.7662135	13.022030	9.830089
## sample_295	6.262138	5.436002	6.411386	9.0820293	9.719569	11.346514
## sample_298	6.473657	4.655712	6.128689	1.4103411	10.033203	12.595996
## sample_404	6.120686	4.531313	4.851804	0.5285713	9.878457	11.796571
## sample_464	6.240665	4.453899	4.337704	8.2980030	9.264222	12.335954
## sample_468	7.408933	5.703131	7.115127	3.6000318	9.729002	12.067132
## sample_531	8.259856	5.122387	6.302295	9.2683338	9.579921	12.379427
## sample_57	7.499240	5.172275	7.389687	7.7189849	10.531459	12.941425

## sample_578	6.522822	3.948321	4.701776	1.5483873	8.985202	13.153655
## sample_579	4.989557	4.919736	4.728786	1.8586176	9.310092	11.738147
## sample_603	7.269763	5.111403	3.885662	2.1806571	9.754126	10.534283
## sample_627	5.670316	5.125345	4.568476	1.6939887	12.456603	12.645393
## sample_638	8.050600	5.079941	4.785085	0.0000000	10.862451	12.351602
## sample_673	11.067172	5.324483	3.636381	3.2594985	9.479386	12.563022
## sample_68	7.062931	5.298141	7.385129	9.6402305	9.778807	11.734290
## sample_69	6.630832	5.811309	5.822878	7.1775292	9.442624	13.469311
## sample_691	5.956656	6.192395	4.683601	0.6427478	8.932244	12.214249
## sample_701	7.392755	4.910373	2.603146	9.8415235	9.941052	12.584758
## sample_735	9.474292	3.492533	3.973988	2.9426459	9.694455	13.916598
## sample_74	2.532866	4.171631	7.322316	5.8223503	11.436993	11.514073
## sample_740	7.794767	4.766023	4.392434	7.7835156	10.233871	12.222804
## sample_75	4.351155	4.017022	4.290233	6.1333642	9.015292	12.053132
## sample_762	7.660787	4.429422	4.394610	2.7334411	9.351754	13.448116
## sample_768	6.886843	5.482942	6.157988	10.8763635	10.349447	13.557464
## sample_793	6.040489	5.699974	2.692092	12.5219807	11.647391	11.122873
## sample_800	5.012381	5.156854	5.129093	7.7897944	9.280205	13.086531
## sample_82	8.031428	4.296751	5.883489	1.3530409	10.244352	9.294577
## sample_95	5.125213	5.358695	5.686260	8.3211028	9.717131	10.767315
##	gene_4999	gene_780	gene_16278	gene_16081	gene_18986	gene_8984
## sample_119	2.9141627	4.747087	10.527076	5.966144	9.590003	7.668013
## sample_275	4.7077180	6.164281	10.628555	5.775306	10.442943	7.541267
## sample_29	5.6932249	4.382321	10.544568	3.470550	11.012352	8.440288
## sample_295	2.4975868	0.000000	9.920426	7.171317	8.832820	10.341652
## sample_298	1.2180992	9.118303	9.897841	7.344172	6.915342	7.975470
## sample_404	5.0186297	6.891164	11.902040	5.621469	10.797467	7.725325
## sample_464	1.0912598	1.091260	10.037972	5.301580	8.133497	10.692162
## sample_468	0.4978423	0.000000	9.954250	7.664739	9.338834	9.240047
## sample_531	0.0000000	0.000000	11.236241	6.535425	8.499395	10.055526
## sample_57	0.7324863	1.577489	10.457371	7.162079	8.209561	9.770662
## sample_578	1.2890672	6.771846	10.748159	5.765978	6.468977	7.452908
## sample_579	2.9549779	4.248785	11.303895	3.742297	10.964066	6.946333
## sample_603	1.6415923	1.468218	10.173315	1.796390	8.918979	7.543125
## sample_627	1.0824982	7.239828	10.412729	5.062394	10.456991	7.339440
## sample_638	2.6345933	4.245998	11.075620	6.182651	10.756515	7.462028
## sample_673	3.5213523	3.521352	9.510822	6.692678	8.333692	8.547199
## sample_68	2.9573389	0.000000	10.163952	6.117302	8.472224	9.273900
## sample_69	2.0581779	4.415732	10.133797	6.307607	7.351354	8.837943
## sample_691	2.5973881	3.319213	10.864434	6.913631	7.404265	8.751460
## sample_701	1.2536263	2.080760	10.613145	3.846253	9.154790	8.137360
## sample_735	2.9426459	5.892644	9.667812	9.113463	7.461324	9.617109
## sample_74	1.7628165	6.104737	11.137881	7.709049	7.837060	7.285892
## sample_740	0.7567671	1.771463	11.032239	5.729920	9.495359	7.776729
## sample_75	1.9308887	5.231440	9.588823	7.756156	7.484291	7.513459
## sample_762	0.5980793	8.088635	9.833304	8.099485	8.069450	8.552554
## sample_768	2.2803328	6.294244	10.213469	5.993015	10.252015	7.957386
## sample_793	2.5195922	4.645875	11.268191	4.581502	9.121746	6.794130

## sample_800	1.2386646	4.981665	12.184556	7.811792	10.667697	7.992123
## sample_82	2.0387873	3.748032	12.378048	5.650357	8.462850	7.514722
## sample_95	5.7553149	4.194662	11.559482	5.408545	10.680896	6.751584
##	gene_5476	gene_14897	gene_11445	gene_11372	gene_17103	gene_19987
## sample_119	9.236682	6.801547	6.898160	11.593186	10.486111	7.946994
## sample_275	7.531225	8.047609	4.342484	11.391544	9.348907	8.470024
## sample_29	8.215116	9.238163	3.863126	8.412536	10.655003	6.930986
## sample_295	8.238414	10.599327	2.088549	12.359181	9.184684	8.173292
## sample_298	7.677466	7.432667	2.993385	13.099009	8.246712	7.159084
## sample_404	7.990166	7.542243	2.657571	11.175518	9.467769	8.109679
## sample_464	7.767834	6.896587	3.061569	12.717167	9.312479	7.816478
## sample_468	8.344979	11.230111	1.957729	13.173179	8.823511	7.995767
## sample_531	8.948995	6.264317	2.027048	12.096995	9.860095	7.739706
## sample_57	7.993176	8.163710	1.215989	12.577849	9.509051	7.705328
## sample_578	9.316143	7.559798	0.000000	11.601246	9.437738	7.628074
## sample_579	6.417076	6.660894	5.762437	10.934774	8.135966	8.181431
## sample_603	8.703121	8.508789	4.871282	12.563926	9.418679	8.854360
## sample_627	8.308758	9.451889	1.316320	11.702913	9.512021	8.334979
## sample_638	9.843879	8.679434	4.265669	11.106118	9.650809	8.087044
## sample_673	6.264598	7.925103	3.889250	12.101191	9.425424	7.873272
## sample_68	8.085828	10.430400	1.324119	13.038732	8.919111	7.381889
## sample_69	8.900520	7.339316	2.341616	11.386326	9.263203	7.477475
## sample_691	6.659982	5.195509	2.031148	13.317709	9.130933	7.973560
## sample_701	7.156447	8.151478	3.799201	12.274760	9.394242	8.978599
## sample_735	5.531319	8.017727	4.541893	13.086823	9.300545	8.276106
## sample_74	10.798075	9.248357	5.618776	9.253394	9.733015	7.797940
## sample_740	7.400060	8.282495	3.708187	11.772822	9.429061	8.385763
## sample_75	8.125574	4.370904	2.219339	12.994739	9.341050	7.388241
## sample_762	6.010784	7.823393	2.200285	12.677136	10.109687	8.367489
## sample_768	9.213221	7.964832	2.541614	10.938653	9.667268	8.671318
## sample_793	8.116479	8.069648	2.771188	11.718178	8.649554	9.271223
## sample_800	8.032354	7.900674	3.650581	11.232319	8.209843	8.553165
## sample_82	6.851024	6.927577	2.851599	9.812979	10.806759	7.930856
## sample_95	8.159755	5.774452	3.271381	10.283887	9.211477	8.094305
##	gene_17372	gene_7104	gene_10377	gene_196	gene_16316	gene_14006
## sample_119	9.206318	7.695166	9.041845	8.938836	8.350537	10.27327
## sample_275	8.734916	7.774794	9.707677	8.914654	9.184744	11.26672
## sample_29	6.432779	6.007801	8.412536	7.737531	9.719029	9.98034
## sample_295	9.203218	7.274523	9.156763	8.608994	11.021563	11.69238
## sample_298	10.020230	7.704436	8.689432	7.720402	9.277090	11.08586
## sample_404	9.469776	7.044110	8.777423	9.166040	9.116297	11.21901
## sample_464	8.352538	7.414364	9.641376	8.733131	9.469123	10.07095
## sample_468	9.878137	5.644918	8.509965	7.931470	12.440516	12.30915
## sample_531	8.263081	7.539601	9.067948	8.077494	10.108747	11.02641
## sample_57	8.881827	8.309203	8.542084	8.385384	9.998576	10.69294
## sample_578	9.140850	9.239515	9.389900	9.847237	9.291403	11.42814
## sample_579	8.810392	6.959446	8.905158	9.701341	9.572890	11.04239
## sample_603	8.775880	7.350418	9.212438	10.525315	9.334325	10.84289

## sample_627	9.036935	8.667956	8.926204	9.978996	8.963292	12.79733
## sample_638	9.337278	8.754450	8.036410	9.163202	8.877818	11.11496
## sample_673	9.347845	8.625870	10.259202	9.831412	9.378687	11.45406
## sample_68	9.924578	6.523208	8.666657	8.458415	11.583576	12.13940
## sample_69	7.579889	9.616376	9.479637	9.666996	9.196959	11.29868
## sample_691	9.366182	7.072481	9.638868	8.543643	9.919523	11.76654
## sample_701	8.146670	9.685336	9.706208	9.137163	9.430392	12.04467
## sample_735	8.000952	8.464529	10.383672	9.775223	8.725366	12.03995
## sample_74	5.657300	7.210184	7.487494	7.639551	8.790365	11.06651
## sample_740	8.567226	8.531370	9.685599	8.141617	9.771785	10.45415
## sample_75	8.654375	7.023632	8.912084	8.186337	9.749256	10.84314
## sample_762	8.529856	9.964448	9.674602	8.987221	9.706884	11.82886
## sample_768	8.311471	10.399406	8.056535	9.544696	9.036262	11.31780
## sample_793	8.798070	7.342750	8.805641	8.954970	9.328731	12.76136
## sample_800	9.425065	7.753892	9.158466	10.310283	9.128984	10.83329
## sample_82	8.237430	5.604999	8.348241	10.558756	9.032955	10.66156
## sample_95	8.381608	6.089189	9.552973	9.477615	10.101280	10.89248
##	gene_10385	gene_15041	gene_7126	gene_9472	gene_4128	gene_1944
## sample_119	5.483445	12.10624	10.883606	7.231231	10.971076	2.5974120
## sample_275	5.169809	12.11537	9.902625	6.140574	10.097940	1.1169635
## sample_29	5.765216	12.09904	11.108975	7.859230	9.971644	0.7051375
## sample_295	4.718285	11.31317	10.192638	7.182811	9.804639	3.1538864
## sample_298	4.761259	11.15610	10.706125	6.233305	10.858812	2.4089020
## sample_404	5.652512	11.90820	9.980554	6.851137	9.729140	1.8698319
## sample_464	5.012461	11.80660	10.770813	5.728565	10.563482	1.4307670
## sample_468	5.995858	11.61720	10.469347	6.895387	10.194646	4.7079941
## sample_531	4.994367	11.63099	10.804551	6.083430	10.759347	2.3507801
## sample_57	5.554171	11.41554	9.874960	5.825033	9.919383	1.5774893
## sample_578	5.273535	11.79852	10.434691	6.205940	8.696463	0.0000000
## sample_579	4.722034	11.62455	9.829434	6.391060	9.534995	2.4601659
## sample_603	5.791705	11.93445	10.746187	6.491569	9.263107	1.4682184
## sample_627	3.656874	11.62320	10.867557	6.763823	9.887059	3.0141767
## sample_638	6.366558	11.38198	10.846415	6.260600	9.954350	2.2949883
## sample_673	5.291865	12.28073	10.337789	5.718488	10.070215	2.4028040
## sample_68	4.915005	11.58481	10.290595	6.503438	9.908832	3.9341107
## sample_69	4.999757	11.48701	10.189046	4.751453	9.148019	1.4896971
## sample_691	5.147653	12.00213	10.645821	5.690649	10.086786	3.0029905
## sample_701	4.327148	12.30547	10.355043	5.632856	9.767233	0.5474498
## sample_735	5.565545	12.03628	9.510903	5.111837	9.382553	1.6912661
## sample_74	5.204364	10.77878	10.621319	7.052188	9.836902	1.0453032
## sample_740	6.846268	12.00176	10.452365	5.187352	10.409062	1.7714631
## sample_75	4.530951	12.23174	10.615308	4.892274	9.685001	0.3576083
## sample_762	6.018921	12.19643	10.254438	5.935891	9.352684	2.2002846
## sample_768	5.482216	11.96266	10.952064	7.109904	10.641483	2.6927396
## sample_793	4.448498	12.05155	10.760246	5.828126	10.133373	1.2967217
## sample_800	5.294430	12.25829	9.760226	6.007038	9.408504	3.3948845
## sample_82	4.911452	12.69590	10.441616	5.810908	8.824707	0.0000000
## sample_95	4.706453	12.26485	9.763363	6.492866	9.883935	2.9731337

##	gene_15202	gene_8328	gene_17227	gene_4921	gene_12054	gene_3198
## sample_119	13.04596	0.0000000	7.853696	11.879131	6.734683	7.286567
## sample_275	15.18085	1.4612142	7.212200	10.543544	6.738579	4.770850
## sample_29	14.18503	0.3952822	10.216479	11.462656	3.470550	4.779964
## sample_295	14.58129	0.0000000	7.819751	9.201864	9.359190	6.045777
## sample_298	13.20636	0.7339617	8.901356	9.324867	4.189255	3.362414
## sample_404	13.92011	0.0000000	7.734351	10.346879	6.309587	3.300065
## sample_464	14.85335	1.4307670	9.214295	9.740042	2.134780	3.328363
## sample_468	13.52694	0.4978423	7.979368	10.879407	10.195914	6.745708
## sample_531	14.65813	1.3436352	9.010315	9.280277	1.343635	4.789541
## sample_57	14.03373	0.0000000	9.043142	9.934686	2.312927	4.286999
## sample_578	14.02450	7.8846101	10.737628	11.689443	0.000000	3.593151
## sample_579	13.87680	2.0006851	7.488740	10.927845	7.640238	3.615534
## sample_603	14.31663	2.4837771	8.119107	10.374496	2.483777	1.936176
## sample_627	12.91426	0.8033924	7.580349	10.248343	3.313522	4.613084
## sample_638	12.68922	0.8330925	7.823966	10.840282	2.369606	5.046216
## sample_673	13.76341	1.7580473	7.110875	9.425424	4.219517	5.141846
## sample_68	13.87360	3.4295356	7.436129	10.102632	8.436666	6.101591
## sample_69	14.23058	6.4031774	9.069224	10.370448	6.403177	4.184201
## sample_691	13.95681	0.8812731	9.886965	9.257393	4.981497	5.116011
## sample_701	13.84218	5.6280182	6.977944	10.249623	2.899272	3.648086
## sample_735	15.43517	0.9197596	7.954301	8.966900	2.942646	4.334347
## sample_74	12.27405	0.0000000	8.167097	11.510250	1.220206	3.365763
## sample_740	14.18292	1.6178157	7.902652	11.175718	6.346102	5.091966
## sample_75	14.02147	0.3576083	8.243698	9.721164	4.310827	2.383884
## sample_762	13.90539	5.6675118	7.055445	9.773597	3.357017	7.534575
## sample_768	12.86463	4.4736895	7.860926	10.447362	4.093780	4.452661
## sample_793	13.54389	1.2967217	6.477810	9.987037	1.968386	2.846152
## sample_800	13.76924	4.8174340	9.165314	10.972585	4.782220	3.195174
## sample_82	14.22852	0.0000000	7.639941	11.276671	4.002414	1.844305
## sample_95	14.22393	0.0000000	7.854600	11.490816	8.039988	3.375318
##	gene_598	gene_5923	gene_15391	gene_1621	gene_8482	gene_3065
## sample_119	2.5247658	9.453577	6.279315	12.77860	8.121860	9.483830
## sample_275	1.7388975	7.687502	4.499833	13.35571	9.880433	10.097611
## sample_29	0.9601410	8.599969	6.265913	12.28053	11.677777	9.388196
## sample_295	3.7096435	11.212509	6.495530	14.07767	9.230049	10.949484
## sample_298	5.1081397	6.121836	5.977459	12.69498	8.567374	9.533552
## sample_404	2.0346735	9.194649	4.446025	13.45071	9.821040	9.704225
## sample_464	0.6464392	9.253573	4.337704	13.68769	8.761594	9.735265
## sample_468	2.1031955	10.331275	4.638091	13.57654	8.967033	10.528903
## sample_531	0.0000000	9.611464	2.488926	14.40995	8.346474	9.234355
## sample_57	1.5774893	10.406960	2.928825	13.29798	8.704160	9.561773
## sample_578	2.2779550	5.876463	6.095344	15.00582	8.583380	9.199925
## sample_579	0.0000000	9.292071	4.972417	14.12080	9.463659	9.605917
## sample_603	2.0635720	6.037037	6.052481	12.55243	9.404456	9.646294
## sample_627	2.7199938	10.673874	6.462434	12.50401	9.738437	9.412993
## sample_638	2.2162996	11.105948	5.091010	13.21398	8.902372	9.721515
## sample_673	0.0000000	10.359409	4.687996	14.59072	8.683851	10.208978

## sample_68	1.8606053	10.637630	4.132692	13.61620	8.542277	10.703186
## sample_69	3.7716955	7.880373	7.060253	13.53645	9.461250	9.101640
## sample_691	2.7253485	8.144995	7.892379	14.27867	9.752605	9.262017
## sample_701	2.0807600	10.876233	6.295108	13.59490	9.670708	9.902618
## sample_735	0.9197596	9.302584	5.009513	13.78598	10.166339	9.955127
## sample_74	2.4650080	8.136063	2.935252	11.74220	10.446473	10.881198
## sample_740	1.4458856	7.346407	6.191310	13.06746	8.840089	8.896508
## sample_75	6.9101210	5.284466	8.703045	14.33807	8.586480	9.460489
## sample_762	2.3531539	9.850265	5.304719	13.40457	8.657222	10.590138
## sample_768	2.0754984	9.966067	5.573384	11.62217	9.638461	9.864745
## sample_793	0.7893537	9.993745	5.320694	13.13604	9.619981	9.812232
## sample_800	2.9633076	8.699552	4.817434	13.33134	9.222261	9.878531
## sample_82	1.8443053	9.712312	5.442599	13.14267	10.043670	9.055608
## sample_95	1.8174595	9.148695	5.613732	13.89459	9.313132	9.697213
##	gene_17834	gene_2325	gene_362	gene_6679	gene_19594	gene_3265
## sample_119	8.082862	4.301741	5.661878	8.829995	9.492190	9.870436
## sample_275	9.216273	3.730477	6.175990	8.271496	9.533170	9.529958
## sample_29	9.603380	5.444485	8.201698	11.273213	8.167318	10.027795
## sample_295	8.323167	4.653427	12.121226	12.573877	10.918640	9.210140
## sample_298	8.650675	5.338407	7.534225	12.173393	10.968062	10.246242
## sample_404	8.237855	4.873744	8.869791	7.882374	9.331730	9.809085
## sample_464	8.015543	1.705403	8.034326	9.765565	9.738133	10.803873
## sample_468	10.303792	5.120601	12.521412	10.338636	11.097216	9.273294
## sample_531	8.311980	4.078208	9.732027	10.425038	9.748456	10.204950
## sample_57	6.907696	4.813571	7.844699	9.892134	10.169073	9.392951
## sample_578	8.322185	4.269542	9.668532	11.097433	10.001493	10.014007
## sample_579	8.504227	5.388899	6.526413	7.962347	9.340471	10.016056
## sample_603	9.195148	5.677415	6.606833	9.199491	8.610113	9.886936
## sample_627	9.174304	5.478703	8.609980	10.013197	10.083253	9.646678
## sample_638	9.163856	5.658077	5.343273	9.544609	10.019313	9.824380
## sample_673	9.197372	3.842104	8.613638	10.085353	10.164744	9.581242
## sample_68	9.731740	4.163281	12.590744	11.195594	11.122796	9.818811
## sample_69	7.298466	2.781779	8.059739	11.982355	10.492274	9.939651
## sample_691	8.730640	3.708022	6.753979	11.290635	10.247251	10.336138
## sample_701	10.119213	4.175868	8.443043	10.586549	10.115018	10.337901
## sample_735	9.273775	1.225028	7.206477	11.840750	11.275403	9.835896
## sample_74	9.536226	5.788041	4.729656	10.643802	10.291930	10.008233
## sample_740	8.559335	4.819857	8.924424	10.223085	9.801637	9.860938
## sample_75	8.798190	4.113950	6.062269	10.787282	10.200322	10.571838
## sample_762	9.445081	3.736410	6.125831	10.571753	11.319977	10.781220
## sample_768	9.567492	5.386445	4.686775	9.778654	11.242650	10.264759
## sample_793	9.288130	5.168317	5.211495	8.598961	9.221925	10.193920
## sample_800	9.215679	3.996208	8.626976	9.197416	10.473716	9.819180
## sample_82	9.314674	5.049905	4.441397	8.322784	8.953757	10.804954
## sample_95	9.664246	4.568099	5.624320	6.672185	8.293177	9.724118
##	gene_8400	gene_15835	gene_5890	gene_9550	gene_5357	gene_3194
## sample_119	9.979597	10.649840	7.127035	1.7559131	3.987212	5.719972
## sample_275	11.595770	10.773840	7.541267	1.1169635	3.103984	3.450974

## sample_29	7.626870	10.684863	7.983718	8.7191218	3.831543	4.315385
## sample_295	10.937565	11.345738	7.894915	2.3736760	1.259544	3.988376
## sample_298	11.427250	10.737560	8.118318	0.9962441	3.215648	2.496283
## sample_404	12.444844	10.606785	7.493479	1.2187192	4.003980	4.190346
## sample_464	9.681655	10.488684	8.654357	4.5264695	2.465269	3.156089
## sample_468	10.699477	11.708670	6.751236	3.3313613	2.758943	5.584253
## sample_531	9.759590	10.055526	7.930637	2.6149925	2.838246	3.119257
## sample_57	10.039056	9.879020	7.596130	3.0489859	2.493237	2.106817
## sample_578	11.594077	10.419560	8.752284	0.5667665	2.760178	3.198636
## sample_579	12.528532	11.024828	7.777334	1.0879380	3.476550	5.913777
## sample_603	8.730215	10.364868	7.408584	0.0000000	3.814981	4.495363
## sample_627	12.922454	10.253197	7.619149	1.0824982	3.014177	7.140462
## sample_638	10.619661	10.273935	7.206155	0.8330925	3.375470	6.353244
## sample_673	10.851601	9.915670	8.460091	0.0000000	2.641823	5.486573
## sample_68	10.998936	11.386978	7.436129	3.9691689	2.250780	4.163281
## sample_69	10.816536	9.915488	7.747079	0.0000000	3.327917	3.041471
## sample_691	11.981446	10.445812	8.735400	2.6627980	3.611940	4.804446
## sample_701	9.653451	10.866282	7.539531	0.0000000	4.695949	5.362740
## sample_735	9.616290	10.271568	7.766184	0.0000000	3.492533	4.771199
## sample_74	9.096127	9.796092	9.276471	0.3402774	4.919116	4.192762
## sample_740	9.741882	10.406917	7.843230	0.0000000	3.589212	4.671571
## sample_75	11.496799	10.549409	8.662558	0.0000000	2.033582	5.165128
## sample_762	11.405514	11.031494	8.369728	0.5980793	6.369372	7.899852
## sample_768	11.389395	10.895727	7.305688	0.0000000	3.412077	5.879424
## sample_793	11.155476	10.716579	7.934440	0.0000000	3.387955	6.460017
## sample_800	11.783113	10.676715	7.811792	1.8951475	2.344658	4.056401
## sample_82	9.378960	10.998005	8.031428	2.0387873	4.634338	6.230699
## sample_95	13.224659	10.943042	7.820479	0.7834986	3.099918	5.183645
##	gene_20514	gene_17914	gene_801	gene_5138	gene_103	gene_11025
## sample_119	7.474274	11.061844	10.401797	8.869948	10.308180	8.290033
## sample_275	7.965652	11.102717	10.784398	8.571032	10.461602	5.104647
## sample_29	7.514493	13.441505	10.670152	6.549312	8.953614	2.923492
## sample_295	7.342199	10.154324	10.559444	10.111331	9.416238	5.640019
## sample_298	6.842174	10.645298	10.192120	9.434710	9.854574	5.768161
## sample_404	8.361400	10.446308	10.174763	8.597244	10.332093	5.949659
## sample_464	8.525282	11.324197	10.643532	8.466554	9.992259	4.545802
## sample_468	7.089159	9.880484	9.867815	11.151683	10.079311	4.805607
## sample_531	7.657840	11.991068	11.246016	8.555509	9.504545	6.324060
## sample_57	8.542084	11.666317	10.297467	9.229537	10.017115	6.837262
## sample_578	2.653885	11.028645	11.685866	7.799916	10.090152	5.292789
## sample_579	8.050747	9.824666	10.577618	9.457512	10.574073	4.539246
## sample_603	7.417557	10.896295	10.317718	10.081257	10.278031	4.538557
## sample_627	7.326106	10.188799	11.013804	9.449293	10.429585	5.317405
## sample_638	8.611877	12.812572	10.035156	9.119950	10.253458	4.238451
## sample_673	6.531116	10.823805	10.821909	8.750807	9.702865	8.474870
## sample_68	8.187852	10.268788	10.232277	11.140606	9.984461	5.286238
## sample_69	4.147868	11.389244	11.655388	8.689453	10.548225	5.581098
## sample_691	8.397602	9.308735	10.631341	10.137939	10.059669	6.412570

## sample_701	4.020449	10.236983	11.078184	9.424590	10.158774	5.182720
## sample_735	5.976765	9.934895	11.913757	9.859367	10.489025	8.013401
## sample_74	3.289333	13.027457	10.098308	9.296136	10.538354	6.602323
## sample_740	2.704363	10.286997	10.843960	9.650273	10.918602	4.498149
## sample_75	4.984635	11.304346	9.920253	9.153539	10.527301	5.775593
## sample_762	6.228230	10.885574	11.485226	9.631339	10.168183	7.866327
## sample_768	6.917861	10.916753	10.697715	9.927532	9.935606	6.370949
## sample_793	6.170173	10.610019	10.834345	8.401810	10.194055	6.451997
## sample_800	7.194757	9.872264	11.308555	10.190973	10.558115	6.061368
## sample_82	7.727199	10.852764	11.293748	8.551193	9.962827	6.537959
## sample_95	8.364375	10.418485	10.178291	8.055716	10.386186	6.923030
##	gene_6209	gene_19214	gene_5825	gene_19520	gene_8716	gene_2389
## sample_119	11.38211	0.0000000	8.343954	9.931231	4.4266136	2.8561292
## sample_275	11.67113	0.6640277	5.805772	11.341408	1.4612142	3.7926472
## sample_29	12.83656	0.3952822	7.867822	10.831798	2.6685049	4.4462893
## sample_295	12.05170	2.9081020	4.484003	11.313966	1.9215887	4.4228049
## sample_298	10.76311	0.4131608	9.864360	12.301450	4.5976206	3.7655878
## sample_404	11.22356	1.4698339	5.859960	11.593984	1.4698339	6.2182251
## sample_464	11.37083	0.0000000	9.237449	11.029115	1.7054029	7.6437263
## sample_468	11.45472	6.6949637	3.000955	11.324698	1.4050666	3.8677384
## sample_531	11.36295	0.0000000	9.287396	11.752565	6.6280605	6.9060367
## sample_57	11.53387	1.2159892	8.949733	11.234500	6.6753502	4.9433025
## sample_578	12.72145	2.1271710	8.428779	10.898662	4.6474066	7.1251035
## sample_579	11.86601	1.3224474	6.774036	11.851577	0.4596414	4.2769805
## sample_603	12.06487	1.2710670	3.445462	10.748260	1.9361762	4.8537562
## sample_627	10.58429	1.5174773	3.991236	11.922361	1.3163196	3.9570417
## sample_638	11.71107	0.8330925	7.429700	11.662971	1.3578335	3.3387952
## sample_673	10.92910	0.5621814	7.994133	11.388900	2.2665767	6.3491547
## sample_68	12.15137	4.0368110	4.462877	11.113521	0.4603756	5.7750347
## sample_69	12.21977	0.5381408	6.379872	12.432573	4.8817136	4.5609595
## sample_691	10.67559	0.3569326	7.843858	12.585324	5.2813051	5.4666924
## sample_701	12.29636	0.5474498	6.821927	10.764333	0.5474498	5.0970081
## sample_735	11.67083	0.0000000	6.352344	10.382710	7.1845283	7.5904299
## sample_74	12.64672	1.2202061	6.771423	11.573051	3.0784049	1.5168724
## sample_740	10.92706	0.4274989	6.829951	11.645830	2.5430009	0.7567671
## sample_75	11.53258	0.0000000	7.550731	10.997800	9.4974527	3.2386035
## sample_762	11.69064	0.0000000	8.394167	10.551151	5.1875975	5.2277026
## sample_768	11.23439	0.4021764	4.949474	11.298590	2.0754984	2.0754984
## sample_793	11.20439	0.4480552	4.603276	12.628949	0.0000000	4.3434647
## sample_800	10.09578	0.7483754	5.337522	11.724608	4.8518090	4.6320820
## sample_82	11.54207	1.6195069	7.605805	10.461367	2.0387873	2.5016939
## sample_95	11.29703	0.0000000	8.239097	11.552049	0.7834986	0.7834986
##	gene_11215	gene_15777	gene_4982	gene_9203	gene_6618	gene_1937
## sample_119	9.354232	12.44723	9.514049	12.023498	1.3137683	7.610183
## sample_275	9.884007	13.68564	10.720868	10.064918	4.8893859	7.965652
## sample_29	8.907963	11.31929	8.736284	11.130023	8.5105427	7.148649
## sample_295	9.292326	12.86409	12.653353	10.419107	0.9482270	7.441965
## sample_298	8.958245	12.96246	11.583816	10.660156	11.1980398	7.547943

## sample_404	9.039525	13.30788	10.402639	10.437128	4.0814419	7.707118
## sample_464	9.575562	13.32901	10.939351	10.297799	7.3051589	8.311412
## sample_468	8.704578	12.13848	10.719714	10.150610	1.4050666	7.261785
## sample_531	9.345392	13.38150	11.080691	10.130017	3.3543393	7.432367
## sample_57	8.801479	12.60858	10.752790	10.175687	7.5510468	7.777130
## sample_578	9.409429	14.33276	9.934963	10.556784	0.9726927	6.804699
## sample_579	9.061258	14.04509	10.697871	10.977015	4.4846795	7.765163
## sample_603	9.149198	12.86328	9.284512	10.866290	6.0292541	7.609888
## sample_627	8.462355	12.54793	10.220233	11.419081	4.7591825	7.050066
## sample_638	8.344580	12.04551	9.375554	11.355968	4.1853195	7.238453
## sample_673	8.711990	13.44402	9.903520	10.619633	4.7144159	9.701495
## sample_68	8.996541	12.73605	10.224593	10.550266	2.7309877	7.298209
## sample_69	9.462175	13.22379	10.541677	10.841635	5.4870908	7.662683
## sample_691	8.546894	13.74469	10.628783	10.202589	7.4373526	7.944823
## sample_701	8.740985	13.51327	10.823208	11.190979	9.1905531	7.557954
## sample_735	9.394340	14.58635	9.861422	9.702197	8.9011475	8.605053
## sample_74	9.447786	11.77178	8.336908	11.148235	1.3761237	7.349861
## sample_740	9.595220	13.13142	10.256822	11.258672	0.0000000	7.228607
## sample_75	9.633958	12.85996	9.506409	10.132525	11.8786487	7.143291
## sample_762	9.303313	14.43936	11.053281	10.922703	12.9786908	8.104877
## sample_768	8.621550	12.80261	11.140542	11.075833	2.5416140	8.124938
## sample_793	8.689551	13.39932	8.648246	11.156399	1.4961044	7.770082
## sample_800	9.399746	13.48611	10.145996	11.224146	9.1957438	7.154018
## sample_82	8.837644	14.33146	7.574775	10.789974	0.6023619	7.852161
## sample_95	9.404258	13.80479	9.994127	10.549929	4.0132300	7.797279
##	gene_17772	gene_5026	gene_4094	gene_18677	gene_9400	gene_15707
## sample_119	4.523004	3.6878557	12.70359	7.765813	13.13934	10.050447
## sample_275	4.607514	1.1169635	11.97518	8.351120	11.85559	9.915779
## sample_29	3.341929	5.6395129	11.69735	8.218587	12.25340	10.909496
## sample_295	5.164119	0.0000000	11.41293	8.337747	12.75355	9.684147
## sample_298	5.729553	1.7316612	11.84459	8.773822	11.94483	9.635711
## sample_404	4.611839	0.5285713	12.28519	8.922052	12.93996	9.346514
## sample_464	3.748075	3.4822287	11.35348	7.515889	11.53565	9.779550
## sample_468	5.327220	0.0000000	11.86315	9.644619	13.06842	9.522219
## sample_531	4.526244	4.7625508	11.25027	8.854476	12.03243	9.854515
## sample_57	4.470374	4.8135709	11.62643	9.879020	12.49249	9.246935
## sample_578	6.135399	1.2890672	12.16610	7.947841	12.28021	10.031660
## sample_579	4.358410	0.4596414	12.87463	8.507208	12.85577	9.949815
## sample_603	5.278327	1.4682184	12.26871	8.582451	11.67815	10.010864
## sample_627	4.698369	4.6983685	12.69777	8.028238	13.16038	9.680008
## sample_638	5.424274	3.8625605	12.13692	8.219033	12.03797	9.076754
## sample_673	4.714416	4.4880843	11.98509	7.007812	12.02291	9.469723
## sample_68	4.462877	3.7037763	11.52367	9.017624	12.60579	10.214125
## sample_69	4.967597	3.3914515	11.82389	7.610242	12.65436	9.984718
## sample_691	5.429596	3.2357117	11.72444	8.156392	12.23573	9.588216
## sample_701	6.432483	0.9432837	12.48795	7.229338	12.12094	9.597831
## sample_735	5.330469	4.0918437	10.92823	6.263031	11.45088	9.092406
## sample_74	3.726177	3.0784049	12.17862	7.766562	12.22560	10.991600

## sample_740	6.099968	2.1533514	12.79393	6.951506	11.82915	9.397687
## sample_75	6.121755	1.4264258	12.29561	8.751875	11.20848	9.980939
## sample_762	6.689607	2.6174866	11.99348	7.752996	12.07097	10.073700
## sample_768	5.306663	3.2750518	12.36533	7.611040	12.23256	9.666698
## sample_793	5.915499	2.6920924	12.57805	7.235268	12.86950	9.897328
## sample_800	5.431432	2.1373396	12.45978	7.999803	12.52264	9.548627
## sample_82	4.092816	2.2101395	13.68504	7.647415	12.44071	9.068198
## sample_95	4.405761	0.4443490	12.60898	9.390369	12.93461	8.894757
##	gene_8093	gene_4731	gene_18519	gene_20466	gene_1114	gene_1960
## sample_119	11.29009	3.5844214	9.581074	4.714795	10.378468	2.5247658
## sample_275	11.00681	1.9716999	9.585691	4.972307	10.048296	2.5047729
## sample_29	11.55516	11.0165297	10.267536	7.198877	12.800049	0.0000000
## sample_295	11.61027	0.0000000	9.469227	5.597034	10.362678	0.5506052
## sample_298	11.36695	0.0000000	10.206770	2.408902	10.312123	1.8689633
## sample_404	10.83115	1.4698339	9.960195	3.743019	9.747198	1.2187192
## sample_464	11.65477	0.0000000	10.181127	3.621267	9.727590	0.0000000
## sample_468	11.29069	3.2710521	9.610803	5.494796	9.838246	1.7959331
## sample_531	11.64051	1.6090457	9.770639	5.803503	9.527774	1.0180644
## sample_57	10.75776	1.5774893	9.766285	1.577489	10.006957	1.2159892
## sample_578	12.33794	0.0000000	10.245778	7.818486	9.462551	1.9588056
## sample_579	10.82536	3.3227358	9.631395	3.966624	9.343808	2.0006851
## sample_603	10.65333	1.0427144	10.588921	5.706992	9.295099	2.1806571
## sample_627	11.39313	0.0000000	10.056746	5.858093	12.130165	1.0824982
## sample_638	11.19969	8.9179956	8.689484	7.111490	10.732854	2.5727684
## sample_673	12.35031	4.4571280	9.122885	3.889250	9.663569	1.5389851
## sample_68	11.32828	2.9573389	10.474902	7.661807	9.910310	2.2507797
## sample_69	11.70312	8.2938808	11.058675	5.727626	9.287412	1.4896971
## sample_691	12.48266	0.3569326	9.840024	3.855172	9.321450	1.6983518
## sample_701	11.32781	0.0000000	10.049318	2.080760	10.272408	0.5474498
## sample_735	11.20521	5.0872592	9.181513	5.298138	10.019132	1.8778620
## sample_74	10.46966	5.3061091	11.322103	8.642665	10.456344	1.3761237
## sample_740	11.30247	1.9103105	9.293715	3.629974	10.760387	1.6178157
## sample_75	10.63739	0.0000000	10.753601	3.711010	8.872804	1.2669364
## sample_762	11.53403	1.3453964	10.114380	5.207791	9.869406	2.6174866
## sample_768	11.07433	3.9208268	9.699980	5.831360	10.280272	1.7006173
## sample_793	11.45925	0.0000000	8.725990	6.465973	12.026451	1.0652276
## sample_800	11.35236	0.7483754	10.071369	6.848310	9.412756	1.8951475
## sample_82	10.74407	0.0000000	10.240028	2.210139	10.033010	2.7441180
## sample_95	11.33941	0.0000000	9.720836	2.759241	9.963192	0.0000000
##	gene_15775	gene_18370	gene_5240	gene_1961	gene_3237	gene_18025
## sample_119	13.40185	0.9195308	11.099065	8.737751	7.099158	11.577278
## sample_275	12.90593	1.4612142	5.241100	10.381078	3.909476	9.023973
## sample_29	13.25860	0.3952822	10.715739	7.992638	5.621155	11.764104
## sample_295	12.62136	0.0000000	5.983561	10.019924	5.873506	7.955598
## sample_298	12.08410	0.0000000	9.826925	9.923433	8.754016	9.736728
## sample_404	12.43704	0.5285713	4.416434	9.844574	7.271836	8.738450
## sample_464	13.31291	1.9360254	9.409211	10.100150	6.618093	10.066399
## sample_468	12.35436	0.0000000	7.658868	9.421446	8.030286	8.614923

## sample_531	12.57112	1.8331330	9.223263	9.427131	8.710076	10.681607
## sample_57	12.59399	1.2159892	9.374622	9.497187	7.610855	10.710703
## sample_578	12.76808	0.0000000	9.232606	10.149036	7.986678	9.939920
## sample_579	12.48500	0.4596414	6.910793	9.924275	3.782199	8.877799
## sample_603	12.85811	1.0427144	8.250459	9.768196	3.953042	9.843954
## sample_627	12.74810	0.4569113	9.461070	9.960477	5.514419	10.544887
## sample_638	12.72735	0.3339961	10.878909	9.255418	5.959255	10.418243
## sample_673	12.07275	1.5389851	7.757330	9.810167	7.613812	9.824604
## sample_68	12.75518	0.4603756	8.253304	9.821265	6.853771	8.417755
## sample_69	12.26764	0.5381408	8.555456	10.763461	7.351354	9.429442
## sample_691	12.77230	0.3569326	6.278371	10.753459	7.469691	8.330007
## sample_701	11.64672	0.9432837	7.901573	9.714987	5.191128	9.818852
## sample_735	12.23003	2.3256455	7.587087	11.104428	8.152690	8.970753
## sample_74	13.29367	0.6153221	10.808819	8.773762	6.510323	10.351293
## sample_740	12.06304	0.4274989	9.687409	9.874803	5.077303	10.905951
## sample_75	12.50418	0.6439485	8.266281	9.878883	7.688915	9.830840
## sample_762	12.00705	1.3453964	8.874898	9.773995	7.342466	9.480784
## sample_768	12.51641	0.0000000	7.884671	8.763531	5.831360	9.512035
## sample_793	12.35131	1.0652276	7.673874	9.848477	6.295859	9.470258
## sample_800	12.49979	0.7483754	8.096151	10.285714	6.567231	10.084078
## sample_82	12.62714	0.6023619	7.299080	8.988287	5.589557	8.400060
## sample_95	12.60856	0.4443490	3.947301	10.071784	2.973134	8.911059
##	gene_4130	gene_5539	gene_10516	gene_14439	gene_7533	gene_15837
## sample_119	10.379443	4.940998	8.494192	8.281003	12.616765	12.049825
## sample_275	10.779128	6.837741	8.873896	7.995332	10.933255	10.949126
## sample_29	9.524702	10.891176	9.635747	9.226022	11.721791	11.433199
## sample_295	9.411325	6.696369	8.789537	8.564938	11.437071	9.995908
## sample_298	9.753451	6.778235	8.768346	7.884952	10.904838	10.836374
## sample_404	10.261566	8.074591	8.458960	8.361128	11.206099	11.263316
## sample_464	10.490390	3.061569	10.029411	8.150788	10.574083	10.868707
## sample_468	9.373305	1.161049	7.953219	8.198199	11.916719	10.174976
## sample_531	9.653393	3.676978	9.413129	7.940490	10.730309	10.914385
## sample_57	9.167842	3.048986	8.898659	8.588909	11.198807	11.005786
## sample_578	10.055337	4.561809	9.388865	7.147062	10.709704	10.979632
## sample_579	10.140293	7.312647	8.457254	8.038732	10.930065	11.103694
## sample_603	9.412707	7.045432	8.437057	7.715797	11.724834	11.173796
## sample_627	9.530309	9.959149	8.222505	7.708180	11.597582	11.113000
## sample_638	10.057897	14.310528	8.539992	7.789390	11.802601	11.118682
## sample_673	9.542508	10.824569	9.802452	8.121725	10.075827	10.835893
## sample_68	9.704697	5.227518	8.833722	8.252807	11.278931	10.192971
## sample_69	9.248267	4.504608	9.152954	8.689156	11.246218	11.859080
## sample_691	10.512187	5.322364	9.330283	8.300298	10.366136	10.005540
## sample_701	10.288462	7.277334	8.683854	8.001937	11.332976	10.974135
## sample_735	10.372550	8.022684	9.758240	8.623764	8.576609	10.270003
## sample_74	9.348870	6.099152	9.404282	7.002331	12.642702	13.373939
## sample_740	10.019577	10.019577	8.125672	8.173877	11.328619	11.779067
## sample_75	9.217652	3.829931	9.286204	7.375613	11.714456	10.814807
## sample_762	10.468838	4.463407	9.151303	6.541857	9.909256	11.330665

## sample_768	10.166226	7.525521	8.941329	7.139654	11.324867	11.535455
## sample_793	10.759031	12.000275	8.962361	8.302415	11.425646	10.407140
## sample_800	10.352904	8.701909	8.720631	8.315811	10.472335	10.981346
## sample_82	10.406248	6.180046	9.001211	7.952718	11.458468	11.845074
## sample_95	10.330704	10.157486	8.166705	8.830445	11.019480	11.072387
##	gene_18614	gene_20273	gene_8148	gene_3616	gene_2567	gene_917
## sample_119	3.433867	6.338792	7.556284	7.732473	9.238940	8.718389
## sample_275	1.971700	5.835608	8.535116	5.507249	9.248668	9.900862
## sample_29	5.719350	4.828774	9.631741	3.648235	8.778176	8.990731
## sample_295	3.603324	6.808282	8.163972	8.396429	9.180068	9.569698
## sample_298	9.776543	3.834276	7.793382	8.121000	8.831855	10.952938
## sample_404	3.233750	6.707911	8.399709	4.899422	9.475267	9.539863
## sample_464	7.726927	5.632704	8.542885	6.268861	9.014439	9.643416
## sample_468	5.235520	7.426357	7.507002	7.722405	8.486553	8.944285
## sample_531	6.012687	6.254662	7.657840	5.477933	8.323640	9.809894
## sample_57	6.204833	5.821830	8.130040	5.429733	8.316345	10.081949
## sample_578	7.863901	5.303730	8.888898	5.403874	8.118001	9.805176
## sample_579	2.129975	5.992410	8.376481	3.662468	9.213722	9.574312
## sample_603	4.972173	6.601594	8.281916	6.920924	8.980574	8.847737
## sample_627	3.417785	7.359214	8.876750	6.721728	9.501596	9.027530
## sample_638	2.216300	7.081286	8.698559	4.647758	9.536908	8.847950
## sample_673	5.122246	5.122246	9.345259	5.648540	8.922769	11.092889
## sample_68	5.870828	6.387273	8.226393	8.195328	8.315131	9.456363
## sample_69	6.650441	5.125403	8.170276	6.377363	8.262410	10.272618
## sample_691	12.908351	5.457506	9.191745	8.524640	8.798977	9.825371
## sample_701	4.975305	5.135395	8.448769	6.519743	8.662373	9.127675
## sample_735	3.704739	5.048907	9.167039	8.345321	8.135637	11.232469
## sample_74	5.618776	7.336623	8.215664	5.298820	9.225142	8.040202
## sample_740	7.661044	6.002453	7.797000	5.443524	8.844219	9.757485
## sample_75	10.314663	5.176391	8.110546	5.559073	8.680613	9.677725
## sample_762	5.992510	6.049563	8.422523	8.361641	8.586438	9.640653
## sample_768	5.329913	7.494992	8.973057	6.453285	9.208390	8.186397
## sample_793	4.263658	6.541236	9.592343	5.317930	8.368685	8.375157
## sample_800	5.737495	6.695799	8.348710	6.391204	9.409024	9.449695
## sample_82	2.210139	6.298748	9.395705	5.888673	9.521565	8.753538
## sample_95	0.000000	6.301972	8.109455	3.814191	9.447548	9.636405
##	gene_10861	gene_940	gene_6116	gene_8892	gene_13365	gene_16969
## sample_119	6.816843	6.634673	9.233358	10.574802	9.777048	8.037239
## sample_275	7.412714	7.842482	10.155678	9.275975	10.505186	8.888418
## sample_29	10.021063	7.151686	9.148072	10.263105	12.089894	7.362040
## sample_295	8.976839	8.000794	10.830816	8.710118	11.221243	8.431343
## sample_298	8.182295	6.577488	10.010192	9.371112	10.153590	9.871996
## sample_404	7.140768	7.398308	10.486463	9.677189	10.304602	8.774596
## sample_464	7.373987	5.216401	10.056475	8.798695	9.905640	8.595612
## sample_468	8.816936	9.208239	11.058161	9.414833	10.075452	8.350289
## sample_531	7.802568	4.619454	9.730287	9.223263	9.872772	9.038549
## sample_57	7.881585	5.437461	9.850359	9.651129	10.293679	9.036143
## sample_578	6.796351	7.280548	10.607062	8.166580	10.959198	9.111430

## sample_579	6.234509	9.572179	10.002492	9.180809	10.188873	8.866478
## sample_603	5.915617	5.391953	9.831139	9.739291	10.721638	7.115855
## sample_627	7.013094	6.753952	10.289293	10.088205	10.222928	8.689352
## sample_638	6.665393	7.321080	10.183437	10.115265	10.721996	8.248620
## sample_673	8.713631	6.446948	10.640887	9.121650	11.181978	9.901746
## sample_68	8.621198	7.958518	11.660482	9.224889	11.302102	7.863932
## sample_69	7.915915	7.999848	10.305971	9.387212	11.344301	8.480762
## sample_691	8.952820	6.930335	9.238746	8.477026	10.054726	12.744441
## sample_701	6.983210	6.945900	10.237545	9.423622	10.574811	9.086874
## sample_735	10.427302	6.398826	10.661769	8.925166	11.117312	9.745938
## sample_74	6.809543	6.543607	8.263894	10.190183	9.505411	6.535524
## sample_740	7.340277	6.121567	10.745683	9.111441	10.742495	8.216169
## sample_75	8.049620	4.997396	9.394486	8.184945	10.947520	10.736808
## sample_762	8.061150	8.473450	9.723634	8.771856	10.849882	9.279696
## sample_768	6.630158	9.470142	9.973468	10.091673	10.871913	7.802439
## sample_793	6.591133	5.446501	9.979254	9.736654	9.456864	7.871443
## sample_800	7.892440	8.106866	10.507121	10.032569	10.262365	8.504704
## sample_82	7.011082	3.748032	8.588568	10.343763	8.772466	8.938236
## sample_95	6.032132	8.111339	10.302959	9.370855	9.490690	8.636164
##	gene_19455	gene_2635	gene_19543	gene_11827	gene_11501	gene_2763
## sample_119	1.1300742	2.7325731	12.55587	10.587674	10.358365	4.5042327
## sample_275	1.4612142	3.6655178	12.62031	10.456806	10.086017	4.3003068
## sample_29	0.0000000	1.3650206	11.44550	8.190847	9.001417	0.0000000
## sample_295	0.0000000	3.4275391	10.54244	9.375433	10.261437	0.5506052
## sample_298	3.1087921	0.9962441	11.59294	9.934480	10.601632	0.7339617
## sample_404	1.8698319	2.4395699	12.81391	10.689893	9.637666	3.4241831
## sample_464	0.6464392	0.6464392	11.11874	10.852483	9.855463	2.4652693
## sample_468	0.4978423	4.5895298	11.09075	9.227209	9.823886	0.0000000
## sample_531	1.0180644	0.5970305	11.15941	10.373561	10.868838	0.0000000
## sample_57	0.0000000	0.7324863	11.85617	10.005091	9.474588	3.8963881
## sample_578	0.0000000	3.2723517	11.09103	10.713017	10.504700	0.5667665
## sample_579	1.5241640	3.8210162	12.07837	9.269442	9.962351	0.0000000
## sample_603	2.6554434	3.7020036	12.04170	10.528425	10.565902	2.4837771
## sample_627	0.4569113	3.3135216	11.13220	8.740125	10.770581	2.7199938
## sample_638	1.4974846	1.9506550	10.99577	9.603767	10.250369	3.8099392
## sample_673	3.2594985	3.9349127	11.88405	9.541586	11.282266	1.9481523
## sample_68	0.8089204	3.7037763	11.11580	11.108126	10.200825	0.0000000
## sample_69	0.5381408	0.0000000	11.51831	11.098480	10.168785	0.0000000
## sample_691	2.6627980	1.9285030	10.73391	9.553074	10.044790	0.6427478
## sample_701	4.1385524	2.2301108	11.88980	9.592666	10.958161	3.2880484
## sample_735	3.1009776	1.8778620	12.14592	11.663313	9.788377	0.5319678
## sample_74	7.3342465	1.5168724	10.08210	8.452044	10.338903	0.8463128
## sample_740	0.0000000	0.7567671	11.54989	10.261695	10.307167	2.8494789
## sample_75	0.0000000	0.3576083	11.08572	10.659389	10.024849	0.3576083
## sample_762	8.0331366	1.8352773	10.93545	10.541736	10.968912	5.6230943
## sample_768	0.4021764	1.5504082	11.54878	9.154107	10.853731	2.9544383
## sample_793	4.9979291	3.7421787	11.37922	8.790293	11.287031	2.4249221
## sample_800	3.4852465	1.6039289	11.98645	10.618009	9.421364	2.8317151

## sample_82	0.6023619	4.0483234	10.94727	9.259635	10.594968	6.3269579
## sample_95	1.2884176	3.7679514	12.56374	9.895974	9.882255	3.4245997
##	gene_7375	gene_1847	gene_376	gene_10118	gene_1839	gene_6957
## sample_119	3.4730076	8.241435	2.0938644	5.442821	5.924397	10.295964
## sample_275	4.0686880	7.485894	4.3834555	5.100721	6.517123	10.543544
## sample_29	1.1767057	7.362540	1.3650206	4.694747	8.223157	9.661240
## sample_295	4.2573728	8.323167	2.9947788	3.657469	5.962211	8.856158
## sample_298	5.2903175	6.911368	2.9320234	5.597463	6.202002	9.907836
## sample_404	3.0142839	7.450337	4.5586595	4.224897	5.541490	10.326137
## sample_464	2.7339833	8.861410	2.8516391	4.755069	11.080864	9.327788
## sample_468	3.6948913	7.796260	2.5716525	5.356538	5.714498	9.374203
## sample_531	5.1847167	7.849593	2.7309442	4.282025	10.862901	9.598166
## sample_57	6.5688059	8.701871	1.2159892	6.055000	10.929059	9.511668
## sample_578	5.1942132	8.706486	0.9726927	4.339800	5.816131	8.839739
## sample_579	2.4601659	7.260637	6.6284266	4.508657	5.559302	8.853428
## sample_603	1.4682184	8.528251	1.6415923	5.329934	4.853756	9.776968
## sample_627	2.7199938	7.651590	2.7993567	3.957042	6.173175	8.533711
## sample_638	2.0447438	7.850262	2.6345933	4.431596	4.877484	10.073673
## sample_673	2.5272455	8.164329	2.4028040	3.979175	5.486573	8.348529
## sample_68	4.4380732	8.690103	2.2507797	4.957692	6.569628	9.223075
## sample_69	3.7231326	7.531350	3.3279173	5.333814	8.170276	9.574219
## sample_691	4.9556336	7.253214	3.9886938	5.080551	6.956649	9.168712
## sample_701	3.7002399	7.646408	2.2301108	5.077422	5.227121	9.405084
## sample_735	3.1740708	6.990773	3.1009776	7.282820	8.265690	9.016184
## sample_74	2.8842072	7.858708	1.0453032	4.234126	7.369484	9.779161
## sample_740	1.6178157	6.975424	2.9813230	5.295694	6.078044	9.360329
## sample_75	4.9587731	6.899961	1.0875986	5.617689	5.274016	8.173747
## sample_762	7.8557468	9.088709	3.4275928	6.136403	6.497927	8.262804
## sample_768	0.0000000	8.305528	0.7163325	5.064827	6.567965	8.396429
## sample_793	0.4480552	7.245695	2.9174318	4.687245	5.239581	8.468587
## sample_800	4.1141838	7.941147	2.3446581	3.570281	6.567231	10.034441
## sample_82	0.6023619	8.065314	1.0259501	5.993178	4.603923	8.662384
## sample_95	1.2884176	7.351328	5.4328392	5.480233	4.673805	10.444477
##	gene_15305	gene_19176	gene_11396	gene_12134	gene_18379	gene_12182
## sample_119	8.855211	4.5415334	10.161296	8.529427	7.258849	14.41191
## sample_275	10.378533	0.6640277	9.303459	9.931407	7.037086	12.76319
## sample_29	9.082490	7.6007604	9.295728	9.901630	8.552319	14.15618
## sample_295	11.361626	0.0000000	8.542219	9.744701	1.515511	12.93608
## sample_298	10.080351	0.0000000	9.185819	9.310624	9.038675	12.57796
## sample_404	10.720107	2.1826287	8.883764	9.290528	4.736805	12.63738
## sample_464	9.039344	8.3185835	10.230837	10.278891	7.299996	12.98410
## sample_468	11.590264	0.0000000	8.209770	8.853122	1.795933	12.76975
## sample_531	8.751849	6.8936413	9.515702	9.186924	5.164239	12.03032
## sample_57	9.371744	7.3953713	9.755285	9.674674	8.531806	12.35843
## sample_578	10.797362	0.5667665	9.859247	9.072899	9.232606	13.63862
## sample_579	10.233620	3.9666243	9.675093	9.988699	6.009316	13.08351
## sample_603	9.082120	1.6415923	10.355946	9.915906	9.642472	13.41665
## sample_627	9.127981	2.9460571	10.418401	9.209088	8.204395	13.64215

## sample_638	9.011988	4.6114734	9.557868	8.867393	8.774329	13.62196
## sample_673	10.604266	2.5272455	10.068160	9.164208	8.816315	12.35425
## sample_68	10.379595	0.4603756	8.872844	9.062950	2.557582	12.88847
## sample_69	10.168221	2.0581779	9.627691	9.134506	8.708287	12.47053
## sample_691	11.558497	6.2731466	10.331925	8.924815	8.755211	12.79322
## sample_701	9.570072	3.8462526	10.046797	9.275724	7.147185	13.05408
## sample_735	9.531040	2.8564479	9.806650	8.854660	8.977148	12.53173
## sample_74	8.320200	3.4733194	9.338257	10.057125	8.185668	13.88674
## sample_740	8.856282	2.9813230	8.900680	9.493289	8.737751	13.68796
## sample_75	11.955763	9.6871079	9.279037	9.536164	10.342441	12.87057
## sample_762	9.579009	5.2491002	9.812851	9.206284	7.333307	13.68611
## sample_768	9.630875	1.8366917	10.534848	9.504392	6.944472	13.15509
## sample_793	8.432546	3.7023805	10.873021	10.160477	9.035269	12.65761
## sample_800	9.896137	6.4425189	9.424223	10.325418	7.605657	13.00213
## sample_82	9.317019	7.1108852	10.507894	8.532823	9.138858	13.41391
## sample_95	10.664643	2.7592411	8.974498	9.943057	4.899765	12.69113
##	gene_19994	gene_11388	gene_14008	gene_14391	gene_8621	gene_14513
## sample_119	8.708173	3.5844214	4.301741	3.7843574	8.390689	3.309656
## sample_275	7.687502	12.5368664	4.462072	4.2120712	9.099698	3.909476
## sample_29	9.139229	8.3045065	5.064426	8.9434906	10.761776	6.889401
## sample_295	7.900541	0.0000000	5.087310	1.9215887	9.027826	6.075907
## sample_298	7.667261	0.7339617	4.189255	6.0442214	7.848385	4.636607
## sample_404	7.951052	10.5874119	4.736805	4.1549558	8.598894	3.091209
## sample_464	7.852205	0.6464392	2.960419	4.1666110	9.892830	3.156089
## sample_468	8.540240	0.0000000	5.494796	1.7959331	9.555782	5.823495
## sample_531	7.948712	3.9404957	4.707000	4.6784044	9.167764	4.557986
## sample_57	7.798290	3.0489859	5.716876	5.8715776	8.910607	4.185756
## sample_578	8.204733	6.5076424	2.414433	5.9456826	8.404397	3.649569
## sample_579	7.669785	12.0925028	4.304635	5.3888990	9.152551	1.858618
## sample_603	8.325476	14.5489363	4.249536	4.2495364	8.841083	2.063572
## sample_627	7.594325	0.0000000	3.957042	4.5909553	8.942685	3.886101
## sample_638	8.361817	6.2874058	3.809939	4.7986571	8.205089	3.480213
## sample_673	6.885245	3.0262866	2.266577	3.0262866	8.236756	4.291493
## sample_68	7.407667	0.0000000	4.751667	1.7029235	9.166625	5.804446
## sample_69	7.940020	9.6402739	4.475584	2.6836965	8.943719	4.504608
## sample_691	8.053464	2.4569638	3.359226	4.0139803	8.190057	5.420169
## sample_701	7.940038	10.3651742	2.230111	3.0675527	8.774820	4.138552
## sample_735	7.093264	5.8817772	4.200803	0.9197596	9.007462	7.343488
## sample_74	10.182941	4.6401697	4.442592	3.1231364	8.918008	3.473319
## sample_740	8.113148	9.3776335	2.849479	5.9710722	7.641243	4.015578
## sample_75	8.552293	1.2669364	4.390386	1.9308887	9.425508	4.113950
## sample_762	8.216217	1.6110307	3.357017	2.9406930	8.479691	5.682020
## sample_768	8.639106	3.4120769	2.893323	3.2750518	8.856821	3.688762
## sample_793	7.881952	11.0124988	2.846152	1.6712934	8.182454	2.692092
## sample_800	7.900674	11.0834195	2.525919	2.1373396	7.517394	4.671107
## sample_82	8.992431	10.4335333	3.855222	1.0259501	9.541265	3.802628
## sample_95	7.892318	13.1246254	4.545999	2.0860704	8.442728	2.596721
##	gene_1390	gene_8533	gene_5700	gene_14685	gene_16968	gene_4603

## sample_119	0.9195308	6.797700	9.642443	7.787321	5.355460	1.1300742
## sample_275	3.0024145	7.735773	10.319051	8.512207	4.634587	1.4612142
## sample_29	9.9438730	6.866203	10.767001	8.328751	2.804880	0.3952822
## sample_295	0.9482270	5.850449	9.791694	10.596730	5.554868	7.7087597
## sample_298	6.0296494	6.724964	9.417323	10.324529	6.219825	1.4103411
## sample_404	0.9145645	6.938768	10.202112	7.716250	4.435442	1.6836515
## sample_464	2.3094673	5.458188	9.964780	10.013727	5.070887	0.0000000
## sample_468	0.0000000	6.064620	9.639411	11.568897	6.073475	9.8007446
## sample_531	1.0180644	5.721783	9.637695	9.825464	6.527349	0.0000000
## sample_57	0.0000000	5.415272	9.442029	9.793424	5.981085	0.7324863
## sample_578	2.4144332	6.354858	10.351050	9.311212	5.302323	5.9900518
## sample_579	0.8077671	5.637509	10.402319	7.621114	2.000685	1.0879380
## sample_603	0.7712095	7.318832	9.781030	7.802594	3.621454	1.0427144
## sample_627	3.9912361	6.224397	10.043574	7.455089	4.142855	1.9929859
## sample_638	2.7508209	7.127705	9.343517	7.769653	4.916706	0.0000000
## sample_673	2.5272455	6.081220	10.014676	9.455107	4.356355	1.2806000
## sample_68	0.4603756	6.327747	9.949783	10.741071	6.185956	6.2443543
## sample_69	1.2364620	6.584011	11.154774	9.967875	5.901357	0.0000000
## sample_691	2.0311481	5.429596	10.659149	10.811792	6.891225	4.3280466
## sample_701	0.5474498	5.573138	9.857401	8.473328	5.418750	0.0000000
## sample_735	6.1498142	5.720339	10.526147	10.011479	7.053720	1.6912661
## sample_74	0.8463128	8.263894	9.440626	8.796877	5.631139	0.6153221
## sample_740	3.8180938	6.469134	9.338975	8.895463	5.333198	0.0000000
## sample_75	1.7006173	4.864404	9.603504	10.469235	6.334781	1.7006173
## sample_762	1.3453964	5.945182	9.529729	9.723634	6.606712	4.2463093
## sample_768	1.9610306	8.250242	9.650052	8.816238	4.919135	0.0000000
## sample_793	0.0000000	6.275657	10.385701	6.779864	4.669186	2.9853547
## sample_800	2.1373396	6.164110	10.210330	8.950156	6.753030	5.3375220
## sample_82	0.6023619	6.106009	10.384600	7.858615	5.527217	1.3530409
## sample_95	2.3124328	6.625517	10.259484	7.059182	3.062415	0.4443490
##	gene_18476	gene_13926	gene_12828	gene_4005	gene_18641	gene_18354
## sample_119	11.352739	5.874116	9.065868	3.1254197	8.866277	4.212056
## sample_275	13.502496	6.874133	4.165848	2.6461857	9.780431	4.770850
## sample_29	9.950780	5.986445	2.053667	7.3926094	6.076786	1.816108
## sample_295	14.302425	6.510316	5.496657	1.7327902	10.717522	2.497587
## sample_298	10.887266	4.729694	5.720508	0.9962441	10.721791	9.330079
## sample_404	12.756563	7.328460	4.585491	2.4395699	10.177519	3.164239
## sample_464	12.269621	5.632704	3.807530	2.9604190	9.105982	7.783542
## sample_468	14.311606	6.116951	5.864887	0.8671848	11.420423	8.724391
## sample_531	10.149684	5.357010	5.100999	1.8331330	9.695922	5.707699
## sample_57	11.675684	5.903814	5.298882	1.5774893	9.632012	6.230478
## sample_578	10.951161	5.660121	3.649569	0.9726927	8.915022	8.008412
## sample_579	13.745370	6.311503	2.555325	4.4354886	8.128247	1.087938
## sample_603	11.541000	6.553559	1.271067	1.6415923	8.250459	2.389649
## sample_627	11.515872	4.873611	3.811492	1.3163196	10.288866	4.450512
## sample_638	11.406269	7.198504	4.516148	2.9584715	10.286627	2.572768
## sample_673	11.887407	4.932694	3.636381	2.2665767	9.502317	1.948152
## sample_68	14.187050	5.417154	5.794710	0.4603756	11.056096	8.745489

## sample_69	11.024953	4.445972	4.184201	3.1917837	9.319410	6.624235
## sample_691	12.267700	5.195509	6.028386	4.3678183	10.182171	10.252240
## sample_701	10.320575	5.501468	6.626893	2.0807600	10.065551	4.865399
## sample_735	11.411273	6.103789	6.065882	5.6832641	10.218563	5.196548
## sample_74	10.189194	5.554914	6.316931	1.5168724	9.200899	1.645102
## sample_740	11.317944	6.843017	4.015578	3.1589348	9.269734	3.158935
## sample_75	12.223869	4.654854	5.961672	4.0418738	10.204106	8.306572
## sample_762	11.481345	6.146900	5.945182	5.3233485	10.016892	6.342228
## sample_768	11.392162	6.671690	4.757066	0.7163325	10.528737	3.652337
## sample_793	9.813400	6.155506	3.171799	0.4480552	9.258413	2.424922
## sample_800	12.785967	6.367876	6.050310	4.1697406	10.152906	3.798901
## sample_82	9.840964	7.395474	4.296751	3.2940766	9.178102	1.353041
## sample_95	12.839777	7.203769	2.203671	1.9579889	9.338268	2.596721
##	gene_18167	gene_4673	gene_19181	gene_3566	gene_6473	gene_17541
## sample_119	9.822740	10.690880	4.323320	3.812704	4.465948	9.216161
## sample_275	9.652171	9.716723	1.461214	6.209418	4.998918	10.662766
## sample_29	11.317887	10.843474	10.645100	7.011183	3.098251	10.637413
## sample_295	8.724452	10.124600	2.497587	3.956772	2.717408	10.735328
## sample_298	10.564835	9.126031	3.266247	3.069444	3.495618	9.793893
## sample_404	10.485568	9.601968	3.644364	7.312202	2.657571	10.091197
## sample_464	9.472569	10.113534	4.724547	5.112738	1.936025	10.176198
## sample_468	9.365216	9.623671	2.571653	3.024515	2.235329	9.999098
## sample_531	10.595798	8.878244	4.243502	4.575772	2.838246	9.506579
## sample_57	9.845181	9.722901	4.744048	4.871444	3.359071	9.981696
## sample_578	11.633944	9.820613	4.406788	4.060333	2.760178	9.991295
## sample_579	12.026689	9.944331	4.160727	4.815166	3.322736	9.831219
## sample_603	8.401319	10.411649	4.249536	5.786818	3.133218	10.540118
## sample_627	9.966152	12.019685	2.719994	3.178731	3.313522	10.039495
## sample_638	10.968480	10.297123	3.986283	4.151672	5.643114	9.911440
## sample_673	11.623324	11.458704	3.329554	6.414195	5.423904	10.648187
## sample_68	9.305959	9.742544	3.861340	4.970205	2.730988	10.403822
## sample_69	11.654376	9.616376	3.391452	6.059827	3.118592	9.678185
## sample_691	9.343432	9.487462	3.676696	2.785320	2.898247	10.380180
## sample_701	9.791318	11.163669	3.418298	3.700240	3.593975	9.928126
## sample_735	10.693191	10.465566	4.927337	5.496405	5.314396	10.844643
## sample_74	10.107387	11.415240	3.783038	4.639551	3.078405	9.734368
## sample_740	12.048671	11.363598	2.261049	3.720322	4.268839	8.991451
## sample_75	9.776813	9.894475	2.845691	2.629916	3.362147	10.065941
## sample_762	10.398648	10.756590	4.529078	4.568652	3.427593	10.442985
## sample_768	9.780236	12.083536	3.614969	5.663347	5.341399	10.339817
## sample_793	8.547936	11.020112	2.214653	4.343465	3.926237	10.541174
## sample_800	10.680307	10.206184	2.525919	3.867719	2.831715	10.234147
## sample_82	11.665260	12.172199	2.038787	5.932784	3.570645	10.286858
## sample_95	9.859473	9.905769	1.487126	5.727204	3.375318	9.842570
##	gene_8639	gene_12300	gene_14325	gene_18134	gene_6741	gene_6042
## sample_119	11.185916	8.2513167	6.707539	5.098453	8.111047	8.683360
## sample_275	11.130435	6.2553969	5.588175	5.622295	8.314506	10.389255
## sample_29	10.302879	8.0209577	4.856194	10.832748	10.727053	9.278259

## sample_295	13.223596	8.3084392	9.665634	6.303980	10.794660	10.970688
## sample_298	12.941970	1.2180992	5.298974	6.466697	9.651857	9.194582
## sample_404	11.737923	6.1298291	5.001991	5.189651	7.758137	9.424760
## sample_464	11.558004	3.0615688	6.678818	5.581339	8.798695	10.162026
## sample_468	12.084619	3.5501496	10.663887	5.013328	10.917402	10.635890
## sample_531	11.417773	1.8331330	6.079171	5.883477	8.797441	9.355587
## sample_57	11.517364	2.4932374	6.959921	4.793407	8.711006	9.105107
## sample_578	12.247026	4.9238670	8.674609	6.901893	9.470401	11.426621
## sample_579	11.715254	6.8178790	6.386344	5.674122	7.621114	10.138374
## sample_603	11.592588	3.0120501	9.649066	5.192407	7.521090	9.111910
## sample_627	12.118367	6.3216864	10.955308	6.158503	8.568104	9.732763
## sample_638	11.688019	5.4758534	8.538173	6.281042	8.648864	8.963124
## sample_673	11.447181	6.6726092	10.825332	5.982047	9.526752	10.230309
## sample_68	12.619557	5.8141197	9.241499	5.364019	10.186473	10.198052
## sample_69	12.011238	3.8641762	7.876664	6.948998	10.410780	10.952421
## sample_691	11.825738	4.1795429	4.044237	5.063701	9.613766	10.022770
## sample_701	11.225587	6.3934179	8.803563	5.480892	8.465815	9.950502
## sample_735	9.989707	5.3143956	9.017588	4.514009	10.826691	11.154470
## sample_74	11.505767	2.9352523	5.531880	9.211397	9.207405	8.173757
## sample_740	11.675185	4.6909590	8.299931	5.099968	8.620029	9.109640
## sample_75	11.624265	0.3576083	4.235268	5.732049	11.360831	9.724526
## sample_762	11.491162	8.4838359	6.363083	6.962780	9.995273	9.139526
## sample_768	12.997273	7.4452963	9.814679	6.702214	9.056437	9.374857
## sample_793	11.446313	4.3691175	12.107449	4.584566	7.629459	9.981325
## sample_800	11.293782	6.0354304	5.299670	3.911394	8.706613	9.978167
## sample_82	12.002942	6.8389266	5.976416	5.087327	7.792719	6.582321
## sample_95	11.400543	5.6348285	6.210011	4.545777	6.751584	9.912264
##	gene_7515	gene_11155	gene_20358	gene_1210	gene_19789	gene_20490
## sample_119	9.362071	10.920353	7.737964	12.074289	9.557940	4.2575990
## sample_275	9.856074	9.770811	7.037086	9.976464	9.658444	0.0000000
## sample_29	9.295004	8.796831	7.781596	10.253824	7.898632	0.0000000
## sample_295	9.279436	9.499184	6.861149	8.682640	9.366320	0.0000000
## sample_298	9.877642	10.521885	5.361863	10.309113	9.515365	6.6121313
## sample_404	9.472268	9.420110	7.155759	9.736013	10.046019	0.0000000
## sample_464	10.040303	9.896257	6.354674	10.412348	10.199267	1.4307670
## sample_468	8.399445	9.444147	7.017555	10.307076	9.665998	0.0000000
## sample_531	10.295092	10.141188	7.250052	9.827093	10.256315	0.0000000
## sample_57	8.957433	9.418515	7.175345	10.211548	9.944405	0.7324863
## sample_578	10.323145	9.776964	7.011317	9.922855	9.851754	1.9588056
## sample_579	9.804830	9.536175	7.239303	9.058216	10.202724	0.4596414
## sample_603	9.129311	9.411212	7.591507	9.669016	10.013825	0.7712095
## sample_627	9.623887	9.223688	8.415894	10.376386	9.028561	0.4569113
## sample_638	9.189829	9.806231	7.747334	12.395232	9.996671	7.9539345
## sample_673	9.185614	10.445729	7.481598	12.032790	10.423872	0.9656181
## sample_68	8.991224	9.823786	6.475245	10.012555	9.663156	0.4603756
## sample_69	9.417093	9.365623	6.114034	9.903279	9.279037	2.8736360
## sample_691	9.603391	10.045555	7.096357	10.441741	9.473290	2.6627980
## sample_701	9.542660	9.659225	7.360707	10.950782	9.983763	6.7433265

## sample_735	9.775958	10.238942	6.113113	11.572662	10.064891	1.2250281
## sample_74	10.400943	9.255907	7.552300	11.558751	8.741898	2.8312692
## sample_740	8.904836	11.106327	7.477815	11.725417	9.586187	2.0369253
## sample_75	9.498576	9.763800	6.933549	9.206018	9.601416	1.7006173
## sample_762	9.912335	10.653615	7.803627	12.628227	9.892980	2.3531539
## sample_768	9.378345	9.999070	8.475438	8.024829	9.630875	4.5349891
## sample_793	9.390311	10.242995	7.989224	9.066961	10.154957	1.2967217
## sample_800	8.956094	9.702531	7.279276	10.121546	9.750008	2.1373396
## sample_82	9.570072	8.677068	7.751685	9.172930	10.229480	1.3530409
## sample_95	9.554357	9.784167	7.428863	10.817479	10.005316	0.0000000
##	gene_3182	gene_18825	gene_4769	gene_69	gene_7470	gene_18541
## sample_119	7.9452565	9.152797	8.986948	6.923114	10.229660	3.511114
## sample_275	6.6904031	10.020147	7.715852	4.889386	9.891128	3.198856
## sample_29	2.7383139	10.062572	7.631468	2.053667	9.804391	4.268973
## sample_295	2.0885487	10.275577	7.813807	4.542716	9.721954	2.373676
## sample_298	5.7999369	9.631494	8.469552	6.290222	9.362417	3.899843
## sample_404	5.5952978	9.873023	7.691757	5.412541	9.731514	2.846935
## sample_464	5.5480852	10.019229	8.757830	4.377485	9.908188	2.960419
## sample_468	0.4978423	9.883005	7.879399	1.161049	9.347984	4.707994
## sample_531	4.7895415	10.056218	8.659764	3.556442	10.514211	3.617968
## sample_57	3.6900483	10.944156	8.446839	4.335076	9.616292	3.449205
## sample_578	5.4851724	9.734401	8.610279	5.194213	9.844976	3.272352
## sample_579	3.5243897	10.308953	6.985318	3.524390	9.830029	4.248785
## sample_603	8.0476911	9.167047	7.807149	4.017414	10.086893	4.302165
## sample_627	4.4988572	10.642945	7.236263	6.773733	9.928588	4.180991
## sample_638	6.3112651	9.667081	8.008955	7.268425	9.302440	4.626766
## sample_673	5.4239042	10.608236	7.988730	7.655323	9.297992	3.580013
## sample_68	1.5259185	10.050896	8.001481	4.101432	9.896151	3.378304
## sample_69	5.1254031	10.275775	8.874533	7.065131	10.745750	4.254231
## sample_691	5.2282838	9.904815	9.213734	6.040739	9.801973	4.994258
## sample_701	6.6601813	9.747842	7.898789	3.648086	9.789814	3.846253
## sample_735	6.7845171	10.649014	8.886270	7.363154	10.267395	3.602528
## sample_74	1.8716860	9.940532	6.989412	6.432497	9.838618	5.097809
## sample_740	8.0049732	9.768366	7.975624	5.120850	10.102015	4.728976
## sample_75	5.9086136	9.228518	7.574344	5.766752	9.651220	3.913100
## sample_762	10.8761486	11.019764	9.474548	6.276828	9.579009	3.990646
## sample_768	6.4614680	9.460300	7.450609	7.818410	9.776010	4.342661
## sample_793	6.1029709	9.461342	7.472650	4.687245	9.679498	5.225607
## sample_800	7.0243973	10.256763	8.396070	4.324623	10.083173	3.570281
## sample_82	3.8026275	10.108760	7.535019	3.215787	9.302886	3.748032
## sample_95	4.2774864	10.030405	7.948163	2.834125	9.663604	4.405761
##	gene_16540	gene_8829	gene_5655	gene_18249	gene_17742	gene_8994
## sample_119	1.4766927	7.686865	10.20953	4.521509	9.102207	1.7559131
## sample_275	2.1720714	6.562435	11.83381	8.422969	8.235579	3.0024145
## sample_29	2.3496767	7.334640	11.25006	6.466950	9.422393	8.1417853
## sample_295	2.2381448	6.371453	11.09727	7.752039	8.914292	3.4885407
## sample_298	2.1096614	7.587433	11.00158	7.357385	9.387634	1.4103411
## sample_404	2.6575714	5.659853	11.60407	6.554113	8.736957	3.0142839

## sample_464	1.0912598	7.559775	12.05104	6.640664	9.159730	1.9360254
## sample_468	2.3563410	6.073475	11.20842	8.579029	8.643449	4.0582126
## sample_531	3.2801546	8.537712	12.26508	5.857819	9.312039	1.3436352
## sample_57	1.5774893	7.806846	11.96901	6.320636	8.692675	0.0000000
## sample_578	4.0355711	8.566998	11.54474	6.563983	9.755438	1.9588056
## sample_579	1.3224474	6.454163	11.84791	7.692595	8.659557	2.1299753
## sample_603	2.3896494	4.705331	10.82333	4.753567	8.469060	2.6554434
## sample_627	3.7726986	6.328391	11.49870	6.352075	8.496986	2.1220763
## sample_638	2.3696061	5.533791	10.64476	5.590267	8.433139	2.5081751
## sample_673	0.9656181	6.885245	11.03674	9.821273	9.370404	0.9656181
## sample_68	2.1320832	6.279689	11.24448	8.608388	8.552423	2.8857696
## sample_69	1.2364620	5.986115	12.49537	7.486417	9.018565	2.3416158
## sample_691	2.7253485	7.571116	11.40763	6.166643	9.749784	2.1269398
## sample_701	0.0000000	9.345951	11.35779	7.538786	8.574590	2.3654686
## sample_735	3.1740708	7.045367	12.36114	6.804622	9.618747	2.7647920
## sample_74	1.0453032	5.554914	10.45743	3.707801	8.305720	2.0674666
## sample_740	2.6259484	5.417100	11.48191	5.952632	8.666071	1.7714631
## sample_75	1.8203219	6.086359	11.30146	5.422580	9.235268	0.8827604
## sample_762	3.8435417	6.738997	11.16626	7.444568	9.664593	1.3453964
## sample_768	2.0754984	6.653381	10.80634	7.001870	8.878661	2.3727850
## sample_793	3.0502056	5.182855	10.82510	4.389064	9.114180	1.6712934
## sample_800	2.8317151	5.313054	11.60847	7.698114	8.458743	2.9633076
## sample_82	3.2940766	3.632233	10.39241	2.370499	9.380084	2.9516251
## sample_95	1.0579701	6.808398	12.17993	7.056486	8.651217	3.0379269
##	gene_19454	gene_14814	gene_1934	gene_10590	gene_15557	gene_4984
## sample_119	9.1376547	8.901669	5.135395	1.6230240	1.1300742	1.4766927
## sample_275	9.9663829	9.234585	6.341826	3.5260939	3.4509737	1.1169635
## sample_29	7.9692771	8.156204	6.095990	8.8431076	1.5315686	1.8161081
## sample_295	1.5155106	9.523074	5.752120	8.0647966	4.5136613	0.0000000
## sample_298	9.2732291	10.454608	5.337932	6.8041568	1.9943263	2.5786731
## sample_404	9.4936794	9.582118	7.362759	3.7430194	2.0346735	1.8698319
## sample_464	6.8478591	8.744582	5.689909	2.6058736	3.1560891	0.6464392
## sample_468	0.4978423	9.544682	5.739084	9.3875780	3.6482468	0.4978423
## sample_531	9.1729000	8.841637	6.030334	3.4248952	1.3436352	1.3436352
## sample_57	8.6810943	8.629018	5.642791	9.9071205	2.6534957	0.7324863
## sample_578	7.2715844	9.249815	6.243353	3.5344476	2.9518116	3.3424694
## sample_579	6.5146194	10.111710	7.108389	5.5707392	2.7286824	1.8586176
## sample_603	10.1133645	9.551651	5.325872	1.2710670	2.8088175	2.6554434
## sample_627	9.0099154	10.406003	5.202990	2.1220763	1.3163196	1.6939887
## sample_638	7.7065306	9.676297	4.852204	0.8330925	1.7417905	0.3339961
## sample_673	4.2195170	8.323001	7.154848	5.4398292	2.4028040	2.5272455
## sample_68	2.0027025	9.174157	4.459182	6.6365695	3.5268576	1.0893626
## sample_69	4.1842009	8.289719	6.402253	3.5106835	1.2364620	2.7817793
## sample_691	8.2270142	10.230825	7.128510	8.5811290	4.1795429	4.4989529
## sample_701	9.1407071	8.949269	3.808735	3.4792694	0.5474498	2.4892089
## sample_735	7.4613239	7.922840	5.886460	0.9197596	3.7047392	3.3099910
## sample_74	7.5045012	8.550501	3.442585	4.7440480	1.8716860	2.5976980
## sample_740	3.9844891	8.547419	5.615034	2.9813230	5.2956936	5.1064823

## sample_75	8.6250220	9.990558	5.203738	5.0350168	1.9308887	3.8859062
## sample_762	4.9202742	9.620082	4.643798	2.0293115	3.2045790	3.9432556
## sample_768	9.6179468	9.834585	5.101764	2.0754984	1.9610306	3.5766068
## sample_793	6.3091707	9.776734	4.958764	1.0652276	0.4480552	2.5195922
## sample_800	7.3710186	9.049761	6.351513	3.1951743	2.5259185	0.7483754
## sample_82	7.4174981	9.468211	6.976593	0.0000000	0.0000000	1.3530409
## sample_95	8.4531595	9.699092	8.353345	5.7063289	1.9579889	1.2884176
##	gene_20505	gene_17777	gene_8511	gene_3641	gene_19379	gene_6175
## sample_119	9.245417	2.8561292	1.7559131	8.591874	0.6730136	4.698380
## sample_275	8.401861	3.0024145	1.4612142	10.294414	1.1169635	7.049870
## sample_29	9.155266	3.3860936	0.3952822	6.854469	0.3952822	5.770224
## sample_295	8.396630	2.0885487	0.5506052	10.478588	2.2381448	9.073344
## sample_298	8.235646	0.7339617	3.4956184	9.610840	0.9962441	8.526444
## sample_404	8.843629	3.0142839	0.5285713	9.644862	1.4698339	6.580301
## sample_464	9.304762	2.8516391	2.1347795	9.176739	8.4734541	7.315430
## sample_468	8.330939	1.6137202	2.3563410	10.334039	1.1610492	7.273357
## sample_531	9.083027	3.4248952	1.8331330	7.715276	5.2444647	7.729288
## sample_57	9.411526	2.3129265	2.7977186	7.789690	7.6446278	7.728016
## sample_578	8.880086	1.5483873	3.0388751	8.520489	1.5483873	6.202927
## sample_579	8.939744	1.7010611	1.0879380	10.304249	8.3098083	5.604510
## sample_603	9.444518	1.4682184	1.0427144	7.246817	1.4682184	3.919750
## sample_627	8.984672	0.0000000	0.4569113	7.965103	1.0824982	6.738930
## sample_638	9.174302	2.6938772	1.4974846	8.402368	0.0000000	5.500968
## sample_673	8.173877	3.8892496	3.1082903	9.108012	2.4028040	7.939174
## sample_68	8.408211	1.7029235	1.7029235	9.879548	1.7029235	7.863907
## sample_69	7.961114	2.9599927	1.7050490	8.179307	0.9291850	7.127097
## sample_691	7.897482	3.8269145	3.3192133	9.001034	0.8812731	8.442525
## sample_701	8.712702	0.5474498	0.9432837	8.047113	0.0000000	4.842375
## sample_735	7.843839	3.4342145	5.6322973	8.152690	2.0430645	6.344449
## sample_74	9.290019	3.0784049	1.9728764	8.540233	0.8463128	5.163515
## sample_740	9.054564	11.9610372	1.0247452	6.834319	0.0000000	5.682270
## sample_75	8.526112	1.9308887	2.9544011	8.207088	0.6439485	7.029861
## sample_762	8.810334	4.0365297	3.3570171	8.944241	2.4913657	6.514234
## sample_768	9.382520	2.8933234	0.9741622	8.788187	1.7006173	6.000279
## sample_793	8.925017	1.2967217	0.7893537	7.460775	0.0000000	5.346748
## sample_800	9.341968	1.8951475	3.0838941	8.764145	1.2386646	6.723913
## sample_82	9.358607	1.3530409	0.0000000	7.526937	1.0259501	5.316960
## sample_95	8.749997	2.2036710	1.2884176	9.605346	0.4443490	6.163658
##	gene_1004	gene_862	gene_15614	gene_11189	gene_18164	gene_15224
## sample_119	9.745930	10.107910	5.695359	9.403176	9.313035	12.69223
## sample_275	9.402219	10.595146	4.462072	9.546763	10.901433	15.03326
## sample_29	9.367694	9.592231	8.382875	8.954356	8.604650	12.66732
## sample_295	10.184590	11.939080	3.709644	9.207544	10.547405	13.44182
## sample_298	12.633520	12.630884	5.338407	10.293943	10.241959	13.53873
## sample_404	9.192468	10.248509	6.285239	8.887705	10.541465	13.37583
## sample_464	9.434182	12.163046	6.218932	9.900530	10.120147	14.38727
## sample_468	9.472876	11.995997	5.546592	8.486553	9.690669	12.48672
## sample_531	9.532777	11.585502	5.930191	10.915917	8.945789	13.40502

## sample_57	9.771754	11.473066	6.618214	9.586810	9.466517	13.64050
## sample_578	9.989934	11.586422	5.054957	9.426682	9.892499	13.63867
## sample_579	8.568382	10.437877	6.234509	9.278554	11.079638	13.18034
## sample_603	8.871878	10.887975	3.814981	8.824313	9.904781	14.47882
## sample_627	10.188799	10.946490	3.313522	9.437837	9.721347	12.39397
## sample_638	9.403014	11.283343	5.296226	8.373196	8.697652	12.09079
## sample_673	8.761933	13.346486	2.939339	9.259256	10.161144	13.43032
## sample_68	11.542345	13.289860	4.580887	8.901072	9.987676	12.88575
## sample_69	10.913727	13.017214	5.233367	10.432176	9.760194	12.63347
## sample_691	10.077390	13.043451	2.381256	9.600786	10.673353	13.69300
## sample_701	9.705411	10.660388	3.288048	9.641830	9.312835	13.97384
## sample_735	11.275147	14.460277	3.100978	10.117955	10.649409	14.67145
## sample_74	9.931916	11.877963	6.148666	8.574302	8.821232	10.85223
## sample_740	10.092017	13.352568	7.483380	9.724343	9.291334	14.34759
## sample_75	10.226641	12.819662	5.263489	9.867793	10.119084	13.80520
## sample_762	10.503965	13.731977	4.322361	9.278468	9.045885	13.70737
## sample_768	10.296951	11.308498	4.173055	8.771301	9.062521	12.64631
## sample_793	8.459661	9.850293	2.323601	9.282125	9.845733	13.59437
## sample_800	9.684764	13.305805	3.298482	10.148591	10.561364	13.16723
## sample_82	7.364047	10.686719	4.002414	8.588568	9.482911	13.87655
## sample_95	8.090494	9.261505	4.523549	9.131170	11.147014	13.81332
##	gene_4244	gene_6356	gene_7329	gene_17566	gene_1528	gene_14932
## sample_119	8.416016	9.097519	1.8775872	2.597412	3.4730076	8.255529
## sample_275	8.659186	10.409232	0.6640277	5.629636	4.7396321	8.348533
## sample_29	8.661899	8.730942	0.9601410	2.929469	0.0000000	7.529337
## sample_295	9.054553	11.297008	2.3736760	7.906139	1.9215887	8.374579
## sample_298	9.114856	10.093985	2.4089020	4.080351	5.1890496	8.853550
## sample_404	8.473401	9.704225	0.5285713	5.412541	4.0039800	8.484138
## sample_464	9.039344	9.643416	0.6464392	5.004097	5.2596190	7.764911
## sample_468	8.753608	10.744531	2.1031955	7.453740	1.1610492	8.754984
## sample_531	8.741518	8.777354	1.0180644	3.201947	2.6149925	8.451607
## sample_57	8.465860	9.451603	2.6534957	3.359071	4.0768669	8.465860
## sample_578	9.151915	10.078645	0.0000000	4.968902	1.2890672	7.574442
## sample_579	8.632530	10.686746	0.8077671	6.532273	0.0000000	8.728818
## sample_603	7.990711	9.121093	0.4365883	3.621454	1.2710670	8.603574
## sample_627	8.695444	10.065807	0.0000000	4.210327	2.5468327	8.724961
## sample_638	8.722777	9.836964	0.6050202	3.755326	3.0966835	8.311839
## sample_673	8.988594	9.646516	1.2806000	3.889250	4.3261778	7.284051
## sample_68	8.534984	10.955621	4.3072099	7.916483	2.0027025	8.695347
## sample_69	8.833662	10.655227	1.4896971	4.254231	3.9926416	7.562739
## sample_691	9.266625	9.670019	2.5973881	3.962956	4.4442032	8.237784
## sample_701	8.034474	9.368286	0.0000000	3.593975	1.7256539	8.600549
## sample_735	8.639254	9.012454	3.0239827	3.373439	8.1322015	7.000856
## sample_74	9.449433	9.111470	0.3402774	3.328061	4.0610417	9.460368
## sample_740	7.205197	9.914200	0.0000000	3.504074	2.4549918	8.381292
## sample_75	8.983575	9.747369	2.3039561	3.647671	4.1139505	8.339516
## sample_762	9.330043	10.369259	0.0000000	4.206721	2.7334411	7.489928
## sample_768	8.712809	10.799856	0.4021764	4.173055	2.1815479	8.680391

## sample_793	8.538076	8.112043	1.4961044	2.846152	0.0000000	9.069898
## sample_800	8.172932	9.727044	0.0000000	5.453985	7.0016109	8.656443
## sample_82	8.924824	8.172838	2.5016939	3.691288	1.3530409	8.190857
## sample_95	8.383195	9.497906	0.4443490	4.951131	0.7834986	8.605194
##	gene_15942	gene_19727	gene_18205	gene_18207	gene_1033	gene_3413
## sample_119	8.366213	9.782287	9.876638	9.800316	8.472630	7.892106
## sample_275	9.493949	8.477944	10.244554	10.536179	9.252668	8.619002
## sample_29	6.534673	9.691283	10.165133	9.066466	11.784872	6.180727
## sample_295	8.356280	8.371341	10.429354	10.231641	9.409353	8.549396
## sample_298	10.676028	9.000516	11.367671	9.784160	7.594877	9.005552
## sample_404	8.203617	8.874914	10.633949	10.601446	9.441384	7.489936
## sample_464	8.487153	9.074516	10.889496	9.950015	8.596540	8.613296
## sample_468	7.835829	8.081617	9.784658	10.358695	8.895184	7.567804
## sample_531	9.199559	8.969752	11.346004	9.129984	8.021674	8.417405
## sample_57	9.557981	9.276336	10.244958	9.532434	6.558718	7.556123
## sample_578	7.476487	8.835779	11.127846	9.939920	6.230927	8.341577
## sample_579	8.464934	8.957787	10.843450	11.320502	8.854600	7.661785
## sample_603	7.925103	9.016474	10.590905	10.367185	10.381392	7.923007
## sample_627	9.607443	8.740101	11.431007	11.194388	7.939056	6.556806
## sample_638	8.486505	8.554098	10.399449	10.032638	10.347898	8.039276
## sample_673	7.174117	8.924709	11.056238	10.745893	9.983108	8.377757
## sample_68	7.898414	8.493435	10.098177	10.181811	8.408211	9.138994
## sample_69	6.095078	9.221377	11.598476	9.902598	8.914280	8.550247
## sample_691	9.540080	9.755642	11.926022	11.116474	6.491372	8.910532
## sample_701	9.289620	7.787210	9.717323	10.233680	10.934207	7.974856
## sample_735	8.259409	8.067112	9.719088	10.209356	9.307668	9.303603
## sample_74	6.036233	8.314184	10.998738	10.646910	7.865306	7.588632
## sample_740	7.417979	8.092662	9.985629	10.307952	6.382337	7.702235
## sample_75	8.773363	8.738055	11.563883	10.197217	6.573397	8.148222
## sample_762	8.998858	8.007448	10.467269	10.617752	7.413738	8.544640
## sample_768	9.179927	8.531038	10.630549	11.061364	8.615648	7.224282
## sample_793	7.832099	8.973743	11.193217	11.653109	11.447631	7.632108
## sample_800	9.182298	8.865563	10.577495	9.850116	7.154018	6.723913
## sample_82	10.800067	8.602714	11.629370	11.431116	11.401301	7.303817
## sample_95	8.288192	9.116144	10.381121	10.677085	9.441403	8.996055
##	gene_11031	gene_17353	gene_15173	gene_19615	gene_18255	gene_20413
## sample_119	6.503784	8.100152	9.281691	6.103420	10.662855	6.436077
## sample_275	11.497707	8.414271	9.520597	6.140574	11.537112	6.152475
## sample_29	9.800827	8.972744	8.951780	6.786074	10.129863	7.617644
## sample_295	7.942028	10.254875	8.299954	0.948227	10.771828	6.045777
## sample_298	12.296133	7.470919	9.078153	7.711302	9.252985	4.495279
## sample_404	10.630731	8.629841	9.411697	5.203229	11.153824	6.518976
## sample_464	12.166596	9.076073	9.422344	7.931181	10.602690	6.019655
## sample_468	6.756743	10.101661	8.759106	0.000000	11.185631	5.915009
## sample_531	11.285368	8.981342	9.698595	8.380158	10.448519	5.320463
## sample_57	9.953095	9.177763	10.072226	8.183536	9.613858	6.398267
## sample_578	8.313482	9.306834	9.469422	9.209339	10.467911	5.421802
## sample_579	10.429260	9.502241	9.540545	5.931624	11.407374	6.526413

## sample_603	5.898624	9.393886	9.697539	5.792257	10.271031	6.779181
## sample_627	9.976865	9.298152	9.169644	6.510391	9.814429	6.180605
## sample_638	9.610514	9.017073	9.039388	7.183070	10.754336	6.853260
## sample_673	10.284778	8.483328	8.281629	7.279638	11.532497	5.908880
## sample_68	9.808747	10.005723	8.940243	1.525919	11.454628	6.077700
## sample_69	10.952749	8.314515	8.993190	9.817698	9.226655	4.836778
## sample_691	9.761028	7.952462	8.194210	7.051274	10.924835	4.875136
## sample_701	7.024708	9.369292	8.986496	9.615762	11.609917	6.286606
## sample_735	10.248485	8.850484	8.887629	6.543765	11.887000	5.124618
## sample_74	10.295907	7.671895	9.318543	8.570823	10.012136	7.085765
## sample_740	7.426676	8.810475	9.710159	8.539419	11.002724	6.156860
## sample_75	6.098253	7.900716	9.273803	7.671102	10.402138	5.165128
## sample_762	9.813675	8.443959	8.691681	9.132939	11.454130	6.305219
## sample_768	8.918508	9.064255	8.884561	7.864910	9.836109	6.644137
## sample_793	9.212214	9.323820	9.345787	6.535583	11.491537	6.750888
## sample_800	11.602170	8.634397	9.728201	7.009246	10.642142	6.414619
## sample_82	4.441397	10.248663	8.796202	4.002414	10.218236	7.651138
## sample_95	9.448866	8.845838	9.510088	4.523549	11.145868	6.469863
##	gene_3979	gene_13784	gene_20409	gene_8353	gene_6661	gene_14299
## sample_119	11.436956	2.190962	7.028669	9.349640	4.2575990	5.253501
## sample_275	9.233183	4.739632	7.300426	6.944285	5.1250849	5.629636
## sample_29	8.991627	5.412815	7.741791	7.220378	9.1648039	6.611082
## sample_295	8.466529	3.488541	5.729300	6.715633	8.5600959	4.852214
## sample_298	10.572710	3.163209	5.094202	8.623036	5.8670439	5.108140
## sample_404	9.019985	3.538476	6.418077	7.507589	3.8353481	6.092897
## sample_464	9.308626	3.407339	6.044580	9.397165	2.7339833	6.251410
## sample_468	9.231171	3.073358	5.297298	6.660124	8.3511071	4.752781
## sample_531	9.974042	2.838246	6.693835	9.246581	4.0782085	5.842692
## sample_57	10.556391	4.019186	6.026003	9.190886	5.0334803	5.698616
## sample_578	8.835188	2.760178	7.135011	9.191637	3.5344476	5.922978
## sample_579	9.149694	3.524390	6.234509	7.336283	4.4354886	5.439673
## sample_603	9.694466	2.483777	6.411069	9.204684	3.4915068	5.846421
## sample_627	10.337031	3.201116	6.080903	8.180511	5.5144190	4.718926
## sample_638	11.322756	1.357833	7.103319	8.573920	3.8364897	5.467384
## sample_673	9.796280	2.116132	7.284051	7.655323	4.1043450	5.423904
## sample_68	8.647458	3.618027	5.550430	6.766794	8.6076928	4.511247
## sample_69	9.441687	6.617609	6.570349	7.626841	7.5798891	6.104586
## sample_691	10.036338	4.847275	6.319511	8.436691	3.1006245	5.332447
## sample_701	9.619998	2.985865	6.004973	8.688701	3.8918080	5.209239
## sample_735	8.650503	5.791749	6.194416	9.401016	0.9197596	6.279687
## sample_74	9.423965	4.593252	7.897845	6.110303	7.2858923	6.812960
## sample_740	9.762076	3.669605	6.877278	6.613309	3.2135961	5.230929
## sample_75	10.024059	3.322130	6.826777	8.989973	2.7881010	5.839426
## sample_762	9.084620	3.204579	5.883755	8.849418	3.6206920	5.933105
## sample_768	9.592941	2.280333	6.057110	7.740300	5.4408492	6.471955
## sample_793	10.743799	1.496104	6.585673	7.472650	3.6614418	6.591133
## sample_800	8.786531	3.933393	6.282247	7.884158	4.7461246	5.700248
## sample_82	9.716761	1.353041	6.838927	8.857119	4.1778779	5.620278

## sample_95	9.200906	2.973134	5.994435	5.726123	2.8341246	6.577293
##	gene_17433	gene_98	gene_13121	gene_9048	gene_7079	gene_2040
## sample_119	12.105856	11.423909	12.122585	8.740611	2.9699517	5.791382
## sample_275	10.231989	9.761127	10.638110	5.541868	6.5078866	6.002725
## sample_29	10.355538	10.333513	11.051440	9.647701	5.9574712	8.705028
## sample_295	10.583685	10.577552	8.988767	5.126225	7.1240386	5.766725
## sample_298	10.761734	9.790115	10.099229	8.386216	1.5799525	4.867175
## sample_404	10.984888	10.105175	10.324653	3.694535	5.7456380	5.803932
## sample_464	10.975038	10.481860	10.409953	8.346412	3.0615688	8.619530
## sample_468	10.448653	10.549679	9.656453	5.571807	7.5520465	5.297298
## sample_531	11.164593	10.250867	10.891207	8.074757	3.4921737	8.797441
## sample_57	12.356251	9.629604	11.090066	7.663579	3.4492048	8.722350
## sample_578	10.371112	9.753832	10.015652	8.594201	4.9689018	4.829352
## sample_579	10.994898	10.420992	9.316051	4.846363	6.8789460	6.175816
## sample_603	11.262999	10.750925	10.909338	6.722384	2.6554434	3.621454
## sample_627	12.118730	11.737171	9.599844	8.076826	3.5150312	5.029859
## sample_638	12.355409	11.287857	10.527692	8.724555	4.1002292	5.903841
## sample_673	9.526752	11.479795	8.693846	6.516174	3.3963650	6.470394
## sample_68	10.002027	10.453785	10.053967	6.020364	7.5705995	5.607165
## sample_69	10.960625	9.476894	10.501300	7.934699	4.1106055	5.544114
## sample_691	10.352882	10.608698	9.117874	5.864312	1.8180325	5.614513
## sample_701	10.539459	10.793506	9.473172	6.124301	6.6862764	6.320320
## sample_735	7.500929	9.809320	9.912377	5.482087	6.1588460	6.296151
## sample_74	11.797139	10.339794	10.964204	10.200972	0.8463128	6.450159
## sample_740	10.870249	10.247026	10.814983	7.510922	2.6259484	6.257837
## sample_75	10.558048	9.890205	11.174994	7.305998	1.7006173	4.997396
## sample_762	9.542705	10.408500	9.684544	7.638646	4.6222920	6.710985
## sample_768	10.666384	11.934262	10.119460	6.582558	5.4301314	5.329913
## sample_793	11.591452	12.963927	9.139554	6.655087	2.6084071	3.336883
## sample_800	11.231503	9.788229	10.221055	5.877553	4.8853840	5.622740
## sample_82	10.404599	10.815503	9.523091	4.804498	1.8443053	4.508619
## sample_95	10.670081	9.854423	10.396262	2.413540	5.9944349	4.864478
##	gene_1866	gene_20523	gene_2220	gene_11735	gene_6283	gene_14124
## sample_119	9.589447	10.897611	8.871779	11.86828	9.555663	8.330697
## sample_275	8.802045	10.085247	11.672686	11.16610	9.441556	9.309585
## sample_29	9.160840	10.794911	10.930383	12.24453	9.726562	6.479693
## sample_295	9.140224	9.481473	9.096799	11.34625	9.979397	11.287383
## sample_298	10.200286	9.722642	7.831681	11.26100	9.412417	10.881190
## sample_404	9.633655	10.385086	10.924100	10.71405	9.515647	9.535567
## sample_464	9.835071	9.955775	9.175330	12.01110	9.194944	8.480318
## sample_468	8.517705	9.530281	8.665578	10.57757	10.131651	11.591228
## sample_531	10.452746	9.476789	8.968114	11.58358	9.165191	8.240424
## sample_57	8.922457	10.331936	9.388928	10.89087	8.955510	9.118905
## sample_578	9.100986	9.558666	10.822499	12.74985	8.610279	7.703288
## sample_579	9.174267	10.008275	11.666792	12.10521	9.628662	8.772803
## sample_603	9.352375	10.282127	13.073486	11.28032	9.566518	6.520622
## sample_627	9.107664	10.056245	10.855717	12.03259	9.783047	9.407463
## sample_638	8.847950	10.349337	12.636783	11.13799	9.724183	8.505248

## sample_673	8.281629	9.976979	10.246812	11.47522	9.649087	9.587877
## sample_68	8.466105	9.985529	7.765144	10.96759	9.729825	11.901267
## sample_69	8.413476	10.329471	9.960041	12.20286	9.095692	8.475255
## sample_691	10.239897	9.517053	8.356192	11.81926	8.934732	9.909875
## sample_701	9.749390	9.646821	12.246269	11.35982	9.733837	7.643084
## sample_735	8.579972	9.756753	10.919623	11.72732	8.161706	8.966900
## sample_74	8.727485	9.941703	9.006461	11.00287	10.505742	8.529887
## sample_740	9.468221	9.954371	11.128497	11.83175	9.126650	7.906806
## sample_75	8.668555	9.594608	7.128860	12.20371	9.130555	9.048626
## sample_762	9.660019	9.906942	11.657305	12.57833	8.926071	9.186070
## sample_768	9.565045	9.968854	12.153517	10.69156	10.093800	9.428343
## sample_793	9.724904	9.862761	11.792782	12.83310	10.286558	7.773278
## sample_800	9.305227	10.243079	10.290053	12.34644	9.016056	9.361485
## sample_82	9.518002	10.431372	8.598246	10.72256	9.671642	7.555034
## sample_95	9.373211	10.021924	10.732880	11.01017	9.451842	8.855980
##	gene_7918	gene_9763	gene_10050	gene_11564	gene_7014	gene_5693
## sample_119	4.953503	7.421661	3.3096564	1.1300742	6.385693	13.43602
## sample_275	7.007577	7.715372	2.8931680	2.3480057	7.347790	14.10520
## sample_29	3.575349	6.508094	4.8759913	0.0000000	3.588637	11.18886
## sample_295	9.609448	7.569286	2.2381448	7.6532177	9.581997	13.35277
## sample_298	7.511128	8.301309	1.4103411	2.1096614	8.637487	12.29270
## sample_404	7.795734	7.262057	2.1826287	0.5285713	7.105657	14.13447
## sample_464	5.670186	8.565319	0.6464392	3.9724721	6.218932	12.19262
## sample_468	9.796070	6.145366	2.9247080	6.1254920	8.703149	13.31702
## sample_531	6.173071	7.242698	1.8331330	2.4889263	7.915401	11.41750
## sample_57	6.661650	7.242431	3.0489859	1.5774893	7.889656	11.99776
## sample_578	4.897555	8.139357	3.7038538	4.6474066	6.589223	12.29851
## sample_579	3.860685	7.440313	1.8586176	2.0006851	5.840091	14.58385
## sample_603	4.429462	7.258915	3.2977637	0.0000000	7.135206	13.12849
## sample_627	5.670316	7.199692	3.8861014	2.1220763	7.971544	13.03330
## sample_638	4.773727	8.446240	6.2969969	1.4974846	7.506113	12.44845
## sample_673	5.032052	8.984014	4.7909023	2.5272455	8.234429	11.78185
## sample_68	8.805986	6.831269	2.3604486	6.8253532	8.132880	13.21221
## sample_69	5.638786	9.787692	2.3416158	1.4896971	6.959051	11.14645
## sample_691	7.093148	8.519738	0.3569326	1.2650167	8.879059	12.96667
## sample_701	4.721411	9.164256	4.6959493	1.5088850	6.821927	12.83012
## sample_735	9.188510	9.766696	4.0143553	4.2008025	8.442512	12.55746
## sample_74	4.875603	7.623823	3.4025175	1.0453032	6.980322	10.82138
## sample_740	6.496502	8.023982	5.1906267	1.0247452	6.561575	13.74510
## sample_75	5.874849	8.452015	1.4264258	1.5700257	7.020513	12.00042
## sample_762	6.357943	8.743020	3.2828104	3.4948636	8.147317	12.75874
## sample_768	5.404845	8.594664	5.9335490	1.3826673	7.770353	13.14237
## sample_793	3.531169	7.843425	4.5593878	0.7893537	7.342750	12.77440
## sample_800	5.485198	6.614546	0.7483754	1.8951475	8.296632	12.98698
## sample_82	4.135970	5.244468	1.6195069	2.2101395	8.567047	13.94554
## sample_95	8.364752	6.930087	0.4443490	0.7834986	6.104393	14.84186
##	gene_6900	gene_13935	gene_18976	gene_10705	gene_9544	gene_20504
## sample_119	10.157928	12.78578	9.225504	8.872693	0.0000000	5.148629

## sample_275	8.696248	13.07072	9.181843	9.761127	2.5047729	5.348473
## sample_29	6.306691	13.20914	8.639272	6.927256	2.3496767	5.753463
## sample_295	11.977162	12.53712	9.206411	9.357147	0.5506052	3.945505
## sample_298	10.582415	12.06067	10.016669	6.804157	6.7607000	5.094202
## sample_404	9.032228	12.67945	9.204420	9.525210	1.8698319	6.695646
## sample_464	8.226287	12.88073	9.113365	7.423897	0.0000000	2.309467
## sample_468	12.634931	12.70276	8.932165	10.044271	3.7400857	1.161049
## sample_531	9.859299	13.10483	9.548664	6.862203	0.0000000	1.833133
## sample_57	9.849325	12.38936	9.317435	7.668275	7.3137447	2.797719
## sample_578	8.332990	13.11160	9.124811	8.173812	0.0000000	2.539109
## sample_579	8.270024	12.92465	9.649707	8.867405	1.7010611	5.886584
## sample_603	7.945930	12.35512	9.371742	9.165271	0.4365883	1.042714
## sample_627	9.776320	12.59396	9.098866	10.541648	0.0000000	3.515031
## sample_638	10.054374	12.07293	8.943412	8.756189	1.0299829	6.484952
## sample_673	9.328251	12.87830	8.502410	8.310063	0.5621814	2.402804
## sample_68	11.687866	12.81722	8.917990	9.384337	3.8235354	4.251348
## sample_69	8.384495	13.07451	8.706727	7.662683	0.0000000	4.742464
## sample_691	10.515513	12.10553	10.223060	6.933643	7.9927628	1.698352
## sample_701	7.672751	12.82151	9.024065	8.598834	1.2536263	2.603146
## sample_735	11.151080	13.29443	8.246755	8.223273	1.4769001	1.691266
## sample_74	10.057125	12.01804	9.531050	6.927719	1.0453032	1.762817
## sample_740	8.664849	12.99417	8.385763	7.312384	0.4274989	5.270133
## sample_75	10.112505	12.26699	9.068416	6.494579	6.6070906	2.033582
## sample_762	10.882834	13.08852	9.008961	8.400760	5.8064117	2.200285
## sample_768	9.044189	12.69832	9.369257	9.163033	1.5504082	6.528315
## sample_793	9.696060	12.51086	8.806814	8.929335	0.0000000	1.065228
## sample_800	10.953913	12.87706	9.047909	8.162698	3.9962079	6.217814
## sample_82	8.956768	12.93899	8.377757	9.387916	0.0000000	5.179324
## sample_95	8.893663	12.96753	9.675107	8.945239	1.4871261	6.170898
##	gene_9807	gene_8245	gene_4303	gene_18150	gene_8246	gene_18058
## sample_119	9.500712	10.578731	6.876431	9.113088	12.167173	8.979325
## sample_275	7.960413	10.280922	6.431793	10.644965	9.028826	10.639693
## sample_29	7.402347	10.716819	3.863126	8.676419	9.704934	10.972398
## sample_295	7.065378	10.035418	6.790225	9.910010	10.108903	9.619132
## sample_298	9.264264	9.762844	8.748465	9.872547	11.909946	9.480295
## sample_404	8.784779	10.225232	6.301516	10.454063	10.435743	10.993370
## sample_464	8.617060	9.652565	5.846220	10.297799	11.678596	9.611453
## sample_468	6.147044	10.000831	7.904430	9.726898	9.606995	8.466525
## sample_531	8.888488	9.619948	6.823546	9.571199	12.770146	9.717166
## sample_57	8.596372	9.454326	7.086858	9.857577	11.345017	10.047192
## sample_578	8.130195	10.572378	6.429249	10.357948	12.560328	9.669385
## sample_579	7.089276	9.812676	5.119817	10.762556	8.543933	10.258142
## sample_603	7.520430	10.174202	6.239518	10.275973	13.237017	8.331817
## sample_627	7.296182	10.851936	5.170822	9.574688	11.432563	8.471468
## sample_638	8.519695	9.820016	7.335113	9.389070	12.994751	9.374422
## sample_673	7.942227	11.132320	6.589386	10.878511	11.558937	9.450203
## sample_68	6.921282	10.239921	8.022495	9.772618	12.705053	8.109507
## sample_69	9.076880	9.756430	8.392279	10.213967	10.476341	9.551801

## sample_691	8.236349	10.461111	8.653304	10.122763	10.858339	8.995490
## sample_701	7.020813	10.717651	6.972647	11.059277	10.719235	9.594390
## sample_735	8.231644	10.241066	8.815217	11.201395	11.505439	8.884909
## sample_74	8.502303	10.380364	6.895848	8.013725	9.729403	9.796092
## sample_740	8.897740	9.733724	6.855965	10.384384	11.427674	9.697032
## sample_75	8.465191	9.415438	8.697193	10.013504	10.946285	9.113805
## sample_762	8.736936	10.390943	7.924468	10.326609	9.966671	10.392596
## sample_768	8.506288	10.173065	7.053242	9.844709	10.990182	8.099442
## sample_793	7.205217	11.757102	7.741716	10.755088	12.409632	9.539669
## sample_800	8.187308	9.628870	7.538655	10.240648	8.159271	10.855702
## sample_82	7.633104	10.636887	6.673019	10.573543	8.723787	11.220136
## sample_95	7.592031	9.807580	4.277486	10.788653	7.066961	10.150179
##	gene_466	gene_1763	gene_4437	gene_6139	gene_14594	gene_8260
## sample_119	0.9195308	6.134266	5.086894	9.231221	1.4766927	9.431393
## sample_275	1.1169635	7.695660	6.351462	9.020730	2.3480057	9.036874
## sample_29	0.0000000	7.524401	5.979257	8.818141	2.3496767	9.388874
## sample_295	1.9215887	7.749829	5.526056	9.480534	2.9947788	9.597660
## sample_298	0.4131608	6.479030	5.149162	8.458714	2.7305530	10.515030
## sample_404	0.5285713	7.280075	6.325593	9.949904	3.1903462	9.370823
## sample_464	1.4307670	6.982937	6.432275	7.801928	3.8075301	10.183226
## sample_468	3.7839071	7.366366	5.203608	10.072695	4.6799755	9.956646
## sample_531	0.5970305	7.729288	7.220601	8.128598	2.8382459	9.487131
## sample_57	1.2159892	6.992587	6.442829	8.614482	3.6141444	8.829022
## sample_578	0.0000000	7.910817	5.453834	6.706876	5.6280503	7.742720
## sample_579	0.8077671	6.666231	5.868166	9.696782	3.3888370	8.287458
## sample_603	0.0000000	7.570205	5.416032	10.140267	3.3486998	9.495551
## sample_627	0.4569113	6.387369	4.739195	10.260661	2.2405586	10.222481
## sample_638	0.0000000	6.706240	4.715267	10.101306	2.2162996	9.770530
## sample_673	0.5621814	7.916590	5.501824	8.565460	2.7479674	8.772974
## sample_68	2.0027025	7.404469	5.365469	9.452497	3.9691689	9.830355
## sample_69	1.4896971	7.545389	7.379040	7.425308	3.7231326	8.488029
## sample_691	1.0858325	6.791020	5.932196	8.623841	3.9886938	9.869757
## sample_701	0.0000000	6.585898	5.573138	8.310917	1.9141052	9.909556
## sample_735	1.8778620	7.000856	6.918172	6.630724	3.1009776	9.185938
## sample_74	1.8716860	6.480554	4.827494	9.026465	2.3937731	9.282030
## sample_740	1.0247452	6.078044	6.055779	7.205197	0.7567671	9.515891
## sample_75	1.0875986	6.564849	5.713608	8.379062	3.6796828	10.282776
## sample_762	2.3531539	8.291470	6.238082	7.431021	5.5138034	9.348348
## sample_768	1.3826673	7.994229	4.668652	9.863750	1.8366917	9.940807
## sample_793	0.4480552	6.711316	4.419087	9.476171	4.2047589	10.182345
## sample_800	0.7483754	6.914852	6.856787	9.936737	3.0838941	9.493899
## sample_82	0.0000000	8.213702	5.004052	9.651335	2.3632552	9.792894
## sample_95	0.0000000	7.217890	6.328111	10.104114	2.9731337	9.535550
##	gene_321	gene_11811	gene_13939	gene_8612	gene_8311	gene_7538
## sample_119	0.0000000	11.205836	8.796640	0.0000000	9.716675	10.336640
## sample_275	5.3064773	10.179872	8.290470	7.0241867	11.571871	8.140175
## sample_29	6.4330947	8.636991	11.069107	4.2451990	12.260279	10.805615
## sample_295	2.6116681	10.578863	5.356408	3.4275391	11.466041	8.503949

## sample_298	0.9962441	8.973551	5.314562	2.8008895	10.096610	8.730555
## sample_404	2.5526968	11.067824	9.017524	5.4568193	11.702926	8.631450
## sample_464	1.4307670	10.583675	7.880171	2.9604190	10.371679	7.990104
## sample_468	0.4978423	10.866089	1.405067	3.2081270	11.787641	8.694567
## sample_531	1.8331330	10.216467	5.855521	2.6149925	10.266564	8.896980
## sample_57	2.1068173	10.438927	2.928825	3.9590932	10.450572	8.900659
## sample_578	0.0000000	10.373202	1.289067	1.5483873	8.920764	7.672991
## sample_579	3.2675957	9.930529	9.068335	7.3396363	12.026949	8.617449
## sample_603	6.3182915	9.674641	7.958280	0.0000000	12.925124	9.058411
## sample_627	0.8033924	9.962386	5.478703	0.4569113	12.296360	8.747649
## sample_638	2.5081751	10.816720	9.214713	0.0000000	13.441336	9.665695
## sample_673	1.7580473	10.145461	10.033822	0.9656181	14.646705	7.508270
## sample_68	2.0027025	11.093061	2.730988	2.6468999	11.523675	8.742963
## sample_69	3.2614556	9.871605	6.248785	0.9291850	10.069101	8.405805
## sample_691	2.5288963	10.182866	3.235712	2.2167963	9.976092	8.607323
## sample_701	2.3654686	9.722849	6.935059	0.9432837	11.724522	8.183893
## sample_735	3.7047392	9.625278	3.548572	1.8778620	11.332143	7.428092
## sample_74	1.2202061	10.534993	3.166507	0.8463128	11.233566	11.357481
## sample_740	2.6259484	10.303632	3.745764	1.0247452	10.352010	9.219711
## sample_75	2.2193389	9.697899	1.700617	0.3576083	10.673866	9.571859
## sample_762	0.0000000	10.381532	1.835277	0.5980793	9.324273	7.473340
## sample_768	0.7163325	11.070020	6.363476	0.9741622	12.708336	8.925201
## sample_793	1.4961044	8.942210	10.341852	0.0000000	12.314544	8.795033
## sample_800	2.8317151	10.008961	8.673362	4.2232372	9.760226	9.170431
## sample_82	0.0000000	9.496428	10.261507	0.0000000	9.015727	9.358607
## sample_95	7.3256113	10.723064	8.410977	5.4802328	11.858168	8.593118
##	gene_17525	gene_20215	gene_1573	gene_9147	gene_5622	gene_2826
## sample_119	8.241435	8.228626	8.193910	9.370515	10.289120	5.022399
## sample_275	8.950885	9.207742	8.166826	10.238978	9.717726	5.879238
## sample_29	9.886424	9.864155	7.640606	10.950512	8.170476	5.465217
## sample_295	8.809102	8.297824	8.267686	10.350409	11.178198	6.045777
## sample_298	8.819010	8.312892	8.928119	10.104022	10.232841	5.361863
## sample_404	9.140252	9.205500	7.644686	10.481557	9.141384	4.637715
## sample_464	9.217039	8.445652	8.596540	10.202050	11.262459	3.864632
## sample_468	8.678695	8.399445	7.536123	11.275426	10.945107	5.596592
## sample_531	8.578909	6.486776	9.084391	9.137883	12.639940	2.027048
## sample_57	8.435858	6.363919	8.492598	9.781547	12.003346	2.493237
## sample_578	7.812325	8.032476	8.422721	9.899784	9.102249	6.085153
## sample_579	8.870879	10.395898	8.120487	10.227760	9.736669	6.042548
## sample_603	8.615321	5.541732	7.829735	10.407523	8.620509	7.559431
## sample_627	8.112221	10.705762	7.554850	9.780604	9.434905	5.982323
## sample_638	9.227934	8.486505	8.372064	10.041591	10.216661	6.371515
## sample_673	8.438459	8.408427	8.318701	9.734681	10.523905	6.770300
## sample_68	8.016223	8.419341	7.794760	9.933791	10.980532	5.650997
## sample_69	9.487836	8.755836	8.933083	10.015917	10.938867	3.391452
## sample_691	8.704671	9.871919	8.390384	9.454367	9.096583	5.438958
## sample_701	8.990427	8.319279	8.118318	10.025873	9.106089	8.475198
## sample_735	8.640867	7.577013	7.857770	9.396252	9.567952	6.501791

## sample_74	10.808176	11.252251	8.237439	12.158780	10.692842	5.571149
## sample_740	7.797000	6.025549	8.141617	10.256409	11.036277	6.238198
## sample_75	7.864601	8.055169	8.459657	10.889017	8.853631	6.086359
## sample_762	7.010176	7.193476	8.890599	9.438688	9.627598	8.991596
## sample_768	8.493567	9.761122	7.763969	10.575388	10.567766	7.049740
## sample_793	9.366630	6.826892	8.411100	9.425881	9.171792	7.160124
## sample_800	8.815124	9.460876	8.596890	10.688058	9.739720	6.514559
## sample_82	9.093050	9.198602	8.619310	10.443762	8.645680	5.298083
## sample_95	9.297506	9.237571	8.098106	10.199685	8.124447	5.514782
##	gene_10905	gene_8486	gene_4442	gene_3037	gene_19266	gene_16019
## sample_119	3.168498	8.135648	4.188741	8.790844	9.195835	1.9897930
## sample_275	3.928427	4.889386	4.572508	10.596236	9.412155	1.1169635
## sample_29	2.449614	7.529337	3.924309	11.483770	8.247301	0.0000000
## sample_295	3.236600	8.474091	12.864070	5.884898	9.067106	2.6116681
## sample_298	2.830783	8.623036	4.135830	9.801954	8.341199	2.4089020
## sample_404	4.326386	5.826604	6.447632	10.850640	9.622359	0.5285713
## sample_464	1.439783	7.335774	4.561424	6.819259	9.276110	3.5534219
## sample_468	4.719550	7.254018	14.546219	4.938723	8.995267	1.1610492
## sample_531	1.853756	8.639507	3.031519	5.338852	9.195781	2.3507801
## sample_57	1.881900	9.256329	3.762157	8.455023	8.579983	2.1068173
## sample_578	3.131310	9.446719	5.250010	3.649569	9.282514	2.9518116
## sample_579	3.031130	6.066978	3.966624	10.385420	8.701774	1.3224474
## sample_603	4.163112	8.059204	3.579434	6.779181	8.802785	7.2059691
## sample_627	4.594961	8.395290	5.140664	7.046011	9.185426	2.1220763
## sample_638	4.596327	9.124672	4.414075	9.126689	9.377250	0.3339961
## sample_673	3.700706	6.603594	5.161186	5.455580	8.783935	3.5213523
## sample_68	4.018857	8.191568	13.279535	6.307347	9.239322	2.0027025
## sample_69	2.466053	9.245046	4.967597	11.285680	9.853389	1.4896971
## sample_691	3.201791	7.108138	5.332447	9.287509	8.356192	1.2650167
## sample_701	4.949465	8.202090	4.616758	8.348166	8.011434	5.6678496
## sample_735	5.089528	3.434215	5.780090	8.319035	8.418270	6.2961508
## sample_74	2.355298	8.685232	5.595158	10.183263	10.048814	3.3280609
## sample_740	4.896189	9.119512	4.015578	10.487448	8.860569	5.9470793
## sample_75	2.464041	7.720039	3.711010	9.263274	8.349507	1.5700257
## sample_762	5.717380	8.614371	5.806412	5.187598	9.479746	4.3589377
## sample_768	4.592834	9.558294	5.105808	5.621284	9.536599	4.8564778
## sample_793	6.323019	6.155506	3.575917	6.170173	9.146059	1.2967217
## sample_800	4.343109	8.159271	4.372764	12.534595	8.788751	2.5259185
## sample_82	3.857284	4.749979	4.334297	1.353041	6.969933	4.3342965
## sample_95	4.560403	2.680279	4.477580	11.720013	8.799881	0.7834986
##	gene_11970	gene_1649	gene_10274	gene_8908	gene_17648	gene_2968
## sample_119	9.867028	11.49158	12.74988	11.37066	11.57362	6.375401
## sample_275	8.477143	11.64320	12.37613	11.19803	11.91728	6.944285
## sample_29	9.178209	12.29263	13.31880	12.43885	11.80471	2.738314
## sample_295	8.588242	11.92277	12.34250	10.59456	11.06164	6.488081
## sample_298	9.218871	11.74594	12.08894	11.15043	11.67354	10.214841
## sample_404	8.645867	11.61855	11.55059	11.10278	11.61285	6.917814
## sample_464	8.815082	11.37083	11.90153	11.78342	12.50150	6.229840

## sample_468	8.691695	12.40511	12.28779	10.96725	10.83127	6.842124
## sample_531	9.459036	12.22192	12.41426	12.46707	12.07025	6.590181
## sample_57	8.951661	10.81450	12.63117	11.72218	11.52968	6.809016
## sample_578	7.857925	12.15214	11.91726	11.34544	12.17978	6.603573
## sample_579	7.724562	11.99229	11.67790	11.11326	12.26926	4.769052
## sample_603	9.395401	12.25636	12.42900	11.21812	11.40906	5.474530
## sample_627	8.526441	11.23554	12.11933	10.38204	11.82053	5.638297
## sample_638	9.545615	11.04784	11.89500	11.84306	11.87483	6.484952
## sample_673	7.266299	11.96666	11.33484	11.09980	12.54864	6.101383
## sample_68	8.944651	13.20165	12.00941	11.07407	10.84910	7.527156
## sample_69	8.764858	10.96683	11.83893	11.09610	11.79977	7.616902
## sample_691	8.449491	11.58655	10.62827	10.61362	11.75779	10.518279
## sample_701	8.167774	11.17609	11.51574	10.96916	11.81118	5.600838
## sample_735	7.626687	12.76817	10.92592	10.69978	12.38695	10.009815
## sample_74	9.387192	12.59884	13.69309	11.95459	11.51747	6.677085
## sample_740	9.232174	12.79435	11.87808	11.36906	11.14045	7.177968
## sample_75	9.211504	10.38707	11.53162	11.50016	12.43961	7.224224
## sample_762	8.387543	11.75574	11.02120	10.33868	11.80649	7.871653
## sample_768	9.572982	10.89401	12.03761	10.63843	10.95042	6.558154
## sample_793	9.213099	11.05460	12.26833	10.37149	12.22234	4.559388
## sample_800	8.680553	11.75103	12.09434	11.24248	12.12563	8.070840
## sample_82	8.157210	12.19119	12.25345	10.66202	11.90771	5.708698
## sample_95	8.464987	11.30342	11.72525	11.17384	11.92204	5.537368
##	gene_2679	gene_1904	gene_7326	gene_19027	gene_6259	gene_11690
## sample_119	8.538682	6.184707	2.2819356	9.265062	8.606246	9.781055
## sample_275	8.708394	5.646569	1.7388975	9.773703	8.959382	8.761215
## sample_29	8.679751	6.544448	2.8048796	10.921268	5.554994	9.488155
## sample_295	7.426734	6.480592	1.2595437	9.914880	9.589850	8.429649
## sample_298	8.716977	5.636454	2.4089020	9.394758	7.224513	9.013432
## sample_404	8.043476	5.132198	2.0346735	9.494564	8.906172	8.554539
## sample_464	8.662430	5.420600	2.6058736	9.508845	7.315430	9.332276
## sample_468	6.789364	7.089159	1.1610492	8.374583	8.720166	8.249265
## sample_531	9.186924	5.842692	1.8331330	9.692352	6.613028	9.583483
## sample_57	8.885242	5.787688	2.1068173	9.608977	7.148690	8.918517
## sample_578	8.396177	6.453218	1.7681631	9.403291	7.919441	9.496520
## sample_579	7.623867	5.701532	2.2486259	9.523727	8.680820	8.710431
## sample_603	8.089466	5.584857	2.1806571	9.380182	8.535139	10.254060
## sample_627	7.667537	7.236263	2.2405586	8.755131	9.780604	9.991734
## sample_638	8.044973	7.206155	2.1330703	9.544105	11.896877	10.576399
## sample_673	8.932017	5.439829	1.2806000	9.735487	9.381783	9.535225
## sample_68	8.560016	5.775035	1.7029235	9.116695	8.291889	8.768905
## sample_69	7.956080	7.015593	2.3416158	10.060574	8.360882	9.470344
## sample_691	10.421171	5.312208	3.9629561	9.426395	7.307902	8.546354
## sample_701	7.408517	6.983210	0.9432837	9.645159	8.533855	9.554556
## sample_735	9.288252	5.423336	0.9197596	10.300936	9.040821	8.182315
## sample_74	7.560455	5.815552	5.2767200	10.335636	6.886148	9.520238
## sample_740	9.209660	6.790459	2.0369253	9.585541	9.421827	9.864774
## sample_75	9.506409	7.331419	2.3838840	9.760529	6.976066	8.960210

## sample_762	8.066692	6.939473	1.8352773	10.280887	10.721902	8.956614
## sample_768	7.975922	7.964832	2.0754984	9.053820	9.643108	9.172100
## sample_793	7.789997	5.493907	3.1122831	9.325458	8.604372	9.632077
## sample_800	8.390233	4.981665	2.1373396	9.579723	8.345698	8.995487
## sample_82	7.909251	4.307363	1.6195069	9.305252	7.443499	9.115920
## sample_95	7.924801	4.733696	2.5967205	9.181324	7.964918	8.422111
##	gene_14654	gene_16499	gene_6387	gene_17050	gene_9463	gene_7165
## sample_119	10.760005	10.630768	0.6730136	5.601349	2.1909625	7.793129
## sample_275	10.369630	10.106053	0.6640277	6.706641	5.8207687	8.379248
## sample_29	10.018103	11.603450	0.0000000	8.110870	4.4252780	8.254764
## sample_295	11.163367	9.971400	0.0000000	7.175964	1.5155106	10.658595
## sample_298	10.565778	9.831133	0.9962441	7.033247	3.6934647	9.009269
## sample_404	10.000014	10.268823	2.6575714	5.150288	5.3605083	7.475660
## sample_464	10.747757	10.732507	1.4307670	6.251410	3.3283626	7.753625
## sample_468	10.249149	10.124742	0.4978423	7.093518	2.4680098	8.811651
## sample_531	10.328843	11.238082	3.1192566	6.566969	2.0270479	7.896908
## sample_57	10.408362	10.416069	3.6141444	5.903814	1.5774893	7.759182
## sample_578	9.769825	10.517069	0.0000000	7.084755	0.0000000	7.930867
## sample_579	10.385010	10.294368	0.4596414	7.541995	3.7012718	8.185153
## sample_603	10.848153	10.551372	0.0000000	8.429683	0.7712095	6.928939
## sample_627	10.933078	9.725797	0.0000000	6.864174	1.0824982	8.361369
## sample_638	10.755430	10.035514	0.0000000	7.053263	2.3696061	8.188440
## sample_673	10.330872	10.490370	0.5621814	8.034018	4.1043450	9.490897
## sample_68	10.707730	10.392167	1.5259185	7.220330	0.8089204	9.495961
## sample_69	9.867421	9.897138	0.0000000	5.975788	3.1185923	9.082519
## sample_691	10.954923	10.459104	1.5678375	7.137196	3.6119396	9.221215
## sample_701	10.612721	10.834187	0.5474498	9.234319	0.5474498	8.423784
## sample_735	10.109256	10.332708	0.5319678	6.874981	6.8019217	9.613828
## sample_74	10.066816	10.415066	0.0000000	5.672425	1.2202061	8.379010
## sample_740	10.569296	10.252346	0.0000000	4.690959	2.7043629	7.477815
## sample_75	10.253150	10.719825	1.2669364	7.014244	1.7006173	7.973565
## sample_762	10.716631	10.352915	0.0000000	11.020126	2.4913657	9.166859
## sample_768	10.893286	10.648286	1.8366917	7.502689	2.0754984	9.018484
## sample_793	10.986582	10.239956	0.0000000	8.552143	2.2146533	7.396742
## sample_800	9.892013	10.502394	0.7483754	7.365080	4.8853840	8.088963
## sample_82	11.153109	10.185867	2.0387873	7.590370	2.6280036	7.299080
## sample_95	10.247548	10.107897	0.4443490	6.577293	4.4775802	6.761219
##	gene_18848	gene_8748	gene_1700	gene_4850	gene_5465	gene_9425
## sample_119	5.111908	9.144487	10.953374	9.428814	10.492374	12.570636
## sample_275	6.665705	9.214853	11.147269	8.842740	10.647072	9.357675
## sample_29	4.729439	10.511001	9.501051	8.802953	10.295298	13.219067
## sample_295	4.484003	9.113841	10.830816	9.710794	9.661972	9.410949
## sample_298	5.923306	7.552185	10.861529	8.813598	9.567994	10.904973
## sample_404	6.830762	9.259619	10.951758	8.845336	10.572814	9.415171
## sample_464	4.842944	8.385957	10.096504	8.139296	10.108368	10.326182
## sample_468	4.093467	8.767317	10.963380	9.926930	9.661602	10.392189
## sample_531	4.762551	8.218244	10.114757	8.618698	9.596257	10.659987
## sample_57	5.623176	8.542084	10.437554	8.641019	10.129579	11.063368

## sample_578	6.164728	8.937853	10.907003	7.930643	9.951186	9.547658
## sample_579	6.248818	9.063285	10.463024	8.574102	10.394291	10.053885
## sample_603	7.604664	8.567747	10.506516	8.997778	10.564216	11.922807
## sample_627	5.820542	8.539500	10.850484	9.186174	10.577618	12.407783
## sample_638	6.187815	8.486505	10.586296	9.164990	10.134786	13.048398
## sample_673	6.315026	8.348529	10.177731	8.763959	9.887644	11.169925
## sample_68	4.334347	8.172897	10.790935	9.861455	9.733653	9.492124
## sample_69	5.834355	7.145647	11.028079	8.293881	10.454659	9.169755
## sample_691	5.249726	8.606283	10.907078	8.091070	10.133977	10.144493
## sample_701	6.786178	8.699951	11.223085	8.134996	10.328372	10.923758
## sample_735	7.841036	8.012753	11.562915	7.363154	10.774400	8.520131
## sample_74	6.476250	9.540355	9.859007	9.750377	11.022728	11.020730
## sample_740	6.898281	7.355554	10.867047	8.306185	11.318532	10.903844
## sample_75	6.930217	7.608018	10.499826	8.249630	10.162580	9.067848
## sample_762	8.993052	8.831637	10.908505	8.131842	10.822730	9.710877
## sample_768	6.894660	8.973057	11.266476	9.217807	10.372016	11.711443
## sample_793	6.162859	8.748153	11.223410	8.833548	10.484360	12.163873
## sample_800	6.865226	8.318303	10.423883	8.534770	10.297868	10.300089
## sample_82	6.909029	9.126872	11.110810	9.006772	11.053926	9.914751
## sample_95	6.058291	9.070526	10.809390	9.091147	10.742023	9.524119
##	gene_8569	gene_15233	gene_4404	gene_945	gene_15380	gene_19890
## sample_119	2.524766	11.447280	8.435962	6.714410	0.9195308	4.6817697
## sample_275	4.256867	12.075095	10.033891	8.100788	1.4612142	0.6640277
## sample_29	8.074206	10.164341	10.920799	9.335683	3.5109999	0.0000000
## sample_295	5.838783	12.610360	9.391495	6.322027	5.7292999	0.9482270
## sample_298	2.730553	11.763939	9.715832	7.296274	1.9943263	6.3617719
## sample_404	3.424183	11.528933	9.966589	8.291811	0.0000000	0.9145645
## sample_464	4.526470	11.937322	9.563757	7.651863	2.4652693	4.4538994
## sample_468	8.983370	12.450002	9.202195	5.219653	6.9709227	0.0000000
## sample_531	4.078208	11.368272	9.863274	7.453616	2.3507801	3.2801546
## sample_57	4.185756	10.589595	9.573094	7.605961	3.0489859	2.4932374
## sample_578	4.035571	11.473031	8.332990	6.371723	2.5391092	5.0961353
## sample_579	5.224878	11.567196	10.685099	7.027419	0.0000000	4.2487853
## sample_603	2.288920	11.387527	8.231927	7.856376	0.0000000	0.4365883
## sample_627	4.613084	10.875020	8.656014	7.221935	1.0824982	0.0000000
## sample_638	2.508175	10.942170	9.128706	8.017683	1.0299829	8.1780971
## sample_673	4.955029	11.532726	7.861496	8.299190	1.2806000	0.0000000
## sample_68	8.107463	12.169145	9.266876	6.380778	5.8427526	0.0000000
## sample_69	3.566803	10.425552	9.289498	9.059480	2.5784316	7.1083154
## sample_691	4.201720	12.071503	8.913054	4.480931	2.8428984	4.4254862
## sample_701	2.603146	11.669173	9.160620	6.730844	2.7087611	0.0000000
## sample_735	4.989402	12.540065	8.882188	7.633177	1.2250281	1.8778620
## sample_74	1.762817	9.683947	9.094022	7.977646	1.2202061	3.2893328
## sample_740	2.361291	12.258781	8.883926	7.595668	1.4458856	0.0000000
## sample_75	3.679683	11.794322	8.328199	5.867501	0.8827604	3.5471043
## sample_762	3.034128	12.824583	8.030303	7.634927	5.2077914	6.2478668
## sample_768	3.496680	11.151188	8.633279	7.906097	0.4021764	6.6065533
## sample_793	2.519592	11.726576	7.618804	6.812947	0.4480552	0.0000000

## sample_800	5.453985	11.652092	10.186956	7.762953	1.2386646	3.5702808
## sample_82	1.353041	11.665260	7.530984	7.812806	0.0000000	0.0000000
## sample_95	3.472267	11.595071	9.444392	7.425837	1.2884176	1.6617040
##	gene_5228	gene_9747	gene_4247	gene_4860	gene_6195	gene_2241
## sample_119	4.5957304	2.448293	5.3301030	9.102989	9.109684	10.325665
## sample_275	2.8931680	3.371740	13.7208748	8.938909	6.679255	8.957685
## sample_29	3.2493391	2.436215	2.5951456	6.401173	9.772541	8.726654
## sample_295	1.2595437	2.611668	0.0000000	9.043159	5.634631	9.325101
## sample_298	0.7339617	4.189255	0.0000000	9.525137	8.873401	10.252985
## sample_404	1.4698339	2.755358	5.2873652	9.460538	4.494403	9.359205
## sample_464	0.0000000	2.733983	0.0000000	9.576632	9.082029	9.295707
## sample_468	0.0000000	3.498481	0.4978423	9.543884	7.101209	9.458583
## sample_531	1.0180644	2.730944	0.0000000	8.833551	9.890272	9.574113
## sample_57	3.8963881	3.614144	0.0000000	9.023106	10.097058	9.671166
## sample_578	0.5667665	3.806664	13.5742742	10.903023	7.838372	9.081896
## sample_579	0.0000000	1.087938	10.1659622	9.668456	5.147726	8.827448
## sample_603	0.0000000	0.000000	6.4524043	10.241435	6.160620	9.976364
## sample_627	4.2672586	2.122076	7.5142643	10.222034	7.871209	9.718797
## sample_638	4.5645977	2.044744	2.2949883	8.563028	9.435743	10.278484
## sample_673	3.3295542	3.396365	4.3261778	8.944627	8.026258	9.290328
## sample_68	0.0000000	2.250780	0.4603756	8.339547	6.664796	9.677136
## sample_69	2.3416158	3.391452	5.3815326	7.888658	8.728679	9.291581
## sample_691	0.3569326	3.676696	0.0000000	9.740821	8.391806	9.890808
## sample_701	0.0000000	2.230111	1.2536263	10.333536	5.027265	8.736311
## sample_735	4.2691155	6.383497	6.9958237	9.477904	7.824189	7.835413
## sample_74	2.8842072	3.032260	7.0801140	7.310276	8.850312	9.936230
## sample_740	1.0247452	2.454992	3.3170580	9.427617	7.579112	8.455451
## sample_75	0.0000000	3.913100	0.0000000	9.650212	8.116349	10.703860
## sample_762	3.2828104	4.591883	6.5303580	8.792061	7.824895	8.831637
## sample_768	2.6191547	2.954438	2.3727850	9.818237	6.181375	9.322210
## sample_793	3.1717993	3.171799	7.0375030	10.537286	7.527328	9.024212
## sample_800	0.0000000	4.851809	6.8902521	8.594354	8.067660	9.153306
## sample_82	0.0000000	2.951625	2.9516251	10.100610	5.257426	9.771671
## sample_95	1.2884176	2.596721	12.5959288	8.915377	5.374208	8.710796
##	gene_15063	gene_11894	gene_19999	gene_15338	gene_19220	gene_15310
## sample_119	10.457124	5.196182	9.851334	3.1254197	12.176186	14.845122
## sample_275	10.550862	6.330661	8.044416	1.4612142	11.969725	9.905265
## sample_29	10.716554	4.763316	9.550397	2.0536674	11.070383	9.161633
## sample_295	9.876879	1.921589	8.766628	0.9482270	11.558081	9.581997
## sample_298	10.382981	5.338407	8.133728	0.7339617	11.754136	10.454608
## sample_404	10.525961	5.769238	8.766572	1.2187192	11.270050	9.786038
## sample_464	10.265697	9.274793	7.944484	1.4307670	12.637748	9.211548
## sample_468	10.417694	1.161049	9.489511	0.8671848	11.021452	9.361606
## sample_531	9.396763	10.461684	8.096536	2.0270479	12.057985	9.537760
## sample_57	9.098158	11.178777	8.645792	0.7324863	11.970196	9.184340
## sample_578	9.697251	7.460808	6.484568	1.2890672	11.982191	9.114828
## sample_579	10.839904	6.836947	8.555490	2.2486259	11.773011	9.597557
## sample_603	11.246307	3.133218	8.907347	2.2889196	11.527247	9.750580

## sample_627	11.188873	5.560706	8.339961	3.0792245	11.248853	10.531879
## sample_638	10.086866	5.305759	9.270545	2.8056020	10.626330	10.756731
## sample_673	10.130815	3.934913	7.002466	0.9656181	11.654363	9.764228
## sample_68	9.662489	2.557582	8.614625	1.5259185	11.150255	9.466364
## sample_69	8.989346	4.072346	7.555824	1.7050490	11.215278	9.914137
## sample_691	10.239897	4.955634	7.310449	0.3569326	11.444420	10.055106
## sample_701	10.838558	4.561442	7.137760	2.6031456	11.968804	9.319103
## sample_735	9.531040	4.541893	2.325645	0.5319678	12.295858	8.576609
## sample_74	9.796955	3.166507	10.465056	1.9728764	9.733917	9.963772
## sample_740	9.879232	4.015578	7.862600	2.1533514	12.102156	10.436295
## sample_75	10.128175	3.965997	8.374179	0.3576083	11.225695	9.878180
## sample_762	9.139526	4.358938	7.530570	1.0196308	11.973719	9.588666
## sample_768	10.273166	3.980811	9.543453	2.3727850	10.849468	10.825897
## sample_793	11.956924	5.853245	8.061803	2.9174318	11.519189	10.230369
## sample_800	10.204522	5.622740	8.532122	2.8317151	12.290169	9.825463
## sample_82	11.706556	7.265465	8.638193	2.0387873	11.275770	9.086189
## sample_95	10.882597	5.735920	8.507069	1.8174595	11.803029	9.693447
##	gene_2365	gene_14605	gene_9140	gene_7667	gene_3694	gene_4477
## sample_119	1.1300742	10.448870	6.295782	11.10120	6.522560	4.2350071
## sample_275	1.1169635	9.833148	7.624803	10.92900	7.433202	6.5976861
## sample_29	0.7051375	10.627671	7.642882	10.41689	2.668505	4.5662976
## sample_295	0.5506052	11.072267	5.026897	11.35801	4.780840	2.3736760
## sample_298	1.2180992	10.923186	6.756276	11.08188	6.495030	8.4409273
## sample_404	0.5285713	9.563677	7.731346	11.17980	7.288266	7.6286497
## sample_464	0.6464392	11.157126	6.767800	10.37079	3.807530	7.4889006
## sample_468	2.1031955	9.933730	4.058213	11.04622	3.985018	3.6948913
## sample_531	1.6090457	10.458007	7.406443	9.97071	3.676978	11.5692908
## sample_57	2.1068173	10.765460	6.363919	10.20264	7.615725	9.7629932
## sample_578	1.5483873	11.504680	6.922531	10.67235	10.372158	11.5342396
## sample_579	1.3224474	9.650382	6.708256	11.18682	5.195635	3.2102642
## sample_603	1.0427144	10.550352	6.783810	10.97484	10.602430	3.9530417
## sample_627	0.8033924	10.717942	6.007606	11.35107	6.252870	5.0298592
## sample_638	1.4974846	10.468980	6.357833	10.76904	10.384341	1.4974846
## sample_673	0.9656181	10.590952	7.402543	10.75842	6.991703	3.5213523
## sample_68	1.7029235	9.877821	5.122449	10.68342	5.703981	5.2561804
## sample_69	1.2364620	10.242245	8.291802	10.24201	8.723797	5.4724748
## sample_691	1.0858325	11.005435	6.513701	11.18455	7.340740	9.1799240
## sample_701	0.5474498	10.569561	4.795201	11.26108	10.035418	1.2536263
## sample_735	0.9197596	10.241066	7.736328	10.86848	9.799248	5.8817772
## sample_74	1.7628165	10.180295	5.579195	10.98843	3.166507	4.5447574
## sample_740	1.0247452	10.796916	7.941629	10.61858	3.852938	0.7567671
## sample_75	1.2669364	10.506813	7.237736	10.48365	5.728993	5.2528849
## sample_762	0.0000000	10.900384	6.464750	10.34650	5.992510	2.0293115
## sample_768	1.1928254	10.688189	6.391371	11.31735	5.879424	2.6191547
## sample_793	0.4480552	10.440423	5.710046	11.27167	9.243755	1.6712934
## sample_800	3.9333932	9.799280	7.133204	10.41074	6.567231	6.6671257
## sample_82	0.6023619	11.132949	6.336241	11.73535	9.091681	0.0000000
## sample_95	0.7834986	9.472496	6.785001	11.03648	3.518397	2.3124328

##	gene_15711	gene_7728	gene_9745	gene_17280	gene_4289	gene_412
## sample_119	12.727365	11.439431	7.043760	4.6983796	5.3824321	7.041747
## sample_275	10.846619	10.044312	5.559853	5.0507150	9.2895140	8.198745
## sample_29	8.015694	11.188471	6.373198	3.6259367	5.0230735	7.565788
## sample_295	8.780418	10.220439	3.818196	1.7327902	8.1802424	5.481732
## sample_298	9.183928	9.223678	4.745119	1.4103411	4.1083572	5.915404
## sample_404	11.121242	10.149125	7.825735	6.5119977	10.6700543	8.766572
## sample_464	10.830705	10.168421	4.755069	0.0000000	6.7375381	7.838019
## sample_468	8.143934	10.446101	4.638091	0.8671848	7.7361854	4.879368
## sample_531	10.870026	8.997349	3.280155	1.3436352	7.8237112	7.890684
## sample_57	11.856667	9.288732	5.981387	2.1068173	5.9353442	7.705328
## sample_578	11.185916	9.346833	2.859174	3.5931506	7.4127307	7.058955
## sample_579	10.315342	10.497133	4.666200	5.4145077	10.8416823	6.933112
## sample_603	11.464770	10.766521	8.646289	6.8427400	0.4365883	7.509949
## sample_627	12.026271	10.846109	6.556806	4.2390772	3.6918133	6.341707
## sample_638	11.394034	10.572065	8.430963	4.9028309	0.8330925	7.468412
## sample_673	10.373463	9.358932	5.618353	4.6610769	2.4028040	7.188322
## sample_68	8.887391	9.995131	5.550544	2.9573389	8.4413507	5.916441
## sample_69	10.865950	9.216826	4.967597	4.0723459	2.3416158	8.042524
## sample_691	11.032170	9.992485	5.958481	2.5288963	6.4502875	6.540047
## sample_701	11.710686	11.709889	6.524307	2.0807600	4.6167576	6.075774
## sample_735	11.545158	10.867202	2.251931	5.0872592	5.8926436	8.298750
## sample_74	11.548027	8.207898	5.829116	0.8463128	0.0000000	7.369484
## sample_740	10.500553	10.764399	5.529081	5.9789804	3.2662468	7.678149
## sample_75	10.306996	9.507525	5.041808	3.2809712	6.1732754	6.865572
## sample_762	12.638026	10.411225	6.600768	5.3047191	1.0196308	7.951664
## sample_768	10.270914	11.150775	5.275029	4.6502993	3.1236331	7.836783
## sample_793	11.570183	13.146860	4.490942	1.6712934	1.2967217	6.680915
## sample_800	11.220862	9.887879	4.056401	4.5507160	10.6939922	7.807419
## sample_82	12.243308	10.681722	6.929661	7.9123565	5.4932662	7.848923
## sample_95	10.218575	10.022923	6.440570	6.3790109	9.6066810	8.226990
##	gene_10442	gene_447	gene_5078	gene_2420	gene_14173	gene_19911
## sample_119	6.685548	2.795663	8.212817	8.379149	8.598250	7.061139
## sample_275	8.066601	6.606365	8.416738	7.684552	8.450877	6.937415
## sample_29	8.868622	2.595146	6.705591	8.276259	5.243121	6.300935
## sample_295	7.370836	8.844536	7.702332	7.089996	7.673062	7.843299
## sample_298	7.662547	5.702248	7.994353	8.620095	6.962133	5.418888
## sample_404	7.535376	9.083727	8.376846	7.946865	8.425929	7.242622
## sample_464	7.546748	9.309911	8.292809	7.838019	8.139296	6.575972
## sample_468	7.769263	10.715121	6.917826	6.247878	7.144832	8.573390
## sample_531	7.686844	10.735522	7.547527	8.290079	8.549600	6.715097
## sample_57	7.591193	11.978496	7.591193	7.755876	8.408032	6.665990
## sample_578	8.181003	10.424628	8.718070	7.337889	8.430791	5.286042
## sample_579	7.541995	8.124375	7.584917	7.225371	8.603600	6.897228
## sample_603	9.190793	9.502597	7.710676	7.607537	8.336560	7.617695
## sample_627	8.492514	7.574722	7.352653	7.610921	7.066197	7.916984
## sample_638	9.098858	4.285077	7.776762	7.307529	7.231384	7.719074
## sample_673	7.556537	6.198182	8.039254	6.686024	8.975000	8.832994

## sample_68	7.727253	12.464962	7.775083	7.071301	7.315340	7.792315
## sample_69	9.059480	6.535621	7.747079	7.084490	8.921561	6.663373
## sample_691	7.501320	9.569687	8.300774	8.032349	7.556138	6.959898
## sample_701	8.079469	7.779627	7.794695	7.065044	8.517453	6.761844
## sample_735	7.798329	8.127613	9.855185	7.240906	9.966745	8.650503
## sample_74	9.093320	2.318866	6.547715	8.323798	5.579195	6.053665
## sample_740	8.111350	2.849479	8.432592	7.723579	9.532742	5.662761
## sample_75	8.117866	8.998726	7.672736	8.027845	8.205383	7.426491
## sample_762	7.682594	6.049563	8.957717	8.091356	8.178346	8.196145
## sample_768	8.126250	4.593796	7.398564	7.757550	7.450609	6.533331
## sample_793	7.968402	6.227389	7.637393	7.712458	7.339503	7.943007
## sample_800	7.615696	8.422040	7.892440	7.155527	7.988264	7.208088
## sample_82	8.605942	8.765611	8.221234	8.297228	7.598112	7.111803
## sample_95	7.338522	4.694044	8.115096	7.577912	9.128394	7.335301
##	gene_17709	gene_5561	gene_2850	gene_9379	gene_17165	gene_11018
## sample_119	10.916365	4.524120	8.589651	8.659172	5.462399	7.971118
## sample_275	10.337410	7.885483	7.620528	8.474773	4.017548	8.852167
## sample_29	10.463810	7.705563	9.211570	6.236064	5.290362	6.897059
## sample_295	10.321883	2.535082	6.956010	8.644613	4.292009	7.734174
## sample_298	10.078591	1.701105	8.613296	8.774916	8.449165	8.698673
## sample_404	10.179723	6.738903	7.775722	8.214407	2.439570	9.000175
## sample_464	9.450593	3.876340	8.804179	9.222510	6.775261	8.782117
## sample_468	10.570159	2.885008	6.989763	8.625497	2.924708	7.744524
## sample_531	10.179324	4.358284	8.760401	8.125951	5.100999	8.203265
## sample_57	9.608977	2.423793	9.129167	8.547195	6.431817	8.500527
## sample_578	8.981085	3.909111	8.887434	8.282602	7.149757	8.613826
## sample_579	9.994014	7.636269	8.056860	8.599418	4.955071	7.872804
## sample_603	10.147930	0.000000	7.518315	8.687575	7.952118	7.746723
## sample_627	10.395266	3.448148	8.417469	8.419042	6.007606	8.284153
## sample_638	10.072628	4.324818	9.240415	7.851893	7.487390	8.657265
## sample_673	10.074556	5.982480	9.609157	8.452262	7.571070	10.189689
## sample_68	10.354547	1.894527	6.830129	8.932492	3.429536	8.097674
## sample_69	9.317367	4.820480	9.860421	8.225800	6.507221	9.799516
## sample_691	9.728941	2.375818	8.129170	8.480431	7.360584	8.529470
## sample_701	9.765543	2.195663	8.190743	9.277871	5.828774	9.011222
## sample_735	9.764908	5.801355	7.724214	8.195013	7.763226	9.778156
## sample_74	10.318012	6.875989	8.636896	8.460784	5.205717	7.681274
## sample_740	9.736642	2.583399	8.169529	8.546088	6.577291	9.196155
## sample_75	9.227840	1.374121	8.356954	8.018078	6.715331	7.587065
## sample_762	9.560480	4.264927	7.936614	8.554527	7.021624	8.953239
## sample_768	10.217376	2.446097	7.700724	8.718327	6.592203	9.292934
## sample_793	10.387274	1.547351	7.092440	9.566246	4.263658	7.768469
## sample_800	10.487448	6.251935	7.655159	8.396070	5.755764	8.810761
## sample_82	10.438386	2.114967	8.462850	8.709573	8.117389	7.308548
## sample_95	10.453044	7.810617	7.255831	8.469398	1.288418	8.133743
##	gene_11558	gene_15364	gene_14520	gene_14352	gene_1342	gene_20294
## sample_119	10.810290	1.6230240	6.933667	10.96543	7.068090	8.879994
## sample_275	10.635990	2.6461857	6.351462	10.50344	8.558340	10.361966

## sample_29	10.708153	3.2963840	9.033874	11.27064	7.557847	9.115655
## sample_295	10.129399	0.5506052	4.828809	11.30366	9.825347	8.223505
## sample_298	10.360233	0.9962441	4.729694	10.84810	7.918321	7.468225
## sample_404	10.068792	2.8469345	8.155177	10.60801	8.140319	9.418245
## sample_464	10.913794	2.3094673	1.705403	10.45014	7.598998	7.647804
## sample_468	9.980382	1.1610492	4.614015	10.56702	9.700806	8.207664
## sample_531	10.960393	0.5970305	3.940496	10.84411	7.731964	9.497401
## sample_57	10.704422	2.3129265	4.671005	10.83345	8.039977	7.885629
## sample_578	10.065739	0.0000000	2.277955	11.34517	9.013660	7.064150
## sample_579	10.248378	3.0882603	5.322808	11.03131	8.174027	9.213092
## sample_603	10.851195	2.2889196	3.133218	12.12043	7.230923	7.756183
## sample_627	10.173515	1.6939887	3.313522	12.59161	8.053280	7.656925
## sample_638	10.767978	1.2032012	3.669322	12.84777	8.170601	7.552900
## sample_673	9.983108	2.1161318	7.741359	11.24989	9.050861	8.095624
## sample_68	10.268788	0.4603756	4.251348	10.93111	8.724425	8.476799
## sample_69	11.853629	2.0581779	4.288012	11.16162	8.730032	7.639977
## sample_691	10.026274	2.5973881	4.406516	11.87336	9.567969	7.592255
## sample_701	10.325778	2.3654686	2.899272	11.77460	8.859457	7.307064
## sample_735	10.278322	2.3256455	6.522931	11.86302	8.856145	7.079058
## sample_74	11.813008	1.8716860	6.680831	10.92429	7.038041	8.710183
## sample_740	10.462277	4.2688386	3.589212	10.60678	7.966765	7.707007
## sample_75	10.955715	0.8827604	4.310827	10.80054	8.418025	7.924296
## sample_762	10.459913	4.6812525	4.870473	11.62482	8.770076	8.612478
## sample_768	10.743345	1.8366917	7.270080	12.20482	8.788950	8.408763
## sample_793	10.634348	2.0967679	3.112283	12.47661	7.812838	8.027989
## sample_800	9.361485	0.0000000	7.411909	10.51586	7.604953	9.320649
## sample_82	10.348938	3.0451459	2.038787	12.10404	9.612048	9.586693
## sample_95	10.245410	2.5080483	8.599838	10.34237	7.666962	10.249682
##	gene_19306	gene_13798	gene_19025	gene_10684	gene_7241	gene_15976
## sample_119	10.194966	0.3754009	10.414728	1.4766927	3.220485	0.3754009
## sample_275	8.661269	7.3425992	9.395975	1.7388975	3.730477	1.1169635
## sample_29	10.076107	6.2300185	9.957656	0.0000000	2.053667	2.0536674
## sample_295	9.077072	0.5506052	9.626786	2.4975868	6.560903	0.0000000
## sample_298	9.555335	7.4271201	10.633903	1.7316612	3.617545	0.9962441
## sample_404	8.371075	4.7605510	9.868248	1.4698339	4.224897	2.9330438
## sample_464	9.019054	2.8516391	9.071535	1.0912598	5.728106	0.6464392
## sample_468	8.503035	0.0000000	9.300693	2.8442049	4.879368	4.0220748
## sample_531	8.779040	1.3436352	10.466403	1.0180644	4.735046	1.0180644
## sample_57	9.200651	3.3590713	9.988287	1.2159892	3.614144	2.3129265
## sample_578	9.224506	4.8293523	10.523689	0.9726927	4.077303	1.5483873
## sample_579	9.165806	4.8463629	9.918832	1.0879380	2.728682	1.5241640
## sample_603	9.557075	0.4365883	10.130751	2.2889196	3.953042	1.6415923
## sample_627	11.917786	1.0824982	10.250997	1.0824982	7.667537	0.4569113
## sample_638	9.253406	0.3339961	9.598443	0.3339961	2.909293	1.0299829
## sample_673	8.310063	4.5183967	10.016669	1.2806000	9.212513	1.7580473
## sample_68	8.503945	0.8089204	9.734926	2.2507797	2.885770	2.6468999
## sample_69	9.212433	5.4124869	10.173315	0.9291850	6.291044	2.2068307
## sample_691	9.569687	0.3569326	9.803334	2.3013828	4.602588	2.1269398

## sample_701	8.960002	2.8071694	9.038063	1.7256539	3.593975	2.6031456
## sample_735	7.775024	2.4485838	8.614850	2.1912468	7.763226	1.4769001
## sample_74	10.090245	5.7949694	10.458253	1.5168724	3.078405	0.8463128
## sample_740	8.757427	1.6178157	9.743044	0.0000000	5.662761	3.7823771
## sample_75	9.444851	0.6439485	9.612863	0.3576083	9.778199	1.5700257
## sample_762	9.213792	1.3453964	9.516638	1.6110307	4.394610	1.8352773
## sample_768	9.595940	1.9610306	10.234027	1.8366917	6.413303	0.7163325
## sample_793	10.845521	0.4480552	9.823872	2.2146533	3.050206	0.4480552
## sample_800	9.190716	3.1951743	9.837306	1.6039289	5.498059	1.2386646
## sample_82	8.658686	0.6023619	9.862805	1.3530409	2.038787	1.8443053
## sample_95	8.576858	4.5459992	9.718985	0.7834986	2.203671	2.9053124
##	gene_12135	gene_13782	gene_10698	gene_11458	gene_7648	gene_1675
## sample_119	9.643515	3.1737351	3.0754299	2.9699517	7.590056	12.63656
## sample_275	10.751519	4.5725077	3.1988556	0.6640277	6.199127	11.69422
## sample_29	8.232258	1.5315686	2.9294691	8.1207044	8.567979	12.60942
## sample_295	10.368277	3.4275391	2.8159033	0.5506052	6.279505	12.12123
## sample_298	9.947134	2.9933845	2.4089020	8.0313952	7.338264	12.67888
## sample_404	10.619009	3.3634515	2.5526968	1.2187192	6.539708	11.90404
## sample_464	10.612620	3.2447957	2.7339833	3.1560891	7.438077	12.11807
## sample_468	10.236397	0.0000000	1.7959331	1.7959331	4.919207	11.01083
## sample_531	10.062464	4.5579859	2.7309442	2.3507801	8.088407	12.52129
## sample_57	10.151562	3.6141444	0.7324863	3.0489859	7.195023	11.49108
## sample_578	10.077991	4.8773860	3.1986356	6.5228216	8.765107	12.58777
## sample_579	11.440864	2.3582837	2.6446177	3.9315791	6.318302	12.11147
## sample_603	10.850921	1.7963901	2.9474417	2.7341568	5.855255	12.21224
## sample_627	10.217073	1.9929859	2.5468327	3.0792245	6.096698	12.18706
## sample_638	10.164894	2.6938772	2.0447438	2.4405536	7.879283	12.44049
## sample_673	10.254710	3.1858824	1.5389851	0.0000000	7.574684	12.25551
## sample_68	10.382040	1.8606053	2.5575816	3.8613401	6.334482	11.86057
## sample_69	10.192391	3.7231326	3.5106835	4.1478684	8.032597	11.89034
## sample_691	10.922332	2.9515691	2.3812557	5.9054255	7.234511	12.81299
## sample_701	11.070778	0.9432837	2.3654686	3.8918080	7.179690	12.11495
## sample_735	11.284199	5.0489075	2.1912468	1.4769001	8.296705	12.01825
## sample_74	9.514357	0.8463128	4.1927623	0.3402774	8.549477	13.16390
## sample_740	10.401413	2.8494789	2.8494789	0.7567671	7.763053	12.70500
## sample_75	9.845912	3.7416719	3.1499422	2.7881010	8.411604	12.05084
## sample_762	11.568350	3.9432556	2.4913657	1.3453964	7.374083	12.64475
## sample_768	10.533437	3.3678323	3.7589640	3.7928241	6.528315	12.64537
## sample_793	11.566554	1.4961044	2.7711884	2.4249221	6.471903	12.90856
## sample_800	11.562829	3.4852465	2.1373396	6.4030751	6.898499	11.68195
## sample_82	9.634504	1.0259501	3.8026275	5.2595738	6.702034	12.57541
## sample_95	11.051943	2.4135399	1.9579889	1.2884176	5.526119	11.67012
##	gene_16116	gene_17528	gene_381	gene_11757	gene_17510	gene_12299
## sample_119	3.173735	9.462719	5.510218	9.765765	9.979597	2.8561292
## sample_275	6.648999	9.755285	8.335542	8.657097	7.611932	6.1759896
## sample_29	6.927256	10.727325	1.176706	10.040290	5.920415	3.7992428
## sample_295	5.306455	11.361368	5.126225	9.041886	10.328159	7.6565464
## sample_298	5.674418	10.018047	5.738541	9.735045	8.303840	1.4103411

## sample_404	5.076533	10.235320	10.460876	8.628227	6.689467	9.6543979
## sample_464	6.558773	9.660648	2.605874	9.264222	7.776117	4.3774847
## sample_468	7.696613	11.655848	5.656751	9.005714	9.124442	8.1914837
## sample_531	6.543375	9.587631	3.280155	9.659807	6.254662	1.0180644
## sample_57	4.943302	9.812654	4.512796	9.287204	6.800486	2.3129265
## sample_578	5.852633	10.654779	5.096135	8.491849	11.318509	1.5483873
## sample_579	6.227302	9.802408	8.440209	8.120487	9.912105	5.6483418
## sample_603	3.445462	11.301273	8.846631	9.506807	7.583557	0.7712095
## sample_627	3.417785	11.287430	5.046216	9.280975	7.965103	2.3500446
## sample_638	2.693877	11.058100	6.762708	9.242898	8.551055	2.7508209
## sample_673	4.932694	10.681221	7.023754	9.401229	9.595893	3.0262866
## sample_68	7.126415	11.093305	4.307210	8.898801	8.356197	8.4843919
## sample_69	7.069992	10.570273	4.033044	8.695761	8.004934	4.3848442
## sample_691	6.059074	10.518004	6.197922	8.409561	10.862909	2.6627980
## sample_701	3.648086	10.844721	5.191128	9.652623	7.619574	3.9788363
## sample_735	2.764792	10.226111	6.312431	7.841036	10.486785	4.3967555
## sample_74	9.480382	10.638029	3.573411	9.200246	6.743044	2.4650080
## sample_740	4.611780	10.139986	6.927185	9.643295	11.455045	0.7567671
## sample_75	4.137200	11.194695	6.508018	9.056269	11.744573	1.4264258
## sample_762	3.034128	10.615859	5.304719	8.849418	9.147388	9.8709888
## sample_768	4.198541	10.512000	4.612877	9.708320	8.666760	4.7570659
## sample_793	1.827494	10.594810	4.964667	9.413607	9.977165	0.4480552
## sample_800	3.798901	10.460548	8.268275	9.135976	9.639977	7.0243973
## sample_82	2.210139	11.045459	6.298748	8.190857	9.153370	1.6195069
## sample_95	5.686260	9.593272	8.844706	8.618496	6.625517	7.3092944
##	gene_1625	gene_16494	gene_8606	gene_10181	gene_5308	gene_17060
## sample_119	9.293354	10.410960	6.835722	2.524766	10.471696	5.959269
## sample_275	8.802045	9.541114	8.433911	7.896562	9.908778	5.505208
## sample_29	9.218454	10.471523	7.235545	2.159242	10.663718	3.150397
## sample_295	5.896199	9.831239	7.875061	4.513661	10.734540	9.189286
## sample_298	9.628475	10.044572	9.532260	5.338407	10.215654	8.299290
## sample_404	8.425929	9.611792	7.758137	6.395505	10.072950	3.233750
## sample_464	10.137619	9.547502	9.868608	4.490641	10.532288	5.439517
## sample_468	4.957980	9.520601	7.169825	5.427321	10.319221	9.872896
## sample_531	10.061763	10.158761	9.381322	1.609046	10.406630	6.918315
## sample_57	9.470559	9.547816	8.042261	3.830833	9.848290	7.509751
## sample_578	9.072899	9.224506	8.112419	6.507642	11.585747	7.989412
## sample_579	8.791248	9.274695	7.379188	7.169564	10.497882	5.006500
## sample_603	10.941319	9.971299	7.916692	8.190644	10.261507	6.198459
## sample_627	8.793190	9.966152	6.480604	4.348664	10.040002	5.670316
## sample_638	9.588715	9.723294	8.137299	6.334736	10.349911	5.145115
## sample_673	9.141262	9.417403	9.492807	5.699960	11.256645	8.490991
## sample_68	8.053448	9.470193	8.668938	5.241920	10.430400	9.610012
## sample_69	9.499597	9.350716	8.781258	3.191784	10.628291	5.028817
## sample_691	10.014188	9.395463	9.450323	6.841319	10.813364	9.296554
## sample_701	9.788309	8.390444	7.348507	7.439532	10.918125	6.907600
## sample_735	10.458724	8.329074	10.474183	6.643648	10.950826	6.836101
## sample_74	10.079285	10.497263	6.642935	1.220206	10.301085	4.528227

## sample_740	10.095199	9.246988	8.218791	6.078044	10.901274	6.592838
## sample_75	9.409484	9.219696	10.037259	1.266936	10.450500	8.077873
## sample_762	8.797068	8.866985	8.351695	7.678987	11.724765	7.391845
## sample_768	9.104413	10.369565	6.951984	5.451488	9.752006	5.798403
## sample_793	10.751134	9.353839	7.569871	3.112283	11.190072	7.196391
## sample_800	8.486360	9.530172	8.812944	6.733164	9.846924	6.705231
## sample_82	11.297318	9.880361	6.963960	7.149148	10.948550	3.215787
## sample_95	8.439735	9.637712	8.122585	7.292800	10.160666	3.767951
##	gene_8955	gene_13875	gene_7254	gene_18996	gene_2665	gene_7196
## sample_119	9.822740	10.229251	3.0754299	10.269805	5.330103	10.428789
## sample_275	10.149010	10.591363	2.6461857	9.352783	7.607619	10.585845
## sample_29	9.906461	9.762269	3.6977738	10.658667	7.814281	9.747872
## sample_295	10.127595	9.498127	3.9013110	9.200725	7.810822	10.080085
## sample_298	9.568625	9.335531	1.2180992	9.176507	6.679016	10.629675
## sample_404	10.221490	10.515660	3.3000651	9.745711	6.841935	10.870550
## sample_464	9.903088	9.532163	0.0000000	9.525538	6.969979	10.082282
## sample_468	10.848138	10.104166	1.7959331	8.759106	7.950818	10.909076
## sample_531	9.776553	9.534921	1.3436352	9.782445	6.770340	9.708353
## sample_57	9.945372	9.772671	0.0000000	9.073571	7.571260	9.762993
## sample_578	9.145797	10.046469	0.0000000	10.478658	8.812203	10.966505
## sample_579	10.541029	10.553226	1.3224474	9.143008	7.897991	11.440475
## sample_603	10.384071	10.511357	0.0000000	9.710982	7.435337	10.739637
## sample_627	10.194769	9.972449	0.4569113	9.581018	7.000552	10.639576
## sample_638	10.360529	9.612699	1.2032012	9.028633	5.186984	10.867086
## sample_673	10.502742	9.536245	0.0000000	10.111057	8.590643	10.025929
## sample_68	10.398209	9.810644	3.6180272	9.410076	7.168492	10.187402
## sample_69	8.973864	10.740725	0.0000000	10.252973	8.083979	10.023449
## sample_691	9.483507	9.844383	0.8812731	9.505401	7.111073	10.530923
## sample_701	9.525601	10.180009	0.0000000	10.036064	6.967342	10.914684
## sample_735	8.393618	10.157612	1.6912661	10.738329	6.960106	9.807168
## sample_74	10.699087	9.514047	2.5976980	10.054971	8.786025	10.141750
## sample_740	9.350453	11.039872	0.7567671	9.692228	8.599653	10.977530
## sample_75	9.768927	10.443938	6.9952810	8.808182	8.600136	10.933890
## sample_762	10.417199	10.332719	5.3047191	9.627598	7.305469	11.369276
## sample_768	9.910283	10.343141	0.7163325	9.361523	6.190103	11.046967
## sample_793	10.589707	10.038700	0.0000000	10.105267	7.832099	11.131793
## sample_800	9.603877	9.809800	2.8317151	9.029257	6.939038	10.983286
## sample_82	10.699815	9.606115	0.0000000	9.445874	7.350373	11.814478
## sample_95	10.715344	10.736275	2.4135399	9.237571	8.232191	11.207014
##	gene_6474	gene_8525	gene_6178	gene_11095	gene_16182	gene_7272
## sample_119	2.0938644	7.512345	5.556503	3.220485	7.321838	9.903745
## sample_275	4.4233025	5.541868	2.774945	4.675082	8.898843	8.987892
## sample_29	3.9539450	6.007801	2.929469	3.150397	5.972031	9.520373
## sample_295	6.2708466	2.717408	4.513661	3.488541	7.080114	8.402586
## sample_298	2.3158850	4.022661	2.578673	3.452569	7.401878	9.478956
## sample_404	5.0381903	5.270925	2.316812	4.154956	7.999250	8.501853
## sample_464	2.1347795	2.851639	2.851639	3.407339	7.537995	8.774688
## sample_468	5.9538915	4.838392	2.103195	5.691677	4.127881	8.694567

## sample_531	2.3507801	4.426594	1.833133	3.119257	6.672241	9.288578
## sample_57	2.7977186	3.614144	3.048986	2.312927	8.690365	8.824819
## sample_578	5.0756952	4.117853	4.532111	2.539109	8.248763	8.040405
## sample_579	4.9375092	3.375860	2.129975	4.600775	8.723709	8.194427
## sample_603	4.4259091	4.645488	1.042714	3.579434	8.652533	8.293338
## sample_627	4.0892354	5.711930	3.811492	3.561302	10.987307	8.875605
## sample_638	3.8881685	5.145115	2.294988	4.078149	9.622482	9.075364
## sample_673	5.1802712	4.457128	1.538985	3.979175	8.603070	9.193856
## sample_68	6.3004807	5.270305	2.250780	4.580887	5.153258	8.403413
## sample_69	3.7231326	3.391452	5.048655	3.620821	8.755836	8.794539
## sample_691	4.4254862	2.725349	2.301383	4.038822	6.985694	9.347167
## sample_701	3.9788363	4.888061	3.354636	3.288048	8.591949	8.847440
## sample_735	5.0095129	3.243623	4.014355	3.602528	9.246498	8.473608
## sample_74	1.7628165	5.017165	6.180757	2.659719	7.082947	10.355395
## sample_740	4.4390513	5.017120	2.704363	1.617816	9.500177	9.147856
## sample_75	3.5814115	4.702392	6.276348	3.238603	6.115916	9.152115
## sample_762	5.0377336	4.991889	2.733441	2.733441	8.281981	8.834885
## sample_768	4.1985413	3.175892	3.537197	3.175892	9.148395	9.134434
## sample_793	0.7893537	3.818646	5.517036	3.855402	8.552143	9.363443
## sample_800	5.4314324	4.274821	2.137340	3.867719	8.186466	9.240214
## sample_82	4.3342965	5.390086	4.721925	4.177878	6.027963	8.824510
## sample_95	4.8098774	4.405761	3.037927	5.032868	7.737579	8.917530
##	gene_770	gene_11872	gene_7247	gene_16786	gene_13331	gene_8921
## sample_119	5.351264	6.140357	1.313768	4.1887411	11.526881	11.611306
## sample_275	4.945191	8.590849	3.371740	3.6655178	10.357475	10.756248
## sample_29	3.249339	9.987520	1.816108	0.0000000	9.355252	10.469560
## sample_295	4.653427	7.468714	4.422805	3.3638439	9.893186	10.526020
## sample_298	8.338242	8.370404	4.022661	2.2164549	9.472236	10.494866
## sample_404	4.916644	8.541797	3.233750	5.2709251	10.619009	11.059412
## sample_464	6.451038	6.451038	5.712637	0.6464392	10.115005	10.603228
## sample_468	5.559256	7.359143	4.022075	3.0733575	10.823089	10.652531
## sample_531	6.855840	5.057247	5.707699	1.0180644	9.308544	10.761402
## sample_57	6.026003	7.630253	5.716876	2.9288251	9.870889	10.878105
## sample_578	6.337791	8.509102	3.703854	2.2779550	9.871899	9.927291
## sample_579	5.414508	8.030187	3.659045	4.2487853	10.196270	10.670293
## sample_603	5.989689	7.507152	2.063572	2.1806571	10.565902	10.743454
## sample_627	5.627467	7.810874	2.240559	2.7199938	10.663309	10.657765
## sample_638	5.716423	8.226581	2.133070	2.5081751	11.115804	11.210263
## sample_673	5.789677	8.650029	6.306368	1.7580473	10.054007	10.875865
## sample_68	5.325448	8.047309	4.438073	3.7448069	10.287562	11.041201
## sample_69	6.282690	8.991910	4.615187	1.4896971	9.396917	11.289512
## sample_691	6.491372	6.314434	5.116011	0.6427478	9.635818	10.449995
## sample_701	6.143263	7.014445	2.708761	0.0000000	10.154109	10.516281
## sample_735	6.352344	7.439257	5.068211	0.0000000	9.385711	10.781679
## sample_74	5.869052	9.125277	1.972876	2.8312692	10.814158	10.730513
## sample_740	5.529081	8.167799	3.745764	4.9702739	10.079658	10.852764
## sample_75	6.543255	9.450663	4.290233	0.0000000	10.281142	10.482758
## sample_762	6.933431	6.396016	3.494864	0.5980793	10.165598	10.353169

## sample_768	6.734019	7.791443	2.954438	3.3678323	10.960002	10.978103
## sample_793	6.803576	6.831510	2.917432	1.0652276	10.343874	11.062107
## sample_800	5.184090	6.705231	4.169741	4.4193501	10.402309	10.801781
## sample_82	5.353987	4.804498	2.363255	2.7441180	11.101989	9.878368
## sample_95	4.454038	8.531179	2.680279	4.5007511	10.577089	10.770449
##	gene_3766	gene_15502	gene_19859	gene_8427	gene_1362	gene_1195
## sample_119	10.253387	11.45326	8.433481	5.806780	6.513203	10.933123
## sample_275	9.892016	11.35048	9.821080	4.739632	6.361751	10.637050
## sample_29	10.740405	11.15417	9.152074	3.511000	6.822692	10.588349
## sample_295	10.698279	11.18944	8.928122	5.610818	6.085813	9.348945
## sample_298	9.873058	10.99760	8.854582	8.648293	5.799937	9.808905
## sample_404	10.527262	11.08614	9.577112	4.154956	6.243722	11.271084
## sample_464	9.902235	10.95915	8.765349	4.842944	5.817585	10.946395
## sample_468	10.741408	10.99285	9.008025	3.985018	4.257855	10.237867
## sample_531	9.938309	10.80082	8.904719	7.486980	5.164239	11.588869
## sample_57	9.695540	10.68369	9.256329	6.867402	5.437461	10.772018
## sample_578	10.292471	11.49389	9.113575	5.286042	5.268139	11.211615
## sample_579	10.913039	11.95282	9.369408	3.782199	5.975302	11.220826
## sample_603	10.362547	10.90030	8.616619	1.271067	4.276095	10.773527
## sample_627	10.913151	11.68637	8.243940	4.057303	6.650742	9.650883
## sample_638	10.167506	11.72302	8.123889	4.798657	7.127705	11.417441
## sample_673	10.923446	11.13904	9.032575	6.470394	6.018979	10.817727
## sample_68	11.643121	11.24515	8.902206	6.132842	5.378564	10.194818
## sample_69	11.703705	11.56313	9.924241	7.127097	6.521491	10.340084
## sample_691	10.296377	11.32213	9.266625	7.392266	5.687038	9.927030
## sample_701	10.188750	11.41157	8.901341	3.750564	5.587056	10.818422
## sample_735	9.427416	11.05337	9.441361	6.287941	7.327553	11.237807
## sample_74	10.399246	10.98726	7.917020	6.829926	8.531962	10.638997
## sample_740	10.131587	13.53481	7.716497	5.889495	6.496942	11.843705
## sample_75	9.159939	11.54947	10.163285	7.207844	5.690219	11.295815
## sample_762	9.783722	11.40360	8.718321	6.378309	6.780034	10.361955
## sample_768	10.750096	11.73342	8.657601	4.223577	7.525521	10.471411
## sample_793	11.105090	12.04012	8.049957	3.336883	5.123807	10.336172
## sample_800	10.755430	10.48676	8.704263	6.676747	6.256818	11.481118
## sample_82	9.910190	13.03874	8.265634	3.368349	4.002414	11.028023
## sample_95	10.780983	11.50592	9.587873	2.413540	5.514782	11.492439
##	gene_7370	gene_3430	gene_14772	gene_18165	gene_17029	gene_8965
## sample_119	3.6878557	8.021485	9.207485	3.548239	2.6665748	7.244782
## sample_275	7.5547807	7.707808	10.956783	8.181426	1.9716999	7.217880
## sample_29	1.1767057	10.982773	6.971406	1.680819	3.9539450	8.303068
## sample_295	0.0000000	5.289410	11.605544	6.480592	4.3258397	7.085063
## sample_298	10.5335439	10.564835	10.572397	4.782796	3.7299769	6.536847
## sample_404	6.7381193	6.092897	9.695783	6.045360	2.8469345	7.787321
## sample_464	1.0912598	12.911540	10.466372	4.072809	2.4652693	7.236512
## sample_468	0.0000000	6.665990	11.245131	10.286281	1.4050666	7.257906
## sample_531	0.0000000	12.202598	9.612410	4.762551	3.0315187	6.797195
## sample_57	0.0000000	11.748218	9.251639	3.762157	2.6534957	7.477985
## sample_578	6.0541430	7.989412	10.426642	3.593151	0.9726927	7.383920

## sample_579	4.5554542	5.104165	10.789420	6.774036	2.8834253	7.401776
## sample_603	0.7712095	4.888592	9.106372	2.389649	0.7712095	7.886849
## sample_627	7.1135442	6.492593	9.244221	6.573833	1.0824982	7.594325
## sample_638	3.3754704	9.070478	10.202063	4.499629	1.3578335	8.155866
## sample_673	3.3963650	4.143769	11.249605	3.979175	3.5800128	7.410646
## sample_68	0.0000000	6.908969	11.698674	8.138666	1.3241193	6.940225
## sample_69	5.1982702	8.908861	10.016544	4.588331	2.3416158	7.089286
## sample_691	8.7315940	7.689614	10.362766	7.179710	1.0858325	6.141684
## sample_701	4.3161602	2.365469	10.635500	6.133812	1.7256539	7.803660
## sample_735	4.6483274	6.391182	10.710187	6.279687	0.9197596	6.896744
## sample_74	1.9728764	9.603864	9.015384	2.719205	5.2160745	9.586999
## sample_740	1.6178157	7.577550	9.789320	3.984489	2.5430009	7.774458
## sample_75	3.1499422	9.060077	9.613900	9.219016	1.5700257	7.280121
## sample_762	5.7796800	5.430335	10.269524	5.819594	2.0293115	8.927594
## sample_768	4.7570659	5.408454	10.733575	5.472533	2.3727850	8.748834
## sample_793	6.2203590	5.658971	10.190010	3.855402	3.0502056	7.402969
## sample_800	4.6320820	8.398975	9.862814	8.258703	1.6039289	6.367876
## sample_82	0.0000000	6.004868	9.616523	1.353041	0.0000000	7.279963
## sample_95	5.1547250	7.035580	9.740432	3.878215	2.6802793	7.407574
##	gene_15494	gene_7778	gene_361	gene_19242	gene_11966	gene_8934
## sample_119	9.961648	1.4766927	6.954557	11.572217	10.026800	9.633834
## sample_275	9.249900	4.2120712	8.638175	10.513283	9.843280	9.184744
## sample_29	10.155603	5.6932249	7.223423	10.505692	11.387753	9.831592
## sample_295	9.235622	2.4975868	8.138026	10.487016	10.899463	9.183531
## sample_298	9.617556	1.4103411	8.393348	10.883728	10.270903	10.310999
## sample_404	9.707283	1.4698339	8.879311	10.459964	9.832289	10.405471
## sample_464	9.406809	0.6464392	6.889729	11.113254	10.306848	9.815292
## sample_468	9.511670	1.6137202	9.941611	10.533320	10.888211	9.613841
## sample_531	9.762998	1.0180644	7.175275	10.892763	9.865654	10.654797
## sample_57	9.344112	1.2159892	8.241445	10.744472	10.720920	10.667936
## sample_578	9.698086	0.0000000	10.126007	10.913862	8.168993	9.280281
## sample_579	9.704588	1.7010611	9.403381	10.442550	10.048201	9.437353
## sample_603	9.537314	2.4837771	8.602261	10.227050	10.099032	9.299961
## sample_627	9.468677	1.3163196	7.627322	11.001274	10.746069	9.812039
## sample_638	9.717503	0.6050202	6.459555	10.971285	10.668965	10.142452
## sample_673	9.079071	0.5621814	7.080722	10.826461	9.765808	9.061166
## sample_68	9.249158	0.4603756	8.916868	10.992294	10.585986	9.308528
## sample_69	8.906145	3.1185923	9.160953	10.958661	10.658461	8.994472
## sample_691	8.935557	1.6983518	8.030419	10.880203	10.328529	9.393056
## sample_701	9.389285	0.0000000	6.514873	10.189318	10.392178	9.135981
## sample_735	8.644087	1.8778620	8.973313	9.824298	8.535590	8.549392
## sample_74	9.637011	2.3188663	7.527563	10.378934	11.452823	9.660433
## sample_740	9.575142	0.0000000	9.472429	10.840172	9.574489	10.426569
## sample_75	10.001282	4.0418738	7.769236	10.492514	9.537869	10.013113
## sample_762	9.325429	1.0196308	7.653447	10.281478	9.191151	9.432266
## sample_768	9.595940	1.8366917	4.296200	10.634348	10.896938	9.493495
## sample_793	9.611276	0.4480552	7.393622	10.378219	9.491581	9.238546
## sample_800	9.453898	1.2386646	9.047909	11.254574	10.125065	9.723569

## sample_82	8.607859	0.0000000	9.438351	10.222626	8.022828	8.786077
## sample_95	9.626565	4.3558127	9.927305	10.793351	9.803511	9.398100
##	gene_9041	gene_3053	gene_10137	gene_2858	gene_10123	gene_17536
## sample_119	7.130097	9.303566	6.590868	3.584421	5.653387	4.9409983
## sample_275	7.998997	9.496287	5.327630	1.971700	5.646569	2.5047729
## sample_29	6.049585	8.921154	5.412815	9.018175	7.122621	6.7357659
## sample_295	9.290187	8.708515	1.921589	5.126225	7.933826	3.4275391
## sample_298	7.384810	9.642310	4.452220	2.800890	7.087017	2.3158850
## sample_404	7.072835	9.583782	6.761844	2.755358	5.870910	1.8698319
## sample_464	10.614701	9.517772	4.120468	3.686063	5.981442	1.0912598
## sample_468	9.013786	9.194099	3.331361	3.142315	7.453740	2.1224408
## sample_531	8.846462	9.144431	4.121339	3.556442	6.375382	0.5970305
## sample_57	10.017115	9.518189	4.911946	2.493237	6.165488	0.0000000
## sample_578	6.746205	9.254370	3.534448	8.207083	5.213052	2.9518116
## sample_579	5.512635	9.667791	5.966673	2.129975	5.669769	0.4596414
## sample_603	8.194131	9.805733	6.029254	5.391953	5.667418	0.4365883
## sample_627	5.947907	9.149910	5.243562	4.348664	7.786734	1.6939887
## sample_638	8.321270	9.483445	5.352501	3.006029	5.989148	1.3578335
## sample_673	11.574575	9.835930	3.636381	3.580013	5.407801	3.7968261
## sample_68	10.287562	9.453271	3.969169	3.573168	7.034249	2.1320832
## sample_69	4.641558	9.748111	4.692896	4.504608	5.715265	1.7050490
## sample_691	5.352409	9.654026	3.855172	4.134147	6.374229	1.0858325
## sample_701	9.135981	9.073097	4.020449	3.537768	5.719304	0.0000000
## sample_735	10.649409	9.773754	2.561864	6.391182	3.602528	1.2250281
## sample_74	1.645102	8.980108	5.205717	4.150178	6.604021	8.1604059
## sample_740	9.379877	8.873355	5.592322	2.543001	5.720514	0.4274989
## sample_75	6.156306	9.654246	4.137200	2.600103	4.428578	0.3576083
## sample_762	7.167940	9.431193	4.284840	4.991889	5.638054	0.5980793
## sample_768	8.726014	9.365043	5.306663	4.066356	6.694267	2.0754984
## sample_793	10.458078	9.387175	3.619307	4.913287	5.769042	0.7893537
## sample_800	4.632082	9.534147	6.503791	2.137340	5.843793	1.6039289
## sample_82	9.910968	8.944683	5.137463	3.905977	5.558170	2.0387873
## sample_95	7.920514	9.457773	7.023632	2.086070	4.714009	1.8365301
##	gene_1456	gene_8623	gene_73	gene_13751	gene_16136	gene_2024
## sample_119	9.602190	6.170432	10.724463	10.441181	5.048546	3.784357
## sample_275	9.656355	5.149048	8.976220	10.270038	2.348006	2.348006
## sample_29	9.596348	2.436215	7.771225	10.216479	3.098251	0.960141
## sample_295	10.576673	6.866908	8.733934	11.324197	2.815903	3.603324
## sample_298	10.726312	6.765110	9.468864	10.788490	2.800890	3.899843
## sample_404	9.543289	5.412541	8.843629	10.447683	3.300065	1.869832
## sample_464	9.723738	3.748075	8.804179	10.719500	1.430767	3.807530
## sample_468	10.058276	6.082275	9.434728	11.324934	1.161049	3.946974
## sample_531	10.018214	3.617968	9.454826	10.850937	2.938116	3.201947
## sample_57	9.813714	4.335076	9.688617	10.385711	1.215989	3.614144
## sample_578	9.476262	4.779669	7.815409	10.643108	2.760178	4.990900
## sample_579	10.068510	3.966624	8.585391	10.069007	2.883425	1.087938
## sample_603	9.364027	4.871282	9.775223	9.448172	1.641592	3.073906
## sample_627	9.554815	5.339611	9.007826	10.284998	3.732823	3.732823

## sample_638	10.031205	7.097843	10.028334	10.522846	3.181977	3.608998
## sample_673	10.255831	5.659733	8.910751	10.587384	3.934913	5.289757
## sample_68	10.234530	4.487268	8.912371	11.078838	2.462393	5.168417
## sample_69	10.260437	7.117737	8.639630	10.968782	3.391452	5.349896
## sample_691	11.039694	4.635540	8.774757	11.730530	3.798102	5.679159
## sample_701	9.908864	5.296523	8.706341	9.843700	1.508885	4.020449
## sample_735	10.495725	6.604521	7.331553	9.942107	5.423336	7.270370
## sample_74	10.387263	7.637625	9.304048	10.083891	2.318866	4.577266
## sample_740	9.651362	6.163815	8.558015	10.396615	2.916916	4.046020
## sample_75	9.894475	2.303956	9.540595	11.486367	3.679683	4.447301
## sample_762	9.882811	6.247867	8.452447	10.833736	5.060117	4.709853
## sample_768	9.632045	7.021458	8.984091	9.993632	3.792824	4.903728
## sample_793	10.078791	4.119979	9.066961	10.388448	2.519592	3.960382
## sample_800	8.948172	3.650581	9.062661	10.310283	2.137340	4.550716
## sample_82	9.806316	5.542215	9.406760	10.266974	2.363255	3.632233
## sample_95	9.675744	4.477580	8.968278	10.383467	1.288418	0.444349
##	gene_4998	gene_12228	gene_4916	gene_15152		
## sample_119	3.2204847	8.896281	9.052204	8.523613		
## sample_275	4.2568672	8.491268	9.476301	10.160855		
## sample_29	8.1672977	8.619754	8.020958	9.039070		
## sample_295	1.9215887	7.598916	8.890471	9.695512		
## sample_298	0.4131608	9.076383	8.474939	10.727444		
## sample_404	6.3255933	8.181798	9.346514	9.513901		
## sample_464	0.0000000	9.576632	9.455247	11.066479		
## sample_468	1.7959331	7.871807	9.553410	8.841766		
## sample_531	2.1979912	9.725053	8.840023	11.805800		
## sample_57	1.2159892	9.447507	9.342641	10.790454		
## sample_578	1.5483873	8.149564	9.323183	9.754635		
## sample_579	4.9895569	7.796935	10.365852	9.718791		
## sample_603	5.8900551	8.028298	8.633406	8.828803		
## sample_627	5.4171369	8.005406	8.979357	9.134691		
## sample_638	5.8064736	8.341105	8.092567	8.626216		
## sample_673	6.3988805	8.790159	10.529499	12.198734		
## sample_68	4.6899700	7.637494	9.313649	9.299087		
## sample_69	8.1054476	8.400021	8.147459	9.471393		
## sample_691	0.6427478	8.721075	9.086189	10.667892		
## sample_701	8.2597009	8.899949	9.958554	10.196147		
## sample_735	3.5485723	8.288502	9.406713	11.153343		
## sample_74	3.8643149	8.830531	8.777308	8.443254		
## sample_740	5.4629423	8.996329	9.953367	10.011395		
## sample_75	0.0000000	8.956873	8.926335	10.315308		
## sample_762	1.6110307	8.997408	10.196246	11.123391		
## sample_768	5.1844430	8.790283	8.253285	8.430436		
## sample_793	2.4249221	8.787919	10.753568	9.732954		
## sample_800	5.1007885	8.874849	9.989792	10.396498		
## sample_82	0.6023619	8.154590	9.368821	9.642008		
## sample_95	7.2003003	8.148487	10.024419	9.595291		

```
print(labels1)
```

```
##          Class
## sample_29   BRCA
## sample_57   COAD
## sample_68   KIRC
## sample_69   BRCA
## sample_74   BRCA
## sample_75   LUAD
## sample_82   BRCA
## sample_95   PRAD
## sample_119  BRCA
## sample_275  PRAD
## sample_295  KIRC
## sample_298  LUAD
## sample_404  PRAD
## sample_464  COAD
## sample_468  KIRC
## sample_531  COAD
## sample_578  BRCA
## sample_579  PRAD
## sample_603  BRCA
## sample_627  BRCA
## sample_638  BRCA
## sample_673  BRCA
## sample_691  LUAD
## sample_701  BRCA
## sample_735  BRCA
## sample_740  BRCA
## sample_762  BRCA
## sample_768  BRCA
## sample_793  BRCA
## sample_800  PRAD
```

```
# scale the gene expressions
stdgexpProj1 = scale(gexpProj1)
```

(Task A2)

(Part 1 of Task A2) Randomly pick 50 genes and their expressions from “stdgexpProj1”, and do the following to these expressions: apply the “gap statistic” to estimate the number of clusters, apply K-means clustering with the estimated number of clusters given by the gap statistic, visualize the classification results using techniques given by “LectureNotes3_notes.pdf.pdf”, and provide a summary on classification errors. You may use the command `table` and “labels1” to obtain classification errors. Note that the cluster numbering given by `kmeans` will usually be coded as follows:

```
#   Class  label
#     PRAD   5
#     LUAD   4
#     BRCA   1
#     KIRC   3
#     COAD   2
```

When you apply `clusGap`, please use arguments `K.max=10`, `B=200`, `iter.max=100`, and when you use `kmeans`, please use arguments `iter.max = 100`, `nstart=25`, `algorithm = c("Hartigan-Wong")`.

```
library(cluster)
library(ggplot2)
set.seed(123)
fiftygenes = sample(colnames(stdgexpProj1, 50))
fiftygeneset = stdgexpProj1[, fiftygenes]

gap = clusGap(fiftygeneset, FUN = kmeans, K.max = 29, B = 200,
              iter.max = 100)
optimal_k = maxSE(gap$Tab[, "gap"], gap$Tab[, "SE.sim"], method = "Tibs2001SEmax")
print(optimal_k)
```

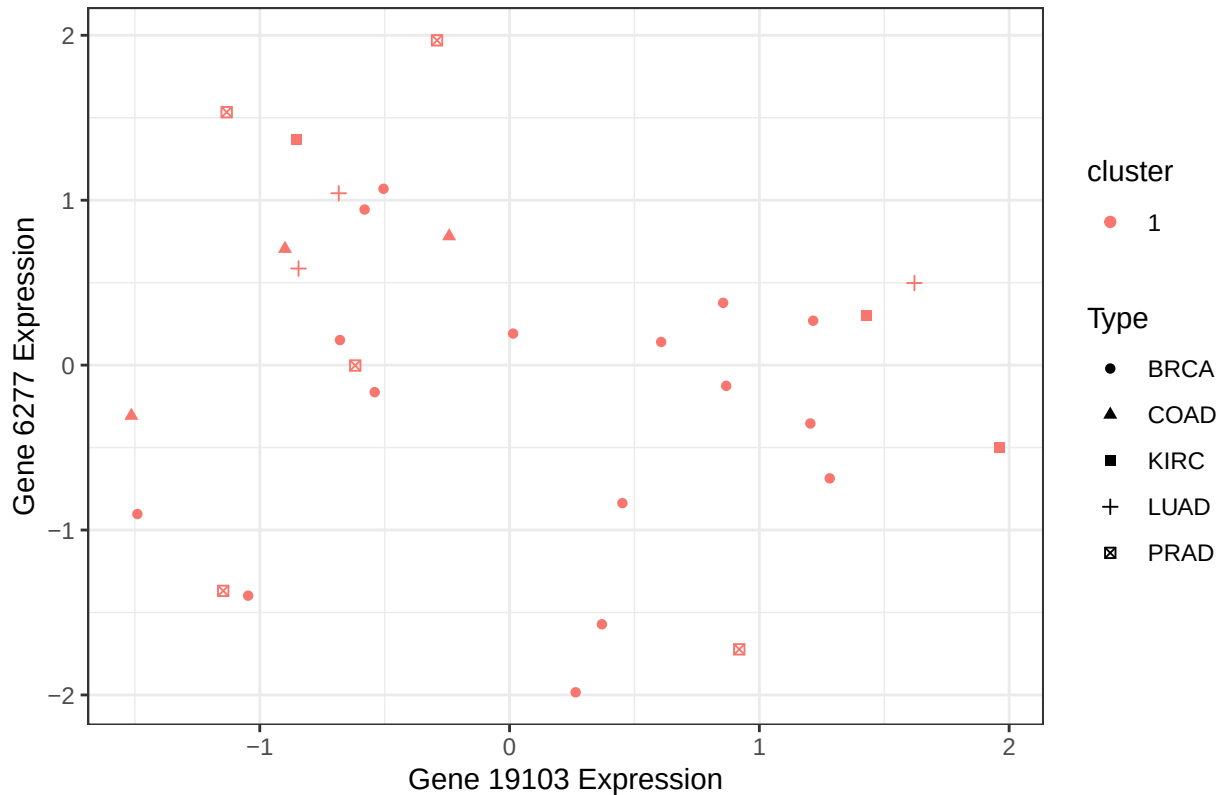
```
## [1] 1
```

```
k_means = kmeans(fiftygeneset, optimal_k, iter.max = 100, nstart = 25,
                  algorithm = c("Hartigan-Wong"))

vis = data.frame(fiftygeneset)
vis$Type = labels1 %>%
  filter(rownames(labels1) %in% rownames(labels1)) %>%
  pull(Class)
vis$cluster = factor(k_means$cluster)

plot = ggplot(vis, aes(x = gene_19103, y = gene_6277)) + xlab("Gene 19103 Expression") +
  ylab("Gene 6277 Expression") + theme_bw() + geom_point(aes(shape = Type,
    color = cluster)) + ggtitle("Clustering results using clusGap to estimate cluster number")
plot
```

Clustering results using clusGap to estimate cluster number



(Part 2 of of Task A2) Upon implementing `kmeans` with k as the number of clusters, we will obtain the “total within-cluster sum of squares” $W(k)$ from the output `tot.withinss` of `kmeans`. If we try a sequence of $k = 1, 2, 3, \dots, 10$, then we get $W(k)$ for each k between 1 and 10. Let us look at the difference $\Delta_k = W(k) - W(k+1)$ for k ranging from 1 to 9. The K^* for which

$$\{\Delta_k : k < K^*\} \gg \{\Delta_k : k \geq K^*\}$$

is an estimate of the true number K of clusters in the data, where $\gg$ means “much larger”. Apply this method to obtain an estimate of K for the data you created in **Part 1**, and provide a plot of $W(k)$ against k for each k between 1 and 10. Compare this estimate with the estimate obtained in **Part 1** given by the gap statistic, comment on the accuracy of the two estimates, and explain why they are different.

```
set.seed(123)
SumSquares <- numeric(10)
km_list <- list()

# run k_means for k values 1 - 10
for (k in 1:10) {
  km = kmeans(fiftygeneset, k, iter.max = 100, nstart = 25,
             algorithm = c("Hartigan-Wong"))
  SumSquares[k] = km$tot.withinss
}
```

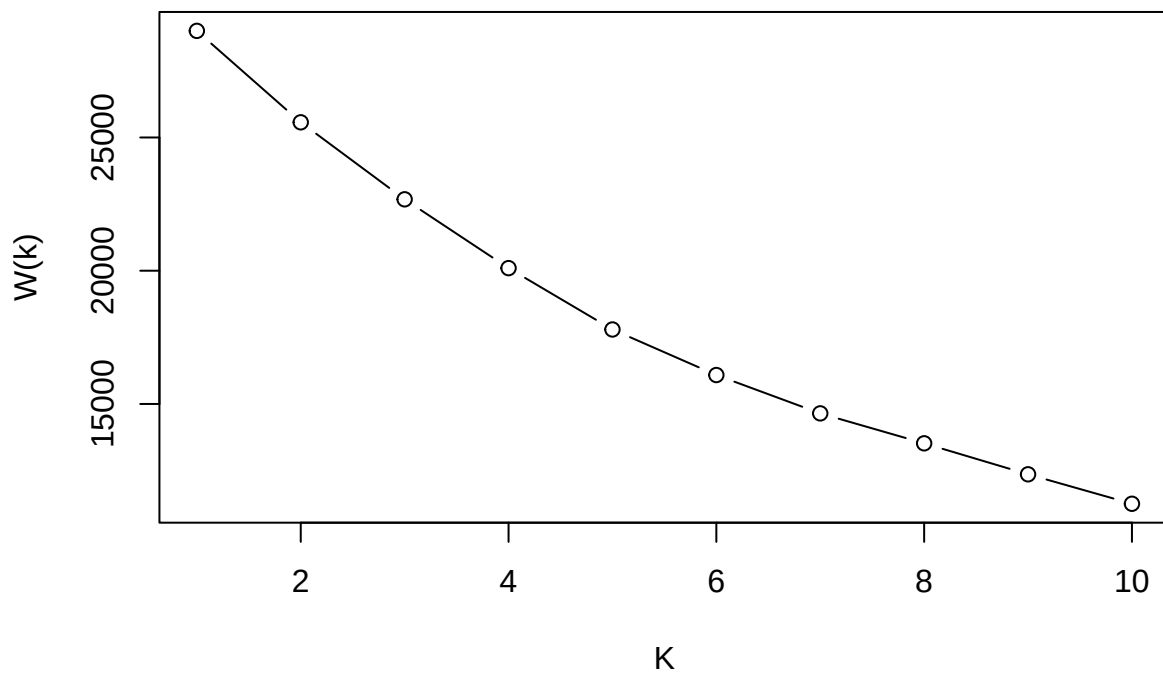
```

# compute the differences Delta k
delta = numeric(9)
for (i in 1:9) delta[i] = SumSquares[i] - SumSquares[i + 1]

K_star = which.max(delta) + 1

plot2 = plot(1:10, SumSquares, type = "b", xlab = "K", ylab = "W(k)")

```



```
plot2
```

```
## NULL
```

```
cat("K* =", K_star)
```

```
## K* = 2
```

```
cat("ClusGap =", optimal_k, "\n")
```

```
## ClusGap = 1
```

```
print(delta)
```

```
## [1] 3430.453 2890.754 2580.995 2303.068 1708.886 1439.001 1124.060 1161.963  
## [9] 1105.759
```

(**Part 3 of Task A2**) Randomly pick 250 genes and their expressions from “stdgexpProj1”, and for these expressions, do the analysis in **Part 1** and **Part 2**. Report your findings, compare your findings with those from **Part 1** and **Part 2**; if there are differences between these findings, explain why. Regard using more genes as using more features, does using more features necessarily give more accurate clustering or classification results?

(**Task A3**) Randomly pick 250 genes and their expressions from “stdgexpProj1”, and for these expressions, do the following: respectively apply hierarchical clustering with average linkage, single linkage, and complete linkage to cluster subjects into groups, and create a dendrogram. For the dendrogram obtained from average linkage, find the height at which cutting the dendrogram gives the same number of groups in “labels1”, and comment on the clustering results obtained at this height by comparing them to the truth contained in “labels1”.

Task B. Classification

For this task, we will use the same data set you would have downloaded. Please use `set.seed(123)` for random sampling via the command `sample` and any other process where artificial randomization is needed.

(**Task B1**) After you obtain “labels.csv” and “data.csv”, do the following:

- Filter out genes (from “data.csv”) whose expressions are zero for at least 300 subjects, and save the filtered data as R object “gexp2”.
- Use the command `sample` to randomly select 1000 genes and their expressions from “gexp2”, and save the resulting data as R object “gexp3”.
- Pick the samples from “labels.csv” that are for cancer type “LUAD” or “BRCA”, and save them as object “labels2”. For these samples, pick the corresponding gene expressions from “gexp3” and save them as object “stdgexp2”

(**Taks B2**) The assumptions of linear or quadratic discriminant analysis requires that each observation follows a Gaussian distribution given the class or group membership of the observation, and that each observation follows a Gaussian mixture model. In our settings here, each observation (as a row) within a group would follow a Gaussian with dimensionality equal to the number of genes (i.e., number of entries of the row). So, the more genes whose expressions we use for classification, the higher the dimension of these Gaussian distributions. Nonetheless, you need to check if the Gaussian mixture assumption is satisfied. Note that we only consider two classes “LUAD” and “BRCA”, for which the corresponding Gaussian mixture has 2 components and hence has 2 bumps when its density is plotted.

Do the following and report your findings on classification:

- Randomly pick 3 genes and their expressions from “stdgexp2”, and save them as object “stdgexp2a”.
- Randomly pick 60% of samples from “stdgexp2a”, use them as the training set, and use the rest as the test set. You can round down the number of samples in the training set by the command `floor` if it is not an integer.

Build a quadratic discriminant analysis model using the training set, and apply the obtained model to the test set to classify each of its observations. You should code “BRCA” as 0 and “LUAD” as 1. If for an observation the posterior probability of being “BRCA” is predicted by the model to be greater than 0.5, the observation is classified as “BRCA”. Report via a 2-by-2 table on the classification errors. Note that the predicted posterior probability given by `qda` is for an observation to belong to class “BRCA”.

Before building a quadratic discriminant analysis model, you need to check for highly correlated gene expressions, i.e., you need to check the sample correlations between each pair of columns of the training set. If there are highly correlated gene expressions, the estimated covariance matrix can be close to being singular, leading to unstable inference. You can remove a column from two columns when their contained expressions have sample correlation greater than 0.9 in absolute value.

(Taks B3) Do the following:

- Randomly pick 100 genes and their expressions from “stdgexp2”, and save them as object “stdgexp2b”.
- Randomly pick 75% of samples from “stdgexp2b”, use them as the training set, and use the rest as the test set. You can round down the number of samples in the training set by the command `floor` if it is not an integer.

Then apply quadratic discriminant analysis by following the requirements given in **Taks B2**. Compare classification results you find here with those found in **Taks B2**, and explain on any difference you find between the classification results.

(Taks B4) Do the following:

- Randomly pick 100 genes and their expressions from “stdgexp2”, and save them as object “stdgexp2b”.
- Randomly pick 75% of samples from “stdgexp2b”, use them as the training set, and use the rest as the test set. You can round down the number of samples in the training set by the command `floor` if it is not an integer.

Then apply k-nearest-neighbor (k-NN) method with neighborhood size $k=3$ to the test data to classify each observation in the test set into one of the cancer types. Here, for an observation, if the average of being cancer type “BRCA” is predicted by k-NN to be greater than 0.5, then the observation is classified as being “BRCA”. Report via a 2-by-2 table on the classification errors. Compare and comment on the classification results obtained here to those obtain in **Taks B3**. If there is any difference between the classification results, explain why.